

Observers Are All You Need

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Abstract

Observer-Patch Holography (OPH) is an observer-consistency framework in which public physics is the stable result of finite observers comparing and repairing the records visible on overlapping screen patches. Time, geometry, matter, and history are treated as readouts of the settled observer-facing structure produced by finite record overlap and repair.

Each observer has access only to local patch data, and overlapping patches must agree on the records both sides can check. Public structure is therefore the quotient normal form of bounded systems with local state, ports or boundaries, readback, records, feedback or repair moves, and public evidence bundles.

The paper gives the program in one pass. Overlap repair produces public records and normal forms. A separate geometric-readout branch then adds modular screen geometry, the three-dimensional spatial chart, Lorentz kinematics, and the Einstein branch with its Newton limit. Receipt-conditional compact sector reconstruction yields the Standard Model charge structure, three colors, and three generations. The same stack carries the record interface, Maxwell and Yang–Mills branches, action-level classical carrier modes with a separate quantum-particle gate, cosmic capacity closure, and the hierarchy/naturality result. The review surface separates theorem claims, branch assumptions, quantitative fixed-point continuations, implementation evidence, and named falsifiers.

The paper’s thesis is that public physical structure is the fixed point of finite observer records, overlap comparison, and repair; spacetime and matter are then recovered as observer-facing readouts of that settled structure.

Keywords: observer patch holography; quantum foundations; emergent spacetime; fixed-point consensus; Standard Model structure; quantum gravity

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What This Paper Contributes

OPH uses familiar mathematics without disguising it. Finite algebras, quantum channels, modular flow, recovery maps, Jacobson’s entropy-to-gravity route, Doplicher–Roberts/Tannaka reconstruction, anomaly algebra, and fixed-point analysis all have lives before this paper. The distinct move is to put them under one operational constraint: finite observers compare only the records visible on overlaps, repair mismatches by declared recovery moves, and accept as public only the normal form that survives those comparisons.

The finite observer-facing patch is the primitive object. The consensus paper proves when patch repair has a unique quotient normal form from a fixed initial quotient state and where global agreement fails. The substrate-neutral analysis in Ref. [1] proves that the stronger comparison of endpoints from different same-boundary sources is equivalent, under observation preservation and completeness, to injectivity of the boundary map on the consistent quotient. It keeps that cross-source criterion separate from same-source confluence, weak normalization, and all-schedule liveness; it does not itself establish injectivity for every physical OPH boundary map. By itself the finite consensus theorem does not prove Lorentzian geometry or Einstein dynamics. The compact SM/GR paper carries the declared geometric-readout branch, controlled BW modular flow, the null-stress bridge, bounded-interval transport, fixed-cap generalized-entropy stationarity, small-ball area variation, remainder control, and timelike coverage into a Jacobson-type Einstein equation. The same compact paper reconstructs compact gauge structure on a cofinal tail carrying its explicit refinement receipt and selects the realized Standard Model quotient on the declared Minimal Admissible Realization (MAR) branch. The particle paper carries the branch into the pixel fixed point, the capacity-electroweak bridge, and the selected-branch hierarchy and Higgs-naturality result. The dark-sector, Yang–Mills, string, and consciousness papers use the same machinery with narrower support boundaries and their own gates.

The technical novelty lies in the coupling. OPH adds a finite observer-facing patch tuple with visible interface algebras and repair instruments; a recovery-derived repair relation with a quotient normal form; a controlled modular-geometric extraction from screen caps; a minimal-admissibility selector that acts after compact gauge reconstruction; and two dimensionless fixed-point closures, $P_\star$ and N_{CRC} , with stated uniqueness tests. Broad slogans about information, entropy, or emergence carry no proof load here. The proof load sits on the maps, branch hypotheses, uniqueness certificates, and falsifiers named in the paper stack. The ontological sentence below the abstract is an interpretive conclusion from that reconstruction chain. It is not an additional axiom used to derive the physics rows.

On that geometric branch, OPH recovers the three-dimensional observer-facing spatial chart, Lorentz kinematics, the Einstein branch, the de Sitter capacity relation for dark energy, the Standard Model quotient, exact charge quantization, three colors, three generations, Maxwell and Yang–Mills equations on their stated branches, and the conditional classical Maxwell, perturbative pure-Yang–Mills, and pure-Einstein carrier modes. Quantum photon/gluon/graviton poles require the separate receipt stated in the compact and shared derivation papers. On the quantitative side it uses two dimensionless fixed points, $P_\star$ and N_{CRC} , plus a selected no- G scale certificate. Those coordinates feed the fine-structure, hierarchy, particle, and cosmology rows.

For gravity, Jacobson supplies the comparison target: Einstein gravity follows from entropy equilibrium or a Clausius relation under local assumptions. OPH uses that target with a different upstream and downstream structure. Upstream, the relevant regions are observer-visible screen caps and collars; their modular flow is extracted from the controlled cap branch. The stationarity statement is fixed-cap generalized entropy stationarity on declared MaxEnt variations, with the collar recovery remainder, bounded-interval kernel, fixed-volume small-ball area identity, and null-

stress bridge supplied on the recovered-core geometric branch. Downstream, timelike-direction completeness upgrades the rest-frame scalar relation to a tensor equation, while local null data leave the metric term open. The cosmic record-capacity fixed point closes that term. A counterexample to any of these branch steps defeats the OPH gravity claim while leaving Jacobson’s original comparison result untouched.

The boundaries matter as much as the outputs. A theorem, a scaling branch, a fixed-point certificate, a phenomenological continuation, and an ontology reading are different kinds of claim. OPH fails as stated if a load-bearing theorem has a counterexample under its assumptions, if the controlled modular branch cannot be physically instantiated, if MAR admits a smaller realized package, if a quantitative closure is nonunique or target-leaking, or if a carrier implementation changes the observable quotient after visible records and repair maps are held fixed.

Canonical Observer Language

An abstract observer patch is the algebraic object

$$\mathcal{O}_i = (\mathcal{A}_i, \rho_i, \mathcal{R}_i, \{(\mathcal{I}_e, \pi_{i,e})\}_{e \ni i}, \mathcal{U}_i, \text{Chk}_i).$$

Here $\mathcal{A}_i$ is the finite or regulated accessible algebra, ρ_i the accessible state, $\mathcal{R}_i$ the exact or approximate record algebra, $\mathcal{I}_e$ the overlap-visible interface algebra, $\pi_{i,e}$ the visible restriction, $\mathcal{U}_i$ the allowed update and repair instruments, and Chk_i the checkpoint data used for continuation.

A support patch, such as a cap $P_i \subset S^2$ or a causal diamond, is a geometric chart of the abstract patch on a geometric branch. A carrier patch is a physical or digital implementation that realizes the same visible interface and record statistics within a declared error. Echoshedral hardware is therefore a reference carrier architecture for a fixed-cutoff implementation surface. The physical claim lives at the quotient of visible records, interface statistics, repair maps, and checkpoint continuation.

The observer patch is therefore the formal access structure used by the proof. A human observer, a horizon cap, and a hardware module can realize or chart it, but none of those examples is the definition. Physical predictions attach to quotient data: visible interface statistics, record readout, repair maps, and checkpoint continuation. A carrier implementation is admissible only when those data remain invariant within its declared error model.

Definition 0.1 (Semantic observer history). *For an observer token O , the observer-readable history is a labeled causal poset*

$$\mathbf{H}_O = (E_O, \preceq_O, \ell_O).$$

The elements of E_O are semantic record events, $\preceq_O$ is physical or informational dependence, and ℓ_O is the observer-visible event label. Event identity is derived from the canonical semantic payload, observer token, visible footprint, and semantic causal parents. It is not derived from worker ID, repair iteration, queue position, retry counter, timestamp, or packet latency metadata. Latency is execution gauge only when it does not alter semantic causality, delivered data, timeout-sensitive operations, or operational clock records.

Definition 0.2 (Observer registry groupoid and namespaces). *Observer identities live in a registry groupoid, not in a flat UUID table. Objects are observer tokens*

$$\text{RID}(O) = (\text{kind}(O), [\text{birth}(O)]_{\mathbf{H}}, \text{lineage}(O)),$$

where $\text{kind}(O)$ lies in the disjoint namespace

$$\{\text{patch observer, cap observer, future observer}\}.$$

The birth event is a semantic event class in H , and lineage is carried by explicit continuation, split, and merge arrows. Local registries on overlapping presentations must preserve observer kind, birth event, lineage arrows, visible observer structure, and checkpoint cuts. The global registry is the descent groupoid or colimit of those local registries when the cocycle and trivial-monodromy checks pass. A split creates child tokens with parent links; it does not silently assign one identity to multiple descendants.

Definition 0.3 (Operational observer clock). *The modular parameter of an observer-facing cap pair is not, by itself, an arbitrary operational clock reading. A clock claim supplies an observer-accessible instrument*

$$\Theta_O : \text{Chk}_O(D) \rightarrow \text{Prob}(\mathbb{R})$$

and a declared affine calibration law. For a corresponding observer O' , the required finite or exact certificate has the form

$$\Theta_{O'}(\Phi_{O*}\rho) \approx (A_{a,b})_*\Theta_O(\rho), \quad A_{a,b}(\tau) = a\tau + b,$$

with an explicit residual bound from state mismatch, calibration error, and record-readout error. Execution counters are provenance; observer record order, modular parameter, and clock uncertainty are separate semantic or operational fields.

Remark 0.4 (Observer-clock naturality gate). *The present paper proves observer-first quotient and readback claims only where the cited theorem surface supplies the required data. Scheduler-independent observer history and clock naturality require a history-augmented quotient relation with history-coherent diamonds, registry descent, state-preserving observer-algebra extraction, support-cap chart naturality, and the operational clock instrument above. A public evidence bundle can falsify the claim by showing counter-dependent semantic event keys, duplicate or namespace-colliding registry identities, broken registry cocycles, non-isomorphic semantic history DAGs across executors, or clock residuals exceeding the declared bound. Implementation field names alone are not the proof.*

This definition leaves the primitive observer size open. The axioms require finite or regulated access at fixed cutoff and permit connected support regions with different finite or regulated algebras. A primitive observer may therefore be a variable finite federation of screen cells. The stronger claim that elementary carriers are isomorphic fixed-capacity cells would require a separate branch: homogeneous UV cellulation, isomorphic local cell algebras, equivariant overlap maps, refinement preservation of the cell type, and a proof identifying one cell or a fixed block of cells with the primitive carrier.

Failure Tests

OPH is wrong as stated if a load-bearing theorem admits a counterexample under its declared assumptions, if the controlled modular/Einstein branch cannot be instantiated under physically realizable conditions, if MAR fails to select the stated realized package on the declared admissible class, if a quantitative closure is nonunique or target-leaking, or if a physical implementation shows branch-essential dependence on hidden carrier presentation after visible quotient data are held

fixed. Concrete falsifiers include failure of a displayed Maxwell, pure-Yang–Mills, or pure-Einstein quadratic kernel to have its stated transverse or TT classical mode under all of that theorem’s action/background/phase hypotheses; failure of a claimed quantum pole after its full quantization and spectral receipt is supplied; gauge-mediated proton decay on the realized Standard Model branch; a fourth light matter generation in the realized low-energy branch; a smaller admissible one-Higgs Standard Model package; nonunique $P_\star$ or N_{CRC} closure; a failed fixed-cap entropy-to-Einstein tensor upgrade; or carrier-dependent observables after the visible interface is held fixed. A failed continuation does not erase the recovered core unless that continuation is one of the stated parents of the core claim.

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1 Introduction

Observer-Patch Holography asks how much of physics is forced once the world is described from the inside. No observer has access to a view from nowhere. In the idealized description there is a shared screen net, often charted by S^2 , but each finite observer has access only to a local patch P_O of that net. The observer carries records inside that patch and can compare only the observables shared with neighboring patches. OPH takes that comparison rule literally: where descriptions overlap, they must agree. Thus an "observer screen" is an operational access cut, not a private physical sphere assigned to the observer.

The formalism used here is deliberately quantum-algebraic. Patches carry algebras and states, record events use the trace/Born rule on their declared operator surface, and the gravity branch uses generalized entropy. OPH's theory-of-everything claim is that the observed effective universe can be recovered from this axiom set and its stated branch conditions if the framework is correct. The reconstruction may use declared standard mathematics and branch hypotheses without treating those ingredients as hidden empirical constants.

The computational interpretation has a sharp boundary. A conventional simulation evolves surrogate universe states,

$$U_t \longrightarrow U_{t+1} \longrightarrow U_{t+2} \longrightarrow \cdots .$$

OPH uses a fixed-point computation, but the equation alone is not the definition of simulation. In the fundamental description there is no global timeline on which the contents of spacetime are updated. The readback-and-repair operator acts on observer-readable world candidates and selects a stable solution

$$\mathcal{T}(W) = W.$$

The companion consensus paper [3], Definition **def:oph-simulation**, requires five independent clauses: recovery-derived endogenous update, nontrivial quotient-readable records, strict overlap-repair descent with a schedule-independent normal form, elimination of a proper candidate basin in the selected boundary/sector fiber, and implementation/clock closure. Its Theorem **thm:oph-simulation** proves that the selected finite OPH packet meets those clauses under the stated boundary and record hypotheses and then derives the fixed-point equation. The same theorem gives an identity-map/constant-functional counterexample, so an ordinary equilibrium is insufficient as a simulation.

The local readout of the selected normal form supplies history for observers who subjectively experience time. Time is therefore internal record order instead of an external update parameter. In OPH, "simulation" means this certified observer-facing self-consistency structure, not a bare fixed point and not a frame renderer. The executable OPH-FPE simulator and its finite-run receipt bundles are a separate public evidence surface for finite OPH experiments [10]; they do not replace the theorem-level definition above.

That certified consistency demand is strong enough to organize the familiar world. The effective bulk is reconstructed from compatible screen data. Locality comes from the way collars control recovery across patch boundaries. Gauge freedom is the freedom to change the local description without changing any shared observable. Records are the stable parts of a patch that can be checked by other patches. Matter is the stable charge and excitation content that survives transport, fusion, and minimal admissible realization.

The first major result is spacetime kinematics. Modular flow on screen caps becomes geometric, and the compact cap-normal theorem identifies a sky direction with $q(\Omega) = (1, \Omega)$, represents an oriented round cap by $n_C = (\cot \alpha, \csc \alpha \mathbf{c})$, proves $\eta(n_C, q(\Omega)) = 0$ exactly on the cap boundary with the interior/exterior sign rule, and gives $n_{gC} = \Lambda_g n_C$. Future unit observer frames then

form $H^3 \simeq \text{SO}^+(3, 1)/\text{SO}(3)$, exactly three-dimensional. A cap determines an H^3 plane/half-space, not a preferred observer point or populated bulk. On the conditional record-population branch, quotient-visible record tokens whose calibrated modular cap responses factor through point-local H^3 sources are uniquely identifiable in exact data and stably localized in finite noisy data. The finite localization radius is $R_H[(L/\alpha)\varepsilon + (2/\alpha)\sigma]$ for exact net minimization, with a positive residual gap required before a unique finite point is reported. Generalized entropy at fixed cap then gives the Einstein relation in the large-scale regime. The cosmological constant is not treated as a vacuum-energy guess. Its dimensionless capacity relation is read from the cosmic record-capacity fixed point N_{CRC} ; with the selected scale certificate this is displayed as

$$\Lambda_{\text{CRC}} = \frac{3\pi}{GN_{\text{CRC}}}.$$

The Gibbons–Hawking entropy and the Planck-2018 late-time de Sitter benchmark give a concrete normalization [24, 25]. With $R_{\text{dS}} \simeq 1.66 \times 10^{26}$ m and $\ell_P \simeq 1.616 \times 10^{-35}$ m, the bare horizon ratio is

$$N_{\text{patch}} = \left(\frac{R_{\text{dS}}}{\ell_P}\right)^2 \simeq 1.05 \times 10^{122}.$$

The entropy capacity is

$$N_{\text{scr}} = S_{\text{dS}} = \frac{A_{\text{dS}}}{4\ell_P^2} = \pi N_{\text{patch}} \simeq 3.31 \times 10^{122},$$

so $\Lambda\ell_P^2 \simeq 2.85 \times 10^{-122}$. OPH uses N_{scr} for the Gibbons–Hawking entropy capacity and N_{patch} for the bare horizon area ratio. The input-free closure is sharper: define the OPH readback map $F(N)$ to be the active cosmic record capacity reconstructed by stable observers inside the universe supplied with capacity N . The cosmic record-closure capacity satisfies

$$N_{\text{CRC}} = F(N_{\text{CRC}}).$$

The fixed-point proof gives uniqueness and stability. The observed branch read-off is $N_{\text{obs}} \simeq 3.31 \times 10^{122}$. With the selected scale certificate, the downstream cosmological display is $\Lambda_{\text{CRC}} = 3\pi/(GN_{\text{CRC}})$.

The second result is the Standard Model structure. On a cofinal tail carrying the explicit compact-gauge refinement receipt—including coherent surjective charge pullbacks, finite block embeddings, tensor realizations, $3 + 1\text{D}$ symmetry, and compatible forgetful fibers—edge sectors reconstruct the bosonic compact-gauge branch. On the realized low-energy branch with the explicit one-generation/one-Higgs matter package, minimal admissible realization then selects the compact gauge quotient

$$\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)/\mathbb{Z}_6.$$

The realized low-energy branch then fixes exact hypercharge, three generations, three colors, integer charge for color singlets, and the absence of gauge-mediated proton decay. On separately stated Maxwell, perturbative/deconfined Yang–Mills, and pure-Einstein action/phase branches, the quadratic reductions have the corresponding classical null carrier modes. Photon, gluon, or graviton particle claims require the additional physical-Hilbert-space and spectral-pole receipts.

The third result is quantitative and sits outside the recovered core. OPH’s constants program has two dimensionless closure coordinates, the local pixel fixed point and the global cosmic record-closure fixed point, plus a selected no- G scale proof record. Its rule is reverse engineering followed by fixed-point closure: observations are allowed to locate the relevant basin, because the OPH

universe is treated as a closed mathematical structure whose branch coordinates are displayed to its internal observers; the published OPH value is then the fixed point picked out by the declared map and uniqueness proof. The exact values $P_\star$ and N_{CRC} must be computed as closure endpoints. A nearby numerical value may mimic one measurement, but a shifted endpoint fails the self-consistency condition for the same simulated world. On the observed branch, the global capacity scale is $N_{\text{CRC}} \simeq N_{\text{scr}} \simeq 3.31 \times 10^{122}$. The local pixel ratio is

$$P := \frac{a_{\text{cell}}}{\ell_\star^2}, \quad \gamma_\star = \frac{\ell_\star \nu_{\text{Cs}}}{c}, \quad B_\star = \frac{3\pi}{\ell_\star^2}.$$

Together, $P_\star$ and N_{CRC} fix $\Lambda_\star \ell_\star^2 = 3\pi/N_{\text{CRC}}$ and $\Lambda_\star a_{\text{cell}} = 3\pi P_\star/N_{\text{CRC}}$. They fix the dimensionless geometry. The selected scale certificate supplies the SI scale product $B_\star = \Lambda_\star N_{\text{CRC}} = 3\pi/\ell_\star^2$. The display $N_{\text{CRC}} \simeq 3.31 \times 10^{122}$ is only a branch-scale approximation; the exact gravity row uses the fixed-point capacity together with the independently supplied scale certificate. The $P_\star, N_{\text{CRC}}$ pair fixes a canonical geometric pixel area and a total geometric cell count on an equal-area screen chart,

$$K_{\text{cell}} = \frac{A_{\text{screen}}}{a_{\text{cell}}} = \frac{4N_{\text{CRC}}}{P_\star} \simeq 8.12 \times 10^{122}.$$

The result supplies a geometric chart count. It does not choose a primitive observer capacity. The quantity $P_\star/4 \simeq 0.408$ nats, or about 0.588 bits, should not be read as $\log \dim \mathcal{H}_{\text{cell}}$ for an autonomous tensor factor. OPH capacity lives on shared cuts, edge centers, and constrained overlap data; adjacent geometric pixels are not assumed to factor into independent Hilbert components. No fixed primitive observer algebra follows from $P_\star$ and N_{CRC} alone. The no- G burden of the scale proof record is the clock hierarchy. The first load-bearing object is the source proof record for $\alpha_U(P_\star)$; after that, the cesium gap $\varepsilon_{\text{Cs}} \sim 10^{-33}$ must be emitted by source-side electroweak transmutation, QCD/hadronic data, flavor and nuclear data, and a cesium hyperfine spectral readout, with no dependency path from measured gravity or any calibrated electroweak or gravity scale. The α_U record is a one-dimensional fixed-point certificate: on the public endpoint branch

$$\alpha_U(P_\star) = 0.041124336195630495, \\ \frac{v}{E_\star} = 2.0199803239725553 \times 10^{-17},$$

and the Krawczyk image of $[0.041123336195630494, 0.041125336195630496]$ lies strictly inside that interval. A source-side diagnostic branch records the nearby value $v(P_{\text{cand}})/E_\star = 2.0198114150099223 \times 10^{-17}$ with the public Thomson endpoint excluded upstream. This solves the local electroweak hierarchy/naturality problem on the selected exact branch and supplies one component of the clock hierarchy; the full SI gravity row requires the full set of cesium-clock source records. The QCD and hadronic weakpoint is narrow. The low-energy Thomson endpoint and the cesium-clock display require confined-quark and nuclear source payloads that are not emitted by the first-principles trunk in this paper. For the Thomson endpoint the missing object is the OPH-QCD hadronic backend: source QCD parameters, a quotient ensemble, finite Euclidean vacuum transfer, Ward-normalized electromagnetic currents, the two-current spectral marginal, same-scheme remainder, systematics, and no-target-leak receipts. The source/root fine-structure audit emits the undressed source/root inverse coupling $\alpha_{\text{root}}^{-1} = 136.9948351646\dots$. Adding the emitted finite-screen unified gauge-width contribution evaluated at the CODATA-derived comparison pixel $P_C \simeq 1.6309682094$, $\alpha_U(P_C) = 0.041124336195630495$, gives the mixed-provenance no-hadron diagnostic

$$A_{\alpha_U}^{\text{fp}} = 137.0359595008\dots$$

The public Thomson endpoint $\alpha^{-1}(0) = 137.035999177(21)$ differs from that mixed diagnostic by the CODATA-minus-diagnostic bookkeeping gap $0.00003967618272\dots$. The full root-to-CODATA bookkeeping difference is $\Delta_{\text{H,cal}}^{\text{fp}} = 0.04116401237835\dots = \alpha_U(P_C)C_{24,Q}$, with $C_{24,Q} = 1.00096478597323262538\dots$; it includes $\alpha_U(P_C)$ and is not the remaining hadronic correction. The declared empirical route would use separate $e^+e^- \rightarrow \text{hadrons}$ spectral input; no same-scheme integrated endpoint payload is emitted here. A source-only endpoint requires the OPH-QCD backend above without Thomson-target leakage. Its two-current spectral measure is only the running- α/HVP marginal; HLbL and rare-decay rows require higher-point and transition spectral sectors from the same source law. For that reason the compact quantitative stress test is the hierarchy row above: it avoids the public Thomson endpoint, G , Λ , W/Z , Higgs mass, hadronic payloads, and the cesium clock.

The same hierarchy exponent has a local/global capacity form:

$$\frac{v}{E_{\text{cell}}} = \left(\frac{N_{\text{CRC}}^{\text{EW}}}{\pi} \right)^{-P_*/12}, \quad N_{\text{CRC}}^{\text{EW}} = \pi \exp \left[\frac{6\pi}{P_* \alpha_U(P_*)} \right].$$

This is the electroweak bridge capacity, about 3.53235×10^{122} on the public endpoint branch. The de Sitter capacity display 3.31×10^{122} remains the cosmological capacity scale. The screen-sieve theorem gives the geometric reading of the factor 12: on the triangulated S^2 screen, $\sum_v (6 - \deg v) = 12$; MaxEnt/refinement stability spreads that curvature charge into twelve unit fivefold defects; edge-center collars make them central ports; A_5 symmetry places them on the icosahedral vertex orbit. Including reversible write/check orientation gives a 24-slot oriented repair register. The hierarchy exponent uses this 24-fold repair normalization. The twelve-port result selects an exposed interface architecture. The internal dimension of a primitive observer algebra belongs to a separate carrier theorem, possibly with MAR acting on the port, record, and repair structure.

The compact proof packages this quantitative surface as a source-certified compression test. A row counts in the accidental-hit estimate only when its declared source map has no dependency path from the measured target or a calibrated proxy. If p_i bounds the conditional accidental-hit probability for row i after earlier accepted rows, then

$$P_{\text{acc}} \leq \prod_i p_i.$$

Twelve one-percent source-certified rows give $P_{\text{acc}} \leq 10^{-24}$; twenty such rows give 10^{-40} . The estimate is an evidential shorthand. The proof load remains the fixed-point maps, uniqueness certificates, source records, and falsifiers.

The value of P comes from giving the same screen cell two readings, with no separate measured constant inserted. From outside, the cell sits a little above the exact self-similar entropy balance point $\varphi = (1+\sqrt{5})/2$. From inside, the same cell appears as the smallest electromagnetic observation step available in the encoded world. The local scale is the value of P where those two readings coincide.

With the P fixed point in place, the same scale organizes the weak-boson benchmark pair, the low-energy electromagnetic endpoint, the Higgs/top quantitative relation, the running quark package, a target-informed weighted-cycle neutrino comparison candidate, and the local cell/edge consistency used by the gravity readout. The gravity normalization itself comes from $\ell_\star^2$, and P cancels in the Newton area-law readout. The support table records the scope of the weak benchmark pair, charged-lepton empirical anchors, the NuFIT 6.1 rejection of the neutrino candidate, its tautological historical basis transport and open charged-basis gate, and the open hadron construction [11].

The fourth result is implementation. The screen-microphysics construction makes patches, overlaps, edge observables, records, repair maps, and observer checkpoints explicit at finite cutoff through a federated patch-carrier architecture. OPH has a regulated observer-and-record surface on which the basic consistency machinery can be written directly.

Axioms in words. The theory uses five axioms. A screen net assigns algebras to connected screen patches. Overlap consistency requires agreement on shared observables. Local MaxEnt selects the least biased local state compatible with the finite constraint data and keeps the realized construction stable under refinement. Recoverable generalized entropy supplies the collar-recovery and focusing structure used by the gravity lane. Minimal admissible realization selects the simplest low-energy sector compatible with the consistency constraints. These axioms define a quantum-algebraic starting point. Quantum mechanics and quantum field theory are treated as effective descriptions carried by the deeper observer-patch architecture, not as structures whose use here depends on first deriving every mathematical ingredient from operational records alone.

The synthesis surface gives the operational picture first, then places the technical proof machinery after the reader has the construction in hand.

2 Main Results

The OPH papers support the following unified picture.

1. **Observers and records become physical objects.** Finite patches, shared overlaps, record algebras, repair maps, and observer checkpoints are represented on the screen as part of the construction.
2. **Spacetime is reconstructed.** On the controlled Einstein-branch geometry readout, the finite cap-normal BW certificate-support-order faithfulness, regularized mixed-GNS modular transport, BW framing, held-out oriented cross-ratio rigidity, and independently normalized geometric 2π -KMS convergence-makes the support-visible cap modular automorphism geometric and supplies Lorentz kinematics. Recoverable generalized entropy together with the null modular bridge, bounded-interval transport, fixed-volume small-ball area variation, admissible fixed-cap stationarity, and timelike tensor upgrade then supplies the Einstein relation. Bare finite consensus supplies only the upstream quotient-normal-form input. The screen-capacity branch ties the dimensionless cosmological-constant relation to the zero-input cosmic record-closure fixed point:

$$N_{\text{CRC}} = F(N_{\text{CRC}}), \quad \Lambda_{\text{CRC}} \ell_{\star}^2 = \frac{3\pi}{N_{\text{CRC}}}, \quad \Lambda_{\text{CRC}} = \frac{3\pi}{GN_{\text{CRC}}}.$$

The last display uses the selected scale certificate $G_{\text{geom}} = \ell_{\star}^2$. Here $F(N)$ is the active cosmic record capacity read back by observers inside the universe supplied with capacity N ; uniqueness and stability follow from the OPH readback fixed-point certificate.

3. **The Standard Model quotient is selected by MAR.** Under the stated coherent transportable-sector refinement conditions, edge sectors reconstruct a bosonic compact-gauge branch. The overlap/holonomy calculus classifies transportable sectors; it does not by itself choose the Standard Model. The realized low-energy branch with the explicit one-generation/one-Higgs matter package and MAR then selects the compact gauge quotient,

exact hypercharge, three generations, three colors, and the structural absence of gauge-mediated proton decay. On the ordinary electromagnetic branch, the unbroken $U(1)_Q$ factor reduces the compact-gauge curvature to $F_Q = dA_Q$; after separately assuming the low-energy Maxwell action and its nonzero kinetic term, variation gives $dF_Q = 0$ and $d*F_Q = g_Q^2 *J_Q$, i.e. Maxwell's equations after canonical electromagnetic normalization.

4. **The local particle scale is computed by fixed-point closure.** The pixel ratio P is fixed by matching a screen cell's outer entropy detuning to its inner electromagnetic observation scale. The public readout is $P \simeq 1.6309682094$, $\alpha^{-1}(0) = 137.035999177(21)$, and $\alpha(0) \simeq 0.00729735256433$. The calculation uses the source anchor and Ward-projected electromagnetic transport to enforce the outer/inner pixel fixed point. A different fine-structure value would move the pixel away from that consistency point. The claim table records the source-side diagnostic trunk and endpoint residual.
5. **The framework has a regulated implementation surface.** The microphysics paper turns the observer language into explicit finite patch carriers, edge observables, records, and repair cycles.
6. **Edge sectors also admit two-dimensional Yang–Mills and worldsheet-style effective descriptions.** The heat-kernel edge partition function reorganizes the same boundary data into the two-dimensional Yang–Mills form, and on the declared large-edge branch this yields a controlled worldsheet effective description. This 2D bridge sits beside the compact paper's controlled compact-gauge repair-gap theorem. The string-vacuum selector uses this bridge only as the effective edge-language input; critical worldsheet closure, Bouchard-Donagi cohomology, operator-safety realization, threshold matching, and moduli locking remain separate gates.
7. **The compact-gauge branch carries the 4D Euclidean Yang–Mills form and isolates the gap theorem.** On the declared controlled compact-gauge branch, with the four-dimensional scaling chart, reflection-positive ordinary vacuum, no additional gauge-invariant relevant dimension-four pure-gauge operator on that branch besides curvature squared, repair completeness, and the compact paper's explicit finite cylinder system in force, OPH proves projective weak-* / GNS extraction of a support-visible cylinder cluster family. Identification of that family with the Euclidean Yang–Mills action additionally uses the renormalized four-dimensional regularity/universality receipt. Weighted observation-fiber resampling is the finite conditional-expectation projector, with the noncircular matrix-recognition criterion of Ref. [1]. A concrete active repair law is identified with that projector only after its independently extracted transition matrix passes the support, equal-fiber-row, and weighted detailed-balance receipt. Identification of Euclidean transfer with the repair generator, vacuum persistence, and passage of a uniform finite-stage gap use the separate finite ground-state-transform, transfer/vacuum, OS-regularity/noncollapse, and uniform-gap receipts. Only on that certified branch is $\Delta_{YM} = \Delta_{rep}$. Weak-* extraction alone is not the Clay-facing axiomatic construction.

3 Physical Implementation and A_5/E_8 Significance

The implementation question is what must exist below the observer-level axioms so they have a physical carrier. OPH's answer is a fixed-cutoff federation of finite patch carriers. Each carrier exposes overlap ports, record registers, repair maps, and checkpoint interfaces. The screen is the observer-facing geometry chart. It is not the carrier itself in the sense of one literal smooth ball.

The implementation is the finite algebraic machinery that makes compare, write, repair, and re-read operations available.

The term “sphere folding” has a restricted technical meaning in this stack. It is the observer-facing spherical presentation of the repaired quotient normal form:

$$\text{Fold}_{S,r}(s) = \chi_{S,r}(n_r(\pi_r(s))).$$

The quotient map removes hidden carrier presentation, the normal-form map performs accepted repair, and the spherical chart displays caps, collars, cuts, edge sectors, and boundary records. Folding is therefore the geometric face of overlap repair on the screen chart. On the consensus surface this repair is defined first as the quotient operator $\text{Rep}_\lambda : Q \rightarrow Q$, with representative-level maps only serving as lifts, so hidden carrier labels do not become additional physical data.

Spherical geometry carries several pieces of the construction at once. An observer-accessible cut has a closed two-dimensional angular chart. Caps and collars on that chart supply the cut data used by modular flow, entropy variation, and overlap comparison. The orientation-preserving conformal group of the same S^2 chart is the celestial-sphere realization of $\text{SO}^+(3,1)$, so the sphere is the kinematic bridge between finite screen cuts and the emergent $3 + 1$ -dimensional Lorentz branch once the controlled cap-pair theorem is satisfied. Finite cellulations of the chart provide the regulator surface on which ports, edge sectors, and local comparison data can be made explicit; they are not by themselves a Lorentz invariant continuum.

The echosahedral carrier is the concrete local realization of this chart data. Its multi-port interface supplies discrete boundary directions, face and edge incidence, exposed readout channels, record slots, and repair channels. Recurrent toroidal subchannels supply local winding and memory structure. The smooth spherical chart describes the observer-facing cut; the echosahedral carrier supplies the fixed-cutoff local machine whose quotient data can present that cut.

The A_5 and E_8 labels mark different roles in that machinery. A_5 -icosahedral symmetry is the local finite-screen language used when a patch carrier is made concrete: it organizes ports, faces, and overlap data with enough discrete symmetry to stabilize observer-facing cuts. E_8 -type language is the exceptional closure language used by the high-symmetry branch. It names the root-lattice and affine exceptional structure in which the icosahedral data can sit. The term is representation-closure language for the branches that call for it.

One finite certificate lives on the E_8 -side of that split. The $E_8/\text{Spin}(8)$ triality sidecar gives exact matrices for an A_8 root subsystem inside E_8 , the subgroup $\text{Alt}(9) \subset W(A_8) \cong \text{Sym}(9)$, its nonsplit $2\text{-Alt}(9)$ lift in $\text{Spin}(8)$, and a positive half-spin image preserving an E_8 lattice. The vector and half-spin presentations have distinct $E_8/2E_8$ orbit fingerprints, $\{9, 36, 84, 126\}$ and $\{120, 135\}$, before $\text{Spin}(8)$ triality fuses them. This is a finite exceptional representation-closure receipt. It is not a recovered-core theorem, a new particle or force, a heterotic edge-CFT proof, or hardware evidence.

This distinction matters for the synthesis paper. The physical claim lives at the observer-visible quotient: finite patches must expose the same records and shared observables after allowed implementation-hiding changes. The microphysics paper gives the concrete reference architecture for that quotient surface, including the federated patch-carrier model, fixed-cutoff edge heat-kernel/Casimir law, central-record/Born–Lüders measurement interface, Bell/CHSH event surface, and checkpoint/restoration package. See Ref. [4], available as the numbered GitHub reference in the bibliography.

Famous equations and structural outputs recovered

The table below maps headline formulas by logical role. Some rows are core theorem results, some are short corollaries of recovered actions, and some are standard limits once the parent OPH branch is established.

Recovered result	OPH role	Famous display	Support boundary
Three-dimensional observer-frame chart, Lorentz kinematics, and invariant light cone	core theorem	$q(\Omega) = (1, \Omega),$ $n_C = (\cot \alpha, \csc \alpha \mathbf{c}), n_{gC} = \Lambda_g n_C,$ $H^3 \simeq \text{SO}^+(3, 1)/\text{SO}(3)$	scaling-limit prime geometric subnet; chart population and neutral bulk are separate gates
Record-populated observer-facing H^3	conditional theorem	$R_i(C, t, O) \rightarrow F(X_i) + e,$ $\alpha \bar{d} \leq F(X) - F(Y) _W,$ $S_i(t) = B_H(\hat{X}_i, r_i)$	point-local calibrated cap responses, compact source domain, bounded error, quantitative cap frame, and positive gap for unique finite output
Einstein equation	core theorem	$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle$	full Einstein-branch theorem stack; dependency discharge in Theorem E0; Λ fixed by the capacity branch
Newton-Poisson gravity	inherited weak-field limit	$\nabla^2 \Phi = 4\pi G \rho, \ddot{\mathbf{x}} = -\nabla \Phi$	weak field and slow motion
Maxwell equations	electromagnetic corollary	$dF_Q = 0, \quad d * F_Q = g_Q^2 * J_Q$	ordinary $U(1)_Q$ branch plus explicit Maxwell action/current hypothesis
Yang–Mills equations	compact-gauge corollary	$DF = 0, \quad D * F = g^2 * J$	compact-gauge connection branch plus explicit Yang–Mills action/current hypothesis
Bekenstein–Hawking / de Sitter area law	global capacity branch	$N_{\text{scr}} = A/(4\ell_P^2) = 3\pi/(G\Lambda)$	screen-capacity closure
Born rule and Lüders update	fixed-cutoff theorem	$\mathbb{P}(E) = \text{Tr}(\rho P_E),$ $\rho _E = P_E \rho P_E / \text{Tr}(\rho P_E)$	central record algebra
Tsirelson bound	fixed-cutoff theorem	$ S_{\text{CHSH}} \leq 2\sqrt{2}$	Bell event surface
Charge quantization	core theorem	$Q = T_3 + Y, \text{ color singlets have } Q \in \mathbb{Z}$	quarks are confined fractional-charge fields
Classical carrier-mode poles	conditional action theorem	$K_X^{\text{phys}} \propto (\omega^2 - c_*^2 \mathbf{k} ^2) \Pi_X$	Maxwell, perturbative pure-Yang–Mills, or pure-Einstein action/phase receipt; quantum particle gate separate

Recovered result	OPH role	Famous display	Support boundary
Coherent-matter scalar susceptibility	conditional scalar theorem	$\chi_\nu^{\text{can}} = e^{-P_\chi/24} = 0.9343006394893864 \dots$	nondegenerate OPH-coherent material source, Scalar Edge-Center Exhaustion, and protected-reserve collar closure; device force and covariant dark stress are separate gates

Particle Support Status

The particle paper treats the closure matrix explicitly. The non-hadron bundle has the following support levels:

Stack-Wide Claim Boundary

The paper stack uses one no-relabeling rule throughout. A finite diagnostic or calibration field does not become a physical observable merely because it is renamed. Capacity bookkeeping needs an independent energy readout before it can count as mass; an archive needs a physical source, propagation, and detector channel before it can count as radiation; a finite repair spectrum needs a continuum operator/readout bridge before it can count as a physical spectrum; and a finite reconstruction threshold needs physical entropy and clock data before it can count as a Page-time claim. The compact paper records this as the source-separation rule behind the support-level table. The χ_ν scalar-channel identity is theorem-grade for nondegenerate OPH-coherent material sources once Scalar Edge-Center Exhaustion is imported. Device-scale force, cosmological abundance, and finite covariant dark stress require their own receipt packages.

Companion Papers

The detailed depth surfaces used throughout this synthesis are:

1. *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems* [1], which gives the substrate-neutral same-source/cross-source quantifier split, robust endpoint and refinement bounds, collar-repair obstructions, and the finite conditional-resampling receipt used by the repair papers.
2. *Recovering Relativity and the Standard Model from Observer Overlap Consistency* [2], which gives the compact technical core for relativity, gravity, receipt-conditional combined-zero-obstruction compact-gauge reconstruction (central or higher-associator strictification plus an allowed trivial-holonomy strict representative), and MAR-selected Standard Model structure.
3. *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics* [3], which carries the finite patch-net fixed-point, defect, quotient, and operator-record consensus package, with the default overlap code treated as a finite constraint code, termination separated from the local-diamond and repair-completeness conditions needed for confluence, and QECC/min-cut, spectral, BFT, and hardware-speedup claims gated by separate certificates.

4. *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH* [4], which carries the regulated federated patch-carrier architecture, the fixed-cutoff edge heat-kernel / Casimir theorem, and the measurement and observer checkpoint/restoration packages.
5. *Deriving the Particle Zoo from Observer Consistency* [5], which carries the particle derivations, masses, couplings, and sector calculations.
6. *Theoretical Bounds on χ_ν in Observer-Patch Holography* and *Observer-Patch Holography and Dark Matter* [6, 7], which carry the coherent-matter same-channel theorem, the protected-reserve susceptibility coefficient, and the separate finite dark-stress proof obligations.
7. *Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics* [8], which isolates the controlled compact-gauge repair mechanism and states the continuum certificate needed before the finite repair gap becomes a Clay-facing four-dimensional Yang–Mills gap.
8. *Observer-Patch Holography as a String-Vacuum Selector* [9], which treats string theory as an effective edge language and applies OPH gates to critical-string candidates, with the Bouchard–Donagi witness conditional on its open certificates.

The sections that follow give the synthesis view. Detailed proof burdens sit with the companion papers and with the late dependency checklist in this paper.

4 Recovered Core: Foundations and Structural Branches

Observer-Patch Holography is a reconstruction program for fundamental physics. Physical data live on a horizon screen S^2 , and the effective world is encoded by the mutual consistency of overlapping patch descriptions. This recovered-core section develops the structural branch: fixed-cutoff overlap repair, the separated cofinal refinement-limit consensus bridge, collar recovery, the separate geometry-readout route to Lorentz and Einstein structure, receipt-conditional compact gauge reconstruction in the bosonic branch, and the realized-branch Standard Model quotient.

4.1 Overview

The recovered core has a simple conceptual order. First, overlap consistency turns local patch states into a constrained gluing problem. The fixed-cutoff branch gives collar decompositions, edge centers, and finite repair dynamics on the declared physical quotient. Second, the continuum geometric branch first reads quotient normal forms as cap, stress, diamond, tetrad, and scale data; it then turns modular flow on caps into Lorentz kinematics, a three-dimensional observer-facing spatial chart, and finally into the Einstein relation through fixed-cap generalized-entropy stationarity plus the null-stress, small-ball, and tensor-upgrade clauses. Bare finite consensus stops at quotient normal forms. Third, the transportable edge-sector branch first classifies fixed-stage zero-obstruction transportable sectors and, on a cofinal tail carrying the compact-gauge refinement receipt, reconstructs some compact internal gauge group in the bosonic branch; Minimal Admissible Realization (MAR), anomaly constraints, and the explicit CKM/weak-sector clauses then select the realized Standard Model quotient with exact hypercharge, structural electroweak force content, the realized color triplet $N_c = 3$, and the generation count $N_g = 3$ on the explicit realized one-generation/one-Higgs package. This branch does not by itself supply a super-Tannakian internalization of fermions/chirality.

The local pixel ratio $P \equiv a_{\text{cell}}/\ell_\star^2$, the cosmic record capacity N_{CRC} , and the selected no- G scale certificate enter only on their declared quantitative branches. P is fixed by the outer/inner closure construction in Section 6. The D6 branch uses the cosmic record-closure fixed point $N_{\text{CRC}} = F(N_{\text{CRC}})$, where $F(N)$ is the active capacity read back by stable observers inside the universe supplied with capacity N , to close the metric-term ambiguity of the Einstein branch on the cosmological-capacity surface. The finite-count representation selects the MAR maximizer of $\log |\Omega_N^{\text{sc}}| - N$, where the subtraction is division by the full screen Hilbert dimension e^N . The selected scale certificate emits $\gamma_\star = \ell_\star \nu_{\text{Cs}}/c$, equivalently $B_\star = 3\pi/\ell_\star^2$, and then $G_{\text{SI}} = c^3 \ell_\star^2/\hbar$; the pixel P cancels in the Newton area-law readout. The late classification section records the formal dependency checklist and separates recovered-core claims from quantitative closures and continuations.

4.2 Model and Axioms

4.2.1 Observers and access model

An observer O is a tuple $(P_O, \mathcal{A}(P_O), \rho_O, R_O)$ where:

- $P_O \subset S^2$ is a connected screen patch (the observer's access region).
- $\mathcal{A}(P_O)$ is the von Neumann algebra associated to P_O .
- ρ_O is the local state, obtained by restricting the global state to $\mathcal{A}(P_O)$.
- R_O is a finite record algebra generated by stable internal record projectors within P_O .

Observers are internal patterns in the global state. Different observers correspond to different patches and their compatible marginals.

4.2.2 Screen, patches, and algebra net

We work in a single static patch with a horizon screen S^2 . Each connected subregion $P \subset S^2$ is assigned a von Neumann algebra $\mathcal{A}(P)$. The net satisfies isotony:

$$P \subset Q \implies \mathcal{A}(P) \subset \mathcal{A}(Q).$$

A global state ω is a positive linear functional on the inductive-limit algebra. Overlap consistency is imposed algebraically: for overlaps $P_1 \cap P_2$, ω restricted to $\mathcal{A}(P_1 \cap P_2)$ is the same from either side.

4.2.3 Five OPH axioms

This paper uses five OPH axioms. The support-visible BW scaling theorem below handles the scaling/BW step. Transportability and the fixed-cutoff bosonic sector category are theorem-level results cited directly where they are used. Refinement/fiber descent and the cofinal MAR-admissible compact-gauge witness are conditional on the explicit compact-gauge refinement receipt stated in Section 6. Other branch-local hypotheses are written out in prose where needed instead of being assigned a separate label family.

1. **Screen Net.** A horizon screen S^2 carries a net of algebras $P \mapsto \mathcal{A}(P)$.
2. **Overlap Consistency.** Local states agree on shared observables for any overlap.
3. **Local MaxEnt and Refinement Stability.** At the UV scale the realized state maximizes entropy subject to a finite family of gauge-invariant local constraints, and the resulting family of

states is stable under coarse-graining so symmetry-allowed relevant operators are not held at zero by unexplained fine tuning.

4. Recoverable Generalized Entropy. A generalized entropy functional exists on caps, obeys quantum focusing on null generators, and comes with the recoverability structure used throughout the manuscript: collar tripartitions have small CMI with controlled recovery maps, with stronger collar/null-strip hypotheses stated explicitly where needed.

5. Minimal Admissible Realization (MAR). On the admissible low-energy branch, the realized sector package is the lexicographically minimal one under the complexity vector $C(\mathfrak{S})$. MAR is an explicit structural-economy axiom on admissible realized branches, not a theorem derived from the preceding four axioms.

4.2.4 Closure values and theorem-local technical data

The quantitative branches discussed here use two dimensionless fixed-point closure coordinates and one selected no- G scale certificate:

$$P \equiv a_{\text{cell}}/\ell_\star^2, \quad N_{\text{CRC}} \equiv \log \dim \mathcal{H}_{\text{tot}}, \quad \gamma_\star \equiv \frac{\ell_\star \nu_{\text{Cs}}}{c}, \quad B_\star \equiv \frac{3\pi}{\ell_\star^2}.$$

Here $P = P_\star$ is the local UV-area ratio fixed on the outer/inner closure branch summarized in Section 6, while N_{CRC} is the unique stable fixed point of the OPH-derived cosmic record-closure readback map. On the observed branch this is the de Sitter entropy capacity usually denoted $N_{\text{scr}} \simeq 3.31 \times 10^{122}$. Capacity-level neutrino side estimates tied to that branch value are legacy bookkeeping terms separate from the rejected weighted-cycle comparison lane. The selected scale certificate supplies $\gamma_\star$, with $B_\star$ as the SI curvature display of the same branch. Once supplied, $\ell_\star^2 = 3\pi/B_\star$ becomes the usual Planck-area display after $G_{\text{SI}} = c^3 \ell_\star^2 / \hbar$. These are quantitative-branch quantities, not extra axioms.

Observation enters this paragraph only as branch localization. In the OPH reverse-engineering reading, the universe is a closed mathematical fixed structure, so approximate measured values may be used to choose the basin in which the internal observers sit. The derivation then requires the relevant readback map to be a self-map on that basin and to carry a Banach contraction, derivative-sign, strict-concavity, or equivalent uniqueness proof. The observed number supplies an initial guess or branch interval; the OPH fixed point supplies the reported value. The exact $P_\star$ and N_{CRC} values must be computed as closure endpoints.

The compact proof uses the same convention as a compression estimate. A quantitative row enters that estimate only when the declared source map has no dependency path from the measured target or a calibrated proxy. If p_i upper-bounds the conditional accidental hit probability of row i after previous accepted rows, then

$$P_{\text{acc}} \leq \prod_i p_i.$$

Twelve one-percent source-certified rows give $P_{\text{acc}} \leq 10^{-24}$, and twenty give 10^{-40} . The numerical bound is secondary to the theorem stack: the same two closure coordinates organize the observer readout, gravity/gauge reconstruction, hierarchy bridge, dark energy, dark-sector budget, product-group proton-decay exclusion, particle inventory, and string-vacuum selector.

The two dimensionless closures determine dimensionless geometry, not an SI scale by themselves. On the de Sitter branch,

$$N_\star = \frac{3\pi}{\Lambda_\star \ell_\star^2}, \quad P_\star = \frac{a_{\text{cell}}}{\ell_\star^2},$$

so

$$\Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}, \quad \Lambda_\star a_{\text{cell}} = \frac{3\pi P_\star}{N_\star}.$$

Equivalently, for the scale product $\mathcal{B}_\ell := \Lambda_\star N_\star$,

$$\mathcal{B}_\ell \ell_\star^2 = 3\pi, \quad \mathcal{B}_\ell a_{\text{cell}} = 3\pi P_\star.$$

These invariants survive the rescaling $\ell_\star^2 \mapsto \lambda^2 \ell_\star^2$, $a_{\text{cell}} \mapsto \lambda^2 a_{\text{cell}}$, and $\Lambda_\star \mapsto \lambda^{-2} \Lambda_\star$, while $\mathcal{B}_\ell \mapsto \lambda^{-2} \mathcal{B}_\ell$. Thus the SI value of $\mathcal{B}_\ell$, the SI area $\ell_\star^2$, and G_{SI} require the selected scale certificate; they are not derived from $P_\star$ and $N_\star$ alone. If the selected no- G scale certificate supplies $\gamma_\star$, equivalently $B_\star$, then

$$\gamma_\star = \frac{\ell_\star \nu_{\text{Cs}}}{c}, \quad B_\star = \mathcal{B}_\ell = \Lambda_\star N_\star, \quad \ell_\star^2 = \frac{3\pi}{B_\star}.$$

The branch certificate used for the SI gravity row is

$$B_\star = 3.60787391468 \dots \times 10^{70} \text{ m}^{-2},$$

hence

$$\ell_\star^2 = 2.61228030237 \dots \times 10^{-70} \text{ m}^2.$$

Rounded capacity displays such as $N_{\text{scr}} \simeq 3.31 \times 10^{122}$ are useful branch labels, but they are not high-precision inputs for the Newton row; with the Planck-2018 curvature benchmark, using only the rounded capacity would shift the Newton row at the 8.4×10^{-4} level.

The capacity normalization uses the de Sitter static-patch entropy [24]. If

$$N_{\text{patch}} = \left(\frac{r_{\text{dS}}}{\ell_P} \right)^2$$

denotes the bare horizon area ratio, then

$$N_{\text{scr}} = S_{\text{dS}} = \frac{A_{\text{dS}}}{4\ell_P^2} = \pi N_{\text{patch}} = \frac{3\pi}{\Lambda \ell_P^2}.$$

Using the Planck-2018 late-time de Sitter scale [25] gives $N_{\text{patch}} \simeq 1.05 \times 10^{122}$, $N_{\text{scr}} \simeq 3.31 \times 10^{122}$, and $\Lambda \ell_P^2 \simeq 2.85 \times 10^{-122}$.

For the recovered core, the controlled scaling/BW step for Lorentz/null-modular/Einstein statements is Theorem 4.2, the support-visible BW scaling theorem. Transportability, the fixed-cutoff bosonic category, fixed-cutoff regulator bookkeeping, quasi-local propagation, endpoint-Lipschitz control, collar decomposition, and related branch-internal statements are cited directly from axioms and earlier theorems. Refinement/fiber descent and the cofinal compact-gauge witness additionally require the Section 6 compact-gauge refinement receipt.

Local MaxEnt at the regulator scale. At the regulator scale ℓ_{UV} , the global state ω maximizes von Neumann entropy subject to:

1. A finite set $\{O_a\}$ of gauge-invariant local operator densities, each supported on a ball of radius $\leq r_0 = O(\ell_{\text{UV}})$.
2. Constraint equations $\langle \sum_x O_a(x) \rangle = C_a$, one homogeneous global sum per density label a , where x runs over the cells of the UV lattice. The constraints are the N_{con} global sums, not one independent constraint per cell, so the number of independent constraints and of Lagrange multipliers is N_{con} (plus the optional global constraints below) on every lattice, independent of the cell count.

3. Optionally, a finite number of global constraints (total energy, charge).

This is the minimal specification needed to derive the local Gibbs form of Lemma 2.6 from Axiom 3 with a cutoff-independent multiplier count.

Clarification (MaxEnt $\neq$ thermal equilibrium). MaxEnt here is **local state selection** given constraints, not "the universe is in thermal equilibrium." Non-equilibrium physics is modeled in the enlarged *local-multiplier* family obtained by replacing the homogeneous multipliers λ_a with slowly varying fields $\lambda_a(x)$, equivalently by imposing per-cell constraints $\langle O_a(x) \rangle = c_a(x)$. That enlarged family has dimension proportional to the number of UV cells, so it is a function-space approximation regime with gradient error terms, not the finite-dimensional family of the refinement argument: the refinement-stable branch and the induced maps $R_{\ell \rightarrow L}$ below are always statements about the homogeneous N_{con} -parameter branch. Controlled violations of exact Markov additivity appear under the explicit mixing hypotheses stated later in Section 2.3. Equilibrium is an approximation regime with explicit error terms.

Rotationally invariant constraint branch. Constraint sets are $\text{SO}(3)$ -invariant on S^2 .

Gauge as overlap redundancy at fixed cutoff. Overlap identifications are not unique; the freedom that leaves overlap observables invariant forms a local groupoid.

Central-defect subbranch. Assume the overlap centers are identified with one fixed abelian unitary coefficient group Z_Σ on which overlap transport acts trivially. Choose unitary overlap implementers with $U_{ji} = U_{ij}^{-1}$. On triple overlaps, when the failure of strict coherence is central, define

$$\Omega_{ijk} := U_{ij}U_{jk}U_{ki} = z_{ijk}, \quad z_{ijk} \in Z_\Sigma,$$

equivalently $U_{ij}U_{jk} = z_{ijk}U_{ik}$. The defect is data of the implementer lift; writing only $\text{Ad}(z_{ijk})$ would erase it because conjugation by a central element is the identity.

Collar double-scaling hypothesis. There exists a UV length ℓ_{UV} such that for any cap C and collar width δ , in the refinement limit $\delta \rightarrow 0$ and $\ell_{\text{UV}} \rightarrow 0$ with $\delta/\ell_{\text{UV}} \rightarrow \infty$, the Markov error satisfies

$$I(A_\delta : D_\delta \mid B_\delta)_\omega \leq \varepsilon(\delta/\ell_{\text{UV}}), \quad \varepsilon(x) \rightarrow 0 \text{ as } x \rightarrow \infty.$$

See Section 2.3 for the collar tripartition definitions and the regulated EC proof that yields this limit from the fixed-cutoff realized presentation.

Geometric-subnet BW lift. The continuum Lorentz statement is asked only on the support-visible extracted prime geometric subnet for caps, together with the controlled collar/refinement limit that carries the Markov and recovery remainders explicitly. The finite cap regulators are type-I algebras, but the scaling-limit observer algebra may leave that class, so the theorem is automorphism-level on the emitted geometric cap pair and the special type-I generator form is $K_C = 2\pi B_C + Z_C$ with Z_C central. The compact paper proves that any cofinal finite cap-normal branch satisfying the support-order, held-out oriented cross-ratio, mixed-modular-matrix, and independently normalized geometric 2π -KMS certificate converges to the support-visible BW cap automorphism. It also proves the cap-normal H^3 chart: $q(\Omega) = (1, \Omega)$, $n_C = (\cot \alpha, \csc \alpha \mathbf{c})$, $n_{gC} = \Lambda_g n_C$, and $H^3 \simeq \text{SO}^+(3, 1)/\text{SO}(3)$, with cap half-spaces but no populated-bulk or preferred-observer-point consequence. The compact paper's geometry-producer packet now produces that certificate from repair normal forms on a computable-receipt branch (spherical incidence, disk/mesh, cross-ratio, and 2π -normalization receipts) and proves that bare confluence underdetermines topology, dimension, framing, and normalization, so the receipts are the irreducible remaining Einstein

branch-entry content; its null-net-standardness and event-manifold packets extend this into derived positive translations and a conditional Lorentzian event manifold (Theorems 4.3c–4.3e).

Derived quasi-local propagation and modular-locality control. At scale ℓ_{UV} , the automorphism group generated by the MaxEnt generator of Lemma 2.6, or more generally by any branch generator lying in the same bounded-support algebraic closure, has a finite Lieb–Robinson velocity v_{LR} : for local operators A, B supported on regions separated by distance d ,

$$\|[A(t), B]\| \leq c\|A\|\|B\| \min(|R_A|, |R_B|) e^{-(d-v_{LR}|t|)/\xi}$$

for $d > v_{LR}|t|$, where $\xi = O(\ell_{UV})$.

This is the branch-internal control statement that turns the quasi-local structure of the local-Gibbs branch into explicit support control for time-evolved operators. It is part of the third-axiom branch unpacked at regulator scale.

Approximate modular covariance on the fixed-cutoff realized branch. Under the fixed-cutoff realized presentation, Lemma 2.6 (local Gibbs), the derived quasi-local propagation bound above, and the collar double-scaling plus mixing hypotheses of Section 2.3, the modular flow $\sigma_t^{\omega, C}$ maps $\mathcal{A}(R)$ into a slightly thickened region algebra:

$$\sigma_t^{\omega, C}(\mathcal{A}(R)) \subseteq \mathcal{A}(R^{+v_{\text{mod}}|t|}) \quad \text{up to error } \eta(d - v_{\text{mod}}|t|),$$

where R^{+s} denotes the s -neighborhood thickening, v_{mod} is a "modular propagation velocity" controlled by v_{LR} and local norm bounds, and d is the distance from R to ∂C .

Heuristic motivation. The modular Hamiltonian $K_C = -\log \rho_C$ is quasi-local by Lemma 4.1a-b (modular additivity localizes it to the collar). The derived Lieb-Robinson control then bounds support spreading under $e^{iK_C t}$. In the double-scaling collar limit ($\delta/\ell_{UV} \rightarrow \infty$), the thickening vanishes in macroscopic units. This is heuristic support for the tangent-limit package used later.

Continuum-limit heuristic. Define the induced region flow

$$f_t^C(R) := \lim_{\ell_{UV} \rightarrow 0} R^{+v_{\text{mod}}|t|}.$$

Then $\sigma_t^{\omega, C}(\mathcal{A}(R)) = \mathcal{A}(f_t^C(R))$ becomes exact in the continuum limit, with error controlled by

$$\eta(\delta) \lesssim 2\sqrt{\ln 2 \cdot c \cdot |\partial C|_{UV}} e^{-\delta/(2\xi)}.$$

The discussion above is heuristic support for the tangent-limit package used in Theorem 4.2.

Refinement-stable MaxEnt branch. Choose any family of coarse-graining channels $\Phi_{\ell \rightarrow L}$ between UV scale ℓ and IR scale L that is compatible with the third OPH axiom, including its refinement-closure clause. Then on the realized branch one may write

$$\Phi_{\ell \rightarrow L}(\omega_\ell(\lambda)) = \omega_L(R_{\ell \rightarrow L}(\lambda)),$$

for an induced map $R_{\ell \rightarrow L}$ on the common finite-dimensional homogeneous multiplier family. The existence of $R_{\ell \rightarrow L}$ as a self-map of that same finite-dimensional family is exactly the closure clause, not a bookkeeping consequence of reusing the labels a : coarse-graining a finite-range Gibbs

family generically generates interactions outside the retained density list, and the axiom asserts that on the realized branch those generated directions vanish or are absorbed into the retained family; the closure defect, the moment-matching I-projection realizing $R_{\ell \rightarrow L}$, and its trace-norm residual bound are constructed in Definition 2.6a and Lemma 2.6b, and what remains assumed rather than proved is that the defect vanishes along the realized branch. Granting closure, the regulator states lie in that shared multiplier family rather than in unrelated state spaces at different cutoffs, and the realized branch is the persistent trajectory or invariant subset selected inside it under the induced refinement maps. This is a state-side persistence statement; it does not by itself upgrade fixed-cutoff edge labels to a transportable refinement-persistent sector colimit.

Fixed-cutoff realized presentation. At a UV scale ℓ_{UV} , local patch algebras are type-I with finite-dimensional Hilbert spaces, and gauge-as-gluing is realized as a boundary action on a lifted finite-dimensional presentation. Section 2.3 proves the resulting fixed-cutoff collar package directly from overlap consistency on this realized presentation, while distinguishing the lifted fixed-point algebra from the sector-preserving collar algebra used in the EC theorem.

External mathematical inputs: SSA and recovery theorems (Petz 1986, 1988; Fawzi and Renner 2015), the fixed-volume area-variation identity for small geodesic balls used in the local Einstein rest-frame relation, and the local quadratic-polarization argument used in the tensor upgrade. The null modular bridge and the small-ball kernel are carried internally on the compact surface rather than imported from a separate EFT small-ball law. For SM contact we also use the Doplicher-Roberts reconstruction (Doplicher and Roberts 1989, 1990) once localized zero-obstruction sectors are assembled in the small-region limit. Full citations appear in the References.

Recovered-core theorem boundary. The Lorentz/null-modular/Einstein recovered core runs through the support-visible automorphism theorem stated in Section 4.2. Fixed-cutoff statements remain regulator statements.

4.2.5 Notation

- ρ_C : reduced state on cap C .
- $K_C := -\log \rho_C^{\mathcal{C}}$: modular Hamiltonian of the reference state.
- B_C : geometric generator of the cap-preserving conformal dilation.
- $S_{\text{gen}}(C)$: generalized entropy on a cap.
- ℓ_{UV} : UV length scale of the refined screen net.
- δ : collar width around a cap boundary.

4.3 Information-Theoretic Tools

4.3.1 Strong subadditivity and Markov states

For any tripartite state ρ_{ABC} ,

$$I(A : C \mid B) := S(AB) + S(BC) - S(B) - S(ABC) \geq 0.$$

Exact Markov states satisfy $I(A : C \mid B) = 0$ and admit a recovery map:

$$\rho_{ABC} = (\text{id}_A \otimes \mathcal{R}_{B \rightarrow BC})(\rho_{AB}).$$

4.3.2 Approximate recovery

If $I(A : C | B) \leq \varepsilon$ (bits), there exists a CPTP recovery map $\mathcal{R}$ with

$$\|\rho_{ABC} - (\text{id}_A \otimes \mathcal{R})(\rho_{AB})\|_1 \leq 2\sqrt{\ln 2 \varepsilon}.$$

4.3.3 Collar refinement and sufficient mechanisms

Fix a cap $C \subset S^2$ with boundary circle. For collar width δ define

$$B_\delta := \{x \in S^2 : \text{dist}(x, \partial C) \leq \delta\}, \quad A_\delta := C \setminus B_\delta, \quad D_\delta := (S^2 \setminus C) \setminus B_\delta.$$

Then $S^2 = A_\delta \cup B_\delta \cup D_\delta$ with A_δ and D_δ interacting only through B_δ . The collar double-scaling hypothesis is the requirement that $I(A_\delta : D_\delta | B_\delta) \rightarrow 0$ in the refinement limit. At fixed regulator scale, the same finite-dimensional realized presentation supplies the patch-net and overlap-gluing data used below. We therefore isolate that realized presentation first, then separate the exact-Markov and quantitative routes.

Finite quotient ensemble theorem surface. Fix a finite regulator r . The physical presentation space is Σ_r , the presentation redundancy groupoid is Γ_r , and the finite physical quotient is

$$Q_r = \Sigma_r / \Gamma_r, \quad \pi_r : \Sigma_r \rightarrow Q_r.$$

The quotient removes nonphysical presentation data: gauge representatives, port relabelings, mesh labels, shard or worker identifiers, queue order, repair schedule identifiers, retry counters, timestamps unless declared semantic, hidden carrier coordinates, and inert ancillary labels. If only settled configurations carry probability, the probability space is the normal-form subset

$$N_r = n_r(Q_r).$$

The map n_r is a normal-form map, not a probability law. Any promoted physical branch inherits this firewall: it must declare the quotient-intrinsic source law or action before a normal form can be read as a selection or prediction claim.

Observable algebras and reference states. In the finite classical case the quotient observable algebra is

$$\mathcal{O}_r = \ell^\infty(Q_r).$$

In the finite quantum case the physical algebra is a declared quotient algebra $\mathcal{A}_r^{\text{phys}}$ with state

$$\omega_r(A) = \text{Tr}(\rho_r A).$$

When the reference object is obtained from a finite lifted carrier, the load-bearing data are not an abstract groupoid cardinality alone. They are a tracially pointed quotient

$$(\mathcal{A}_{r,b}^{\text{phys}}, \tau_{r,b}^0), \quad \mathcal{A}_{r,b}^{\text{phys}} = z_{r,b} B(\tilde{\mathcal{H}}_r)^{G_r} z_{r,b},$$

where $U_r : G_r \rightarrow U(\tilde{\mathcal{H}}_r)$ is the compact gauge action and $z_{r,b}$ is the central projection for the declared boundary or superselection sector. The reference trace is

$$\tau_{r,b}^0(A) = \frac{\text{Tr}_{\tilde{\mathcal{H}}_r}(A)}{\text{Tr}_{\tilde{\mathcal{H}}_r}(z_{r,b})}.$$

If

$$z_{r,b} \tilde{\mathcal{H}}_r \cong \bigoplus_{\alpha} V_{\alpha} \otimes M_{\alpha},$$

with $d_{\alpha} = \dim V_{\alpha}$ and $m_{\alpha} = \dim M_{\alpha}$, then

$$\mathcal{A}_{r,b}^{\text{phys}} \cong \bigoplus_{\alpha} I_{V_{\alpha}} \otimes B(M_{\alpha}), \quad p_{r,\alpha} = \frac{d_{\alpha} m_{\alpha}}{\sum_{\beta} d_{\beta} m_{\beta}}.$$

These are the induced central-sector weights only after the carrier representation and boundary sector have been fixed.

OPH quotient ensemble. An OPH quotient ensemble is specified by a quotient-intrinsic base weight and action

$$m_r : Q_r \rightarrow \mathbb{R}_{>0}, \quad S_r : Q_r \rightarrow \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q) e^{-S_r(q)}, \quad Z_r = \sum_{q \in Q_r} w_r(q), \quad \mu_r(q) = Z_r^{-1} w_r(q).$$

Equivalently, one may state an intrinsic projective prior ν_r on Q_r and set $\mu_r = (n_r)_{\#} \nu_r$. Uniform quotient counting, uniform representative counting pushed to the quotient, groupoid weights, and tracial central-sector weights are different physical claims. The paper must declare which one is being used.

Normal-form projector non-selection. For any map $N : Q \rightarrow Q_{\text{nf}}$, the induced map on laws

$$\mathcal{C}_Q(\mu) = N_{\#} \mu$$

is idempotent:

$$\mathcal{C}_Q^2 = \mathcal{C}_Q.$$

Every law supported on Q_{nf} is fixed. Therefore settlement or canonicalization never selects a unique physical probability law by itself.

Selection-gap corollary. Let $X \subseteq Q_{\text{nf}}$ be a finite set of quotient-normal candidates distinguished by visible invariants. Normal-form data determine X and its quotient-visible invariants, but they do not choose a member of X . If two laws μ, ν are supported on X and concentrate on different candidates, both are fixed by $\mathcal{C}_Q$. Unique sector selection therefore requires source data: an intrinsic action with a unique minimizer, a declared physical ensemble, or a refinement-stable gap certificate. A defect or holonomy classification can classify possible sectors without choosing the physical sector, and a contraction or repair generator can certify convergence toward a declared target without creating the target law.

Finite MaxEnt quotient ensemble. For finite Q , positive m , and quotient observables $F_1, \dots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = - \sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

subject to

$$\sum_q \nu(q) = 1, \quad \sum_q \nu(q) F_a(q) = c_a$$

has the full-support solution, when the feasible full-support surface is nonempty,

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

Boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra with faithful reference state σ_r ,

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\text{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

The finite constraint ledger must name every $F_{r,a}$, its units and support, the target expectation and source, sector or zero-mode treatment, refinement transformation, and proof that no run output or observational output entered the source definition.

Refinement compatibility and RG closure. For $s \succeq r$, let $c_{sr} : Q_s \rightarrow Q_r$ be the physical coarse map. Exact compatibility of weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q' : c_{sr}(q')=q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of q . Then

$$(c_{sr})_{\#} \mu_s = \mu_r.$$

If the one-step defects are

$$\delta_{k+1,k} = \|(c_{k+1,k})_{\#} \mu_{k+1} - \mu_k\|_{\text{TV}},$$

then

$$\|(c_{nr})_{\#} \mu_n - \mu_r\|_{\text{TV}} \leq \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

For exponential-family refinement, exact closure requires the fine conditional free energy

$$G_{sr,\theta}(q_r) = -\log \mathbb{E}_{m_s^0} [\exp[-\theta \cdot F_s(Q_s)] \mid c_{sr}(Q_s) = q_r]$$

to equal $\kappa_{sr}(\theta) + R_{sr}(\theta) \cdot F_r(q_r)$. If the residual is uniformly bounded by ε , the induced total-variation defect is bounded by $\tanh \varepsilon$.

Implementation invariance and representative lifting. If implementations A, B have quotient bijections $h_r : Q_r^A \rightarrow Q_r^B$ satisfying

$$m_r^B(h_r q) = m_r^A(q), \quad S_r^B(h_r q) = S_r^A(q), \quad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then

$$(h_r)_{\#} \mu_r^A = \mu_r^B.$$

For tracially pointed quantum quotients the corresponding equivalence is a trace-preserving quotient equivalence. It is invariant under unitary intertwiners preserving the gauge action and sector, and under inert trivial ancillas $A \mapsto A \otimes I_{\text{anc}}$. It is not invariant under arbitrary changes of gauge-representation multiplicities.

If an implementation stores representatives, a representative-level law must be a conditional lift

$$\tilde{\mu}_r(x) = \mu_r(\pi_r x) \kappa_r(x \mid \pi_r x), \quad \sum_{x : \pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_{\#} \tilde{\mu}_r = \mu_r$. Uniform representative sampling yields orbit-size weights and is physical only if representative counting is the declared base measure.

Quotient-lumpable kernels and sampler correctness. A representative kernel $\tilde{P}(x, y)$ descends to Q_r only when

$$P_Q(q, q') = \sum_{y: \pi(y)=q'} \tilde{P}(x, y)$$

is independent of the chosen representative $x \in \pi^{-1}(q)$. For $w(q) = m(q)e^{-S(q)}$ and proposal $R(q, q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min \left\{ 1, \frac{w(q')R(q', q)}{w(q)R(q, q')} \right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q).$$

Repair-informed proposals must include the Hastings asymmetry term; otherwise the stationary law is generically changed.

Repair generators are not selectors. A repair generator of the form

$$L_{\text{rep}} = \sum_C c_C (I - E_C)$$

is a relaxation or sampling object after a law has been selected. Conditional expectations E_C are defined on $L^2(X_r, \pi_r)$, so the reference law π_r is input. On overlapping collars the expectations need not commute. The correct finite gap certificate is the Poincare constant

$$\kappa_r = \inf_{f \perp 1} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If local fiber rates have a positive lower bound γ_* , then

$$L_{\text{rep}} \geq \gamma_* \kappa_r (I - P_0).$$

Finite repair completeness gives $\kappa_r > 0$ at fixed regulator. A uniform refinement lower bound $\inf_r \kappa_r > 0$ is a separate theorem or receipt.

Finite evidence accuracy. For bounded coarse observables O , if

$$\|\hat{\mu}_s - \mu_s\|_{\text{TV}} \leq \epsilon_{\text{samp}}$$

and the refinement defects sum to ϵ_{ref} , then

$$\left| \mathbb{E}_{\hat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O] \right| \leq 2\|O\|_{\infty}(\epsilon_{\text{samp}} + \epsilon_{\text{ref}}).$$

Continuum-facing observables require a realization map and correlation Cauchy bound in addition to a finite histogram.

Vacuum promotion gate. A stationary sampler is not a physical vacuum. For any faithful target law one can build a positive transfer operator with that law as ground state, so positivity alone is not a selector. Vacuum promotion requires source Euclidean slab data

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

whose conductance $J_r(q, q') = J_r(q', q) \geq 0$, local potential V_r , and slab thickness $a_{t,r}$ are derived without using the target law or sampler output. With connected event graph,

$$(H_r^E f)(q) = \frac{1}{m_r^0(q)} \sum_{q'} J_r(q, q') (f(q) - f(q')) + V_r(q) f(q)$$

is self-adjoint and bounded below on $L^2(Q_r, m_r^0)$; its finite Feynman–Kac semigroup is positivity improving. Perron–Frobenius gives a unique positive normalized ground state Ω_r , and the finite vacuum law is

$$\mu_r^{\text{vac}}(q) = |\Omega_r(q)|^2 m_r^0(q).$$

For $T_r = e^{-a_{t,r}(H_r^E - E_{0,r})}$, the Doob kernel is stochastic and detailed-balanced with μ_r^{vac} . Continuum promotion additionally requires reflection positivity or equivalent reconstruction plus refinement compatibility of the transfer family.

Primordial and cosmological prediction firewall. A screen covariance is not automatically a primordial curvature spectrum. The map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

has a null space unless a radial prior, finite parametric family, or source-stress bridge is declared. OPH primordial promotion requires the source-only stress, single-clock, entropy-repair, curvature-evolution, adiabatic-mode, phase-coherence, screen-to-radial-lift, radial-null-space, and forward-projection receipts. Observable CMB promotion further requires frozen source, solver, likelihood, no-data-use, pooled-reducer, and falsification-rule receipts.

Claim tiers and required receipts. Every ensemble-facing run records immutable ensemble id, claim tier, regulator id, representative schema hash, gauge action hash, canonicalizer hash, base measure, action coefficients, coarse maps, zero-mode projector, amplitude convention, sampler hash, smoothing policy, source provenance, and explicit nonclaims. The seed belongs to the run receipt, not to the ensemble definition. The claim tiers are

- $E0$: seed noise, proposal noise, repair jitter,
- $E1$: conventional reference ensemble,
- $E2$: OPH-native quotient ensemble,
- $E3$: OPH vacuum,
- $E4$: OPH primordial field,
- $E5$: observable cosmological prediction.

The evidence bundle must keep separate receipts for stationary-law schedule invariance, detailed balance of the aggregate kernel, and pathwise partition invariance. Deterministic replay of semantic random streams or a canonical serial chain is useful, but it is not pathwise partition invariance. Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of S_r unless explicitly declared.

Derived regulator realization. Choose a finite UV cellulation at scale ℓ_{UV} and let R be a finite union of cells. Hilbertize each cell's finite local data to a finite-dimensional space $\tilde{\mathcal{H}}_i \cong \mathbb{C}^{n_i}$. The extended algebra before quotienting by overlap redundancy is

$$\tilde{\mathcal{A}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R), \quad \tilde{\mathcal{H}}_R = \bigotimes_{i \subset R} \tilde{\mathcal{H}}_i.$$

Overlap-preserving changes of local trivialization act only on cut data. On each finite-dimensional chart, their unitary image has compact closure, so one may represent the boundary gluing redundancy by a compact group $G_{\partial R} \subset U(\tilde{\mathcal{H}}_R)$. The lifted boundary-invariant endomorphism algebra is

$$A_{\text{inv}}(R) = \tilde{\mathcal{A}}(R)^{G_{\partial R}} = \mathcal{B}(\tilde{\mathcal{H}}_R)^{G_{\partial R}}.$$

This is the fixed-cutoff realized presentation used throughout the collar analysis. The EC theorem works on the invariant-state realization and uses the sector-preserving induced collar algebra there; the entire fixed-point algebra on the unreduced tensor product is outside that theorem surface. When the consensus paper speaks of quotient repair, the physical repair law is defined on this fixed-point / overlap-invariant data and any representative-level map is only a lift of that quotient update. Descent to the gauge quotient is therefore built into the physical-algebra formulation as quotient-level repair: the local map is $\text{locRep}_\lambda : Q \rightarrow Q$, and the global map is the finite quotient normal-form operator

$$\text{Rep}_\lambda = \overline{\text{nf}}_\lambda : Q \rightarrow Q.$$

On the declared fixed-cutoff collar branch, the local repair step itself is read from exact Markov splice or a declared Petz/Fawzi–Renner recovery channel. The declared fixed-cutoff branch adds repair completeness, together with control on the declared Petz domain where that branch is used. The consensus paper proves those clauses on a nontrivial rooted-tree packet-net export: finite packet labels live on a rooted tree, hidden labels are quotient-only, weighted parent-copy repair strictly lowers Φ , normal forms are exactly consistent packet assignments, and the classical full-support Petz channel is CPTP and trace-norm contractive on its positive-support-gap domain. Gauge-invariant observables factor through the quotient normal form on that verified branch. The declared touched-overlap acceptance contract makes Φ the finite-patch Lyapunov functional for primitive repair proposals on the scalar branch, but a proposal becomes a physical step only through the transactional acceptance layer: snapshot-current read sets, unchanged-outside-write validation, preserved boundary/sector/holonomy data, and exact well-founded descent. The read set is validation-complete: it contains every support of every measure term that can change under the write. Overlapping primitive proposals are aggregated by connected conflict component; the restriction-compatible union-collar glue supplies the canonical aggregate payload, while primitive members of that component do not commit independently. On the same fixed-cutoff quantum lift, the terminal expectation functional on each declared physical observable algebra is therefore unique even when microscopic representative lifts differ by gauge labels globally or by sector labels inside one quotient-local glued state. The unqualified coding statement at this stage is only a finite constraint-code statement: the overlap-net codewords are the globally consistent states $C = \Phi^{-1}(0)$. A bare graph does not determine code distance—the same graph can carry constant readouts with distance 1 or repetition constraints with distance $|V|$. Topological-code distance/min-cut, Knill–Laflamme resilience, spectral-gap convergence, BFT wall-clock liveness, and hardware search-work reduction therefore require their own certificates and are not imported by the core overlap graph. The logic is not “termination implies uniqueness”: exact descent gives termination, while the atomic conflict-component diamond on the physical quotient plus repair completeness gives confluence. The theorem therefore

gives one schedule-independent terminal quotient normal form from a fixed initial quotient state. A stronger same-boundary conclusion requires a preserved boundary/sector map and at most one consistent quotient extension in that boundary fiber. The layered functional carrier in the consensus paper proves the finite multi-edge, multi-step witness for $H_B \wedge H_{\text{fib}}$; the functional selected-fiber theorem gives the rooted obstruction-check form by constructing the single candidate from the boundary/root data and then checking all non-tree, sector, and holonomy obstructions. On that branch, multiple same-boundary candidate interiors can be present, inconsistent candidates are eliminated, and all surviving candidates share one quotient normal form. Normal-form hashes are public equality receipts after quotienting, not selectors among physically distinct minimizing endpoints. If accepted schedules terminate at different observer-facing quotient normal forms without a declared holonomy or higher-gauge obstruction, that repair law is outside the OPH consensus theorem. The neutral result of Ref. [1] packages this quantifier split exactly: under observation preservation and completeness, agreement modulo a silent equivalence of normal endpoints reached from all same-boundary sources is equivalent to injectivity of the induced boundary map on the consistent quotient. This cross-source criterion, same-source confluence, weak normalization, and all-schedule liveness are separate obligations. The named layered and functional carriers discharge the cross-source premise only on their stated domains. For separated cofinal refinement systems, the consensus paper adds an inverse-limit bridge: if finite-stage restriction maps commute with the quotient normal-form maps and holonomy maps, and if compatible families are visibly separated on cofinal finite stages, then finite normal forms and holonomy obstructions assemble into unique refinement-limit classes with finite-stage witnesses for nonzero holonomy. It also adds the RG-facing comparison: when a chosen coarse-graining channel shadows the finite-stage normal forms and obstruction maps with declared defects ε^n and ε^h , reconciling first and then coarse-graining gives the same macroscopic law readout as coarse-graining first and then reconciling, up to $\max\{\varepsilon^n, \varepsilon^h\}$; exact naturality is the zero-defect case. Exact normal-form naturality is discharged by a repair-morphism witness, and holonomy naturality by the corresponding cochain map. For distributed implementations, the same paper proves that a worker run presents one finite universe only when it starts from one global carrier and each event projects to a legal monolithic repair path, a physical stutter, or a certified rollback to an earlier committed projection. The required certificate names the global graph, initial state, partition, cut interfaces, observer registry, run/config/code hashes, linearized commits, restart roots, final monolithic normal form, and recomputed readout. Executor order is separate from observer history. Observer-clock naturality additionally requires a history-augmented quotient with semantic event keys independent of worker ID, repair iteration, queue position, timestamp, and retry metadata; a global observer registry descended from local registry groupoids with disjoint patch/cap/future namespaces and explicit lineage arrows; a state-preserving observer-algebra extraction functor from terminal augmented normal forms; a support-visible cap chart map natural with that extraction; and an operational clock instrument whose readout is calibrated up to the declared affine reparameterization and residual bound. Until those objects or proof-producing finite certificates are supplied, modular-time/readback naturality is a named gate, not a consequence of implementation bookkeeping. The same quotient boundary applies to neutral-bulk readout. For each shard s , raw record rows Σ_s must first be quotient by the declared presentation groupoid Γ_s and then sent through the terminal normal-form map n_s , producing

$$X_s = n_s(\Sigma_s/\Gamma_s).$$

Interface maps $\tau_{ts} : U_{st} \rightarrow U_{ts}$ between these terminal charts must satisfy the identity, inverse, cocycle, and zero-cycle-holonomy checks before rows from different shards are compared. A global

neutral readout is the quotient

$$Q_{\text{vis}} = \left(\bigsqcup_s X_s \right) / \sim_\tau,$$

not the concatenation of shard-local tables. Channel features $F_{s,c}$ descend to Q_{vis} only when their domains and values are transported by the same atlas. For a complete declared channel set $\mathcal{C}_\star$, the neutral distance is

$$d_{\text{neu}}(x, y) = \left(\sum_{c \in \mathcal{C}_\star} w_c d_c(F_c(x), F_c(y))^p \right)^{1/p}.$$

This is always stated first as a quotient-visible pseudometric. It becomes a metric only after quotienting zero-distance feature-collision classes or proving that the channels jointly separate points. Pairwise available-channel comparison is explicitly excluded: missingness is handled only by complete cases, a fixed quotient-visible missing symbol, or a train-only imputation map labelled as an imputed-representation metric. Presentation changes—gauge representatives, port relabelings, observer order, schedules, and shard partitions—preserve the metric only when they induce a bijection of Q_{vis} together with channel isometries; refinement claims additionally require a cofinally vanishing distance tail modulus. Finite Euclidean claims require the double-centered Gram certificate

$$B = -\frac{1}{2}H(D^{\circ 2})H \succeq 0$$

and noisy runs must report negative spectral mass, rank/effective-rank data, held-out stress, and planted/shuffled controls. Statistical certificates split independent generative shard batches before preprocessing; chart alignment, scaling, imputation, weights, graph construction, dimension choice, and thresholds are train/validation objects, and a test batch, seed, boundary condition, trajectory family, duplicate, or descendant appearing in more than one split blocks the claim. These theorems establish at most a common finite quotient metric or pseudometric. Identifying it with a physical Riemannian bulk or with Lorentzian/Einstein spacetime (via the conditional event-manifold packet of Theorem 4.3e, under its population/separation/frame/cocycle receipts) requires the separate modular/geometric branch below. The same consensus surface is classified explicitly as a finite-state exact theorem package plus this controlled inverse-limit bridge: decidable normal-form computation with the Lyapunov step bound, automatic approximate stability only through the collar-local splice and record controls, a conditional fair-block contraction branch for long-run noisy approximate consensus, and a verified rooted-tree packet-net subdomain. Computational universality for growing patch-net families belongs to the consensus paper’s expressive-power boundary and is not a dependency for the recovered-core branches.

Fixed-cutoff packet closure map and invariant simplex. On any declared fixed-cutoff packet quotient Q whose repair relation is terminating and confluent, the normal-form map

$$N : Q \rightarrow Q_{\text{nf}}$$

is well-defined and schedule-independent. It induces an affine OPH closure map on the probability simplex over packet states,

$$\mathcal{C}_Q : \Delta(Q) \rightarrow \Delta(Q), \quad \mathcal{C}_Q \left(\sum_{q \in Q} p_q \delta_q \right) = \sum_{q \in Q} p_q \delta_{N(q)}.$$

Since Q is finite, $\Delta(Q)$ is a nonempty compact convex set and $\mathcal{C}_Q$ is a continuous affine self-map. Its image $\Delta(Q_{\text{nf}})$ is invariant, and $\mathcal{C}_Q^2 = \mathcal{C}_Q$. This is an actual closure map on the fixed-cutoff packet-closed quotient branch. It is not the full Appendix/habitat closure-map theorem for arbitrary OPH

state-and-law data; the first operation not internalized at that level is the general habitat lift from this finite packet quotient to the full compact-convex observer-supporting sector.

Finite quotient ensemble theorem surface. The closure map above is an idempotent normal-form projector. It sends any input law to terminal quotient normal forms, and every law supported on those normal forms is a fixed point. Hence the physical probability law at regulator r must be supplied as an additional quotient-intrinsic object. Let

$$Q_r = \Sigma_r / \Gamma_r$$

be the finite physical quotient, or let $N_r = n_r(Q_r)$ be the quotient-normal-form subset when only settled configurations are intended to carry probability. The declared OPH quotient ensemble is specified by an intrinsic base weight and action

$$m_r : Q_r \rightarrow \mathbb{R}_{>0}, \quad S_r : Q_r \rightarrow \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q)e^{-S_r(q)}, \quad Z_r = \sum_{q \in Q_r} w_r(q), \quad \mu_r(q) = Z_r^{-1}w_r(q).$$

Equivalently, one may state an intrinsic projective prior ν_r on Q_r and set $\mu_r = (n_r)_\# \nu_r$. The choice of m_r , S_r , or ν_r is load-bearing: uniform quotient counting, uniform representative counting pushed to the quotient, finite-groupoid weights, and trace weights of a declared quotient algebra are generally different.

Finite MaxEnt form. For a finite Q , positive m , and quotient observables $F_1, \dots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = - \sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

over the affine constraint surface

$$\sum_q \nu(q) = 1, \quad \sum_q \nu(q) F_a(q) = c_a$$

has a unique full-support solution, when the feasible set has one, of the form

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

This is the finite-dimensional Lagrange-multiplier argument with strict concavity of $\mathcal{H}_m$; boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra the corresponding density operator with faithful reference state σ_r is

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\text{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

Refinement and projective compatibility. For $s \succeq r$, let $c_{sr} : Q_s \rightarrow Q_r$ be the physical coarse-graining map. Exact compatibility of the weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q' : c_{sr}(q')=q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of q ; necessarily $\alpha_{sr} = Z_s/Z_r$. The proof is just the equality

$$(c_{sr})_\# \mu_s(q) = Z_s^{-1} \sum_{q' : c_{sr}(q')=q} m_s(q') e^{-S_s(q')}$$

compared with $Z_r^{-1}m_r(q)e^{-S_r(q)}$. For the selected-prior route, if

$$c_{sr} \circ n_s = n_r \circ c_{sr}, \quad (c_{sr})_{\#} \nu_s = \nu_r,$$

then functoriality gives $(c_{sr})_{\#}(n_s)_{\#} \nu_s = (n_r)_{\#} \nu_r$. Along a chain, finite-stage compatibility defines a unique projective-limit probability measure on the cylinder σ -algebra. If the one-step defects are

$$\delta_{k+1,k} = \|(c_{k+1,k})_{\#} \mu_{k+1} - \mu_k\|_{\text{TV}},$$

then total-variation contraction gives

$$\|(c_{nr})_{\#} \mu_n - \mu_r\|_{\text{TV}} \leq \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

Implementation invariance and representative lifting. If two implementations A, B carry quotient bijections $h_r : Q_r^A \rightarrow Q_r^B$ with

$$m_r^B(h_r q) = m_r^A(q), \quad S_r^B(h_r q) = S_r^A(q), \quad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then $(h_r)_{\#} \mu_r^A = \mu_r^B$. If an implementation stores representatives $\pi_r : \Sigma_r \rightarrow Q_r$, a representative-level law must be a conditional lift

$$\tilde{\mu}_r(x) = \mu_r(\pi_r x) \kappa_r(x \mid \pi_r x), \quad \sum_{x: \pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_{\#} \tilde{\mu}_r = \mu_r$. Uniform representative sampling weights quotient states by orbit size and is physical only if representative counting is the declared base measure. A representative Markov kernel must also be quotient-lumpable: the transition probability to an orbit cannot depend on which representative of the source orbit is used.

Sampler correctness. A production sampler is a separate interface from a deterministic settler:

$$\text{settle}(q) \rightarrow n_r(q), \quad \text{sample}(\mu_r) \rightarrow q.$$

For $w(q) = m(q)e^{-S(q)}$ and a proposal kernel $R(q, q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min \left\{ 1, \frac{w(q')R(q', q)}{w(q)R(q, q')} \right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q),$$

and hence $\mu P = \mu$. Irreducibility and aperiodicity give uniqueness and convergence on the finite state space. If a repair-informed proposal is $q' = R_{\text{repair}}(q) + \eta$ with density g , the Hastings ratio must include

$$\frac{g(q - R_{\text{repair}}(q'))}{g(q' - R_{\text{repair}}(q))},$$

dropping this term generally changes the stationary law. For a lazy reversible finite chain with spectral gap γ ,

$$\|P^n(x, \cdot) - \mu\|_{\text{TV}} \leq \frac{1}{2} \sqrt{\mu(x)^{-1} - 1} (1 - \gamma)^n,$$

and the usual autocorrelation bound follows from the nontrivial spectral radius. Exact small regulators should therefore enumerate Q_r , w_r , P_r , detailed-balance error, stationarity error, refinement residuals, lumpability/orbit-size checks, and the exact spectral gap.

Gaussian screen continuation. For a real reduced screen field space V_r after background and dipole modes are removed, let K_r be positive definite and impose fixed expected quadratic release energy,

$$\mathbb{E}[q] = 0, \quad \mathbb{E}[q^\top K_r q] = d_r A_r, \quad d_r = \dim V_r.$$

The unique maximum-entropy density relative to the declared Euclidean volume is

$$p_r(q) = \frac{\sqrt{\det K_r}}{(2\pi A_r)^{d_r/2}} \exp \left[-\frac{1}{2A_r} q^\top K_r q \right], \quad \text{Cov}(q) = A_r K_r^{-1}.$$

This is the relative-entropy proof against the displayed Gaussian. If instead every sample obeys $q^\top K_r q = d_r A_r$, the MaxEnt law is uniform on the ellipsoid, not Gaussian. Under a linear coarse field map $C_{sr} : V_s \rightarrow V_r$, exact Gaussian refinement is the covariance criterion

$$A_r K_r^{-1} = C_{sr} (A_s K_s^{-1}) C_{sr}^\top.$$

For harmonic truncation, choose real modes $Y_{\ell m}$, $2 \leq \ell \leq L$, set

$$K_L Y_{\ell m} = \kappa_\ell Y_{\ell m}, \quad \kappa_\ell = [\ell(\ell+1)]^{1+\theta/2},$$

and sample independent coefficients

$$a_{\ell m} \sim \mathcal{N}(0, A/\kappa_\ell).$$

Projection from L' to L then discards high modes and is exactly refinement-compatible. For irregular meshes, covariance should be primary under coarse-graining:

$$\Sigma_r = C_{sr} \Sigma_s C_{sr}^\top, \quad K_r = A_r \Sigma_r^{-1}.$$

Equivalently, when eliminating fine variables from a precision matrix

$$K_s = \begin{pmatrix} K_{cc} & K_{cf} \\ K_{fc} & K_{ff} \end{pmatrix},$$

the coarse precision is the Schur complement

$$K_r = K_{cc} - K_{cf} K_{ff}^{-1} K_{fc},$$

not the principal block K_{cc} .

Vacuum and primordial promotion gates. A stationary sampler is not a physical vacuum: the identity transition kernel leaves every measure stationary. A finite vacuum theorem needs a physical transfer operator. If T_r is real symmetric, positive definite, and strictly positive, Perron–Frobenius gives a simple largest eigenvalue $\lambda_{0,r}$ with positive eigenvector Ω_r . Then

$$H_r = -a_t^{-1} \log(T_r/\lambda_{0,r})$$

is positive semidefinite, Ω_r is the unique zero-energy state, and the Doob kernel

$$P_r(x, y) = \frac{T_r(x, y) \Omega_r(y)}{\lambda_{0,r} \Omega_r(x)}$$

is stochastic and detailed-balanced with

$$\mu_r(x) = \frac{\Omega_r(x)^2}{\sum_z \Omega_r(z)^2}.$$

Continuum promotion further requires reflection positivity or an equivalent reconstruction certificate plus refinement compatibility of the transfer family. Likewise, a screen covariance does not by itself determine a primordial curvature spectrum: for finitely many C_ℓ , the thin-shell map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

has an infinite-dimensional null space unless a radial prior or finite parametric family is declared. Conventional free-scalar and lattice-gauge baselines may therefore be recorded, but those baselines carry closed OPH-native-vacuum and primordial-promotion receipts unless these gates are supplied.

4.3.4 Source-only primordial bridge theorem

The finite screen covariance theorem is a screen theorem. To read it as a source-only primordial curvature spectrum one must add the following bridge objects and receipts. The dependency chain is

$$\boxed{\text{nf}(\mathcal{Q}_r) \rightarrow (\chi_r, u_r, h_r) \rightarrow \text{BackgroundCurvatureStatus}_r \rightarrow (g, T_I) \rightarrow q_\zeta = \zeta_\rho \rightarrow \text{single clock} \rightarrow \text{freeze-out} \rightarrow \text{coherent}} \rightarrow \text{coherent}$$

No successful fit of a screen C_ℓ or a TT spectrum may replace one of these hypotheses.

Definition (background curvature branch label). The primordial bridge carries a finite record

$$\text{BackgroundCurvatureStatus}_r = (\text{BranchType}, \kappa, I_K, I_{\Omega_K}, \text{Basis}, \text{Top})$$

where

$$\text{BranchType} \in \{\text{FLATEXACT}, \text{FLATASSUMED}, \text{OPENCURVED}, \text{CLOSEDCURVED}, \text{UNRESOLVED}\}.$$

FLATEXACT is allowed only after a direct theorem or conditional CMH receipt. FLATASSUMED may run flat kernels but its outputs remain assumption-conditioned. UNRESOLVED blocks physical promotion. The H^3 observer-facing atlas is not this record; this record comes from the clocked FLRW slice and its spatial Levi–Civita holonomy or declared branch input.

Definition (source stress readout). On a reconstructed branch with relational metric g_{ab} , let

$$S_{\text{src}} = \sum_I S_I[g, \Psi_I], \quad T_{ab}^I = -\frac{2}{\sqrt{-g}} \frac{\delta S_I}{\delta g^{ab}}, \quad T_{ab}^{\text{tot}} = \sum_I T_{ab}^I,$$

where I ranges over every active sector, including ordinary matter, radiation, neutrinos, OPH anomaly or repair sectors, and any auxiliary source kept in the run. At finite regulator this is a quotient-visible map

$$\text{StressRead}_r : \text{nf}(\mathcal{Q}_r) \longrightarrow \{T_{r,ab}^I\}_I.$$

All source stress components must be read from the same global normal form as the collar scalar.

Source-stress closure theorem. Assume each S_I is relationally diffeomorphism invariant, the source equations hold, sector exchanges satisfy

$$\nabla_a T_I^{ab} = Q_I^b, \quad \sum_I Q_I^b = 0,$$

and StressRead_r descends to the physical quotient and converges. Then

$$\nabla_a T_{\text{tot}}^{ab} = 0.$$

On the recovered Einstein branch,

$$G_{ab} + \Lambda g_{ab} = 8\pi G T_{ab}^{\text{tot}}.$$

The evidence bundle must also report the non-circular comparison

$$E_T^{ab} = T_{\text{src}}^{ab} - \frac{G^{ab} + \Lambda g^{ab}}{8\pi G};$$

using the geometric right-hand side alone is a geometry diagnostic, not a source-only primordial receipt.

Total-energy frame and curvature covector. When T^a_b has a unique future timelike unit eigenvector u^a , define

$$T^a_b u^b = -\rho u^a, \quad h_{ab} = g_{ab} + u_a u_b,$$

and decompose

$$T^{ab} = \rho u^a u^b + p h^{ab} + \pi^{ab}, \quad u_a \pi^{ab} = 0, \quad \pi^a_a = 0.$$

Let $\Theta = \nabla_a u^a$ and let σ_{ab} be the shear. Total energy conservation gives

$$\dot{\rho} + \Theta(\rho + p) + \sigma_{ab} \pi^{ab} = 0, \quad \dot{f} := u^a \nabla_a f.$$

On an expanding branch, define

$$p_{\text{eff}} = p + \frac{\sigma_{ab} \pi^{ab}}{\Theta};$$

then $\dot{\rho} + \Theta(\rho + p_{\text{eff}}) = 0$.

Theorem (exact total-curvature evolution). Let

$$\dot{\alpha} = \Theta/3, \quad \zeta_a = \nabla_a \alpha - \frac{\dot{\alpha}}{\dot{\rho}} \nabla_a \rho,$$

and

$$\Gamma_a^{\text{eff}} = \nabla_a p_{\text{eff}} - \frac{\dot{p}_{\text{eff}}}{\dot{\rho}} \nabla_a \rho.$$

Then, exactly and without a long-wavelength approximation,

$$\mathcal{L}_u \zeta_a = -\frac{\Theta}{3(\rho + p_{\text{eff}})} \Gamma_a^{\text{eff}}.$$

Hence a barotropic effective source, $p_{\text{eff}} = p_{\text{eff}}(\rho)$, conserves ζ_a .

Collar scalar corollary. For relational rods Lie-dragged by the total-energy flow, the collar-volume Jacobian satisfies

$$J_X = \lambda \sqrt{\det \sigma_{AB}}, \quad \frac{d}{d\tau} \frac{1}{3} \log J_X = \frac{\Theta}{3}.$$

Relative to a frozen background,

$$q_\zeta = \frac{1}{3} \log \frac{J_X}{\bar{J}_X}.$$

On the uniform-total-density cut,

$$q_\zeta = \zeta_\rho + \text{constant}.$$

After removing the monopole and dipole,

$$q = \Pi_{\ell \geq 2} q_\zeta = \Pi_{\ell \geq 2} \zeta_\rho, \quad Z_q = 1.$$

This is the source-only quotient scalar on the OPH screen; by itself it has type SCREENCURVATURE, not PRIMORDIALCURVATURE.

Definition (rank-one single-clock branch). Let

$$\mathcal{F}(x) = (\rho, p_{\text{eff}}, \{\rho_I, p_I, Q_I\}_{I=1}^N)$$

be the scalar source map on the source moduli space. A branch is rank-one single-clock when $\text{rank } d\mathcal{F} = 1$, $d\rho \neq 0$, every level set of ρ is connected, all sector velocities agree with u^a , and orthogonal momentum transfer vanishes or is interval-bounded. At finite cutoff this requires a quotient-visible clock χ_r and a factorization

$$\mathcal{F}_r = \widehat{\mathcal{F}}_r \circ \chi_r.$$

Theorem (single-clock adiabaticity). On a rank-one single-clock branch,

$$p_{\text{eff}} = p_{\text{eff}}(\rho), \quad S_a^{IJ} = 3(\zeta_a^I - \zeta_a^J) = 0,$$

and the intrinsic nonadiabatic pressure and transfer covectors obey

$$\Gamma_a^I = 0, \quad \Xi_a^I = 0.$$

Therefore $\mathcal{L}_u \zeta_a = 0$. The proof is the chain rule through the single clock: every source scalar is a function of the same χ , and connected ρ -fibers remove residual branch labels.

Theorem (entropy repair gap). Let the linearized scalar repair/evolution block be

$$\delta Y_{n+1} = L_r \delta Y_n + \eta_n,$$

and let $P_\perp$ project away gauge directions, constrained zero modes, and the adiabatic growing direction. If there is a positive metric H_r such that

$$\|P_\perp L_r P_\perp\|_{H_r} \leq e^{-\gamma_r} < 1,$$

then

$$\|P_\perp \delta Y_n\|_{H_r} \leq e^{-n\gamma_r} \|P_\perp \delta Y_0\|_{H_r} + \sum_{j=0}^{n-1} e^{-(n-1-j)\gamma_r} \|P_\perp \eta_j\|_{H_r}.$$

In the unforced case the orthogonal entropy, decaying, and repair scalar modes vanish. Eigenvalues alone do not certify this theorem for nonnormal repair matrices; a norm bound or a discrete Lyapunov inequality

$$L_\perp^\top H L_\perp \preceq e^{-2\gamma} H$$

is required.

Theorem (growing-mode coherence). Let $X(k)$ collect the scalar initial-condition variables for curvature, radiation, baryons, CDM/anomaly, neutrinos, and repair-sector scalars, and let $v_+(k)$ be the normalized adiabatic growing vector. If the single-clock and repair-gap theorems hold through readout and no independent stochastic source acts afterward, then

$$X(k) = v_+(k) \zeta(k), \quad \mathcal{P}_X(k) = \mathcal{P}_\zeta(k) v_+(k) v_+(k)^\dagger.$$

Thus $\text{rank } \mathcal{P}_X(k) = 1$, source-side isocurvature fractions vanish, and all species share a deterministic growing-mode phase. Finite runs must report λ_2/λ_1 , $\beta_{\text{iso}}(k)$, decaying-mode fraction, and phase-convention defects as source diagnostics.

Theorem (screen-to-primordial bridge). Assume the geometric record and causal-area metric receipts, the collar identity $q_\zeta = \zeta_\rho$, source-stress closure, rank-one normal form, positive repair gap, freeze-out, adiabatic growing-mode, isocurvature, and phase-coherence receipts all pass. Also assume the physical scale-bridge theorem: a source embedding with `source_embedding_hash`, a physical mode basis with `physical_mode_basis_id`, a calibrated comoving coordinate χ , mode-normalization and density-of-states receipts, and a no-post-hoc-calibration receipt, and a `BackgroundCurvatureStatusr` that is not UNRESOLVED. Then the low-mode-removed OPH screen scalar is the source-only primordial curvature field restricted to the declared shell or radial window:

$$q(\hat{n}) = \Pi_{\ell_{\text{src}} \geq 2} \int d\chi W(\chi) \zeta_\star(\chi \hat{n}).$$

For a thin shell, $q(\hat{n}) = \Pi_{\ell_{\text{src}} \geq 2} \zeta_\star(R_\star \hat{n})$. The forward spectrum is exactly

$$C_\ell^q = 4\pi \int_0^\infty d\ln k \Delta_\zeta^2(k) |\Psi_\ell^{\text{BranchType}}(k)|^2, \quad \Psi_\ell^{\text{BranchType}}(k) = \begin{cases} \int dr W(r) j_\ell(kr), & \text{BranchType} = \text{FLATEXACT} \\ \int dr W(r) \Phi_{\ell k}^{(-)}(r), & \text{BranchType} = \text{OPENCURVED} \\ \int dr W(r) \Phi_{\ell k}^{(+)}(r), & \text{BranchType} = \text{CLOSEDCURVED} \end{cases}$$

and for a thin shell

$$C_\ell^q = 4\pi \int_0^\infty d\ln k \Delta_\zeta^2(k) \left| \Psi_\ell^{\text{BranchType}}(k; R_\star) \right|^2.$$

At finite regulator the continuum integral is represented by the finite spectral measure

$$d\nu_r(k) = \sum_j w_{rj} \delta(k - k_{rj}),$$

so the actual finite evidence receipt is

$$C_{\ell_{\text{src}}, r}^q = 4\pi \sum_j w_{rj}^{\log k} \Delta_{\zeta, r}^2(k_{rj}) |\Psi_{\ell_{\text{src}}, r}^{\text{BranchType}}(k_{rj})|^2.$$

Every radial node, window, and mode bin is bound to the geometry hash, source-embedding hash, scale-certificate hash, boundary-condition hash, and mode-basis identifier, and the evidence bundle must emit `RADIAL_KERNEL_GEOMETRY_COMPATIBILITY_RECEIPT`. The observed multipole L_{CMB} is not ℓ_{src} ; it is the output index of the line-of-sight transfer solver. Cap eigenmodes, inverse cap-opening angles, and relabeled unit strings remain diagnostic by the dimensional, angular, cap-mode, and unit-string no-go propositions.

Physical scale dependency. The k_{rj} grid in the preceding equation is not a free producer field. It is consumed only after

`PHYSICAL_SPATIAL_K_RECEIPT`

recomputes the comoving k -intervals from the physical geometry, scale certificate, scale factor, finite eigenvalues, normalization, and refinement residuals. The radial kernel and source-screen projection additionally require

`SCREEN_TO_PHYSICAL_K_ASSOCIATION_RECEIPT`.

The source covariance is a finite positive operator on the common physical mode projectors:

$$C_{\zeta, r} \succeq 0.$$

Homogeneity and isotropy on the safe band require

$$P_I C_{\zeta,r} P_I = \mathcal{P}_\zeta(k_I) P_I + E_I$$

with a declared residual bound, together with an off-diagonal leakage report

$$\sum_{I \neq J} |P_I C_{\zeta,r} P_J|^2.$$

The same physical projector family is used by primordial covariance and anomaly response:

COMMON_PRIMORDIAL_ANOMALY_MODE_BASIS_RECEIPT.

Mode freezeout and common initial surface are separate gates:

PHYSICAL_MODE_FREEZEOUT_MAP_RECEIPT, PHYSICAL_FREEZEOUT_SURFACE_RECEIPT.

The latter contains initial data and normal derivatives on one common spacelike surface.

Radial prior, null space, and residual receipt. The map above is a forward projection, not a free inversion. With finite bins p_j for Δ_ζ^2 ,

$$A_{\ell j} = 4\pi \int_{\text{bin } j} d \ln k \, b_j(k) |\Psi_\ell(k)|^2, \quad C^q = A p.$$

The evidence bundle must declare the basis, support, prior p_0, Q , positivity conditions, singular spectrum of $AQ^{-1/2}$, effective rank, null basis, resolution kernels, prior sensitivity, and every forward residual

$$r_\ell = C_\ell^q - \hat{C}_\ell.$$

Only after the radial prior, null-space report, positivity test, source-only ancestry proof, finite mode-normalization receipt, common primordial/anomaly mode-basis receipt, and forward residual pass may the run emit A_s , n_s , and α_s . Otherwise the typed output is SCREENCURVATURE or CONDITIONALRADIALCONTINUATION. If the geometry packet is imported, the strongest possible claim tier is CONDITIONAL_PHYSICAL; OPH_NATIVE_PHYSICAL requires the native CosmoGeomRead_r theorem.

TOTAL_STRESS_CLOSURE_RECEIPT	$T_I^{ab}, Q_I^a, E_T, \nabla_a T^{ab}$
SINGLE_CLOCK_NORMAL_FORM_RECEIPT	$\chi_r, \text{rank } d\mathcal{F}, \text{ connected fibers}$
ENTROPY_REPAIR_GAP_RECEIPT	$L_\perp, H, \gamma, \text{ forcing/refinement bounds}$
CURVATURE_EVOLUTION_RECEIPT	$\zeta_a, \Gamma_a^{\text{eff}}, \text{ evolution residual, freeze bound}$
ADIABATIC_MODE_RECEIPT	$S_{IJ}, \Gamma_I, \Xi_I, \text{ growing-mode projection}$
ISOCURVATURE_BOUND_REPORT	$\beta_{\text{iso}}(k) \text{ and source covariance}$
PRIMORDIAL_PHASE_COHERENCE_RECEIPT	$\lambda_2/\lambda_1, \text{ phase and decaying-mode defects}$
SCREEN_TO_RADIAL_LIFT_RECEIPT	$W(r), R_\star, \Psi_\ell(k), \text{ source hashes}$
RADIAL_NULL_SPACE_REPORT_RECEIPT	singular values, null basis, resolution kernels
FORWARD_PROJECTION_RESIDUAL_RECEIPT	$r_\ell, \text{ aggregate norms, quadrature and solver errors.}$

Source-provenance and pooled-reducer firewall. Let the source artifact DAG have parent-to-child edges and let M be the set of measurement, residual, likelihood, posterior, fitted-parameter, or measurement-calibrated nodes. Define

$$\tau(v) = \mathbf{1}_{v \in M} \vee \bigvee_{p \in \text{Parents}(v)} \tau(p).$$

Then $\tau(v) = 1$ exactly when v has a forbidden measurement ancestor. The proof is induction over a topological ordering of the DAG. Every promoted source node for $\eta_R, \Gamma_{\text{rec}}, A_\zeta, q_{\text{IR}}, \ell_{\text{IR}}, B_A(k, a), \rho_A(a)$, and N_{CRC} must have $\tau = 0$. Unknown source metadata fails closed.

For nonlinear estimates, shards may not average nonlinear quantities. Each row must emit additive sufficient statistics in a commutative monoid $(\mathcal{M}, \oplus)$, for example

$$\sum w_i, \quad \sum w_i x_i, \quad \sum w_i x_i x_i^\top$$

or, for weighted response regression,

$$X^\top W X, \quad X^\top W y, \quad y^\top W y.$$

Associativity, commutativity, complete coverage, and duplicate removal make the pooled statistic partition-invariant; the deterministic estimator is then evaluated only after pooling.

Three-dimensional homogeneous lift. A rotationally invariant screen covariance about one observer is not a statistically homogeneous three-dimensional primordial field by itself. A frozen lift

$$\mathcal{L}_r : V_r^{\text{screen}} \rightarrow \mathcal{H}_r^{(3)}$$

must define

$$\Sigma_r^{(3)} = \mathcal{L}_r \Sigma_r^{\text{screen}} \mathcal{L}_r^*$$

and certify positivity, refinement convergence, rotation covariance, translation covariance or an equivalent multicenter consistency theorem, and explicit null-space treatment. If the limiting covariance commutes with translations and rotations, Fourier diagonalization gives

$$\langle \zeta_a(\mathbf{k}) \zeta_b^*(\mathbf{k}') \rangle = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{k}') P_{ab}(|\mathbf{k}|),$$

with $P_{ab}(k)$ positive semidefinite.

Transfer-ready initial data. The source-to-transfer map must emit a complete initial state

$$Y(k, \eta_i) = \mathcal{M}_{\text{init}}(k; \mathfrak{B}_{\text{src}}) \xi(k),$$

including metric, photon, polarization, baryon, neutrino, anomaly, recipient, and auxiliary response variables. Near a singular early-time point, regular Frobenius modes solve

$$Y' = \left(\frac{M_{-1}}{\eta} + M_0 + O(\eta) \right) Y, \quad Y = \eta^s (v_0 + v_1 \eta + \cdots), \quad M_{-1} v_0 = s v_0.$$

The branch must classify regular adiabatic growing modes, allowed isocurvature modes, decaying modes, gauge modes, and singular interaction modes. On the adiabatic branch,

$$S_{IJ} = 3(\zeta_I - \zeta_J) = 0$$

for all active sectors. The finite evidence receipt must bound the initial Hamiltonian and momentum constraints and the propagated residual

$$\epsilon_{\text{constraint}} = \sup_{k, \eta} \|\mathcal{C}(k, \eta) Y(k, \eta)\|.$$

Receipt contract. Every ensemble run must record the immutable ensemble id, claim tier, regulator id, representative schema hash, gauge action hash, quotient canonicalizer hash, base measure definition, action coefficients, coarse-map hashes, zero-mode projector hash, amplitude

convention, sampler kernel hash, partition-invariant random-event schema, smoothing policy, source provenance, and explicit nonclaims. The seed belongs to the run receipt, not to the ensemble definition. Claim tiers are:

- $E0$: seed noise, proposal noise, repair jitter,
- $E1$: conventional reference ensemble,
- $E2$: OPH-native quotient ensemble,
- $E3$: OPH vacuum,
- $E4$: OPH primordial field,
- $E5$: observable cosmological prediction.

Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of S_r unless explicitly declared. For bounded coarse observables O , if the implemented law satisfies $\|\hat{\mu}_s - \mu_s\|_{\text{TV}} \leq \epsilon_{\text{samp}}$ and the refinement defects sum to ϵ_{ref} , then

$$\left| \mathbb{E}_{\hat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O] \right| \leq 2\|O\|_{\infty}(\epsilon_{\text{samp}} + \epsilon_{\text{ref}}),$$

which is the finite evidence accuracy statement attached to this theorem surface.

Theorem 2.3 (EC from regulated overlap gluing). For a collar B_{δ} around a cap boundary Σ , there is a canonical decomposition

$$H_{B_{\delta}} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{G_{\Sigma}} = \bigoplus_{\alpha} (H_{b_L^{\alpha}} \otimes H_{b_R^{\alpha}}),$$

and the sector-preserving collar algebra

$$A_{\text{EC}}(B_{\delta}) := \bigoplus_{\alpha} (B(H_{b_L^{\alpha}}) \otimes B(H_{b_R^{\alpha}}))$$

with

$$Z(A_{\text{EC}}(B_{\delta})) = \bigoplus_{\alpha} \mathbb{C} \mathbf{1}_{\alpha},$$

such that $\mathcal{A}(A_{\delta}B_{\delta})$ acts only on $H_{b_L^{\alpha}}$ and $\mathcal{A}(B_{\delta}D_{\delta})$ acts only on $H_{b_R^{\alpha}}$ within each block.

Proof. Split the collar into half-collars B_L and B_R meeting on $\Sigma = \partial C$. By the realized regulator presentation above, the physical collar Hilbert space is the diagonal invariant subspace $(\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{G_{\Sigma}}$. Decompose each side into irreps:

$$\tilde{\mathcal{H}}_{B_L} = \bigoplus_{\alpha} (V_{\alpha} \otimes H_{b_L^{\alpha}}), \quad \tilde{\mathcal{H}}_{B_R} = \bigoplus_{\beta} (V_{\beta}^* \otimes H_{b_R^{\beta}}).$$

Then

$$\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R} = \bigoplus_{\alpha, \beta} (V_{\alpha} \otimes V_{\beta}^*) \otimes (H_{b_L^{\alpha}} \otimes H_{b_R^{\beta}}).$$

By Schur's lemma,

$$(V_{\alpha} \otimes V_{\beta}^*)^{G_{\Sigma}} \cong \begin{cases} \mathbb{C}, & \alpha = \beta, \\ 0, & \alpha \neq \beta. \end{cases}$$

Therefore the invariant subspace is

$$H_{B_\delta} = \bigoplus_{\alpha} (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

as claimed. The induced sector-preserving collar algebra is

$$A_{\text{EC}}(B_\delta) = \bigoplus_{\alpha} (B(H_{b_L^\alpha}) \otimes B(H_{b_R^\alpha})),$$

so the center is generated by the block projectors. Adjacent region algebras act on the left or right factor only because the gluing action is supported on Σ . QED.

Remark. On the central-defect subbranch, replace G_Σ by its central extension. The sector label α then ranges over irreps of the extension; the decomposition is unchanged.

We refer to the decomposition in Theorem 2.3 as **edge-center completion (EC)**.

Exact Markov route. If $I(A_\delta : D_\delta \mid B_\delta) = 0$, the HJPW structure theorem yields a blockwise factorization of the state over *some* direct-sum/tensor decomposition of $\mathcal{H}_{B_\delta}$; that decomposition is state-dependent, and HJPW does not by itself identify it with the preselected EC factors b_L^α, b_R^α of Theorem 2.3. The exact identity used for literal Markov-modular equalities below is the **EC-aligned** normal form

$$\rho_{A_\delta B_\delta D_\delta} = \bigoplus_{\alpha} p_{\alpha} (\rho_{A_\delta b_L^\alpha} \otimes \rho_{b_R^\alpha D_\delta}), \quad I_{\omega}(A_\delta : D_\delta \mid B_\delta) = 0,$$

and the identification of the HJPW factors with the EC factors is a genuine additional state condition, the **Markov-split alignment hypothesis (MSA)**. It is not implied by exact Markovity: with four qubits A, B_L, B_R, D and $\rho = \Phi_{AB_R} \otimes \Phi_{B_L D}$ (Bell pairs across the cut, one center block), one has $I(A : D \mid B) = 0$ but $I(A : B_R) = 2 \ln 2$, so the state is exactly Markov yet not of the displayed EC-aligned form, which would force $I(A : B_R) = 0$. All literal Markov-modular equalities below therefore assume exact Markovity *plus* MSA, or an explicitly stated idealized limit that supplies both; the compact paper states MSA formally and records this counterexample. MSA is checkable state by state: the compact paper proves it equivalent to a splitting of $\log \rho$ into commuting one-sided terms supported on Ab_L^α and $b_R^\alpha D$, to the existence of a ρ -preserving conditional expectation onto either one-sided edge algebra (a Takesaki commuting square over the collar center [30]), and to blockwise vanishing one-sided mutual information $I(Ab_L^\alpha : b_R^\alpha D) = 0$. On the MaxEnt branch this reduces the derivation of MSA to a sharp condition on the effective Hamiltonian: every cross-cut coupling must act through the central sector label, as it does in lattice-gauge-type regulators where the interface energy is a function of the conserved flux. On that declared **central-interface branch** the compact paper derives both MSA and exact collar Markovianity as theorems (descent of one-sided invariants and boundary charges through the edge-center quotient), and it adopts the central-interface normal form as an explicit axiom-level branch clause stated with its MaxEnt axiom, so the literal Markov-modular identities hold on the declared branch without a separate state hypothesis. Whether overlap-consistent repair *forces* that clause remains open, and off the declared branch MSA is carried as an explicit hypothesis.

Interpreting collar refinement as the inductive limit of these regulators with $\delta/\ell_{\text{UV}} \rightarrow \infty$, Theorem 2.3 supplies the kinematic edge-center decomposition. Exact Markovity is one idealized route. The following lemma and axiom provide the quantitative decay route used when the manuscript keeps the approximation explicit.

Lemma 2.6 (MaxEnt with local constraints implies local Gibbs form). Under the local MaxEnt branch of Axiom 3, with its homogeneous global-sum constraints

$\langle \sum_x O_a(x) \rangle = C_a$, and the finite-dimensional regulator realization above, the MaxEnt state has the Gibbs form

$$\omega = \frac{e^{-H_{\text{eff}}}}{\text{Tr } e^{-H_{\text{eff}}}}, \quad H_{\text{eff}} = \sum_x \sum_a \lambda_a O_a(x) + (\text{global terms}),$$

where the sum runs over UV cells x and density labels a , with one cell-independent multiplier λ_a per label. The effective Hamiltonian H_{eff} is quasi-local with range $O(\ell_{\text{UV}})$, and the multiplier count matches the constraint count N_{con} plus the number of retained global terms on every lattice.

Proof. On a finite-dimensional algebra, the unique state maximizing $S(\rho) = -\text{Tr}(\rho \log \rho)$ subject to linear constraints $\text{Tr}(\rho O_i) = c_i$ is given by Lagrange multipliers:

$$\rho = \frac{e^{-\sum_i \lambda_i O_i}}{\text{Tr } e^{-\sum_i \lambda_i O_i}}.$$

Strict concavity of von Neumann entropy ensures uniqueness. Here the constrained operators are the N_{con} global sums $O_a := \sum_x O_a(x)$ plus the finitely many global terms, so the exponent is $\sum_a \lambda_a \sum_x O_a(x) + (\text{global terms})$: a sum of local terms with one shared multiplier per density label. Had the constraints instead been imposed per cell, $\langle O_a(x) \rangle = c_a(x)$, the multipliers would be cell-dependent fields $\lambda_a(x)$ and the family would grow with the lattice; that enlarged family belongs to the non-equilibrium approximation regime of the clarification above and is not the branch used in the refinement argument. QED.

Definition 2.6a (closure defect and induced refinement map). Fix successive regulator scales ℓ (finer) and L (coarser), let $\{\omega_L(\lambda')\}$ be the homogeneous global-sum exponential family at scale L , and let $\Phi_{\ell \rightarrow L}$ be an admissible coarse-graining channel. For fine-scale multipliers λ set $\sigma := \Phi_{\ell \rightarrow L}(\omega_\ell(\lambda))$ and define the **closure defect**

$$\varepsilon_{\ell \rightarrow L}(\lambda) := \inf_{\lambda'} D(\sigma \parallel \omega_L(\lambda')),$$

with D the quantum relative entropy. When the infimum is attained at a unique λ^* , set $R_{\ell \rightarrow L}(\lambda) := \lambda^*$.

Lemma 2.6b (I-projection residual bound). If the constrained operators at scale L together with the identity are linearly independent, then $\lambda' \mapsto D(\sigma \parallel \omega_L(\lambda'))$ is smooth and strictly convex; for faithful σ the minimizer $\lambda^* = R_{\ell \rightarrow L}(\lambda)$ exists, is unique, and is characterized by moment matching $\langle S_c \rangle_{\omega_L(\lambda^*)} = \langle S_c \rangle_\sigma$ for every constrained operator S_c ; and

$$\|\sigma - \omega_L(R_{\ell \rightarrow L}(\lambda))\|_1 \leq \sqrt{2\varepsilon_{\ell \rightarrow L}(\lambda)},$$

so the family reproduces every bounded expectation up to $\|B\|\sqrt{2\varepsilon_{\ell \rightarrow L}(\lambda)}$. The refinement-closure clause of Axiom 3 is exactly the statement $\varepsilon_{\ell \rightarrow L}(\lambda) = 0$ along the realized branch, in which case $\sigma = \omega_L(R_{\ell \rightarrow L}(\lambda))$ and the induced refinement map is this moment-matching I-projection.

Proof. Writing $\log \omega_L(\lambda') = -\sum_c \lambda'_c S_c - \log Z_L(\lambda')$, the objective is $-S(\sigma) + \sum_c \lambda'_c \langle S_c \rangle_\sigma + \log Z_L(\lambda')$, and the Hessian of $\log Z_L$ is the Duhamel (Kubo–Mori) covariance matrix of the constrained operators, positive definite under the stated independence; this gives strict convexity, uniqueness, and the moment-matching first-order condition, while attainment for faithful σ is standard finite-dimensional exponential-family duality [28]. The displayed residual bound is the quantum Pinsker inequality $D(\rho \parallel \tau) \geq \frac{1}{2} \|\rho - \tau\|_1^2$ [29] evaluated at λ^* , and $\varepsilon_{\ell \rightarrow L}(\lambda) = 0$ forces $\sigma = \omega_L(\lambda^*)$ by strict positivity of relative entropy off the diagonal. QED.

Acceptance check. A finite two-lattice test implementing this package—the constraint/multiplier count against the displayed N_{con} -dimensional family on successive lattices, the moment-matching projection, the residual bound, a generic decimation channel with strictly positive closure defect, and the exactly closed transverse-field subfamily where $R_{\ell \rightarrow L}$ acts as the identity—is included with the repository sources (`code/maxent/`).

Exponential mixing hypothesis. There exist constants c and correlation length $\xi = O(\ell_{\text{UV}})$ such that

$$I_\omega(A_\delta : D_\delta \mid B_\delta) \leq c |\partial C|_{\text{UV}} e^{-\delta/\xi}, \quad |\partial C|_{\text{UV}} \sim \frac{\text{length}(\partial C)}{\ell_{\text{UV}}}.$$

This is the standard fixed-cutoff clustering/mixing condition used for local Gibbs states. One sufficient realization on a chosen fixed-cutoff collar family is a Dobrushin-type uniqueness estimate for the corresponding local Gibbs specifications, or a spectral-gap estimate uniform across that family. Used here, it is a collar-local recoverability condition on the selected branch; it does not imply infinite-volume uniqueness and it does not decide whether the separately constructed refinement-limit gauge-sector colimit is trivial or nontrivial.

Theorem 2.5 (Local Gibbs + mixing implies collar refinement). Under Lemma 2.6 and the exponential mixing hypothesis above, the collar double-scaling hypothesis holds in the limit $\delta \rightarrow 0$, $\ell_{\text{UV}} \rightarrow 0$ with $\delta/\ell_{\text{UV}} \rightarrow \infty$.

Proof. The mixing bound above has polynomial growth in $|\partial C|_{\text{UV}}$ and exponential decay in δ/ℓ_{UV} . In the double-scaling limit the exponential dominates, so $I_\omega(A_\delta : D_\delta \mid B_\delta) \rightarrow 0$. QED.

This bound is the quantitative hinge for constructive gluing.

4.3.5 Concrete UV realization: quantum link models

Section 2.3 internalizes the regulator package: the fixed-cutoff type-I algebra and boundary fixed-point structure are the realized form of the screen net plus overlap gluing. Quantum link models are included here only as an explicit microscopic example of that structure, not as a separate axiom layer.

UV regulator. Triangulate S^2 at scale ℓ_{UV} , giving vertices v , oriented links ℓ , and plaquettes p . Refinement corresponds to $\ell_{\text{UV}} \rightarrow 0$ with increasing lattice size.

Degrees of freedom. Attach to every oriented link ℓ a **finite-dimensional** Hilbert space $\mathcal{H}_\ell$. In ordinary Wilson lattice gauge theory, the continuum/refinement-limit edge description is modeled by $L^2(G)$ (infinite-dimensional for continuous G), but the microscopic OPH regulator is

instead a **quantum link model** with finite-dimensional link Hilbert spaces that preserve gauge symmetry in operator form [47]. Optionally attach matter Hilbert spaces $\mathcal{H}_v$ at vertices. Then:

$$\tilde{\mathcal{H}}_{\text{total}} = \bigotimes_{\ell} \mathcal{H}_{\ell} \otimes \bigotimes_v \mathcal{H}_v,$$

finite-dimensional on any finite lattice. This is a concrete realization of the extended type-I presentation used in Section 2.3.

Boundary gluing as Gauss-law invariants. Define a local gauge transformation group G_v at each vertex v acting on incident links (and matter at v). Physical states satisfy:

$$|\psi\rangle \in \mathcal{H}_{\text{phys}} \iff U(g_v)|\psi\rangle = |\psi\rangle \quad \forall v, g_v \in G_v.$$

Equivalently: $\mathcal{H}_{\text{phys}} = \tilde{\mathcal{H}}_{\text{total}}^{\prod_v G_v}$.

Region algebras. For any region $R \subset S^2$, define an extended Hilbert space $\tilde{\mathcal{H}}_R$ from the links/vertices in R . The **boundary gauge group** $G_{\partial R}$ acts on the cut degrees of freedom (the "half-links" ending on ∂R). Define:

$$\mathcal{A}_{\text{inv}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R)^{G_{\partial R}}.$$

This is the concrete quantum-link instance of the lifted fixed-point presentation used in Section 2.3. The same gauge constraints then induce the sector-preserving EC algebra on collars.

Why EC follows immediately. Take a cap C and a collar B_{δ} around ∂C . Because the *only* coupling between inside and outside is through the boundary gauge constraint, the collar Hilbert space decomposes into superselection blocks labeled by boundary irreps:

$$\mathcal{H}_{B_{\delta}} \cong \bigoplus_{\alpha} (H_{b_L^{\alpha}} \otimes H_{b_R^{\alpha}}),$$

with center generated by the projectors P_{α} . This is precisely the Schur-lemma mechanism of Theorem 2.3. The labels α are the familiar edge-mode / electric-flux labels appearing whenever one factorizes gauge theories across an entangling cut [48]. Exact Markovity depends on the state and is not forced by the decomposition alone; nor does exact Markovity align the HJPW factors with these edge-mode labels, which is the separate Markov-split alignment hypothesis of Section 2.3.

Dynamics and MaxEnt. The natural Hamiltonian is a 2+1D lattice gauge Hamiltonian on the screen worldvolume: plaquette ("magnetic") terms, electric terms on links, vertex Gauss terms as constraints, plus local matter couplings. In quantum link form this is finite-dimensional per link while behaving like gauge theory in the continuum limit. Then the MaxEnt assumption becomes concrete: the MaxEnt state is a Gibbs state $\rho \propto e^{-\sum_i \lambda_i O_i}$ with quasi-local O_i , precisely the local-Gibbs regime.

Geometry and G . This microphysics naturally supplies the emergent geometric objects:

- **Edge entropy / area operator:** $L_C = \sum_{\alpha} (\log d_{\alpha}) P_{\alpha}$ becomes "log of boundary irrep dimension" in the gauge link model.
- **Newton constant G :** the conversion factor between edge entropy density per boundary UV cell and macroscopic geometric area.

Thus area is an operator living in the center of the boundary algebra, because in gauge systems the center is where the cut labels live.

Scope of this example. The quantum-link realization makes the fixed-cutoff matrix/fixed-point bookkeeping concrete. Modular flow on caps becomes geometric conformal dilation with the 2π KMS normalization through the support-visible BW scaling theorem on the prime geometric subnet. Viable architectures for this include holographic quantum error-correcting codes [49] and quantum double / string-net Hamiltonians [50], but any use of QECC distance, min-cut resilience, or Knill–Laflamme correction requires the corresponding code subspace, logical-operator, error-family, and recovery certificate.

4.3.6 Conformal-modular fixed-point microphysics

On the local finite-constraint MaxEnt branch of the third axiom, the logarithm of the selected state is a quasi-local UV generator, and the refinement-stable branch lies inside one common finite-dimensional multiplier family. At each regulator stage the patch and cap algebras are type-I matrix algebras. The finite regulator class is not refinement-closed, so the scaling-limit observer algebra may leave that class. The fixed-cutoff analysis proves a local-Gibbs/refinement-stable branch with propagation and endpoint-Lipschitz control. The prime geometric cap pair with geometric modular flow is obtained in the support-visible sense: the target algebra is the extracted geometric subnet, the support-visible modular matrix elements converge under regularization, and weak-*/GNS extraction plus support-readable modular covariance and ordered cut-pair rigidity gives the geometric cap automorphism.

Local finite-constraint MaxEnt branch. The constraint family $\mathcal{C}$ is generated by finitely many gauge-invariant local densities $\{O_a(x)\}$ of UV range $O(\ell_{UV})$, constrained through their homogeneous global sums $\langle \sum_x O_a(x) \rangle = C_a$, with the same finite label set retained under refinement by the refinement-closure clause of the third OPH axiom. This clause unpacks that axiom at regulator level and adds no postulate beyond it; the closure clause itself is the substantive assumption.

Theorem 2.6 (Local constraints imply a local-Gibbs form). If the MaxEnt constraints are expectations of finitely many quasi-local operators $\{O_a\}$ with bounded support size at scale ℓ_{UV} , then the entropy maximizer is

$$\omega \propto \exp\left(-\sum_a \lambda_a O_a\right),$$

so the MaxEnt generator $H_{\text{MaxEnt}} = -\log \omega$ is a UV-range quasi-local sum. This is exactly the local-Gibbs form used later.

Proof. Standard exponential-family result: maximum entropy subject to linear constraints $\langle O_a \rangle = c_a$ yields the Gibbs state with Lagrange multipliers λ_a . QED.

Derived propagation control on the same branch. Because H_{MaxEnt} is a finite-range or quasi-local sum on a finite type-I regulator net, standard Lieb–Robinson estimates [13] apply to the automorphism group it generates, or to any branch generator lying in the same bounded-support algebraic closure. Thus there is a finite propagation velocity v_{LR} and constants C, ξ such that for local observables A_X, B_Y ,

$$\|[\tau_t(A_X), B_Y]\| \leq C \|A_X\| \|B_Y\| \min(|X|, |Y|) e^{-(d(X,Y) - v_{LR}|t|)/\xi}.$$

Because changing a bounded interval endpoint only adds or removes an $O(|\Delta v|)$ collar of local terms once the central endpoint piece is separated off, the same local branch also gives bounded-interval endpoint-Lipschitz matrix elements,

$$|\langle \psi, (K[I'] - K[I])\phi \rangle| \leq C_{\psi, \phi, I_{\max}} |I' \Delta I|,$$

used later in the null-modular bridge. So quasi-local propagation and endpoint-Lipschitz control are not external regularity selectors; they are the local-constraint MaxEnt branch written in dynamical form.

Refinement-stable multiplier branch. By the refinement-closure clause of the third OPH axiom—a substantive renormalization condition, since coarse-graining a finite-range Gibbs family generically generates interactions outside any fixed finite density list—the same finite constraint family is preserved under coarse-graining, and the regulator states lie in one common finite-dimensional multiplier family rather than in unrelated state spaces at different cutoffs. The “refinement-stable” language used later therefore means persistence along the stable or fixed branch of this multiplier family. This state-side notion is enough to compare realized states across cutoffs. `FixedCutoffBosonicSectorCategory` constructs the fixed-stage tensor-generated category and its forgetful fiber. `RefinementFunctorAndFiberDescent` constructs the refinement pullbacks and compatible fibers only on a cofinal tail carrying the compact-gauge refinement receipt. The cofinal compact-gauge witness uses the same receipt, while the Standard Model selection step is the later MAR application to sector packages. The third OPH axiom supplies the realized state branch; it does not supply the matrix-algebra embeddings, compact-group surjections, or tensor-realization data needed for the sector colimit.

Scaling limit and algebraic type. The regulator presentation gives a family of finite type-I algebras

$$\mathcal{A}_\ell(C) \cong \mathcal{B}(\mathcal{H}_{C,\ell}).$$

A scaling limit of this family need not be type I. In the continuum-QFT case of interest one expects the local limit algebra $\mathcal{A}_\infty(C)$ to be non-type-I, typically type III. Accordingly the fundamental modular datum in the limit is the automorphism group of the pair $(\mathcal{A}_\infty(C), \omega_\infty^C)$, not a density matrix inside $\mathcal{A}_\infty(C)$.

Corollary 2.6 (Geometric modular action on certified caps). For any support-visible extracted scaling-limit geometric cap pair $(\mathcal{A}_\infty^{\text{geo},\text{sv}}(C), \omega_\infty^{\text{geo},C})$ emitted by the finite cap-normal BW certificate theorem, let $\alpha_{\lambda_{\widehat{C}}(s)}$ be the automorphism induced by the independently normalized BW-framed cap flow. Then

$$\sigma_t^{\omega_\infty^{\text{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

If the limit cap algebra happens to be type I, this may be written as

$$K_C = 2\pi B_C + Z_C, \quad Z_C \in Z(\mathcal{A}(C))_{\text{sa}}.$$

In the generic continuum case, the automorphism identity is the complete statement.

Proof. This is the compact paper’s finite cap-net BW theorem applied to the extracted prime geometric cap pair. The theorem’s hypotheses include BW framing, support-order faithfulness, cap-family-uniform mixed-GNS convergence, held-out oriented cross-ratio rigidity, independently normalized geometric 2π -KMS convergence, and wrong-normalization controls. QED.

Alternative derivation via net regularity. The same modular-covariance property can also be read off from a scaling-limit support map when the net satisfies the outer-regularity / minimal-support condition used later. This clarifies how the geometric labeling of the support-visible extracted limit net is read.

(NR) Outer regularity / minimal support. For any operator O , the intersection of all connected regions P with $O \in \mathcal{A}(P)$ is again a connected region, denoted $\text{supp}(O)$.

Proposition 2.7 (Modular covariance from net regularity). Under (NR), define for any region $R \subset C$

$$f_t^C(R) := \bigcup_{O \in \mathcal{A}(R)} \text{supp}(\sigma_t^{\omega, C}(O)).$$

Then $\sigma_t^{\omega,C}(\mathcal{A}(R)) = \mathcal{A}(f_t^C(R))$, which is exactly the desired modular-covariance property.

Proof. Since $\sigma_t^{\omega,C}$ is an automorphism of $\mathcal{A}(C)$, and (NR) allows one to read support from the net labeling, the map $R \mapsto f_t^C(R)$ is well-defined and consistent. QED.

Null-surface modular structure. On the same support-visible scaling branch and extracted geometric-subnet branch, the null-surface modular machinery narrows as follows:

- **Fixed-cutoff null-strip bridge.** The null-strip package proves transferred cut-center data, the theorem-local inherited left/right strip-split condition needed for the spatial-collar-type tensor decomposition, exact-or-controlled four-term strip additivity on one inherited strip model, renormalized endpoint-Lipschitz control up to the weak tail generator, and the derived half-sided modular pair whose Borchers–Wiesbrock consequence is an explicit positive null-translation generator on its Stone domain with the affine half-line modular relation; the same half-line family then fixes the generator/charge identification internally.
- **Derived half-sided modular inclusion.** Nested null half-line algebras satisfy half-sided modular inclusion on the declared geometric scaling branch by Corollary 5.2e; Borchers–Wiesbrock then identifies the positive null-translation generator on its Stone domain together with the affine half-line modular relation [17].
- **Weak continuity and finite variation.** The bounded-interval and half-line endpoint-Lipschitz control follow from the local MaxEnt branch and define the weak tail generator at fixed cutoff. The support-visible scaling theorem together with the extracted geometric cap pair supplies the continuum null-generator setting in which that weak-tail data can be matched to the geometric null modular action of the relativistic phase.

Constraint set specification. On the local finite-constraint branch, the “correct fixed-cap constraint set” becomes explicit: constraints are the local conserved charges of the symmetries used in the derivation:

1. **Edge/cap label constraints:** fix the distribution of collar-sector labels, equivalently $\langle L_C \rangle$ for each cap size, giving the area term.
2. **Gauge charges:** fix boundary flux or charge operators.
3. **Geometric (conformal) charges:** fix the expectation of the conformal Killing charges that preserve the cap, i.e. the generator B_C or its microscopic lattice approximation.

MaxEnt therefore selects the unique finite-stage invariant state compatible with those conserved charges. The support-visible BW scaling theorem is the continuum statement: if the limit algebra is non-type-I, the geometric modular action on that branch is outer rather than an inner density-matrix identity.

QNEC internalization. QNEC has rigorous QFT proofs in broad settings [26]. On the support-visible scaling branch with the extracted geometric cap pair in place, the Recoverable Generalized Entropy axiom can be supported internally by:

- EC + MaxEnt derive $S_{\text{gen}} = S_{\text{bulk}} + \langle L_C \rangle$ (Section 5.4).
- On the null-stress and Einstein theorem branch, focusing becomes a semiclassical consequence in the same scaling regime.

Summary. The CMFP package therefore separates the dependency structure into two layers:

- **Internal branch consequences:** the local-Gibbs form, quasi-local propagation, bounded-interval endpoint-Lipschitz control, and the realized state-side refinement branch along which later transport questions are asked, all from the local finite-constraint MaxEnt branch.
- **Support-visible scaling statement:** the scaling limit emits the support-visible prime geometric cap pair and ordered cut-pair rigidity holds on it. On that branch the limit algebra may be non-type-I, the modular action may be outer, and the 2π normalization and later null half-sided-inclusion bridge follow.

The theorem route works on the support-visible quotient. The realized transported cap-local system packages the geometric cap-local test family, the projectively compatible transported marginal family, and the asymptotic transport-equivalence certificate. The regularized transport theorem below replaces the unavailable full-algebra lower spectral floor. Local weak-* extraction and GNS gluing then emit the support-visible scaling-limit cap pair, support-readable modular covariance reads the modular group as a cap-local support map, and ordered cut-pair rigidity collapses the residual cap-preserving conformal freedom to the unique hyperbolic subgroup.

Proposition 2.6a (Recoverability is not modular geometry: common-floor collapse countermodel). Exact or asymptotically exact Markov recovery at finite cutoff does not by itself imply a full-algebra lower spectral floor along refinement. In a two-dimensional matrix algebra, let

$$\rho_n = \begin{pmatrix} e^{-n} & 0 \\ 0 & 1 - e^{-n} \end{pmatrix}.$$

Each ρ_n is faithful at finite n , and tensoring it with any fixed finite exact-Markov collar factor gives a full-rank exact-Markov collar family. Nevertheless $\lambda_{\min}(\rho_n) = e^{-n} \rightarrow 0$, so there is no positive lower bound along the refinement family. On the off-diagonal matrix unit E_{12} , the modular generator carries the logarithmic gap

$$[-\log \rho_n, E_{12}] = n E_{12} + O(1)E_{12},$$

and the unregularized modular transport has no finite common-floor limit on that direction. Thus collar Markovity and finite-stage faithfulness are not enough for the false stronger full-algebra BW lift. The observer-facing theorem uses the support-visible regularized replacement, with the exact-Markov comparison family checked only on fixed collars and made replacement-independent after weak-*/GNS extraction.

Theorem 2.6b (regularized support-visible modular transport). Fix a local collar model of finite dimension $d_{m,\delta}$. Let ρ_n be the transported physical collar marginal, $\hat{\rho}_n$ the exact-Markov comparison marginal on the same collar model, and $\Delta_n = \|\rho_n - \hat{\rho}_n\|_1$. For $a > 0$, set $K_a(\rho) = -\log(\rho + a\mathbf{1})$. For every bounded collar observable O in the support-visible algebra,

$$|\mathrm{Tr} \rho_n O(K_a(\rho_n) - K_a(\hat{\rho}_n))| \leq \|O\| \left(\frac{4\Delta_n}{a} + d_{m,\delta}a + 4\Delta_n |\log a| \right).$$

Consequently, if $a_n \downarrow 0$ is chosen with

$$\Delta_n/a_n \rightarrow 0, \quad d_{m,\delta}a_n \rightarrow 0, \quad \Delta_n |\log a_n| \rightarrow 0,$$

then the regularized support-visible modular matrix elements converge on that fixed collar model.

Proof sketch. On the spectral interval $[a, \infty)$, the logarithm is operator-Lipschitz at the scale used above by its integral representation. Splitting the comparison into the support above a , the a -tail, and the trace-distance error gives the displayed bound. The three displayed conditions make the three error terms vanish. QED.

Definition 2.6c (explicit multiresolution reference branch). The finite-to-continuum bridge is evaluated on a declared reference branch, not on an unstructured sequence of finite plots. A regulator is

$$r = (m, L, b),$$

where m is the refinement depth, L is the finite support or volume scale, and b is a boundary/phase label. The screen branch uses nested icosahedral subdivisions of S^2 ; Euclidean branches use dyadic mesh $a_m = a_0 2^{-m}$, with an additional temporal spacing a_t when a transfer matrix is present. Each added cell, collar, edge-center, or support-visible channel j carries

$$D_j = \bigoplus_{\alpha \in S_j} M_{d_{j,\alpha}}(\mathbb{C})$$

and a faithful detail state τ_j . At regulator r ,

$$\widetilde{M}_r = \bigotimes_{j \in \mathcal{J}_r} D_j, \quad \widetilde{\omega}_r = \bigotimes_{j \in \mathcal{J}_r} \tau_j.$$

A finite-depth local presentation circuit $\Gamma_r : \widetilde{M}_r \rightarrow M_r$ changes from multiresolution coordinates to the OPH patch, port, collar, and gauge coordinates. For $r \preceq s$, write

$$\widetilde{M}_s = \widetilde{M}_r \otimes D_{s \setminus r}, \quad \tau_{s \setminus r} = \bigotimes_{j \in \mathcal{J}_s \setminus \mathcal{J}_r} \tau_j.$$

The canonical refinement and coarse-graining maps are

$$\iota_{rs}(A) = \Gamma_s \left[\Gamma_r^{-1}(A) \otimes \mathbf{1} \right],$$

and

$$Q_{sr}(X) = \Gamma_r \left[(\text{id} \otimes \tau_{s \setminus r})(\Gamma_s^{-1}(X)) \right].$$

The embedded conditional expectation is $E_{sr} = \iota_{rs} Q_{sr}$. The reference state is $\widehat{\omega}_r = \widetilde{\omega}_r \circ \Gamma_r^{-1}$.

Theorem 2.6d (exact modular-compatible reference tower). The data

$$(M_r, \widehat{\omega}_r, \iota_{rs}, E_{sr})_{r \preceq s}$$

form a directed tower. The maps ι_{rs} are unital injective $*$ -homomorphisms with $\iota_{st} \iota_{rs} = \iota_{rt}$. The Q_{sr} are faithful state-preserving completely positive coarse-grainings with $Q_{tr} = Q_{sr} Q_{ts}$. The maps E_{sr} are faithful conditional expectations onto $\iota_{rs}(M_r)$, and

$$\widehat{\omega}_s \circ \iota_{rs} = \widehat{\omega}_r, \quad \widehat{\omega}_r \circ Q_{sr} = \widehat{\omega}_s, \quad \widehat{\omega}_s \circ E_{sr} = \widehat{\omega}_s.$$

Moreover the finite reference modular groups are exactly compatible:

$$\sigma_t^{\widehat{\omega}_s}(\iota_{rs} A) = \iota_{rs}(\sigma_t^{\widehat{\omega}_r}(A)).$$

This theorem uses no full-algebra spectral floor uniform in r ; in bare coordinates it is only the identity $A \mapsto A \otimes \mathbf{1}$, partial expectation against faithful detail states, and conjugation by Γ_r, Γ_s .

Theorem 2.6e (canonical renormalization and correlation Cauchy bound). Let $\mathfrak{A}$ be the C^* -inductive limit with embeddings $j_r : M_r \rightarrow \mathfrak{A}$. There is a unique state $\widehat{\omega}$ with $\widehat{\omega} \circ j_r = \widehat{\omega}_r$, and the maps Q_{sr} induce state-preserving conditional expectations

$$\mathbb{E}_r : \mathfrak{A} \rightarrow j_r(M_r)$$

such that $\|\mathbb{E}_r(A) - A\| \rightarrow 0$ for every $A \in \mathfrak{A}$. The finite-regulator realization of a continuum test observable is

$$\mathcal{R}_r(A) = j_r^{-1}(\mathbb{E}_r A).$$

For $\|A_k\| \leq M$,

$$\left| \widehat{\omega}_r \left(\prod_{k=1}^n \mathcal{R}_r(A_k) \right) - \widehat{\omega} \left(\prod_{k=1}^n A_k \right) \right| \leq M^{n-1} \sum_{k=1}^n \|A_k - \mathbb{E}_r A_k\|.$$

Thus the operator mixing matrix and additive subtraction at finite cutoff are outputs of $\mathcal{R}_r$, not free fits to the target continuum answer. Independent lattice-spacing, volume, and boundary sweeps must publish the corresponding martingale-tail or physical-error envelopes before finite regulator data are read as continuum evidence.

Theorem 2.6f (compact-time modular convergence on fixed local algebras). Let $\rho_{s\downarrow r}^{\text{phys}}$ be the density matrix of the physical state at fine stage s , restricted to the embedded fixed local algebra M_r , and let

$$\epsilon_{s,r} := \|\rho_{s\downarrow r}^{\text{phys}} - \widehat{\rho}_r\|_1.$$

For a regularizer $\eta_s > 0$, define

$$K_{s\downarrow r}^{(\eta_s)} = -\log(\rho_{s\downarrow r}^{\text{phys}} + \eta_s \mathbf{1}).$$

If $\lambda_r = \lambda_{\min}(\widehat{\rho}_r) > 0$, then for every $A \in M_r$ and every $T < \infty$,

$$\sup_{|t| \leq T} \left\| e^{-itK_{s\downarrow r}^{(\eta_s)}} A e^{itK_{s\downarrow r}^{(\eta_s)}} - \sigma_t^{\widehat{\omega}_r}(A) \right\| \leq 2T\|A\| \left[\frac{\epsilon_{s,r}}{\eta_s} + \log \left(1 + \frac{\eta_s}{\lambda_r} \right) \right].$$

Hence $\eta_s \rightarrow 0$ and $\epsilon_{s,r}/\eta_s \rightarrow 0$ give uniform-on-compact-time convergence on each fixed support-visible local algebra. For a countable cofinal local test tower, pointwise $\epsilon_{s,r} \rightarrow 0$ admits a diagonal subsequence and one cutoff schedule. The phrase “finite-stage modular action” below means this transported, regularized local action; it does not mean the unregularized full finite-cap modular group.

Cap-family uniformity clause. For application to the finite cap-net BW theorem, the compact-time estimate is required uniformly over the declared finite nondegenerate cap family and the fixed separating support-visible test tower. The operator-norm estimate then implies the quadratic mixed-GNS Cauchy residual because $\|XD\|_{2,\omega} \leq \|X\|\|D\|_{2,\omega}$. The reference tower supplies the analytic engine; the finite cap-normal certificate adds the support-order, BW-frame, oriented cross-ratio, geometric 2π -KMS, and wrong-scale controls.

Theorem 2.6g (finite positive transfer and Lorentzian handoff gate). On Euclidean repair branches, let $P_{C,r}$ be weighted resampling inside the fibers of the complete repaired visible datum for active collar C . Ref. [1] proves that this finite operator is the orthogonal conditional expectation and gives the exact support, equal-fiber-row, and weighted detailed-balance matrix receipt that recognizes it. A concrete active collar repair kernel is identified with $P_{C,r}$ only when its independently derived transition matrix passes that receipt; constructing the matrix from the target formula is not a verification. Choose $c_{C,r} \geq 0$, and set

$$L_r^{\text{rep}} = \sum_C c_{C,r}(I - P_{C,r}), \quad T_r(a_t) = e^{-a_t L_r^{\text{rep}}}.$$

Then $L_r^{\text{rep}} = L_r^{\text{rep}*} \geq 0$, $0 < T_r(a_t) \leq I$, $T_r(a_t)\mathbf{1} = \mathbf{1}$, and the stationary Euclidean path measure is reflection positive:

$$\mathbb{E}_r[(\Theta F)F] \geq 0$$

for positive-time cylinder functions F . A Lorentzian unitary statement requires this finite positivity plus either exact shell-additive direct-limit semigroup identities or a Mosco / strong-resolvent substitute, including vacuum-projection and intertwiner convergence. Weak-* state convergence alone is not a transfer theorem.

BW-side closure boundary. Proposition 2.6a blocks the stronger unregularized full-algebra conclusion from the proved finite-cutoff package. Theorems 2.6c–2.6g supply the explicit reference-tower certificate needed for the finite-to-continuum reading: continuum correlations come from the conditional-expectation renormalization map, compact-time modular group convergence is only for transported regularized physical marginals on fixed support-visible local algebras, and Lorentzian transfer needs reflection positivity or an equivalent positive-transfer certificate. Theorem 4.2 then uses exactly the observer-facing support-visible content to close the BW/geometric cap-pair automorphism statement. Without the multiresolution regulator certificate, finite regulator data remain regulator evidence. A black-box AQFT/BW reconstruction route would require separately verifying conformal-net properties such as locality, additivity, duality, standardness, positive energy, and suitable modular inclusions.

4.4 Overlap Consistency and Gluing

The constructive part of overlap consistency is the tree-gluing theorem below. The structural part is the origin of the gluing redundancy itself. On the ordinary or central-defect branch, gauge-as-gluing is the finite-regulator overlap redundancy of local chart presentations: once the overlap algebras are realized in finite-dimensional charts, overlap-consistent rechartings of a cut form a compact unitary transition system. The collar theorem of Section 2.3 should therefore be read with its boundary group G_Σ understood as shorthand for this derived compact boundary action, not as a model-specific add-on. On the genuinely noncentral branch, the same weak gluing data are encoded by a compact crossed-module change system, so the fixed-cutoff collar theorem upgrades to a higher-gauge statement rather than failing. The fixed-cutoff topological package is therefore closed on all three branches. A second structural point is UV underdetermination. If each patch Hilbert space is tensored with an inert finite ancillary factor and the observable patch algebras are embedded as $\mathcal{A}(P) \otimes \mathbf{1}$, then the physical observables, collar conditional mutual information, carried Markov errors, and quotient normal forms are unchanged. So the physical UV branch is fixed only modulo gauge or implementation hiding together with such ancillary stabilization, not a unique microscopic presentation; the unique theorem-grade object is the induced family of terminal expectation functionals on the declared physical observable algebras, not a preferred microscopic representative.

4.4.1 Constructive gluing on tree covers

Theorem 3.1 (tree gluing). Let a rooted tree of patches satisfy a tree-ordered overlap structure and a tripartite factorization (A_k, B_k, C_k) at step k . If a target state ρ^* obeys $I(A_k : C_k \mid B_k) \leq \varepsilon_k$, then there exist recovery maps $\mathcal{R}_k$ such that

$$\|\rho_{A_k B_k C_k}^* - (\text{id}_{A_k} \otimes \mathcal{R}_k)(\rho_{A_k B_k}^*)\|_1 \leq \delta_k,$$

with

$$\delta_k = 2\sqrt{\ln 2 \varepsilon_k}.$$

The iteratively glued state $\hat{\rho}$ satisfies

$$\|\hat{\rho} - \rho^*\|_1 \leq \min \left(2, \sum_{k=2}^n \delta_k \right).$$

Proof. Induct on k . The recovery error contracts under CPTP maps, so the errors add. QED.

4.4.2 Gauge-as-gluing and loops

At finite regulator scale, the fixed-cutoff gauge-as-gluing package identifies the overlap-consistency redundancy of local chart presentations. Choose finite-dimensional local presentations of the overlap algebras on a connected cut Σ . Because the overlap algebras are matrix algebras, any overlap-consistent change of chart is inner and is implemented by a unitary on the cut Hilbert space. Fixing a reference chart, the compact closure of the subgroup generated by all such recharting unitaries is the boundary gluing group K_Σ ; before fixing the reference chart, the same data form a compact unitary groupoid.

Proposition 3.2a (Derived gauge-as-gluing at finite regulator). Let a finite regulator chart be chosen for the patches meeting along a connected interface Σ . Then overlap consistency determines a compact unitary transition system on the cut data. On the ordinary or central-defect branch, this transition system reduces to a compact boundary group K_Σ , and when the triple-overlap defect is central its projective composition law lifts to a genuine action of a compact central extension $\widehat{K}_\Sigma$. Gauge is therefore the overlap redundancy itself, not an extra primitive.

Proof. On a finite-dimensional matrix algebra every $*$ -automorphism is inner, so each overlap-consistent recharting is conjugation by a unitary on the cut Hilbert space. The subgroup generated by those unitaries has compact closure inside a finite-dimensional unitary group. If triple-overlap defects are central, the resulting projective composition law lifts to a central extension. QED.

Lemma 3.2b (trees vs loops). If the patch adjacency graph is a tree, one can choose local charts h_i so that the overlap labels satisfy $g_{ij} = h_i^{-1} h_j$ on all edges. If loops exist, the loop holonomy

$$H(\gamma) = g_{i_1 i_2} g_{i_2 i_3} \cdots g_{i_n i_1}$$

is invariant under local frame changes. Nontrivial holonomy is the obstruction to global trivialization. QED.

Corollary 3.2c (Collar consequence on the ordinary or central-defect branch). Let $B_\delta = B_L \cup B_R$ be a collar around a cap boundary Σ , and set $\widehat{K}_\Sigma = K_\Sigma$ on the ordinary branch. Then the EC theorem of Section 2.3 has the derived interpretation

$$H_{B_\delta} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\widehat{K}_\Sigma} \cong \bigoplus_{\alpha} (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

with

$$Z(A_{\text{EC}}(B_\delta)) = \bigoplus_{\alpha} \mathbb{C} \mathbf{1}_\alpha.$$

The right half-collar carries the contragredient action because it sees the inverse transport across the same cut. Exact Markov normal forms used later require the additional state hypothesis $I_\omega(A_\delta : D_\delta \mid B_\delta) = 0$ *together with* the Markov-split alignment hypothesis of Section 2.3, or the explicitly stated idealized recoverability limit supplying both; the block decomposition itself is kinematic and follows from the derived boundary action.

Proof sketch. Decompose the left boundary data into irreps $(V_\alpha \otimes H_{b_L^\alpha})$ of $\widehat{K}_\Sigma$ and the right boundary data into the dual modules $(V_\beta^* \otimes H_{b_R^\beta})$. Then Schur's lemma leaves a singlet only when $\alpha = \beta$, producing the displayed direct sum. QED.

4.4.3 Loop obstruction class (central defect)

On the central-defect subbranch, identify the overlap centers with one fixed abelian unitary coefficient group Z_Σ and assume overlap transport acts trivially on it. Choose unitary overlap implementers normalized by $U_{ji} = U_{ij}^{-1}$ and define central defects z_{ijk} at the implementer level by

$$\Omega_{ijk} := U_{ij}U_{jk}U_{ki} = z_{ijk} \in Z_\Sigma,$$

or equivalently $U_{ij}U_{jk} = z_{ijk}U_{ik}$. This implementer-level relation records the projective multiplier; the automorphism-level expression $\text{Ad}(z_{ijk})$ alone would be the identity and would not determine z_{ijk} .

Then $\{z_{ijk}\}$ is an untwisted Čech 2-cocycle with coefficients in Z_Σ , and its cohomology class $[z]$ is gauge invariant. Central defects do not obstruct the collar block decomposition: they only replace K_Σ by its central extension $\widehat{K}_\Sigma$. The condition $[z] = 0$ is exactly the strictifiability of the projective triangle defect. It is not by itself global path independence: at least one resulting strict edge 1-cocycle must also have trivial represented holonomy around every closed overlap loop. With a nontrivial action on varying centers, the corresponding statement uses the twisted/local-system Čech differential instead. (A full proof for the fixed-coefficient subbranch appears in Section 6.4 below, in the algebra-net language.)

4.4.4 Non-central obstruction (2-group cocycle)

When defects are not central, the natural coefficient data is a crossed module $(H \rightarrow G)$ with an action of G on H by conjugation. Here G is the reconstructed gauge group, and H is the unitary group acting on edge multiplicity spaces, with boundary map $\partial : H \rightarrow G$.

A crossed module is a homomorphism $\partial : H \rightarrow G$ together with an action of G on H such that

$$\partial(g \triangleright h) = g \partial(h) g^{-1}, \quad \partial(h) \triangleright h' = hh'h^{-1}.$$

On a good cover $\{P_i\}$, a weakly coherent gluing is encoded by:

$$g_{ij} : P_{ij} \rightarrow G, \quad h_{ijk} : P_{ijk} \rightarrow H,$$

obeying the 2-cocycle conditions

$$g_{ij}g_{jk} = \partial(h_{ijk})g_{ik},$$

and on quadruple overlaps,

$$h_{jkl}h_{ijl} = (g_{ij} \triangleright h_{ikl})h_{ijk}.$$

Gauge changes act by 1- and 2-cochains in the standard way for crossed-module cohomology.

Write

$$q_\Sigma = [(g, h)] \in \check{H}^2(N_\Sigma, H_\Sigma \rightarrow G_\Sigma)$$

for the full crossed-module gluing orbit. It classifies the equivalence orbit of the pair (g, h) but does not in general determine a unique ordinary G_Σ -valued 1-cocycle class after strictification, because the map below need not be injective. Let

$$\begin{aligned} \iota_\Sigma : \check{H}^1(N_\Sigma, G_\Sigma) &\longrightarrow \check{H}^2(N_\Sigma, H_\Sigma \rightarrow G_\Sigma), \\ [g] &\longmapsto [(g, 1)] \end{aligned}$$

be the natural strict-locus map. Define the separate $\{0, 1\}$ -valued strictification obstruction by

$$\begin{aligned} o_{\Sigma}^{(2)}(g, h) = 0 & \iff q_{\Sigma} \in \text{im}(\iota_{\Sigma}) \\ & \iff [(g, h)] \text{ has a representative } (g^{\text{str}}, 1) \\ & \quad \text{with } g_{ij}^{\text{str}} g_{jk}^{\text{str}} = g_{ik}^{\text{str}}. \end{aligned}$$

and set $o_{\Sigma}^{(2)}(g, h) = 1$ otherwise.

Theorem 3.4 (non-central higher-defect strictification). The higher associator defect of (g_{ij}, h_{ijk}) is removable iff $o_{\Sigma}^{(2)}(g, h) = 0$. In that case a higher-gauge change gives $h_{ijk} = 1$ and a genuine G -valued edge 1-cocycle. Strict endpoint-only transport additionally requires that at least one allowed strict representative have trivial represented holonomy. The full orbit q_{Σ} need not determine a unique ordinary H^1 class.

Proof sketch. Removing the higher associator corresponds to $h_{ijk} = 1$ and $g_{ij}g_{jk} = g_{ik}$. Crossed-module coboundaries preserve the full orbit, so this strictification exists exactly when that orbit meets the image of the ordinary G -valued 1-cocycle locus, which is the definition of $o_{\Sigma}^{(2)} = 0$. The map ι_{Σ} need not be injective: H -valued edge changes can relate strict representatives and alter their ordinary holonomy through ∂ . Thus the invariant residual test is whether at least one allowed strict representative has trivial represented holonomy, as in Theorem 3.4b. QED.

The central-defect case is the abelian truncation with H central and trivial action, which reduces to Section 3.3. A genuinely noncentral class is the point where the fixed-cutoff collar theorem upgrades from the ordinary-group package to the higher-gauge one below.

Corollary 3.4a (Fixed-cutoff higher-gauge EC and transportability). For a finite regulator chart on a connected cut Σ , the genuinely noncentral branch admits a compact crossed-module change system

$$\mathcal{T}_{\Sigma} = C^1(N_{\Sigma}, H_{\Sigma}) \rtimes C^0(N_{\Sigma}, G_{\Sigma}).$$

The physical higher-gauge collar is

$$\mathcal{H}_B^{2g} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\mathcal{T}_{\Sigma}} \cong \bigoplus_{\lambda} (\mathcal{H}_{b_L^{\lambda}} \otimes \mathcal{H}_{b_R^{\lambda}}),$$

with

$$Z(\mathcal{A}_{2g}(B)) = \bigoplus_{\lambda} \mathbb{C} \mathbf{1}_{\lambda},$$

and the full orbit q_{Σ} is invariant under local rechartings and classifies the fixed-cutoff genuinely noncentral gluing datum. Its separate invariant property $o_{\Sigma}^{(2)} = 0$ holds iff the higher associator is removable. The resulting strict representatives can have different ordinary G_{Σ} -valued 1-holonomies, so transportability is the separate existential test of Theorem 3.4b rather than a uniquely defined holonomy attached to q_{Σ} .

Proof sketch. Pair-overlap rechartings are inner, while triple-overlap associators strictify to compact crossed-module data. Finite-dimensional unitary $\mathcal{T}_{\Sigma}$ -modules decompose semisimply, and Schur matching leaves only diagonal left/right blocks. Crossed-module cohomology classifies the higher defect; Theorem 3.4b separately tests the ordinary loop holonomy that remains after strictification. QED.

Theorem 3.4b (TransportabilityFromOverlapGluing). Fix a connected cut Σ and a finite regulator charting nerve N_{Σ} . Transport of a collar charge is defined by the path composite in the overlap recharting groupoid: along $p = (i_0 \rightarrow \cdots \rightarrow i_m)$, the charge block is moved by $U_p = U_{i_{m-1}i_m} \cdots U_{i_0i_1}$. This construction uses the overlap unitary transition system itself, not an added DHR transportability assumption.

After a central or crossed-module strictification, let U^{str} denote a resulting genuine edge 1-cocycle. Write $\mathcal{C}_\alpha$ for the full transported collar-charge block, including its boundary carrier and multiplicity/intertwiner data, and define

$$\text{Hol}_{\alpha, U^{\text{str}}} : \pi_1(|N_\Sigma|, i_*) \longrightarrow U(\mathcal{C}_\alpha), \quad [\gamma] \longmapsto U_\gamma^{\text{str}}|_{\mathcal{C}_\alpha}.$$

On the ordinary branch, transport is strictly path-independent iff every closed overlap loop has trivial holonomy on the collar-sector block. On the central branch, elementary triangle moves accumulate the central cocycle z_{ijk} . Strict path-independent transport exists iff $[z]_\Sigma = 0$ and there is a central 1-cochain strictification for which $\text{Hol}_{\alpha, U^{\text{str}}} = 1$. The first condition removes the projective triangle defect; the second removes the residual ordinary loop action. If $[z]_\Sigma = 0$ but the second condition fails, the lifts form a genuine 1-cocycle and transport is still path dependent.

On the genuinely noncentral branch, path comparisons are H_Σ -valued 2-morphisms in the crossed module $H_\Sigma \rightarrow G_\Sigma$. Strict ordinary transport exists iff $o_\Sigma^{(2)} = 0$ and at least one crossed-module strictification gives a genuine G_Σ -valued 1-cocycle with trivial represented holonomy on $\mathcal{C}_\alpha$. If $o_\Sigma^{(2)} \neq 0$, the sector has higher transport rather than ordinary compact-group DR transport. If $o_\Sigma^{(2)} = 0$ but every allowed strict representative has nontrivial represented G_Σ -holonomy, the higher associator has strictified but ordinary path dependence remains. The full orbit $q_\Sigma = [(g, h)]$ need not determine a unique ordinary H^1 class and is not itself the strictification test. Thus “zero obstruction” below means the combined associator-strictification and trivial-holonomy criterion, not vanishing of triangle or higher-associator data alone. QED.

Triangle-free cycle check. Let $N_\Sigma = C_n$, $n \geq 4$, have no filled 2-simplices. On $\mathcal{C}_\alpha = \mathbb{C}^2$, put identity transport on every oriented cycle edge except one, where the sector action is the non-scalar unitary $V = \text{diag}(-1, 1)$, and use inverse transports on reversed edges. All central 2-data and higher associator data are vacuously trivial, so $[z]_\Sigma = 0$ and $o_\Sigma^{(2)} = 0$. Its ordered cycle holonomy is $V \neq 1$, which no central rephasing removes. For a genuinely noncentral instance, take the crossed module

$$\text{SU}(2) \hookrightarrow \text{U}(2)$$

with conjugation action and the standard action on $\mathcal{C}_\alpha$. The band $\text{U}(2)/\text{SU}(2) \cong \text{U}(1)$ is detected by the determinant, and the cycle has band holonomy $\det V = -1$. An $\text{SU}(2)$ -valued edge change has determinant one and a vertex-frame change only conjugates the based loop, so no allowed strict representative has trivial holonomy. The two routes between the endpoints of the distinguished edge differ, and the combined criterion rejects both the central and noncentral assignments.

4.5 Modular Flow and Lorentz Kinematics

4.5.1 Modular additivity in the Markov collar limit

Consider a collar tripartition $A : B : D$ around a cap boundary, with the EC decomposition of Section 2.3 understood. Define, for a faithful reference state ω ,

$$\Delta K(\omega) := K_{ABD}(\omega) - K_{AB}(\omega) - K_{BD}(\omega) + K_B(\omega).$$

Exact modular additivity is not a consequence of EC alone. It is available only on the exact Markov set, or along a controlled fixed-cutoff family that approaches that set on one fixed collar model. Three quantities must therefore be kept separate:

1. the raw collar conditional mutual information $I(A : D \mid B)$;
2. the constructive Fawzi–Renner comparison error

$$r_{\text{FR}}(\varepsilon) := 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon};$$

3. the fixed-collar exact-Markov replacement modulus

$$\delta_{A:B:D}^{\text{M}}(\varepsilon) := \sup \left\{ \inf_{\sigma \in \mathfrak{M}_{A:B:D}} \|\rho - \sigma\|_1 : I(A : D \mid B)_\rho \leq \varepsilon \right\},$$

where

$$\mathfrak{M}_{A:B:D} := \{\sigma_{ABD} : I(A : D \mid B)_\sigma = 0\}.$$

Lemma 4.1a (EC-aligned exact Markov implies exact additivity up to a central term). If $I(A : D \mid B)_\omega = 0$ and ω satisfies the Markov-split alignment hypothesis of Section 2.3, then the state takes the EC-aligned HJPW block form

$$\omega_{ABD} = \bigoplus_{\alpha} p_{\alpha} \omega_{Ab_L^{\alpha}}^{(\alpha)} \otimes \omega_{b_R^{\alpha}D}^{(\alpha)},$$

and $\Delta K(\omega)$ is central. On the canonical HJPW block model one may take

$$\Delta K(\omega) = 0.$$

Equivalently, there exists a central operator $K_{\partial,ABD}(\omega) \in Z(\mathcal{A}(B))$ such that

$$K_{ABD}(\omega) = K_{AB}(\omega) + K_{BD}(\omega) - K_B(\omega) + K_{\partial,ABD}(\omega).$$

Proof. By alignment, the HJPW blocks are the EC blocks. On each block the modular Hamiltonians of ABD , AB , BD , and B are the logarithms of tensor-product states with the same classical block weight p_{α} . The blockwise logarithms therefore cancel exactly in the combination $K_{ABD} - K_{AB} - K_{BD} + K_B$. If one keeps the blockwise endpoint-label bookkeeping explicit rather than absorbing it into the canonical block identification, the remainder is a direct sum of block constants, hence central in $Z(\mathcal{A}(B))$. Without alignment, exact Markovity still cancels the four logarithms on the state’s own HJPW blocks, but the resulting block constants need not lie in $Z(\mathcal{A}(B))$, so the central form used here is unavailable. QED.

Proposition 4.1b (Controlled exact-Markov replacement on a fixed collar). Because the state space is compact and conditional mutual information is continuous in finite dimension,

$$\delta_{A:B:D}^{\text{M}}(\varepsilon) \longrightarrow 0 \quad (\varepsilon \downarrow 0).$$

Thus small collar CMI converges to the exact HJPW normal form only in this controlled fixed-cutoff sense; the manuscript does not claim a dimension-free one-shot trace-norm bound from small $I(A : D \mid B)$ directly to an exact Markov state. Moreover the limit points are exact Markov states in the CMI-zero sense; identifying their factorization with the preselected EC factors is the Markov-split alignment hypothesis of Section 2.3, which is carried separately and is not produced by small CMI (the Bell-pair counterexample of Section 2.3 has zero CMI at fixed distance from every EC-aligned state).

Theorem 4.1b' (Finite-collar Markov replacement stability). On one fixed faithful finite-dimensional collar model, the qualitative modulus above can be made collar-local and quantitative. If all relevant marginals stay above a chosen floor $\lambda_* > 0$, then there are constants $C_{A:B:D,\lambda_*} > 0$ and $\theta_{A:B:D,\lambda_*} > 0$, depending only on that collar model and floor, such that

$$d_M(\rho) \leq C_{A:B:D,\lambda_*} I(A : D | B)_\rho^{\theta_{A:B:D,\lambda_*}}.$$

This is a fixed-collar Łojasiewicz-type rate for the analytic function $I(A : D | B)$ on the faithful state manifold. It is not a dimension-free stability theorem for arbitrary tripartite quantum systems.

Corollary 4.1c (Carried collar defect operator). Let ω_ε satisfy $I(A : D | B)_{\omega_\varepsilon} \leq \varepsilon$, and choose $\sigma_\varepsilon \in \mathfrak{M}_{A:B:D}$ with

$$\|\omega_\varepsilon - \sigma_\varepsilon\|_1 \leq \delta_{A:B:D}^M(\varepsilon).$$

Define the carried defect operator

$$\mathfrak{D}_{A:B:D}(\omega_\varepsilon, \sigma_\varepsilon) := \Delta K(\omega_\varepsilon) - \Delta K(\sigma_\varepsilon).$$

Then

$$K_{ABD}(\omega_\varepsilon) = K_{AB}(\omega_\varepsilon) + K_{BD}(\omega_\varepsilon) - K_B(\omega_\varepsilon) + K_{\partial,ABD}(\sigma_\varepsilon) + \mathfrak{D}_{A:B:D}(\omega_\varepsilon, \sigma_\varepsilon),$$

where $K_{\partial,ABD}(\sigma_\varepsilon) = \Delta K(\sigma_\varepsilon)$ is central. For every bounded observable X supported on $A \cup B$,

$$|\text{Tr}[X(\omega_\varepsilon - \sigma_\varepsilon)]| \leq \|X\|_\infty \delta_{A:B:D}^M(\varepsilon).$$

If the relevant marginals of ω_ε and σ_ε are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\mathfrak{D}_{A:B:D}(\omega_\varepsilon, \sigma_\varepsilon)\|_\infty \leq 4\lambda_*^{-1} \delta_{A:B:D}^M(\varepsilon).$$

Independently, Fawzi–Renner recovery supplies a constructive comparison state $\omega_\varepsilon^{\text{rec}}$ with

$$\|\omega_\varepsilon - \omega_\varepsilon^{\text{rec}}\|_1 \leq r_{\text{FR}}(\varepsilon), \quad r_{\text{FR}}(\varepsilon) := 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}.$$

This $O(\varepsilon^{1/2})$ term is the finite-stage observable error. The modulus $\delta_{A:B:D}^M(\varepsilon)$ is instead the fixed-collar quantity that justifies replacing the physical state by an exact Markov normal form in the later geometric modular arguments.

Accordingly, the manuscript's later exact collar formulas take one of two forms:

1. literal exactness when the reference state is exact Markov on the relevant collar; or
2. a controlled collar family for which $\delta_{A:B:D}^M(\varepsilon_\delta) \rightarrow 0$, with the constructive Fawzi–Renner remainder and the carried operator defect $\mathfrak{D}_{A:B:D}$ kept explicit at finite stage.

Theorem 4.1d (Finite-stage modular-defect propagation and dimension quarantine). For any fixed downstream branch calculation that uses finitely many collar or strip modular-additivity identities, replacing each exact identity by its controlled finite-stage form changes every bounded downstream modular observable O by at most

$$\mathcal{P}_O\left(\{r_{\text{FR}}(\varepsilon_j)\}, \{\delta_j^M(\varepsilon_j)\}, \{\eta_j^{\text{reg}}\}\right),$$

where $\mathcal{P}_O \rightarrow 0$ as all listed errors vanish. The polynomial-continuity modulus depends only on the declared fixed collar models, support-visible dimensions, bounded observable class, faithful floors or regularization schedules, and bounded modular-time intervals. No BW/Lorentz, null-modular, or Einstein scaling step uses a dimension-free trace-norm stability theorem from small CMI directly to exact Markov normal form.

4.5.2 Theorem: BW_{S^2} on the extracted prime geometric subnet

The collar analysis proves a fixed-cutoff statement on the finite type-I regulator net: the reduced cap state has a literal density matrix, its modular Hamiltonian exists, and its nonadditive part is confined to a shrinking collar up to carried errors. The Lorentz claim is therefore not a fixed-cutoff matrix-algebra identity and not the slogan “finite cells imply Lorentz invariance.” Its target is the support-visible refinement-limit geometric cap pair $(\mathcal{A}_\infty^{\text{geo},\text{sv}}(C), \omega_\infty^{\text{geo},C})$. The branch theorem is the conditional implication

$$\text{FiniteCapBWCertificate} \implies \sigma_t^{\omega_\infty^{\text{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

The finite cap-normal certificate consists of cap-normal density and nondegeneracy, BW framing, support-order faithfulness, cap-family-uniform mixed-GNS modular convergence, support covariance, held-out oriented cross-ratio convergence, independently normalized geometric 2π -KMS convergence, and wrong-normalization separation. The support-visible theorem below uses regularized modular transport, projectively compatible exact-Markov replacement on fixed collars, weak-*/GNS extraction, support-readable modular covariance, and that finite cap-normal certificate; it does not require a type-I continuum algebra or a full-algebra unregularized common spectral floor. Bare finite consensus does not imply the certificate without the separate Einstein branch-entry production theorem.

Fix a cap $C \subset S^2$ and a shrinking collar family $(A_\delta, B_\delta, D_\delta)$ around ∂C . Write

$$\varepsilon_\delta := I(A_\delta : D_\delta \mid B_\delta)_\omega, \quad r_{\text{FR}}(\varepsilon_\delta) := 2\sqrt{1 - e^{-\varepsilon_\delta}} \leq 2\sqrt{\varepsilon_\delta},$$

and, on each fixed faithful collar model,

$$\eta_\delta^{\text{M}} := 4\lambda_*^{-1} \delta_{A_\delta:B_\delta:D_\delta}^{\text{M}}(\varepsilon_\delta),$$

where $\lambda_* > 0$ is the lower spectral bound used to compare modular Hamiltonians. All exact collar formulas in this subsection are therefore to be read either literally at exact Markovity or along a controlled collar family satisfying

$$\delta_{A_\delta:B_\delta:D_\delta}^{\text{M}}(\varepsilon_\delta) \rightarrow 0 \quad (\delta \downarrow 0),$$

with $r_{\text{FR}}(\varepsilon_\delta)$ and η_δ^{M} carried explicitly.

For each cap C , let $\lambda_C(s) \subset \text{Conf}^+(S^2)$ denote the standard cap-preserving conformal one-parameter subgroup, normalized so that the null blow-up near a smooth cut acts by $v \mapsto e^{-s}v$, and let $\alpha_{\lambda_C(s)}$ denote the induced automorphism of the scaling-limit cap net.

Theorem 4.2 (Finite cap-net support-visible BW_{S^2} scaling theorem). Let $\widehat{C} = (C, n_C, p_C^-, p_C^+)$ be a nondegenerate BW-framed cap. For every OPH-realized observer-supporting refinement branch satisfying the five OPH axioms, the derived fixed-cutoff regulator/collar/consensus package, and the finite cap-normal BW certificate above, the support-visible prime geometric cap net admits a weak-*/GNS scaling-limit cap pair $(\mathcal{A}_\infty^{\text{geo},\text{sv}}(C), \omega_\infty^{\text{geo},C})$. Then the scaling-limit modular automorphism group is

$$\sigma_t^{\omega_\infty^{\text{geo},C}} = \alpha_{\lambda_{\widehat{C}}(2\pi t)}.$$

At finite regulator stage the nonadditive part of

$$K_C^{(\delta)} := -\log \rho_C^{(\delta)}$$

is confined to the shrinking collar up to the carried errors $r_{\text{FR}}(\varepsilon_\delta)$, the fixed-collar replacement modulus δ^M , and the regularized support-visible modular transport bound. The support-visible modular limit is independent of the exact-Markov replacement sequence after projective compatibility and GNS quotienting, and support-readable modular covariance turns the limiting modular group into a cap-local support map before the certificate's framed-cap and oriented cross-ratio rigidity clauses are applied. No separate cap-isotropy/SO(2)-equivariance selector, finite-cell Lorentz-invariance premise, or full-algebra unregularized common floor is used. Record/pointer and interface-inert auxiliary registers are outside the extracted geometric subnet. If the scaling-limit cap algebra is type I, the same automorphism identity may be represented by

$$K_C = 2\pi B_C + Z_C, \quad Z_C \in Z(\mathcal{A}(C))_{\text{sa}};$$

otherwise the automorphism identity is the full statement and the action is outer. Consequently, on a branch carrying the multiresolution reference-tower certificate of Theorems 2.6c–2.6f, the transported regularized modular action on every fixed support-visible geometric local algebra converges uniformly on compact modular-time intervals to the geometric cap-dilation action. Without that certificate, this theorem identifies only the scaling-limit cap-pair modular automorphism and does not assert compact-time convergence of finite-regulator modular groups.

Proof. Markov locality localizes the modular defect to the shrinking collar, with carried finite-stage errors $r_{\text{FR}}(\varepsilon_\delta)$ and δ^M . Proposition 2.6a shows why a full-algebra unregularized floor is unavailable in general. Theorem 2.6b supplies the matrix-element replacement on fixed collar models, and Theorem 2.6f supplies the stronger compact-time group estimate only for transported regularized local actions with the stated reference-tower certificate. Projective replacement compatibility and the support-visible convergence audit ensure that the emitted modular limit is independent of the chosen exact-Markov comparison family. Support-visible cap-pair extraction on the local GNS support quotient uses weak-* compactness of finite-stage state spaces and the consensus refinement mechanism to supply local limit states, then applies GNS construction to obtain the scaling-limit cap pair. Support-readable modular covariance reads the modular group as a cap-local support map. The finite cap-normal certificate supplies support-order faithfulness, BW framing, held-out oriented cross-ratio rigidity, and compact-time equicontinuity, so the support flow is the framed cap-preserving conformal flow up to normalization:

$$\sigma_t^{\omega_\infty^{\text{geo},C}} = \alpha_{\lambda_C(\kappa_C t)}$$

for some $\kappa_C > 0$, with the certificate-selected frame understood in λ_C . The certificate then passes the independently normalized geometric 2π -KMS condition to $s \mapsto \alpha_{\lambda_C(s)}$. KMS uniqueness, together with wrong- β separation and nontriviality, fixes $\kappa_C = 2\pi$. Thus

$$\sigma_t^{\omega_\infty^{\text{geo},C}} = \alpha_{\lambda_C(2\pi t)}.$$

If the limit algebra is type I, this automorphism identity is represented by $K_C = 2\pi B_C + Z_C$ with Z_C central; otherwise the automorphism identity is the full statement and the action is outer. QED.

Definition 4.2a (BW-branch observer-relative time reading). On the branch satisfying the hypotheses of Theorem 4.2, OPH uses the modular automorphism parameter t of the extracted cap pair $(\mathcal{A}_\infty^{\text{geo},\text{sv}}(C), \omega_\infty^{\text{geo},C})$ as that cap observer's relative time parameter. This is a declared BW-branch reading of the geometric modular flow derived in Theorem 4.2; it is not an additional proof that arbitrary operational clocks, global time, or the full problem of time have been derived from the axioms.

Scope note. The theorem fixes the 2π normalization and the carried refinement limit on the support-visible extracted geometric cap pair under the finite cap-normal certificate. The UV-side scaffold is the realized transported geometric cap-local system, regularized support-visible modular transport on fixed local collars, support-visible cap-pair extraction on the local GNS support quotient, support-readable modular covariance, BW framing, oriented cross-ratio rigidity, and geometric KMS normalization. No separate cap-isotropy input is used, and the conformal support-map clause is the explicit scaling-limit branch condition rather than a finite-regulator Lorentz premise. Compact-time convergence of finite-regulator modular groups is available only through the transported-local certificate of Theorem 2.6f; a black-box conformal-net BW reconstruction would be an additional validation route, not a hidden premise.

4.5.3 Theorem: cap-normal H^3 reconstruction

Theorem 4.3 (Cap-normal H^3 reconstruction, imported from the compact proof). On the support-visible round-cap BW branch, write the celestial null section as

$$q(\Omega) = (1, \Omega), \quad \Omega \in S^2.$$

For a nondegenerate oriented round cap

$$C(\mathbf{c}, \alpha) = \{\Omega : \mathbf{c} \cdot \Omega \geq \cos \alpha\}, \quad 0 < \alpha < \pi,$$

define

$$n_C = (\cot \alpha, \csc \alpha \mathbf{c}).$$

Then, for the time-first Lorentz form $\eta = -dt^2 + d\mathbf{x}^2$,

$$\eta(n_C, n_C) = 1, \quad \eta(n_C, q(\Omega)) = \frac{\mathbf{c} \cdot \Omega - \cos \alpha}{\sin \alpha}.$$

Thus $\eta(n_C, q(\Omega)) = 0$ exactly on ∂C , with positive sign in the cap interior and negative sign outside. For $g \in \text{Conf}^+(S^2)$ with Lorentz representative Λ_g ,

$$\Lambda_g q(\Omega) = \omega_g(\Omega) q(g\Omega), \quad n_{gC} = \Lambda_g n_C.$$

Consequently

$$\text{Conf}^+(S^2) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1),$$

oriented round caps form the de Sitter normal space

$$\text{Cap}_{\text{round}}^{\text{or}}(S^2) \cong dS_3^{\text{cap}} \cong \text{SO}^+(3, 1)/\text{SO}^+(2, 1),$$

and future unit timelike observer frames form the distinct hyperbolic space

$$H^3 \simeq \text{SO}^+(3, 1)/\text{SO}(3), \quad \dim H^3 = 3.$$

Each cap normal determines the geodesic half-space

$$\mathcal{H}_C^+ = \{u \in H^3 : \eta(n_C, u) \geq 0\},$$

not a preferred observer point.

Proof. This is the cap-normal reconstruction theorem in the compact paper, which supplies the projective-null-cone model, the cap-normal bijection, Lorentz equivariance, the H^3 observer-frame

hyperboloid, the ideal-boundary identification, and the cap/half-space duality. The BW theorem supplies the cap-preserving conformal subgroup and its 2π modular normalization; the cap-normal theorem supplies the explicit projective Lorentz representation and does not rederive BW from bare finite consensus. QED.

Corollary 4.3a (Chart-population boundary). The theorem proves the exact canonical observer-facing Lorentz/ H^3 kinematic chart on the stated branch. It does not, by itself, populate that chart with records, construct a chart-blind neutral bulk, derive a physical curvature radius R_H , produce stress-energy, or close the Einstein branch-entry umbrella. Populated observer-facing records require the conditional cap-response localization certificate imported next; object families, neutral bulk, scale, stress, and Einstein branch entry remain separate receipts. Theorems 4.3c–4.3e below import the compact paper’s geometry-producer, null-net-standardness, and event-manifold packets, which discharge parts of this boundary conditionally on named computable receipts.

Theorem 4.3b (Conditional record-populated H^3 localization). Fix a quotient-visible record token i and clock slice t . On the chart of Corollary 4.3, suppose its record-conditioned modular cap responses

$$R_i(C, t, O) = \omega_{i,O} \left(\sigma_t^{C,O}(M_{C,0,O}) \right), \quad \omega_{i,O}(A) = \frac{\omega_O(P_{i,O} A P_{i,O})}{\omega_O(P_{i,O})},$$

admit the calibrated point-local factorization

$$y_{i,j} = F_j(X_i(t)) + e_{i,j}, \quad F_j(X) = \psi_j \left(\frac{\eta(X, n_j)}{R_H} \right),$$

on a compact source region $\Omega \subset H_{R_H}^3$. Assume the finite response map has quantitative observability

$$\alpha \bar{d}(X, Y) \leq |F(X) - F(Y)|_W \leq L \bar{d}(X, Y), \quad \alpha > 0,$$

and the total error obeys $|e_i|_W \leq \sigma_i$. If $\mathcal{N}_\varepsilon$ is an ε -net of Ω and $\hat{X}_i(t)$ is a residual minimizer up to tolerance τ_i , then the localized record support is

$$S_i(t) = B_H \left(\hat{X}_i(t), R_H \left[\frac{L}{\alpha} \varepsilon + \frac{2}{\alpha} \sigma_i + \frac{1}{\alpha} \tau_i \right] \right).$$

Exact continuous data have a unique minimizer. Finite noisy data produce the displayed localization ball; a unique finite point is licensed only when the certified residual gap $\Delta_{\text{loc},i}(t) > 0$. If that gap fails, the natural output is AMBIGUOUS. This theorem does not derive point locality, resolve unlabeled mixtures or extended sources, identify particle species, produce stress-energy, or establish chart-blind neutral bulk.

Theorem 4.3c (Quotient-intrinsic geometry producer and dimension-selection boundary). On a refinement tower of repaired quotient normal forms, the support-visible incidence complex — vertices the quotient-visible patch classes, simplices the jointly supported patch sets — is computable from the normal form alone and is invariant under repair schedule, gauge, and refinement. If the tower carries the computable receipts (i) spherical incidence (connected closed orientable combinatorial surface with Euler characteristic 2), (ii) disk/mesh nondegeneracy, (iii) modular cross-ratio Cauchy convergence, and (iv) independently normalized geometric 2π -KMS comparison with wrong-normalization separation, then the refinement limit produces the round S^2 , its conformal structure, the oriented round-cap family with cap normals, and every clause of the finite cap-normal BW certificate, with quantitative error envelopes. Conversely, there are finite repair systems with isomorphic rewrite structure whose incidence complexes realize S^2 , T^2 , the 2-skeleton of $\partial\Delta^4$, a wedge of two spheres, and any modular temperature β : repair confluence

selects no topology, dimension, orientation framing, or normalization. The screen S^2 is therefore *derived on the receipt branch and underdetermined off it*; the receipts are the irreducible geometric branch content of Einstein branch entry. (Compact paper, geometry-producer packet; machine receipts in `code/geometry/.`)

Theorem 4.3d (Null-net standardness, derived translations, and the four-translation assembly). On the producer branch, the null half-line/interval blow-up net at a produced cap boundary point lives on one common GNS space, and: (i) the reference vector is cyclic and separating for every half-line and interval algebra used, given the finite-stage faithfulness of the MaxEnt states, the mixed-GNS Cauchy clause, and one half-line cyclicity receipt, with interval cyclicity then derived by the Reeh–Schlieder/Borchers argument rather than assumed; (ii) the half-sided modular inclusion is derived from the produced geometric cap flow, without assuming any translation or dilation unitaries, and Borchers–Wiesbrock yields the positive null-translation generator on its Stone domain; (iii) on the exact-Markov/central-interface branch the reduced interval modular Hamiltonians split into local terms plus uniformly bounded interface blocks (endpoint-Lipschitz), while an explicit four-qubit nearest-neighbour Gibbs witness shows finite-range locality alone does *not* imply this (machine-verified); (iv) the two derived translation groups and the modular dilations generate Möbius covariance and the projective bounded-interval action; and (v) on the modular-intersection receipt branch the per-direction positive generators commute, transform with the null-vector weight under the produced Lorentz action, satisfy the dependent-family linearity relation, and assemble into one four-parameter translation representation with joint spectrum in the closed future cone — a positive-energy Poincaré representation. (Compact paper, null-net standardness packet.)

Theorem 4.3e (Conditional Lorentzian event manifold). Events are coincidence classes of record germs: refinement-compatible families of localized record tokens (Theorem 4.3b) with shrinking certified boxes, an observer-independent relation invariant under schedule, gauge, refinement, and Lorentz chart change. Unconditionally the event space is a second-countable completely regular uniform space. Under the receipts E1 (population density), E2 (certified separation gaps), E3 (rank-four calibrated response frames: three cap responses plus one derived modular clock line), and E4 (Poincaré overlap cocycle), it is a Hausdorff topological four-manifold with a locally bi-Lipschitz atlas; the intrinsic transactional ancestry order gives local causal cones whose null-direction sphere is the produced celestial S^2 , forcing signature $(-+++)$, exactly one time direction, three spatial directions, and a repair-oriented time orientation; H^3 is strictly the fiber of future unit timelike frames over events, with no equivariant section, and is never a spatial slice; tetrads, the produced conformal metric (absolute scale a separate receipt), a $C^{1,1}$ Levi-Civita connection and almost-everywhere curvature readout, and glued scheduler-independent modular clocks follow, with stable causality and global hyperbolicity carried as named assumptions. Explicit countermodels (single-worldline population, label inflation, reconciliation branching) show that $\dim H^3 = 3$ by itself promotes nothing: event dimension, Hausdorffness, and manifoldness are controlled exactly by E1–E4. This packet supplies, conditionally, the locally Lorentzian $d = 4$ regime consumed by the Einstein section. (Compact paper, event-manifold packet; machine receipts in `code/geometry/.`)

4.6 Gravity from Fixed-Cap Generalized-Entropy Stationarity

4.6.1 Cap first law

Section 4.2 fixes the cap modular statement first at the automorphism level. On the extracted prime geometric subnet, the scaling-limit cap modular flow is geometric with the standard 2π normalization,

$$\sigma_t^{\omega_\infty^{\text{geo}, C}} = \alpha_{\lambda_C(2\pi t)}.$$

If the realized scaling-limit cap algebra happens to be type I, one may choose a density matrix $\rho_C^{\omega_\infty^{\text{geo},C}}$ and modular Hamiltonian

$$K_C := -\log \rho_C^{\omega_\infty^{\text{geo},C}},$$

and then for a perturbation $\rho(\varepsilon)$ with $\rho(0) = \omega_\infty^{\text{geo},C}$ the first-law identity reads

$$\delta S_C = \delta \langle K_C \rangle = 2\pi \delta \langle B_C \rangle.$$

This displayed variation is the special type-I/fixed-central-sector form of $K_C = 2\pi B_C + Z_C$, where the central term has no first-order contribution on the chosen sector. In the generic continuum/non-type-I case, Section 4.2 does not supply an inner operator identity; the theorem surface there is the geometric modular automorphism statement, so the cap first-law formula is restricted to this special realization.

4.6.2 Null-surface modular bridge

This subsection extends the fixed-cutoff weak-tail-generator boundary to the derived positive null-translation stage and the exact half-line generator/charge identification. At fixed cutoff one first transfers cut-center data to regulated null strips, then imposes the extra inherited split condition needed for the spatial-collar-type tensor decomposition, obtains exact or controlled strip additivity on one inherited strip model, and derives a weak tail generator for the renormalized half-line family. On the scaling-limit geometric-cap branch of Theorem 4.2, the null half-line blow-up net then inherits geometric dilation and therefore half-sided modular inclusion, so Borchers–Wiesbrock supplies an explicit positive null-translation generator on its Stone domain. Bounded-interval formulas are discharged by the E0.5 affine/projective kernel below, and the tensor upgrade carries the null-invisible metric ambiguity. Accordingly the null modular bridge consists of:

- null-cut center transfer yielding the central cut-label decomposition, together with the extra inherited split condition needed for the spatial-collar-type tensor decomposition;
- exact or controlled four-term strip additivity on one fixed inherited strip model;
- endpoint-Lipschitz control for the renormalized half-line family and the resulting weak tail generator;
- derived half-sided modular inclusion on the null half-line blow-up net, and the resulting Borchers positive translation generator.

The later density-upgrade template is kept below as an explicit downstream template rather than as a fixed-cutoff theorem proved in this subsection. Lemma 5.2f makes the positive null-translation generator itself explicit, and Theorem 5.2g closes the half-line generator/charge identification on that same family: they record the Borchers unitary group, positivity and self-adjointness on the Stone domain, the corrected affine commutator relation, the half-line modular-Hamiltonian identity $K_a = K_0 - 2\pi a P_\Omega$, and the exact identification of P_Ω with the local null-stress charge. Bounded-interval formulas are downstream.

Proposition 5.2a (Null-cut center transfer and inherited split). Fix a regulated null tripartition

$$I_- = (v_1, v_2), \quad J = (v_2, v_3), \quad I_+ = (v_3, v_4),$$

with cuts $\Gamma_- := \{v = v_2\}$ and $\Gamma_+ := \{v = v_3\}$. Under the fixed-cutoff realized presentation, assume that the two null cuts inherit the ordinary or central-defect boundary-redundancy data used in the spatial collar branch, so that in a compatible type-I regulator presentation one has

$$\tilde{\mathcal{H}}_{I_-} \cong \bigoplus_{\alpha_-} W_{\alpha_-} \otimes \mathcal{H}_{i_-^{\alpha_-}},$$

$$\begin{aligned}\tilde{\mathcal{H}}_J &\cong \bigoplus_{\alpha_-, \alpha_+} W_{\alpha_-}^* \otimes \mathcal{H}_{j^{\alpha_-, \alpha_+}} \otimes W_{\alpha_+}, \\ \tilde{\mathcal{H}}_{I_+} &\cong \bigoplus_{\alpha_+} W_{\alpha_+}^* \otimes \mathcal{H}_{i_+^{\alpha_+}},\end{aligned}$$

where opposite sides of each cut carry inverse transport. If the strip algebra is the commutant of the transported cut actions, then

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-, \alpha_+} \mathcal{B}(\mathcal{H}_{j^{\alpha_-, \alpha_+}}), \quad Z(\mathcal{A}(J)) = \bigoplus_{\alpha_-, \alpha_+} \mathbb{C} P_{\alpha_-, \alpha_+}.$$

If, in addition, each multiplicity space factors as

$$\mathcal{H}_{j^{\alpha_-, \alpha_+}} \cong \mathcal{H}_{j_L^{\alpha_-, \alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-, \alpha_+}},$$

with $\mathcal{A}(I_- \cup J)$ acting blockwise only on $\mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-, \alpha_+}}$ and $\mathcal{A}(J \cup I_+)$ acting blockwise only on $\mathcal{H}_{j_R^{\alpha_-, \alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}$, then

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-, \alpha_+} \mathcal{B}(\mathcal{H}_{j_L^{\alpha_-, \alpha_+}}) \otimes \mathcal{B}(\mathcal{H}_{j_R^{\alpha_-, \alpha_+}}),$$

and

$$\mathcal{H}_{I_- \cup J \cup I_+} \cong \bigoplus_{\alpha_-, \alpha_+} \mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-, \alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-, \alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}.$$

Proof. Complete reducibility gives the displayed decompositions. Commuting with the two transported cut actions forces strip operators to preserve the sector pair (α_-, α_+) and act only on the multiplicity space; Schur's lemma kills the off-diagonal intertwiners and leaves the direct-sum algebra above. Taking invariants across both cuts in the glued tripartition again uses Schur's lemma and leaves exactly the matching sector pairs. The left/right split of the multiplicity spaces is additional structure; it is not forced by the center transfer alone. QED.

Corollary 5.2b (Exact or controlled four-term null modular relation on an inherited strip model). On one fixed finite-dimensional strip model satisfying Proposition 5.2a, define

$$\mathfrak{M}_{I_-:J:I_+} := \{\sigma : I(I_- : I_+ | J)_\sigma = 0\},$$

and

$$\delta_{I_-:J:I_+}^M(\varepsilon) := \sup \left\{ \inf_{\sigma \in \mathfrak{M}_{I_-:J:I_+}} \|\rho - \sigma\|_1 : I(I_- : I_+ | J)_\rho \leq \varepsilon \right\}.$$

For any strip state η , let

$$\Delta K_J(\eta) := K_{I_- \cup J \cup I_+}(\eta) - K_{I_- \cup J}(\eta) - K_{J \cup I_+}(\eta) + K_J(\eta).$$

If the strip reference state ω is exact Markov and EC-aligned on the inherited split (the strip form of the Markov-split alignment hypothesis of Section 2.3), then $\Delta K_J(\omega)$ is central, and on the canonical inherited HJPW model one may take

$$\Delta K_J(\omega) = 0.$$

Equivalently, there exists a central operator $K_{\partial, J}(\omega) \in Z(\mathcal{A}(J))$ such that

$$K_{I_- \cup J \cup I_+}(\omega) = K_{I_- \cup J}(\omega) + K_{J \cup I_+}(\omega) - K_J(\omega) + K_{\partial, J}(\omega).$$

If instead $I(I_- : I_+ \mid J)_\omega \leq \varepsilon$, choose $\tilde{\omega}_J \in \mathfrak{M}_{I_-:J:I_+}$, assumed EC-aligned on the inherited split wherever the central identification below is used, with

$$\|\omega - \tilde{\omega}_J\|_1 \leq \delta_{I_-:J:I_+}^M(\varepsilon).$$

Define

$$\mathfrak{D}_J(\omega, \tilde{\omega}_J) := \Delta K_J(\omega) - \Delta K_J(\tilde{\omega}_J).$$

Then

$$K_{I_- \cup J \cup I_+}(\omega) = K_{I_- \cup J}(\omega) + K_{J \cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\tilde{\omega}_J) + \mathfrak{D}_J(\omega, \tilde{\omega}_J),$$

with $K_{\partial,J}(\tilde{\omega}_J) = \Delta K_J(\tilde{\omega}_J) \in Z(\mathcal{A}(J))$. Every bounded observable on $I_- \cup J$ or $J \cup I_+$ then differs from the exact-Markov reference by at most

$$\|O\|_\infty \delta_{I_-:J:I_+}^M(\varepsilon).$$

If the relevant strip marginals are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\mathfrak{D}_J(\omega, \tilde{\omega}_J)\|_\infty \leq 4\lambda_*^{-1} \delta_{I_-:J:I_+}^M(\varepsilon).$$

Independently, Fawzi–Renner gives a recovered comparison state with trace-norm error

$$r_{\text{FR}}(\varepsilon) = 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}.$$

Thus the exact four-term strip relation is available only at EC-aligned exact Markovity or in a controlled strip family on one fixed inherited strip model admitting EC-aligned replacements with $\delta_{I_-:J:I_+}^M(\varepsilon_J) \rightarrow 0$, while $\mathfrak{D}_J$ is carried explicitly at finite stage.

Proof. On the inherited split of Proposition 5.2a, the EC-aligned exact-Markov case is the same HJPW block calculation as in the spatial collar theorem: alignment identifies the HJPW blocks with the inherited blocks, the four logarithms cancel blockwise on the canonical model, and any residual bookkeeping term is central. Without alignment, HJPW factorizes the state only over its own state-dependent split, and the centrality of ΔK_J in $Z(\mathcal{A}(J))$ is not available. The controlled replacement is the same fixed-model compactness argument used for ordinary collars, after relabeling $A, B, D \mapsto I_-, J, I_+$; the operator-norm bound on $\mathfrak{D}_J$ is the corresponding relabeling of the modular-transport estimate. QED.

Definition (Renormalized null modular functional). For a null interval I on generator Ω , let $K_{\partial}(I, \Omega)$ denote the central endpoint-label term singled out by Corollary 5.2b on the inherited strip model, or by its controlled exact-Markov replacement when that model is used as reference. Define

$$\widetilde{K}[I, \Omega] := K[I, \Omega] - K_{\partial}(I, \Omega), \quad \widetilde{K}_a(\Omega) := \widetilde{K}[(a, \infty), \Omega].$$

Proposition 5.2c (Endpoint-Lipschitz null modular families and weak tail generator). For renormalized modular Hamiltonians on one null generator,

$$\widetilde{K}[I, \Omega] := K[I, \Omega] - K_{\partial}(I, \Omega), \quad \widetilde{K}_a(\Omega) := \widetilde{K}[(a, \infty), \Omega],$$

the branch-internal endpoint-control coming from the local finite-constraint MaxEnt branch gives:

$$|\langle \psi, (\widetilde{K}[(a', b'), \Omega] - \widetilde{K}[(a, b), \Omega])\phi \rangle| \leq C_{\psi, \phi, \Omega}(|a' - a| + |b' - b|)$$

for bounded intervals in a compact endpoint window, and

$$|\langle \psi, (\widetilde{K}_{a'}(\Omega) - \widetilde{K}_a(\Omega))\phi \rangle| \leq C_{\psi, \phi, \Omega}|a' - a|$$

for half-lines. Hence

$$f_{\psi,\phi}(a) := \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle$$

is locally Lipschitz and therefore absolutely continuous, and its distributional derivative defines the weak tail generator

$$\langle \psi, q(a, \Omega)\phi \rangle := -\frac{1}{2\pi} \partial_a f_{\psi,\phi}(a).$$

For $a < b$,

$$\langle \psi, (\widetilde{K}_b(\Omega) - \widetilde{K}_a(\Omega))\phi \rangle = -2\pi \int_a^b \langle \psi, q(v, \Omega)\phi \rangle dv.$$

$q(a, \Omega)$ is a weak tail generator rather than a local operator-valued density; its identification with the positive self-adjoint Borchers generator occurs only after Corollary 5.2e and Lemma 5.2f.

Proof. The derived endpoint-control estimate applies to the renormalized family after removal of the central endpoint term. For bounded intervals, the symmetric-difference length is bounded by $|a' - a| + |b' - b|$; for half-lines it is exactly $|a' - a|$. Therefore the matrix-element functions are Lipschitz in the endpoints. A Lipschitz function on a compact interval is absolutely continuous, so $f_{\psi,\phi}(a)$ has an L_{loc}^∞ derivative and obeys the fundamental theorem of calculus. Defining q as the rescaled negative derivative gives the displayed integral relation. QED.

Downstream c.12–c.13 boundary. The fixed-cutoff bridge established above is the end of this subsection's proved content.

Lemma 5.2d (Downstream density-upgrade template). If, in addition, the weak tail generator $q(a, \Omega)$ is weakly differentiable in a and both $f_{\psi,\phi}(a)$ and $q_{\psi,\phi}(a)$ vanish at $+\infty$, then one may define a density

$$\langle \psi, p(a, \Omega)\phi \rangle := -\partial_a \langle \psi, q(a, \Omega)\phi \rangle = \frac{1}{2\pi} \partial_a^2 \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle$$

and obtain the half-line formula

$$\langle \psi, \widetilde{K}_a(\Omega)\phi \rangle = 2\pi \int_a^\infty (v - a) \langle \psi, p(v, \Omega)\phi \rangle dv.$$

This is a downstream density-upgrade step and is not proved from the fixed-cutoff strip arguments above.

Null half-line blow-up net. Fix a smooth cut point and choose affine coordinate v on generator Ω so that the cut sits at $v = 0$. For $a \geq 0$, write

$$H_a := (a, \infty), \quad \mathcal{M}_a(\Omega) := \overline{\bigvee_{a < c < d < \infty} \mathcal{A}((c, d), \Omega)}.$$

By isotony of the null interval net,

$$a \leq b \implies \mathcal{M}_b(\Omega) \subseteq \mathcal{M}_a(\Omega).$$

Corollary 5.2e (Derived half-sided modular inclusion on null half-lines). On the scaling-limit geometric-cap branch of Theorem 4.2, the blow-up modular action near a smooth entangling cut acts on the null coordinate by

$$v \mapsto e^{-2\pi t} v.$$

Therefore, for every $a \geq 0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) = \mathcal{M}_{e^{-2\pi t}a}(\Omega).$$

Hence, for every $a > 0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) \subseteq \mathcal{M}_a(\Omega) \quad (t \leq 0),$$

so the inclusion $\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega)$ is half-sided modular. After the harmless convention change $t \mapsto -t$, this is the standard positive-time half-sided-inclusion form. The half-sided modular inclusion is therefore derived from the null-interval structure, isotony, and the scaling-limit geometric action rather than imported separately.

Proof. Blow up the cap modular flow of Theorem 4.2 near a smooth cut and restrict to the chosen null generator. In the tangent limit the cap-preserving flow becomes the null dilation $v \mapsto e^{-2\pi t}v$. For every bounded interval $(c, d) \subset H_a$, this sends $\mathcal{A}((c, d), \Omega)$ to $\mathcal{A}((e^{-2\pi t}c, e^{-2\pi t}d), \Omega)$, and taking the von Neumann closure of the interval net yields the displayed identity for $\mathcal{M}_a(\Omega)$. If $t \leq 0$, then $e^{-2\pi t}a \geq a$, so $H_{e^{-2\pi t}a} \subseteq H_a$, and isotony gives $\mathcal{M}_{e^{-2\pi t}a}(\Omega) \subseteq \mathcal{M}_a(\Omega)$. QED.

Lemma 5.2f (Positive null-translation generator). For the derived half-sided modular inclusion

$$\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega) \quad (a > 0),$$

let $\Delta_0(\Omega)$ be the modular operator of the standard pair $(\mathcal{M}_0(\Omega), \omega)$ and define $K_0(\Omega) := -\log \Delta_0(\Omega)$. Borchers–Wiesbrock then yields a unique strongly continuous one-parameter unitary group

$$U_\Omega(a) = e^{iaP_\Omega}, \quad a \in \mathbb{R},$$

such that

$$U_\Omega(a)\omega = \omega, \quad U_\Omega(a)\mathcal{M}_b(\Omega)U_\Omega(a)^* = \mathcal{M}_{a+b}(\Omega) \quad (a, b \geq 0),$$

and whose generator P_Ω is positive and self-adjoint on the Stone domain

$$D(P_\Omega) := \left\{ \psi \in \mathcal{H} : \lim_{a \rightarrow 0} \frac{U_\Omega(a)\psi - \psi}{ia} \text{ exists} \right\}.$$

Moreover,

$$\Delta_0(\Omega)^{it}U_\Omega(a)\Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t}a), \quad \Delta_0(\Omega)^{it}P_\Omega\Delta_0(\Omega)^{-it} = e^{-2\pi t}P_\Omega,$$

and on the common invariant analytic core of $K_0(\Omega)$ and P_Ω ,

$$[K_0(\Omega), P_\Omega] = -i2\pi P_\Omega.$$

If $\Delta_a(\Omega)$ denotes the modular operator of $(\mathcal{M}_a(\Omega), \omega)$ and $K_a(\Omega) := -\log \Delta_a(\Omega)$, then

$$\Delta_a(\Omega) = U_\Omega(a)\Delta_0(\Omega)U_\Omega(a)^*, \quad K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*,$$

so

$$K_a(\Omega) = K_0(\Omega) - 2\pi a P_\Omega$$

as a quadratic-form identity on $D(K_0(\Omega)) \cap D(P_\Omega)$. Equivalently,

$$\langle \psi, (K_b(\Omega) - K_a(\Omega))\phi \rangle = -2\pi(b - a)\langle \psi, P_\Omega\phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$. In the canonical normalization of Corollary 5.2b, the weak endpoint derivative from Proposition 5.2c is therefore exactly the Borchers generator. Bounded-interval modular-Hamiltonian formulas are downstream and require the separate interval-preserving branch recorded in Theorem 5.2g.

Proof. Corollary 5.2e gives the required half-sided modular inclusion. Borchers–Wiesbrock then yields the unique strongly continuous unitary group $U_\Omega(a)$, its positive self-adjoint generator

P_Ω , and the affine covariance relation with the modular group; Stone's theorem gives the displayed domain formula. Differentiating

$$\Delta_0(\Omega)^{it} U_\Omega(a) \Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t} a)$$

first at $a = 0$ and then at $t = 0$ on the common analytic core gives

$$[K_0(\Omega), P_\Omega] = -i 2\pi P_\Omega.$$

Because $U_\Omega(a)\omega = \omega$ and $U_\Omega(a)\mathcal{M}_0(\Omega)U_\Omega(a)^* = \mathcal{M}_a(\Omega)$, uniqueness of modular data for the translated standard pair gives the displayed formulas for $\Delta_a(\Omega)$ and $K_a(\Omega)$. Differentiating $K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*$ in a yields $dK_a/da = -2\pi P_\Omega$ as a quadratic-form identity, and integrating from 0 to a gives the stated half-line modular-Hamiltonian relation. QED.

Theorem 5.2g (The half-line generator is the local null-stress charge). In the canonical normalization of the preceding half-line construction,

$$\langle \psi, P_\Omega \phi \rangle = -\frac{1}{2\pi} \frac{d}{da} \Big|_{a=0^+} \langle \psi, \widetilde{K}_a(\Omega) \phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$. In the effective spacetime description of that same renormalized half-line family, the local null-stress charge is defined by the identical endpoint derivative,

$$Q_{kk}(\Omega) := -\frac{1}{2\pi} \frac{d}{da} \Big|_{a=0^+} \widetilde{K}_a^{\text{eff}}(\Omega),$$

so

$$P_\Omega = Q_{kk}(\Omega)$$

as a quadratic-form identity on the common domain. The only continuum input used here is the standard local modular-Hamiltonian form on the effective description of that same half-line family; it is used only to name, in continuum language, the operator fixed by the OPH half-line derivative. Bounded-interval transport through the affine-covariant kernel $g_I(v)$ is discharged by E0.5 below, and reconstruction of a full tensor from the directional charges is subject to the null-invisible metric ambiguity.

Boundary of the null bridge. The null bridge is an explicit theorem branch rather than an automatic fixed-cutoff identity. Its required data are:

- transferred cut-center data alone give the central sector decomposition of the null strip rather than the left/right HJPW split;
- the extra decomposition-inheritance condition of Proposition 5.2a is exactly what upgrades those sectors to the same collar-type block structure used in the spatial branch;
- the fixed-cutoff null-strip bridge is exact for exact Markov strip states on that inherited decomposition, and otherwise uses a controlled limit with the carried defect operator $\mathfrak{D}_J$;
- the local finite-constraint MaxEnt branch gives endpoint-Lipschitz control strong enough to define a weak tail generator for renormalized half-lines, and the scaling-limit geometric-cap branch of Theorem 4.2 then derives half-sided modular inclusion on the half-line blow-up net;
- Borchers–Wiesbrock therefore supplies a genuine positive null-translation generator P_Ω inside the bridge itself, together with its Stone domain and the half-line modular-Hamiltonian relation $K_a = K_0 - 2\pi a P_\Omega$;
- the half-line generator/charge identification is fixed inside the bridge itself, while bounded-interval transport and the later tensor upgrade are downstream.

This is the route from null-strip factorization to the later D5 Einstein branch, with the half-line generator/charge identification proved inside the null modular bridge itself.

4.6.3 Modular energy as stress-tensor charge on null half-lines

The effective-theory input here is the standard stress-tensor representation used to name the operator fixed by the null bridge. In the canonical normalization of the preceding null-half-line construction, the positive null generator is the endpoint derivative of the renormalized half-line modular family:

$$\langle \psi, P_\Omega \phi \rangle = -\frac{1}{2\pi} \frac{d}{da} \Big|_{a=0^+} \langle \psi, \widetilde{K}_a(\Omega) \phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$.

In the effective spacetime description of that same half-line family, the local null-stress charge is defined by the identical endpoint derivative,

$$Q_{kk}(\Omega) := -\frac{1}{2\pi} \frac{d}{da} \Big|_{a=0^+} \widetilde{K}_a^{\text{eff}}(\Omega),$$

so on the common quadratic-form domain

$$P_\Omega = Q_{kk}(\Omega).$$

Thus the null generator is exactly the local null-stress charge in the effective spacetime description; this step is not a separate scaling-limit input. When the effective description admits a local stress tensor, for example in a UV CFT regime or any local Lorentzian regime with local modular Hamiltonians, the same operator is the standard null-stress charge associated with the chosen null generator.

For comparison, in a CFT vacuum on a ball one has the familiar local formula

$$H_\zeta = \int_\Sigma T_{ab} \zeta^b d\Sigma^a,$$

so the present identification is the null-half-line version of that same modular-energy locality rather than an additional EFT postulate.

The downstream boundary is narrower. The E0.5 bounded-interval kernel transports this half-line statement to bounded null intervals with affine-covariant kernel $g_I(v)$, and the null-to-tensor upgrade determines T_{ab} only up to the familiar ambiguity

$$T_{ab} \mapsto T_{ab} + \phi g_{ab}.$$

But the generator/charge identification itself is fixed inside the null bridge.

4.6.4 Localized generalized entropy from Markov + MaxEnt

Using the collar decomposition and the collar double-scaling hypothesis, supplied either as a stated collar-limit condition or derived via Theorem 2.5 from the local-Gibbs form together with exponential mixing, the state takes the Markov normal form. MaxEnt selection maximizes entropy within each edge sector, producing

$$\rho_C = \bigoplus_\alpha p_\alpha \left(\rho_{\text{bulk},C}^\alpha \otimes \frac{\mathbf{1}_{\text{edge}}^\alpha}{d_\alpha} \right).$$

The entropy splits as

$$S(\rho_C) = H(p_\alpha) + \sum_\alpha p_\alpha S(\rho_{\text{bulk},C}^\alpha) + \sum_\alpha p_\alpha \log d_\alpha.$$

Convention: Throughout this paper, "log" denotes the natural logarithm ($\ln$), so entropies are measured in **nats** ($1 \text{ nat} = 1/\ln 2 \approx 1.443 \text{ bits}$). This is standard in thermodynamics and QFT; the Bekenstein-Hawking formula $S = A/4G$ uses nats. When clarity requires it, we write $\log_2$ explicitly for bits.

Define

$$S_{\text{bulk}}(C) := H(p_\alpha) + \sum_\alpha p_\alpha S(\rho_{\text{bulk},C}^\alpha),$$

and the central area operator

$$L_C := \sum_\alpha (\log d_\alpha) P_\alpha.$$

Then

$$S_{\text{gen}}(C) := \text{Tr}(\rho L_C) + S_{\text{bulk}}(C).$$

Newton coupling as an edge-entropy area-law readout.

In the collar double-scaling limit, the edge contribution becomes extensive along the entangling surface $\Sigma = \partial C$:

$$\text{Tr}(\rho L_C) \approx N_\Sigma \cdot \bar{\ell}(t), \quad \bar{\ell}(t) := \sum_\alpha p_\alpha \log d_\alpha,$$

where N_Σ is the number of UV cut elements covering Σ and $\bar{\ell}(t)$ is the **single-cell edge entropy** from the heat-kernel distribution (Theorem 6.20). Similarly, the geometric area is extensive:

$$A(C) \approx N_\Sigma \cdot a_{\text{cell}},$$

where a_{cell} is the area per UV cut element in the emergent metric.

Matching these expressions gives the geometric area-law coupling:

$$G_{\text{geom}} = \frac{a_{\text{cell}}}{4 \bar{\ell}(t)}$$

where:

- a_{cell} is fixed operationally from the UV correlation/mixing length ξ via $a_{\text{cell}} \sim \xi^2$ (from the exponential mixing hypothesis),
- $\bar{\ell}(t) = \sum_R p_R(t) \log d_R$ is computed from the heat-kernel edge distribution with $p_R \propto d_R e^{-t\lambda_R}$.

Explicitly:

$$\bar{\ell}(t) = \frac{\sum_R d_R e^{-t\lambda_R} \log d_R}{\sum_R d_R e^{-t\lambda_R}}.$$

On the OPH gravity branch this formula is not used to back-solve the scale. The selected no- G scale certificate supplies $\gamma_\star = \ell_\star \nu_{\text{Cs}}/c$, equivalently $B_\star = 3\pi/\ell_\star^2$. The local cell readings are

$$a_{\text{cell}} = P \ell_\star^2, \quad \bar{\ell}_{\text{shared}} = P/4,$$

so the Newton area-law readout gives

$$G_{\text{geom}} = \frac{P \ell_\star^2}{4(P/4)} = \ell_\star^2.$$

The SI display is $G_{\text{SI}} = c^3 \ell_\star^2 / \hbar$.

4.6.5 Fixed-cap generalized-entropy stationarity from MaxEnt

MaxEnt selection implies that for variations preserving cap labels (fixed size and charges),

$$\delta S_{\text{gen}}(C) = 0.$$

Using the split above and the first law for the bulk term,

$$\delta S_{\text{gen}}(C) = \delta \langle L_C \rangle + \delta \langle K_{\text{bulk}} \rangle.$$

4.6.6 No-smuggling dependency discharge for the Einstein bridge

Theorem E0 (Einstein bridge dependency discharge). Write OPH5 for the five-axiom basis: the S^2 screen net, overlap consistency, local MaxEnt/refinement, recoverable generalized entropy, and MAR. On a recovered-core, support-visible scaling branch of a cofinal OPH regulator family satisfying quotient confluence and the regularity hypotheses of Sections 2–5, OPH5 supplies the inputs used by the Einstein bridge:

- geometry readout through quotient normal forms via the screen fold $\chi_{S,r} \circ n_r \circ \pi_r$;
- Lorentz/H3 readout from round cap pairs, since $\text{Conf}^+(S^2) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1)$ and $H^3 \simeq \text{SO}^+(3, 1)/\text{SO}(3)$;
- 2π -normalized support-visible BW flow on the extracted prime geometric subnet;
- the null generator/local null-stress charge identity on the same branch;
- the bounded-interval kernel of Lemma E0.5;
- fixed-cap generalized-entropy stationarity from the MaxEnt/refinement theorem;
- the standard fixed-volume small-ball area identity after geometry readout;
- $o(\ell^4)$ remainder control by Lemma E0.6; and
- all local timelike directions by the cap-pair coverage of Lemma E0.7.

Thus the finite consensus reduct supplies quotient normal forms, while OPH5 on the declared recovered-core, support-visible scaling branch supplies the geometric, modular, stress, entropy, and coverage inputs needed by the Einstein bridge. Bare finite consensus without these OPH5 branch closures does not imply the Einstein equation.

Lemma E0.5 (Bounded-interval kernel from null projective covariance). On the null blow-up of the support-visible geometric branch, the half-line generator/charge identity, endpoint-Lipschitz renormalized interval control, and affine/projective covariance give

$$K_{(a,b)}^{\text{null}} = 2\pi \int_a^b \frac{(v-a)(b-v)}{b-a} T_{kk}(v) dv + R_{(a,b)}.$$

The remainder $R_{(a,b)}$ is the carried endpoint/collar term of the same refinement family. In the small-ball limit this is the ball modular kernel used below.

Lemma E0.6 (Diagonal remainder control). If $\eta_\delta \rightarrow 0$ is the carried collar/Markov/recovery error at collar scale δ , then for the small-ball radius ℓ one can choose a diagonal refinement $\delta = \delta(\ell)$ with $\eta_{\delta(\ell)} \leq \ell^5$. Hence endpoint, collar, and long-wavelength derivative remainders are $o(\ell^4)$ and do not alter the Einstein coefficient.

Lemma E0.7 (Cap-pair timelike coverage). On the cap-normal H^3 branch, every future timelike vector v has $r = \sqrt{-\eta(v, v)}$ and $u = v/r \in H^3$. For any unit $s \in u^\perp$, the vectors

$$q_\pm = u \pm s$$

are future null and

$$v = \frac{r}{2}(q_+ + q_-).$$

Thus normalized future null directions in observer skies cover every local timelike direction once the observer-frame chart is supplied. This does not say that two unnormalized projective rays alone select a unique observer frame; the observer normalization or equivalent causal-diamond scale data is also required.

Proposition E2 (Recovered-core stress closure). If the recovered stress tensor is reconstructed from the full compatible family of null and timelike modular charges on the recovered core, then $\nabla^a \langle T_{ab} \rangle = 0$. The overlap-compatible local charge family cancels on opposite faces of infinitesimal recovered-core diamonds; after the cap-pair tensor upgrade this cancellation is the covariant divergence-free condition.

4.6.7 Einstein equation from fixed-cap generalized-entropy stationarity

This subsection is a branch theorem. The finite consensus reduct supplies quotient normal forms; Theorem E0 discharges the OPH5 recovered-core inputs needed by the Einstein bridge: geometry readout, support-visible BW modular covariance, the null-stress bridge, bounded-interval transport, fixed-cap MaxEnt stationarity, the fixed-volume small-ball area variation, carried-remainder control, and the timelike scalar-to-tensor upgrade.

In the d=4 scaling regime, the small-ball bridge is internal to the derived gravity chain once the E0.5 bounded-interval kernel is included. The only extra standard geometric input is the fixed-volume area-variation identity for a small geodesic ball. The modular kernel itself is not imported from a separate EFT law: on the extracted prime geometric subnet, a sufficiently small cap is the tangent causal diamond D_ℓ , whose preserving conformal Killing field is

$$\xi_{D_\ell} = \frac{1}{2\ell} \left((\ell^2 - r^2 - t^2) \partial_t - 2t x^i \partial_i \right).$$

On the t=0 slice its lapse is

$$\xi_{D_\ell} \cdot n = \frac{\ell^2 - r^2}{2\ell}.$$

The bounded-interval kernel comes from combining the D3 geometric cap generator with the E0.5 affine/projective transport of the D4 half-line stress bridge, so the D3 cap-modular theorem and the D4 stress bridge together yield

$$\delta S_{\text{bulk}}(C) = 2\pi \int_{B_\ell} \frac{\ell^2 - r^2}{2\ell} \delta \langle T_{00} \rangle d^3x + \delta \langle E_{C,\ell}^{(\eta)} \rangle.$$

If $\delta \langle T_{00} \rangle$ is approximately constant across the ball and the carried remainder is $o(\ell^4)$, then

$$\delta S_{\text{bulk}}(C) = \frac{8\pi^2 \ell^4}{15} \delta \langle T_{00} \rangle + O(\ell^5 \partial T) + o(\ell^4).$$

Stationarity of generalized entropy therefore gives the observer-covariant scalar relation

$$\delta \left[(G_{ab} + \Lambda g_{ab}) u^a u^b \right] = 8\pi G u^a u^b \delta \langle T_{ab} \rangle$$

for the local diamond rest-frame four-velocity u^a . In the adapted rest frame $u^a = (1, 0, 0, 0)$, this is

$$\delta (G_{00} + \Lambda g_{00}) = 8\pi G \delta \langle T_{00} \rangle.$$

4.6.8 Overlaps supply all timelike directions

This scalar-to-tensor upgrade is internal once the Lorentz branch is in place. Overlapping observers through the same bulk point realize every local timelike four-velocity u . Define

$$Y_{ab} := G_{ab} + \Lambda g_{ab} - 8\pi G \langle T_{ab} \rangle.$$

The rest-frame relation of the previous subsection says

$$u^a u^b \delta Y_{ab} = 0$$

for every such u . In local inertial coordinates,

$$u^a = \gamma(1, v^i), \quad |\vec{v}| < 1,$$

so

$$0 = u^a u^b \delta Y_{ab} = \gamma^2 \left(\delta Y_{00} + 2v^i \delta Y_{0i} + v^i v^j \delta Y_{ij} \right)$$

for all $|\vec{v}| < 1$. The polynomial therefore vanishes coefficientwise, hence

$$\delta Y_{00} = 0, \quad \delta Y_{0i} = 0, \quad \delta Y_{ij} = 0.$$

So the full tensor variation vanishes:

$$\delta Y_{ab} = 0.$$

Along any connected scaling branch of reference states, Y_{ab} is therefore constant. The only purely local freedom left by the null reconstruction is the metric term, and fixing that term on one maximally symmetric reference state gives

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle.$$

Thus the tensor upgrade does not require a separate EFT handoff once the geometric-cap theorem, the null bridge, and the overlap-complete timelike family are in place.

4.6.9 Non-tunable numerical constants

The gravity chain yields specific numerical constants as rigid outputs of the axiom chain.

The 2π KMS normalization. From Theorem 4.2, cap modular flow on the support-visible scaling-limit geometric cap pair is fixed by the KMS condition to the standard 2π normalization. This is the same rigidity that fixes Unruh/Hawking temperature normalization.

The geometric coefficient $\Omega_{d-2}/(d^2 - 1)$. This coefficient appears in both (a) the CFT-ball modular Hamiltonian weight integral and (b) the geometric area-variation identity. It is an exact integral identity:

$$\int_{B_\ell^{d-1}} \frac{\ell^2 - r^2}{2\ell} d^{d-1}x = \frac{\Omega_{d-2} \ell^d}{d^2 - 1}.$$

In $d = 4$:

$$\frac{\Omega_2}{4^2 - 1} = \frac{4\pi}{15} \approx 0.8377580409572781.$$

This is the reason prefactors cancel cleanly when going from $\delta S_{\text{gen}} = 0$ to the Einstein equation (leaving $8\pi G$ with the 2π fixed by Theorem 4.2).

What is predicted. The framework cleanly separates:

- **Non-tunable constants:** 2π (KMS period), $\Omega_{d-2}/(d^2 - 1)$ (geometric coefficient), the existence of the Einstein form.
- **Global closure constants:** G is the conversion between edge entropy and geometric area on the emitted local gravity branch. Λ is not fixed by the local reference-state variation data alone; the capacity closure fixes the dimensionless product $\Lambda_{\text{CRC}} \ell_\star^2 = 3\pi/N_{\text{CRC}}$, while the SI curvature scale also uses the selected OPH scale certificate.

4.6.10 Quantitative Markov error and controlled corrections

The role of this subsection is to keep three distinct quantities separate:

1. the raw collar conditional mutual information

$$\varepsilon_\delta := I(A_\delta : D_\delta \mid B_\delta)_\omega;$$

2. the constructive recovery error

$$r_{\text{FR}}(\varepsilon_\delta) := 2\sqrt{1 - e^{-\varepsilon_\delta}} \leq 2\sqrt{\varepsilon_\delta};$$

3. the exact-Markov replacement error on one fixed faithful collar model,

$$\eta_\delta^{\text{M}} := 4\lambda_*^{-1} \delta_{A_\delta:B_\delta:D_\delta}^{\text{M}}(\varepsilon_\delta),$$

where $\lambda_* > 0$ is the lower spectral bound needed to compare modular Hamiltonians.

The exact identity for the modular defect expectation is unchanged:

$$\langle \Delta K_\delta \rangle_\omega = -I(A_\delta : D_\delta \mid B_\delta)_\omega = -\varepsilon_\delta,$$

with

$$\Delta K_\delta := K_{A_\delta B_\delta D_\delta} - K_{A_\delta B_\delta} - K_{B_\delta D_\delta} + K_{B_\delta}.$$

What changes is the interpretation of later “exact” collar formulas. For each fixed finite-dimensional collar one may choose an exact Markov reference state σ_δ with

$$\|\omega_\delta - \sigma_\delta\|_1 \leq \delta_{A_\delta:B_\delta:D_\delta}^{\text{M}}(\varepsilon_\delta),$$

and a constructive recovered comparison state $\omega_\delta^{\text{rec}}$ with

$$\|\omega_\delta - \omega_\delta^{\text{rec}}\|_1 \leq r_{\text{FR}}(\varepsilon_\delta).$$

These two comparisons do different jobs:

Bounded-observable control. For every bounded observable X supported on $A_\delta \cup B_\delta$,

$$|\text{Tr}[X(\omega_\delta - \omega_\delta^{\text{rec}})]| \leq \|X\|_\infty r_{\text{FR}}(\varepsilon_\delta),$$

and

$$|\text{Tr}[X(\omega_\delta - \sigma_\delta)]| \leq \|X\|_\infty \delta_{A_\delta:B_\delta:D_\delta}^{\text{M}}(\varepsilon_\delta).$$

The first bound is constructive and dimension-free; the second is the fixed-collar route to the exact Markov set.

Modular-additivity control. If the relevant marginals are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\Delta K_\delta(\omega) - \Delta K_\delta(\sigma_\delta)\|_\infty \leq \eta_\delta^M.$$

Since $\Delta K_\delta(\sigma_\delta)$ is central by exact Markovity, the finite-stage modular defect is within η_δ^M of the exact splice or additivity value.

Therefore the manuscript's exact collar identities arise as limits with a carried remainder

$$\eta_\delta := r_{\text{FR}}(\varepsilon_\delta) + \eta_\delta^M,$$

together with the separate long-wavelength derivative remainders present in the small-ball expansion. Exact identities at finite collar width require exact Markovity; otherwise one works with η_δ -controlled approximations and then lets $\eta_\delta \rightarrow 0$ in the controlled collar limit.

Finite-cutoff Einstein remainder bound. In the small-ball rest-frame calculation, write

$$\delta S_C = \frac{8\pi^2 \ell^4}{15} \delta \langle T_{00} \rangle + \delta \langle E_C^{(\delta)} \rangle + O(\ell^5 \partial T),$$

where $E_C^{(\delta)}$ packages the carried collar remainder. Then the derived fixed-cap generalized-entropy stationarity theorem gives

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G \delta \langle T_{00} \rangle + \mathcal{E}_\delta,$$

with

$$\mathcal{E}_\delta := \frac{15G}{\pi \ell^4} \delta \langle E_C^{(\delta)} \rangle + O(\ell \partial T),$$

and, by the bounds above,

$$|\delta \langle E_C^{(\delta)} \rangle| \leq C_C \eta_\delta$$

for some collar-dependent constant C_C on the fixed faithful model. Equivalently, before the controlled collar and small-ball limits, one may write

$$|\mathcal{E}_{\ell,\delta}| \leq C_1 r_{\text{FR}}(\varepsilon_\delta) + C_2 \delta_{A_\delta:B_\delta:D_\delta}^M(\varepsilon_\delta) + C_3 \eta_\delta^{\text{reg}} + C_4 \ell \|\partial T\| + o_\delta(1) + o_\ell(1),$$

with constants depending only on the fixed collar model, the bounded observable class, and the small-ball chart. Thus the exact Einstein relation is the controlled limit, not a finite-cutoff identity inherited from approximate Markovity.

If one wishes to package this carried remainder as an effective anomalous energy density, the natural definition is

$$\delta \langle T_{00}^{\text{anom}} \rangle := \frac{15}{8\pi^2 \ell^4} \delta \langle E_C^{(\delta)} \rangle.$$

Then

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G \delta(\langle T_{00} \rangle + \langle T_{00}^{\text{anom}} \rangle) + O(\ell \partial T).$$

This is a bookkeeping definition of the finite-stage remainder, not an upgrade of the recovered-core theorem. Its significance is exactly that it is controlled:

$$|\delta \langle T_{00}^{\text{anom}} \rangle| \leq \frac{15 C_C}{8\pi^2 \ell^4} \eta_\delta.$$

On a separate dark/anomaly continuation branch, this carried scalar remainder may be promoted to a repair stress only after a quotient-invariant source-localization receipt identifies the same collar

operator with the small-ball Einstein remainder, and after a positive causal packet lift supplies the full tensor. In SI units that continuation has rest-energy normalization

$$T_A^{ab} u_a u_b = \frac{15\hbar c}{8\pi^2 \ell^4} R_{r,\ell} + O(\epsilon_{\text{loc}}),$$

where $R_{r,\ell}$ is measured in nats and ℓ is the proper geodesic small-ball radius. The scalar quantity $I(A : D \mid B)$ alone is not a stress tensor: packet factorization, causal directions, an allocation rule, a rest frame, source-localization map, and transport law are additional continuation data.

The framework carries the Markov error through three distinct controls:

1. $r_{\text{FR}}(\varepsilon_\delta)$ controls bounded observables at one stage;
2. $\delta_{A_\delta:B_\delta:D_\delta}^{\text{M}}(\varepsilon_\delta)$ controls convergence to the exact Markov normal form on a fixed collar;
3. η_δ^{M} controls modular-Hamiltonian replacements once faithfulness is assumed;
4. the Einstein-branch remainder is a carried $O(\eta_\delta)$ term, not a silent exact identity at finite cutoff.

This is the concrete bridge from “axioms about screens” to “precision GR predictions plus a disciplined carried remainder”.

4.6.11 Focusing/QNEC internalization via relative entropy

Once the fixed-cutoff null modular structure of Section 5.2 is combined with the exact half-line generator/charge identification and the E0.5 bounded-interval kernel, focusing constraints follow from information-theoretic principles in the same scaling-limit branch rather than from a separate fixed-cutoff postulate.

Derivation chain. QNEC and focusing are supported within the same branch:

- Fixed-cutoff null modular bridge (§5.2) $\implies$ exact-or-controlled strip additivity and a weak tail generator
- Derived half-sided modular pair + exact half-line generator/charge identification $\implies$ local null-stress charge on half-lines and $[K, P] = i \, 2\pi \, P$
- Relative entropy monotonicity $\implies$ QNEC
- Einstein (Thm 5.1) + Raychaudhuri $\implies$ QFC for S_{gen}

Relative entropy monotonicity argument. The key input is the monotonicity of relative entropy under partial trace, which is pure information theory:

$$S(\rho_{AB} \parallel \sigma_{AB}) \geq S(\rho_A \parallel \sigma_A).$$

For null deformations parameterized by λ , consider nested null regions $R(\lambda) \subset R(\lambda')$ obtained by varying the entangling cut along v . The modular Hamiltonian K_λ generates the modular flow, and relative entropy satisfies convexity:

$$\frac{d^2}{d\lambda^2} S(\rho_\lambda \parallel \sigma_\lambda) \geq 0.$$

Proposition 5.10a (Internal QNEC). Under the fixed-cutoff null bridge of §5.2 together with the downstream density-upgrade and E0.5 bounded-interval conditions, the second null variation of von Neumann entropy satisfies

$$\frac{d^2 S_{\text{bulk}}}{d\lambda^2} \leq 2\pi \langle T_{kk}(\lambda) \rangle,$$

with the 2π normalization fixed by Theorem 4.2.

Proof. Use the derived half-sided modular-inclusion statement of Corollary 5.2e and its Borchers-Wiesbrock consequence from Lemma 5.2f. This gives the translation structure with $[K, P] = -i2\pi P$.

Consider the relative entropy $S(\rho_\lambda \| \omega_\lambda)$ between the state ρ restricted to $R(\lambda)$ and the reference state ω . Monotonicity under restriction to smaller regions ($\lambda' > \lambda$) gives:

$$S(\rho_{R(\lambda)} \| \omega_{R(\lambda)}) \leq S(\rho_{R(\lambda')} \| \omega_{R(\lambda')}).$$

Using the first law $\delta S = \delta \langle K \rangle$ and the geometric half-line modular form supplied by Section 5.2, expand to second order in the deformation. The convexity of relative entropy yields the QNEC inequality. The bound saturates for coherent states in the standard way. QED.

Corollary 5.10b (QFC for generalized entropy). With the central area operator L_C from EC/MaxEnt (Section 5.4), define

$$S_{\text{gen}} = \text{Tr}(\rho L_C) + S_{\text{bulk}}.$$

Given the Einstein branch (Theorem 5.1) and the classical Raychaudhuri identity for null congruences, the Quantum Focusing Conjecture (QFC) follows within the same scaling regime: the generalized expansion Θ_{gen} is non-increasing along null generators.

Proof sketch. The Raychaudhuri equation relates expansion evolution to R_{kk} . Einstein's equation gives $R_{kk} = 8\pi G(T_{kk} - \frac{1}{2}g_{kk}T)$. For null k , this simplifies to $R_{kk} = 8\pi GT_{kk}$. The QNEC (Prop 5.10a) then bounds the bulk entropy production, ensuring $d\Theta_{\text{gen}}/d\lambda \leq 0$. QED.

Significance. This section shows how the Recoverable Generalized Entropy focusing package is internally supported inside the same null-modular branch once the stress-tensor identification and Einstein relation are in place. That package is part of the stated axiom set.

Theorem 5.1 (OPH Einstein-branch closure in the scaling regime). Under OPH5 on a recovered-core, support-visible scaling branch satisfying the E0 dependency discharge above, the fixed-cap generalized-entropy stationarity condition implies the rest-frame first-variation relation

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G \delta \langle T_{00} \rangle.$$

If this relation holds for all local directions and reference states in the locally Lorentzian scaling regime, the timelike scalar-to-tensor upgrade promotes it to the semiclassical Einstein equation modulo the expected Λg_{ab} ambiguity. The D6 capacity branch closes that metric term globally.

Proof sketch. The E0 discharge turns the OPH5 recovered-core branch into the exact hypothesis package required by the Jacobson-type bridge: quotient-safe geometry readout, 2π -normalized BW cap flow, local null stress charge, bounded-interval ball kernel with $o(\ell^4)$ remainder, fixed-cap generalized-entropy stationarity, the fixed-volume small-ball area identity, and all timelike directions. The cap first law identifies δS_C with $\delta \langle K_C \rangle$, while the null bridge together with the E0.5

bounded-interval kernel relates the bulk modular variation to the required null stress-tensor charge on the diamond. Since the reference state is maximally symmetric, the fixed-volume small-ball area identity enters at first order in the perturbation, and the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family yields the displayed rest-frame relation. Lemma E0.7 supplies all local timelike directions, the scalar-to-tensor theorem upgrades the relation to the tensor equation modulo Λg_{ab} , and D6 supplies the global capacity closure for that remaining metric term. QED.

4.6.12 Black-hole structural statements and continuation-level spectroscopy

This subsection mixes two claim tiers and is organized accordingly. The retained structural black-hole package is a four-step chain: fixed-cutoff edge-center collar decomposition on the exact-Markov or idealized exact-recoverability horizon carrier, the small-CMI recoverability reading of interior encoding on that same carrier, Hawking/KMS normalization on the geometric modular branch, and the discrete area spectrum together with the Schwarzschild transition identity $\Delta E = k_B T_H \ln(d'/d)$ for an actual sector change $d \rightarrow d'$. Finite same-boundary records and finite reconstruction thresholds do not by themselves supply physical radiation entropy, exterior time, evaporation flux, a geometric island, a radiative quotient, or an asymptotic waveform readout. Any clean comb structure, QNM selector, Page-type linewidth estimate, PBH burst template, Kerr/LIGO horizon spectroscopy template, or Page-curve/island closure is continuation-level and uses continuation inputs stated below.

Derived structural statements.

Area eigenvalues from edge sectors. The central area operator (Section 5.4) is

$$L_C = \sum_{\alpha} (\log d_{\alpha}) P_{\alpha},$$

where $d_{\alpha} \in \mathbb{N}$ is the dimension of the edge Hilbert space in sector α . With the normalization $\text{Tr}(\rho L_C) = \langle A \rangle / 4G_{\text{geom}}$, the area eigenvalues are

$$A_{\alpha} = 4G_{\text{geom}} \log d_{\alpha} = 4\ell_{\star}^2 \ln d_{\alpha},$$

where $\ell_{\star}^2 = \hbar G_{\text{SI}} / c^3$ is the scale-readout area displayed as the Planck area. Since d_{α} is a positive integer, **areas are discretely spaced** with logarithmic gaps.

Hawking emission energy quantization. For a Schwarzschild black hole with $A(M) = 16\pi G^2 M^2 / c^4$, a transition between sectors $d \rightarrow d'$ changes the area by

$$\Delta A = 4\ell_{\star}^2 \ln(d'/d).$$

The corresponding ADM energy change of the hole is $\Delta(Mc^2) = c^2 \Delta M$, with $\Delta M = \Delta A / (dA/dM)$. This gives

$$\Delta(Mc^2) = \frac{\hbar c^3}{8\pi G M} \ln(d'/d).$$

Using the Hawking temperature $T_H = \hbar c^3 / (8\pi G k_B M)$, whose 2π normalization is fixed by the BW_{S^2} tangent-limit normalization of Theorem 4.2:

$$\Delta(Mc^2) = k_B T_H \ln(d'/d).$$

Continuation-level spectroscopy templates. The remainder of this subsection is not part of the recovered core. It records what follows only if one adds extra discrete-horizon selection rules and,

where stated, standard semiclassical inputs such as evaporation-power models, QNM/transition identifications, or greybody matching.

Integer transitions. If dominant emission steps reduce the edge dimension by an integer factor k (i.e., $d' = d/k$), the emitted spectrum becomes a discrete comb:

$$\Delta E_k = k_B T_H \ln k, \quad \Delta f_k = \frac{c^3}{16\pi^2 G M} \ln k.$$

Structural condition: comb vs. generic discreteness. The log-integer *comb* structure requires the additional dynamical assumption that integer-ratio emission steps ($d \rightarrow d/k$) dominate. If generic transitions between arbitrary integers dominate instead, the set of $|\ln(d'/d)|$ values becomes a dense log-rational set that may appear quasi-continuous after folding in linewidths and astrophysical effects. What follows directly from the axioms is the *discrete area spectrum*; the clean comb pattern follows only with that selection rule.

Continuation-level template (Discrete Hawking spectrum). If the additional integer-transition selection rule is realized, the Hawking emission spectrum consists of discrete lines with spacing $\Delta E_k = k_B T_H \ln k$, where k is an integer characterizing the dominant sector transitions, instead of a continuous thermal profile.

Mass-independent fractional linewidth (continuation-level estimate). Using Page's semiclassical calculation for emission power $P(M) = p_0 \hbar c^6 / (G^2 M^2)$ with $p_0 \approx 2 \times 10^{-4}$, the emission rate is $\dot{N} \approx P/\langle E \rangle$ where $\langle E \rangle = a k_B T_H$ with $a \sim \mathcal{O}(1-10)$. The natural linewidth $\Gamma \sim \hbar \dot{N}$ divided by the level spacing gives:

$$\frac{\Gamma}{\Delta E_k} \approx \frac{64\pi^2 p_0}{a \ln k} \approx 3 - 5\%$$

Within this continuation branch, the emission lines are narrow (few-percent fractional width) and the estimated fraction is mass-independent.

Connection to quasinormal modes (interpretive continuation). The highly-damped Schwarzschild quasinormal modes have asymptotic real part (Motl, 2002):

$$\text{Re } \omega \rightarrow \frac{c^3}{8\pi G M} \ln 3.$$

This matches exactly the $k = 3$ transition frequency $\Delta E_3/\hbar$. If one adopts a Bohr-type identification between quantum transition frequencies and asymptotic QNM frequencies, this selects

$$\Delta A = 4\ell_\star^2 \ln 3 \approx 4.39 \ell_\star^2$$

as the fundamental area quantum.

Scope statement. The area quantization follows from the edge-sector structure (derived). The $k = 3$ selection requires the additional interpretive identification with QNM frequencies (not derived from axioms). The linewidth prediction uses standard semiclassical inputs.

Numerical examples. For $\Delta f_k = (c^3/16\pi^2 G M) \ln k$:

- **M = 30 M_⊙:** $k=2$ at 29.7 Hz, $k=3$ at 47.1 Hz
- **M = 1 M_⊙:** $k=2$ at 891 Hz, $k=3$ at 1412 Hz
- **M = 10¹² kg (primordial):** $k=2$ at 7.3 MeV, $k=3$ at 11.6 MeV

Within the integer-transition continuation, these frequencies track $k_B T_H \ln k$ exactly and are in principle distinguishable from a continuous thermal spectrum.

Continuation-level PBH burst search template.

Within the discrete-horizon continuation, the Hawking comb would provide a distinctive gamma-ray template. A template-level discriminant is **log-integer energy ratios**: if two emission lines are observed at energies E_2 and E_3 , their ratio must satisfy

$$\frac{E_3}{E_2} = \frac{\ln 3}{\ln 2} \approx 1.585$$

exactly, independent of black hole mass. Within that continuation template, the ratio is fixed once the log-integer rule is assumed.

Available instruments and energy coverage. The $k = 2$ line energy $E_2 = k_B T_H \ln 2$ determines which instruments can see a given BH mass:

Instrument	Energy band	BH mass range (k=2 in band)
Fermi GBM (BGO)	0.15–40 MeV	2×10^{11} – 5×10^{13} kg
Fermi LAT	0.1–300 GeV	2×10^7 – 7×10^{10} kg
H.E.S.S.	0.1–100 TeV	7×10^4 – 7×10^7 kg
LHAASO-WCDA	1–15 TeV	5×10^5 – 7×10^6 kg

Detector resolution vs. intrinsic linewidth. The continuation-level linewidth estimate is 3–5% (mass-independent). Published detector energy resolutions:

- Fermi GBM: < 10% (0.1–1 MeV), $\sim 4\%$ at 10 MeV (BGO)
- Fermi LAT: < 10% (1–100 GeV)
- H.E.S.S.: $\sim 15\%$ (TeV)
- LHAASO-WCDA: $\sim 33\%$ (TeV)

Within this template, the comb could be resolvable with GBM/LAT; at TeV energies it would appear as moderately broad bumps rather than sharp lines.

Search protocol. A dedicated OPH-comb search would:

1. Select burst-like candidates (10–120 s time windows, matching existing PBH burst search protocols).
2. Fit each candidate with null model (smooth continuum) vs. OPH comb model (peaks at $E_k = E_0 \ln k$ convolved with detector response).
3. Scan over the single scale parameter $E_0 = k_B T_H$ (equivalently, BH mass).
4. Require at least two lines satisfying log-integer ratio to claim detection.
5. Correct significance for trials (time windows $\times$ sky positions $\times$ E_0 scan).

Observational context. Dedicated PBH burst searches (H.E.S.S., LHAASO) report **no significant bursts**. An OPH-specific comb-template analysis of archival data would:

- Set upper limits on OPH-comb PBH burst rates
- Demonstrate direct testability of the discrete spectrum prediction
- Provide constraints comparable to or stronger than generic PBH burst limits

Data availability. Fermi GBM provides public Time-Tagged Event (TTE) burst data; Fermi LAT provides public photon event lists with documented analysis workflows. H.E.S.S. has a small public test data release.

Continuation-level GW horizon spectroscopy template for Kerr remnants.

The same continuation-level discrete-horizon branch extends to gravitational wave observables. For Kerr black holes, the thermodynamic first law is $\delta M = T_H \delta S + \Omega_H \delta J$, so the entropy change for absorbing a quantum with frequency ω and azimuthal number m is:

$$\delta S = \frac{\hbar(\omega - m\Omega_H)}{k_B T_H}.$$

In the edge-sector framework, $\delta S = \ln(d'/d)$, so the discreteness condition becomes:

$$\hbar(\omega - m\Omega_H) = k_B T_H \ln k, \quad k \in \{2, 3, 4, \dots\}$$

Under those additional discrete-horizon assumptions, this gives the **GW horizon spectroscopy comb**: a continuation-level set of discrete resonant frequencies where the horizon can efficiently absorb or emit energy. A physical QNM or ringdown claim needs a separate exterior-background bridge, a radiative quotient, a finite operator converging to the continuum perturbation generator, future-horizon and future-null-infinity boundary conditions, and an asymptotic readout map. A finite repair spectrum or fitted clock scale is not that bridge.

Kerr line frequencies. For a remnant with mass M and dimensionless spin $\chi = a_*/M$, define the spin correction factor:

$$g(\chi) = \frac{2\sqrt{1-\chi^2}}{1+\sqrt{1-\chi^2}}, \quad \Omega_H(M, \chi) = \frac{c^3}{2GM} \cdot \frac{\chi}{1+\sqrt{1-\chi^2}}.$$

The line frequencies are:

$$f_{k,m}(M, \chi) = \frac{m \Omega_H(M, \chi)}{2\pi} + \frac{c^3}{16\pi^2 GM} g(\chi) \ln k$$

Within this continuation template, once LIGO/Virgo infers (M, χ) for a remnant, the line pattern is fixed by the inferred remnant parameters and the assumed discrete-horizon rule. The line pattern belongs to a continuation template outside recovered core.

Line weights from GR envelope + discretization. In this continuation template, the line strengths are modeled by matching to the known GR greybody absorption spectrum in the semiclassical limit. The discretization rule gives bin width $\Delta\omega_k \approx \omega_T \ln(1 + 1/k)$ where $\omega_T = k_B T_H / \hbar$. The net line weight (absorption minus stimulated emission) is:

$$W_{k,\ell m}^{\text{net}} = \Gamma_{\ell m}^{\text{GR}}(\omega_{k,m}) \cdot \Delta\omega_k \cdot \frac{k-1}{k}$$

where $\Gamma_{\ell m}^{\text{GR}}$ is the standard GR greybody factor and the $(k-1)/k$ factor arises from KMS detailed balance with $e^{(\omega - m\Omega_H)/T_H} = k$.

Universal stacking coordinate. Define the dimensionless rescaled frequency:

$$x := \frac{GM}{c^3 g(\chi)} (\omega - m\Omega_H).$$

Then the predicted line locations collapse to universal constants:

$$x_k = \frac{\ln k}{8\pi} \quad (k = 2, 3, 4, \dots)$$

Numerically: $x_2 = 0.02758$, $x_3 = 0.04371$, $x_4 = 0.05516$, $x_5 = 0.06404$.

Stacking test. Multiple BBH events can be mapped to this universal x coordinate and stacked. If the comb is real, peaks align across events with different (M, χ) ; detector noise does not stack coherently.

Comparison to existing work. Prior area-quantization searches [51] used parameterized models with one free spacing constant. The OPH prediction is more constrained: multiple lines with exact $\ln k$ ratios, plus the $(k-1)/k$ weight hierarchy from detailed balance.

Numerical example (GW170608). Remnant parameters: $M_f \approx 18.0M_\odot$, $\chi_f \approx 0.69$. For $m=2$, the horizon rotation frequency is $m\Omega_H/(2\pi) \approx 719$ Hz. The **thermal comb spacing** (the part that encodes the area quantization) is:

k	$\Delta f_k := \frac{c^3 g(\chi)}{16\pi^2 GM} \ln k$ (Hz)	Relative weight $(k-1)/k$
2	41.6	0.500
3	65.9	0.667
4	83.2	0.750
5	96.5	0.800
6	107.5	0.833

The full physical frequencies are $f_{k,2} = 719 + \Delta f_k$ Hz (i.e., 760–827 Hz), outside LIGO's most sensitive band for this remnant. However, the **stacking analysis** uses the rescaled coordinate $x = GM(\omega - m\Omega_H)/(c^3 g(\chi))$, which maps the thermal spacing to universal constants $x_k = \ln k/8\pi$ regardless of the rotation offset.

Template-matching criterion. After rescaling by (M, χ) , spectral features in this continuation template must satisfy $f_k/f_2 = \ln k/\ln 2$ exactly, independent of remnant parameters. Absence of coherent stacking at the predicted x_k values would challenge the discrete-horizon continuation template, not by itself the derived area-spectrum statement.

4.6.13 Classical mechanics from emergent GR

Once the Einstein equation is established, the framework inherits standard GR consequences. This section makes explicit how classical mechanics emerges.

Stress-energy conservation is automatic. The contracted Bianchi identity is geometric:

$$\nabla^a G_{ab} = 0.$$

Combined with the Einstein equation, this implies:

$$\nabla^a \langle T_{ab} \rangle = 0.$$

Geodesic motion from dust limit. For pressureless classical matter ("dust"), $T^{ab} = \rho u^a u^b$. Conservation yields:

$$\nabla_a(\rho u^a u^b) = 0 \quad \Rightarrow \quad u^b \nabla_a(\rho u^a) + \rho u^a \nabla_a u^b = 0.$$

Projecting orthogonally to u^b using $h^b_c = \delta^b_c + u^b u_c$ kills the first term, giving:

$$\rho u^a \nabla_a u^b = 0 \quad \Rightarrow \quad u^a \nabla_a u^b = 0.$$

This is the geodesic equation: free classical bodies follow spacetime geodesics.

Newtonian limit from weak-field GR. Take the weak-field, slow-motion limit with metric:

$$g_{00} \approx -(1 + 2\Phi/c^2), \quad g_{0i} \approx 0, \quad g_{ij} \approx \delta_{ij}(1 - 2\Phi/c^2),$$

and velocities $|\mathbf{v}| \ll c$. Then $G_{00} \approx 2\nabla^2\Phi/c^2$ (leading order), and $T_{00} \approx \rho c^2$. The Einstein equation reduces to:

$$\nabla^2\Phi = 4\pi G\rho.$$

Geodesic motion reduces to:

$$\ddot{\mathbf{x}} = -\nabla\Phi.$$

These are Newton's gravitational law and Newton's second law. Classical mechanics is recovered as a controlled limit of the emergent GR dynamics.

Precision classical predictions. Once the field equation is fixed to Einstein form, the framework inherits the standard GR precision toolbox (post-Newtonian expansion, lensing, time delay, etc.), with no free "shape" parameters beyond G and Λ .

Selected precision predictions (in the regime where the GR derivation applies):

Light bending by mass M : For impact parameter b ,

$$\Delta\theta = \frac{4GM}{c^2b}.$$

For the Sun with $b \approx R_\odot$: $\Delta\theta \approx 1.751$ arcsec.

Mercury perihelion advance: Per orbit,

$$\Delta\varpi = \frac{6\pi GM}{a(1-e^2)c^2}.$$

Using Mercury's orbital parameters: $\Delta\varpi \approx 42.98$ arcsec/century.

Gravitational redshift: Between two radii in a static potential,

$$\frac{\Delta\nu}{\nu} \approx \frac{\Delta\Phi}{c^2}.$$

For the Sun (surface to infinity): $z \approx 2.12 \times 10^{-6}$.

These predictions are fixed functions of G and known source parameters, and are confirmed observationally to high precision. The framework contains them automatically on the derived Einstein branch.

4.6.14 Precision gravity predictions and experimental bounds

The pure Einstein linearization makes classical dispersion and polarization statements that can be confronted with the tightest available experimental bounds. This section translates those branch-scoped statements into the observables that experiments constrain; it does not infer a quantum graviton pole from the classical equation.

Speed of gravitational waves. On a suitable background, the two transverse-traceless classical modes of the pure Einstein linearization propagate on the same invariant null cone as the Maxwell modes:

$$\frac{c_{\text{GW}} - c_\star}{c_\star} = 0 \text{ exactly on this action/background branch.}$$

Published bound (GW170817 + GRB 170817A multi-messenger):

$$-3 \times 10^{-15} < \frac{c_{\text{GW}} - c}{c} < +7 \times 10^{-16} \quad (90\% \text{ credibility}).$$

For a source at ~ 40 Mpc, this fractional difference corresponds to only a few seconds of propagation-time mismatch across $\sim 10^8$ years of travel.

Einstein quadratic mass parameter. On the additional pure two-derivative Einstein–Hilbert action branch, the transverse-traceless quadratic operator has no Fierz–Pauli hard-mass parameter:

$$\mu_{\text{FP,hard}}^2 = 0.$$

This is an action-content statement, not a graviton-particle mass prediction from diffeomorphism redundancy alone. A quantum mass or pole claim requires Definition 6.18, while diffeomorphism-invariant Stueckelberg/bimetric completions and extra-field theories lie outside the pure-Einstein field-content branch.

Published bound (GW dispersion analysis, PDG 2025):

$$m_{\text{grav,disp}} \leq 1.76 \times 10^{-23} \text{ eV}/c^2 \quad (90\% \text{ credibility; model-dependent dispersion fit}).$$

This corresponds to a reduced Compton wavelength $\bar{\lambda}_C \gtrsim 1.6 \times 10^{16}$ m, i.e., order ~ 1.6 light-years.

No dipole radiation. Many modified gravity theories predict extra channels (scalar/vector) producing dipolar radiation at (-1) PN order. The derived GR limit predicts no such channel.

Published bound (GW170817 inspiral phasing, PDG 2025):

$$-4 \times 10^{-6} < \delta\hat{p}_{-2} < 2 \times 10^{-5} \quad (90\% \text{ credibility}).$$

Tensor polarizations on the selected field-content branch. The pure Einstein linearization contains two tensor polarizations. It does not prove that every broader diffeomorphism-invariant EFT lacks additional scalar, vector, or massive modes. Pure non-tensor hypotheses are disfavored by observational constraints, and mixed tensor-scalar/vector models are tightly constrained.

Equivalence principle tests. Additional null checks from the derived GR structure:

- Universality of free fall (space tests): precision $\sim 10^{-15}$
- Nordtvedt parameter ($\eta = 4\beta - \gamma$): $(0.47 \pm 0.55) \times 10^{-4}$
- Binary pulsar radiative damping (PSR J0737-3039): 0.999963 ± 0.000063

4.6.15 State-observable error propagation and the dynamical bridge

On the additional pure-Einstein action/background branch, the framework states the classical equalities recorded in Section 4.6.14. Finite Markov/recovery control supplies a separate and more limited quantitative statement. This subsection therefore keeps two error ledgers separate. The *state-observable ledger* tracks expectation values of specified bounded local observables in a recovered state; the Markov/recovery machinery controls this ledger. The *dynamical-parameter ledger* tracks spectral and dispersion properties of an evolution generator—propagation speeds, particle-pole masses, and correlator poles. The trace-distance chain alone does *not* control that ledger.

The key quantitative hook. From Theorem 3.1, if the target state satisfies

$$I(A_k : C_k \mid B_k) \leq \varepsilon_k,$$

then, writing ρ for the target state and σ for its recovered comparison state, the recovery map gives the trace-norm bound

$$\|\rho - \sigma\|_1 \leq \delta_k := 2\sqrt{\ln 2 \cdot \varepsilon_k}.$$

State–observable ledger. Trace distance gives immediate bounds on expectation-value errors for the declared operator class: bounded operators supported in the recovered region, compared at the time of recovery. Using the standard dual norm inequality:

$$|\langle O \rangle_\rho - \langle O \rangle_\sigma| \leq \|O\|_\infty \|\rho - \sigma\|_1 = 2\|O\|_\infty D(\rho, \sigma),$$

where $D(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$ is the trace distance. This inequality is exactly as strong as it looks and no stronger: the norm $\|O\|_\infty$ is the Lipschitz constant only for bounded O , and the comparison is an equal-time statement about states.

Exponential decay from the mixing hypothesis. Writing w for collar width, so it is not confused with the recovery error δ_k , the mixing assumption (Section 2.3) provides

$$I_\omega(A_w : D_w \mid B_w) \leq C_{\text{mix}} |\partial C|_{\text{UV}} e^{-w/\xi}.$$

Combining these gives an explicit precision dial for bounded local observables:

$$\delta_{\text{step}} \lesssim 2\sqrt{\ln 2 \cdot C_{\text{mix}} |\partial C|_{\text{UV}}} \cdot e^{-w/(2\xi)}.$$

For an operator normalized by $\|O\|_\infty \leq 1$, an independently chosen absolute expectation-error target τ_O is guaranteed at one step if $\delta_{\text{step}} \leq \tau_O$; the theorem therefore gives the sufficient CMI condition $\varepsilon \leq (\tau_O/2)^2 / \ln 2$. The target τ_O belongs only to the state–observable ledger and cannot be set by importing a gravitational-wave speed tolerance.

What the dial does not certify: dynamical parameters. The gravitational-wave propagation speed and any graviton pole or dispersion-mass parameter are not bounded local observables. They are spectral/dispersion properties of the dynamical generator—poles of retarded correlators, or long-time wave-packet phase and group velocities—so the trace-distance inequality supplies neither their norm nor their Lipschitz constant. Two states can be close in trace distance at one time while being vacua of generators with different dispersion relations, and a small equal-time expectation error does not control phase accumulation over the $\sim 10^8$ -year propagation baseline of GW170817. Consequently, $\delta \lesssim 10^{-15}$ is *not* a theory-side certificate for the multimessenger bound $|c_{\text{GW}}/c - 1| \lesssim 10^{-15}$ of Section 4.6.14, and the recovery error must not be presented as the precision dial for that published bound. The ideal classical relation $c_{\text{GW}} = c_\star$ and the vanishing pure-Einstein hard parameter $\mu_{\text{FP,hard}}^2 = 0$ belong to the additional action-level branch recorded there; neither stability away from that branch nor a quantum graviton pole follows from the trace-distance chain.

Required bridge (dynamical-parameter ledger; open). A sufficient acceptance gate can be stated precisely, but its hypotheses have not been derived from the Markov remainder in this paper. Let $\mathcal{B} = [k_{\min}, k_{\max}]$, with $0 < k_{\min} < k_{\max}$, be the declared experimental wave-number band. Suppose a future construction supplies, for each tensor helicity, a C^1 self-adjoint finite-regulator generator $H_\nu^{\text{TT}}(k)$ and the GR reference generator $H_0^{\text{TT}}(k)$, expressed in angular-frequency units ($\hbar = 1$) on one specified common finite-dimensional helicity fiber or after a declared C^1 unitary identification of the two fibers. Assume that the reference TT tensor-mode eigenvalue $\omega_0(k) = c_\star k$ is simple in the helicity block and is separated from the rest of the spectrum by a uniform gap $g_{\mathcal{B}} > 0$. Define

$$M_{\mathcal{B}} := \frac{1}{c_\star} \sup_{k \in \mathcal{B}} \left\| \partial_k H_0^{\text{TT}}(k) \right\|_\infty$$

and the dimensionless dynamical error

$$\eta_{\text{dyn},\nu} := \max \left\{ \frac{1}{g_{\mathcal{B}}} \sup_{k \in \mathcal{B}} \|H_{\nu}^{\text{TT}}(k) - H_0^{\text{TT}}(k)\|_{\infty}, \frac{1}{c_{\star}} \sup_{k \in \mathcal{B}} \|\partial_k H_{\nu}^{\text{TT}}(k) - \partial_k H_0^{\text{TT}}(k)\|_{\infty} \right\}.$$

For $\eta_{\text{dyn},\nu} < 1/4$, let $\omega_{\nu}(k)$ be the continued isolated eigenvalue and define its group velocity by $c_{\text{GW},\nu}(k) := \partial_k \omega_{\nu}(k)$. The standard isolated-eigenvalue projector estimate and the Hellmann–Feynman formula then give, uniformly over $\mathcal{B}$,

$$\sup_{k \in \mathcal{B}} \left| \frac{c_{\text{GW},\nu}(k)}{c_{\star}} - 1 \right| \leq L_{\mathcal{B}} \eta_{\text{dyn},\nu}, \quad L_{\mathcal{B}} := 1 + 4M_{\mathcal{B}}.$$

Indeed, if $P_{\nu}(k)$ and $P_0(k)$ are the rank-one spectral projectors, then $\|P_{\nu} - P_0\|_{\infty} \leq 2\eta_{\text{dyn},\nu}$, hence $\|P_{\nu} - P_0\|_1 \leq 4\eta_{\text{dyn},\nu}$. Differentiating the isolated eigenvalue therefore bounds the group-velocity change by the derivative perturbation plus $4M_{\mathcal{B}}c\eta_{\text{dyn},\nu}$. This displays a dimensionless constant derived from a specified generator and uniform over the declared band. If $q := L_{\mathcal{B}}\eta_{\text{dyn},\nu} < 1$, then stationary propagation at fixed group velocity over path length D obeys the group-delay bound

$$|\Delta t| \leq \frac{D}{c_{\star}} \frac{q}{1 - q}.$$

The same bound applies to varying propagation only if it holds pointwise along the entire path; on a cosmological path, the redshifted wave number must remain inside $\mathcal{B}$. A wave-packet phase bound likewise requires uniform control of the inverse dispersion $k_{\nu}(\omega)$ on the corresponding frequency band. A model-dependent particle dispersion-mass bound is a separate spectral estimate at $k = 0$; it is not a corollary of the band-limited speed estimate.

What remains open is the OPH-specific bridge. One must

1. construct H_{ν}^{TT} , or an equivalent retarded two-point kernel, from the finite and scaling OPH dynamics;
2. prove a norm-resolvent, quadratic-form, or correlator estimate of the form $\eta_{\text{dyn},\nu} \leq C_{\mathcal{B}}r_N$, where $r_N := \min(2, \sum_{k=2}^N \delta_k)$ is the accumulated trace-norm recovery bound of Theorem 3.1 for the associated N -step construction and $C_{\mathcal{B}}$ is derived rather than assumed; and
3. apply the resulting band-uniform dispersion estimate to the detector band and propagation baseline, including phase accumulation where used by the likelihood.

Only after the second item is proved may one combine the estimates to obtain

$$\sup_{k \in \mathcal{B}} \left| \frac{c_{\text{GW},\nu}(k)}{c} - 1 \right| \leq L_{\mathcal{B}} C_{\mathcal{B}} r_N.$$

No estimate $\eta_{\text{dyn},\nu} \leq C_{\mathcal{B}}r_N$ is proved here. Substituting δ_k , δ_{step} , or any equal-time trace-distance error for $\eta_{\text{dyn},\nu}$ without that bridge is invalid.

Precision summary. The framework provides:

1. The ideal pure-Einstein linearization statements $\mu_{\text{FP,hard}}^2 = 0$ and $c_{\text{GW}} = c_{\star}$, without promotion to a quantum graviton pole.
2. Translation of those ideal equalities into the specific observables experiments constrain (Section 4.6.14).
3. Explicit bounds on how far expectation values of bounded local observables in the recovered state can drift, using the conditional mutual information $\rightarrow$ trace distance $\rightarrow$ observable error chain.

It therefore provides quantitative error control on recovered states and bounded observables, but no present theory-side certificate for the published GW-speed or model-dependent particle dispersion-mass bounds. Dynamical-parameter certification remains in the separate open ledger above.

4.6.16 Dark-sector response from the modular anomaly

This subsection records a D12 continuation benchmark obtained by adding a specific deep-IR response ansatz on top of the modular-anomaly term. It is not a theorem-level closure of dark-matter phenomenology: neither the observational dark-matter identification nor MOND/RAR-like dynamics are derived here.

The modular anomaly term T_{ab}^{anom} derived in Section 5.9 supplies one structural ingredient for a possible dark-sector continuation, without introducing new particle species.

The identification. The anomalous stress-energy contribution

$$\langle T_{00}^{\text{anom}} \rangle = \frac{15}{8\pi^2} \cdot \frac{\delta \langle K_C^{(\text{anom})} \rangle}{\ell^4}$$

is "dark" by construction: it arises from information-theoretic/gravitational structure (modular Markov imperfections), rather than Standard Model fields. Its coupling is gravitational, with no electromagnetic coupling. These properties motivate the dark-sector interpretation; observational identification requires separate phenomenology.

Connection to the cosmological parameter package. The framework makes the dimensionless Λ -capacity product a global capacity parameter; with the selected OPH scale certificate it gives the same D6 static-patch display:

$$\Lambda_{\text{CRC}} = \frac{3\pi}{G_{\text{geom}} N_{\text{CRC}}}, \quad r_{\text{dS}} = \sqrt{\frac{3}{\Lambda_{\text{CRC}}}}.$$

On the observed branch N_{CRC} is read as N_{scr} . This imports the D6 static-patch scale into the continuation surface. Any galaxy-scale continuation built from the anomaly term would therefore be an IR phenomenon, appearing only when accelerations are small and distances are large. The imported scale does not by itself determine the source/response law.

Acceleration benchmark. If one assumes the relevant deep-IR response is controlled only by r_{dS} , c , and the fixed anomaly prefactor, then the benchmark scale must:

1. Vanish if $r_{\text{dS}} \rightarrow \infty$ (infinite capacity, no de Sitter static-patch scale)
2. Be controlled by r_{dS} as the only new IR scale
3. Carry non-tunable coefficients from the derivation

The anomaly enters with prefactor $\frac{15}{8\pi^2}$. The corresponding benchmark acceleration scale constructible from (Λ, c) is:

$$a_0^{(\text{OPH})} := \frac{15}{8\pi^2} \cdot c^2 \sqrt{\frac{\Lambda}{3}} = \frac{15}{8\pi^2} \cdot \frac{c^2}{r_{\text{dS}}}$$

Normalization estimate. Using Planck 2018 Λ CDM parameters ($H_0 \approx 67.4$ km/s/Mpc, $\Omega_\Lambda \approx 0.685$):

- $\Lambda \approx 1.09 \times 10^{-52} \text{ m}^{-2}$

- $r_{\text{dS}} \approx 1.66 \times 10^{26} \text{ m}$
- Therefore:

$$a_0^{(\text{OPH})} \approx 1.03 \times 10^{-10} \text{ m/s}^2$$

For comparison, observational fits to galaxy regularities (RAR/MDAR/MOND phenomenology) quote $a_0 \sim 1.2 \times 10^{-10} \text{ m/s}^2$. This numerical proximity is only a benchmark coincidence unless a separate response law and a controlled galaxy-scale limit are derived.

One illustrative response ansatz. If one further assumes that T_{00}^{anom} is the dominant deep-IR source and that the response organizes into a MOND/RAR-like law, the Newtonian limit could be written as:

$$\nabla^2 \Phi = 4\pi G(\rho_b + \rho_{\text{anom}}),$$

i.e., baryons plus an effective extra density. Under the same extra ansatz the radial acceleration relation (RAR) could be written as:

$$g_{\text{obs}} \approx g_b + \sqrt{a_0 \cdot g_b}, \quad g_{\text{DM}} := g_{\text{obs}} - g_b \approx \sqrt{a_0 \cdot g_b}.$$

With $a_0 = a_0^{(\text{OPH})}$ fixed, that same ansatz would imply:

(i) Baryonic Tully-Fisher relation.

$$V^4 \approx G \cdot M_b \cdot a_0^{(\text{OPH})}$$

where V is the asymptotic rotation velocity and M_b is baryonic mass.

(ii) Flat rotation curves. For a point mass M_b :

$$g_{\text{DM}}(r) = \frac{\sqrt{G M_b a_0^{(\text{OPH})}}}{r} \Rightarrow M_{\text{DM}}(r) \propto r$$

i.e., inferred dark mass grows linearly with radius, producing flat rotation curves.

(iii) Characteristic surface density.

$$\Sigma_0^{(\text{OPH})} = \frac{a_0^{(\text{OPH})}}{2\pi G} \approx 0.25 \text{ kg/m}^2 \approx 120 M_{\odot}/\text{pc}^2.$$

This lies in the range of observed central halo surface densities, but it is not a derived halo theorem.

Scope. What is grounded in the framework developed here:

- The local BW branch fixes the coefficient $\frac{15}{8\pi^2}$ for a source variable proved to equal anomalous local modular energy; identifying raw collar CMI with that variable is a separate D12 matching theorem
- screen capacity fixes the dimensionless de Sitter entropy relation; Λ and r_{dS} in SI units also require the selected scale certificate
- The benchmark scale $a_0^{(\text{OPH})}$ follows only after importing the D6 static-patch scale and assuming that no additional IR scale enters the continuation

Scope boundary for theorem-level closure:

- A controlled nonrelativistic limit from the anomaly term to galaxy observables

- A derived sign and closure analysis for the effective anomaly contribution on the relevant galaxy-scale states
- A derived response law selecting the MOND/RAR functional rather than alternative IR behavior
- A proof that the anomaly term supplies the relevant dominant source on galaxy scales rather than only a benchmark normalization
- Lensing, cluster, and Bullet-Cluster phenomenology
- Cosmological abundance and structure-formation checks
- Environment-dependence and stability control

Cosmology/Boltzmann contract. The static galaxy ansatz above is not an FLRW perturbation kernel. A cosmological dark/anomaly claim must instead expose the variables needed by an Einstein–Boltzmann implementation:

$$\bar{\rho}_A(a), \quad \bar{\rho}_{A,\text{eq}}(a), \quad w_A(a), \quad c_{s,A}^2(k,a), \quad \sigma_A(k,a), \quad Q_A^\mu, \quad B_A(k,a), \quad \Gamma_{\text{rec}}(k,a).$$

The first, source-side no-go is that $\bar{\rho}_A(a)$, $\bar{\rho}_{A,\text{eq}}(a)$, and $B_A(k,a)$ alone do not define a physical Boltzmann source. A physical source must be emitted by a finite covariant collar-packet parent

$$\mathcal{P}_r[X, g] = (C_r, Z_r, A_r, R_r, G_r, \pi_r[X], L_r[X], Q_r, D_r),$$

whose finite packet states split the anomaly and any repair recipient, whose repair generator and reaction channels close the total stress tensor, and whose causal response data set the pressure, sound speed, anisotropic stress, exchange current, and repair rate. The parent must also declare exactly one local source route: fixed-reference anomalous modular energy, or nonlinear CMI stress with a CMI-to-modular-source matching theorem/hypothesis and residual ledger. Raw collar CMI is otherwise a recoverability diagnostic, not the $15/(8\pi^2)$ BW source. B_A is gauge-invariant only when built from the anomaly-frame baryon density $n_b^{(A)} = -u_{A\mu}J_b^\mu$. If the exchange branch is nonzero, explicit recipient stress and an equal-and-opposite exchange current are mandatory; otherwise the repair term is not a closed stress-energy source. A transition spectral number is only $\gamma_{\text{repair step}}$ until the active fiber, physical clock, and common-parent response pole certify a physical Γ_{rec} . Physical likelihood promotion additionally requires immutable source, solver, and likelihood hashes frozen before the data comparison.

Canonical finite covariant collar-packet parent. For a regulator r and background (X, g) , the source object is

$$\mathcal{P}_r[X, g] = (\mathcal{C}_r, g_r, e_r, U_r, Z_r, \omega_r, p_r, f_r, \Phi_r, \mathcal{R}_r, \mathcal{G}_r, \pi_r, \text{Read}_r).$$

Here $\mathcal{C}_r$ is a finite causal cell complex with oriented faces, cell four-volumes, causal adjacency, and finite parallel transports $U_{F \rightarrow c}$; e_r is the tetrad/local-frame data; $Z_r = \bigsqcup_s Z_{s,r}$ is the packet state space for anomaly, recipient, radiation, standard matter, and any interaction sector; ω_r, p_r, f_r are invariant weights, local momenta, and occupations; Φ_r is the face flux; $\mathcal{R}_r$ is the reaction channel list; $\mathcal{G}_r$ is the finite gauge/quotient data; π_r is the restriction/refinement map; and Read_r is the local observable readout. With signature $(-+++)$, every packet used in a kinetic stress branch must satisfy $g_{ab}p_z^a p_z^b = -m_z^2$, $p_z^0 > 0$, and $\omega_z f_{c,z} \geq 0$, with the invariant measure convention declared explicitly.

Scalar-to-stress nonuniqueness. A scalar collar row, even a finite row for $\rho_A(a)$, $\rho_{A,\text{eq}}(a)$, or $B_A(k, a)$, does not determine a stress tensor. Many inequivalent packet parents can have the same scalar density while differing in pressure, sound speed, anisotropic stress, exchange current, gauge-invariant perturbation variables, and causal response. Therefore no evidence bundle may promote scalar rows to physical CMB, lensing, growth, or S_8 inputs unless the parent supplies the missing stress and response variables.

Finite MaxEnt lift. When only finitely many one-cell source readings $e_x(u_i) = T_{ab}^A(x)u_i^a u_i^b$ are available, a stress lift is not unique until the branch supplies a finite selection rule. The admissible MaxEnt lift minimizes no hidden observational residual: it maximizes the declared packet entropy subject to the source readings, mass shells, conserved charges, local frame covariance, and refinement restrictions. The lift is physical only when held-out timelike probes pass a quadraticity/tomography test and the variational stress agrees with the packet-moment stress in the predeclared norm.

Packet stress and exchange closure. For a kinetic parent,

$$T_{I,r}^{ab}(c) = \frac{1}{V_c} \sum_{z \in Z_{I,r}(c)} \omega_z f_{c,z} p_z^a p_z^b, \quad J_{I,r}^a(c) = \frac{1}{V_c} \sum_{z \in Z_{I,r}(c)} \omega_z f_{c,z} p_z^a.$$

More general packet parents may add an internal stress Σ_z^{ab} , but then Σ_z^{ab} is part of the primitive parent data and must obey the same local-frame and refinement checks. The finite transport/reaction equation must imply

$$\text{Div}_r T_{I,r}^{ab} = Q_{I,r}^b, \quad \sum_I Q_{I,r}^b = 0.$$

If the repair exchange is nonzero, a row labelled “recipient” is not enough: the recipient stress T_R^{ab} , exchange current Q_R^b , and closure residual $Q_A^b + Q_R^b$ must be emitted and certified. If exchange is off, the parent instead emits a repair-exchange-off receipt.

Response, rate, and gauge receipts. The finite causal response must provide a finite domain of dependence, subluminal characteristic bound, retarded support residual, and response stability residual. The transition-matrix diagnostic may be called $\gamma_{\text{repair step}}$ until the active fiber, conserved-sector decomposition, physical clock $\Delta\tau_{\text{physical}}$, and common parent response pole are certified. Only then may one write

$$\Gamma_{\text{rec}} = -\text{Re } \lambda_{\text{active}} / \Delta\tau_{\text{physical}}.$$

Gauge checks must be stated in gauge-invariant or rest-frame variables; raw Newtonian/synchronous agreement is a diagnostic, not a gauge-independence receipt.

Receipt hierarchy. The finite parent receipt is the conjunction of primitive evidence receipts, not a producer-declared boolean. The source-side conjunction includes

SOURCE_ROUTE, ENTROPY_UNIT, MODULAR_NONADDITIVITY_IDENTITY, SOURCE_LOCALIZATION, STRESS,
FINITE_PACKET_KINEMATICS, PACKET_MASS_SHELL, TRANSPORT_COVARIANCE, CHANNEL_FOUR_MOMENTUM, FINITE,
VARIATIONAL_MOMENT_STRESS_AGREEMENT, LOCAL_FRAME_COVARIANCE, CARRIER_QUOTIENT_INV,
TOTAL_STRESS_CLOSURE, EXCHANGE_CURRENT_CLOSURE, COSMOLOGICAL_GAUGE_INVARIANCE,
FINITE_DOMAIN_OF_DEPENDENCE, SUBLUMINAL_CHARACTERISTICS, RETARDED_RESPONSE, RESPONSE,
REFINEMENT_CONVERGENCE, CDM_LIMIT_RECOVERY.

Frozen source, solver, likelihood, and data hashes are later promotion receipts. They do not belong to the parent theorem and cannot rescue a failed parent.

The same firewall applies to the promoted CMB-source inputs

$$\eta_R, \quad \gamma_{\text{repair step}} \text{ or certified } \Gamma_{\text{rec}}, \quad A_\zeta, \quad q_{\text{IR}}, \quad \ell_{\text{IR}}, \quad B_A(k, a), \quad \rho_A(a), \quad N_{\text{CRC}}.$$

Each one must have an immutable source-provenance DAG node recording source report, source hash, parents, no-CMB-data flag, and measurement-use flags. Contradictory provenance, such as declaring no CMB data use while also fitting to Planck, fails closed. Reducers must pool additive sufficient statistics globally before any nonlinear estimate of tilt, amplitude, inverse precision, rank, condition number, isocurvature leakage, B_A , ρ_A , or Γ_{rec} ; raw transition diagnostics remain $\gamma_{\text{repair step}}$. Shard-local nonlinear averages and shard-local `any()` likelihood rollups are diagnostic only. N_{CRC} remains the D6 consensus invariant unless a separate additive capacity schema proves disjoint coverage. The cold transported limit is the check case: when exchange, pressure, sound-speed, and anisotropic stress corrections are turned off, the anomaly slot must reduce to a CDM-like component before recombination. Any nonzero-field response, late-time growth suppression, or S_8 -relief branch has to be emitted by the finite-collar parent evaluator through $B_A(k, a)$ and $\Gamma_{\text{rec}}(k, a)$ after the physical-clock and response receipts pass, not fitted as a free environmental kernel to CMB, weak-lensing, SPARC, or cluster data. The compressed H_0 , Ω_m , σ_8 , and S_8 rows are plumbing diagnostics for a low- H_0 , Planck-like branch, not theorem-grade cosmology.

Finite cosmological source, transfer, and frozen prediction. The continuation-level physical cosmology gate is the conjunction

$$\text{CMB_READY} = \text{SOURCE} \wedge \text{SCALE} \wedge \text{PARENT} \wedge \text{KERNEL} \wedge \text{INIT} \wedge \text{TRANSFER} \wedge \text{FREEZE} \wedge \text{LIKE}.$$

SOURCE is the source-only DAG with transitive measurement-taint rejection and globally pooled sufficient statistics; **SCALE** is the physical mode/clock/lift bridge; **PARENT** is the finite covariant dark/anomaly parent with packet kinematics, mass shell, packet stress readout, reaction-channel four-momentum, recipient stress and exchange-current closure for nonzero exchange, total stress closure, local-frame/quotient invariance, cosmological gauge invariance, finite domain of dependence, subluminal retarded response, response stability, refinement, and CDM-limit recovery receipts; **KERNEL** emits physical ρ_A , $\rho_{A,\text{eq}}$, B_A , and certified Γ_{rec} ; the source-only ρ_A row additionally requires the anomaly abundance selector

$$\mathcal{E}_{A,r} = \mathbb{E}_{\mu_{A,r}^{\text{rel}}} [\mathbb{L}_{A,r}]$$

and its no-data-use and refinement receipts; **INIT** supplies regular initial modes and Einstein constraints; **TRANSFER** supplies Boltzmann well-posedness and numerical convergence; **FREEZE** fixes the model generation; and **LIKE** is the official likelihood execution. While this conjunction is false, physical CMB, BAO, lensing, growth, RSD, and S_8 outputs remain diagnostic or conditional, not recovered-core predictions.

This branch is a phenomenological continuation above D6 and D12 inputs. It contains structural ingredients and an IR benchmark, but it does not derive the precise galaxy-scale response from the recovered core.

Falsifiability. A completed continuation must supply the derivation steps above and remain compatible with galaxy, lensing, cluster, Bullet-Cluster, and cosmological data. Failure of that continuation retracts the modular-anomaly dark-sector continuation. It does not retract the recovered core theorem package.

4.6.17 De Sitter holography: static patch vs boundary-at-infinity

A natural question arises: how does this framework relate to the “unsolved problem” of de Sitter holography?

What the usual dS holography problem is. When people say “dS holography is unsolved,” they typically mean that we do not have anything as sharp as AdS/CFT, where the bulk has a timelike asymptotic boundary supporting a well-defined dual CFT with a precise dictionary. For de Sitter, there is no asymptotic timelike boundary in the static patch where one can simply place the dual theory. The classic dS/CFT proposal at future infinity has familiar difficulties, including non-unitarity worries and complex conformal weights.

Static-patch/horizon-screen setup. The framework begins with an observer’s static patch and its horizon screen S^2 , building a net of subregion algebras on that screen. At finite cutoff those algebras are type-I regulators. The Lorentz statement, when invoked, is a support-visible scaling-limit statement about the refinement-limit observer net, and that limit may leave the regulator class. By Theorem 4.5.2, the realized scaling-limit cap modular action on the extracted prime geometric cap pair is geometric and, in the non-type-I case of interest, generally outer.

This is therefore a fundamental fork away from AdS/CFT-style holography:

AdS/CFT	This framework
Codimension-1 boundary at infinity	Codimension-2 horizon screen (S^2)
Single global boundary theory	Observer-dependent patches that overlap
Dual CFT required	Only algebras + consistency conditions
Negative Λ	Positive Λ natural

This aligns with the static-patch/complementarity intuition in the dS literature, where the fundamental description is patch-based and different static patches are related by consistency rules, not by a single global boundary theory.

The mechanism: Λ as global capacity, not local physics. A key structural result is that null modular data reconstruct the stress tensor only up to an additive metric term. This is the statement that vacuum-energy or cosmological-constant shifts are invisible to the local null-data route. The Einstein equation derived from the fixed-cap generalized-entropy stationarity theorem is therefore fixed only up to Λg_{ab} .

Theorem boundary: cosmological-constant / screen-capacity closure stack. On the D5→D6 branch, assume the local Einstein equation is fixed only modulo Λg_{ab} , the cosmic record-capacity fixed point

$$N_{\text{CRC}} = F(N_{\text{CRC}}),$$

its observed-branch de Sitter entropy readout

$$N_{\text{CRC}} = S_{\text{dS}},$$

the standard de Sitter entropy relation

$$S_{\text{dS}} = \frac{A_{\text{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

and the standard de Sitter static-patch formulas

$$r_{\text{dS}} = \sqrt{\frac{3}{\Lambda}}, \quad t_{\Lambda} = \frac{r_{\text{dS}}}{c}.$$

Then:

1. local null data determine the Einstein branch only modulo Λg_{ab} ;
2. with the selected scale certificate, the same branch has the global display

$$G_{ab} + \frac{3\pi}{GN_{\text{CRC}}} g_{ab} = 8\pi G \langle T_{ab} \rangle;$$

3. the same D6 closure fixes the entropy and dimensionless capacity relations

$$S_{\text{dS}} = N_{\text{CRC}}, \quad A_{\text{dS}} = 4GN_{\text{CRC}}, \quad r_{\text{dS}} = \sqrt{\frac{3}{\Lambda}}, \quad t_{\Lambda} = \frac{r_{\text{dS}}}{c};$$

4. the observed cosmic age is a downstream FLRW benchmark rather than an additional theorem output.

Thus the cosmological-constant package is one local/global theorem stack rather than a split local-plus-global story.

Reason. The local null-data route is blind to metric-term shifts, so the Einstein branch is fixed locally only modulo Λg_{ab} . The cosmic record-capacity fixed point $N_{\text{CRC}} = F(N_{\text{CRC}})$, together with its observed-branch de Sitter entropy readout, then yields

$$\Lambda_{\text{CRC}} \ell_{\star}^2 = \frac{3\pi}{N_{\text{CRC}}}, \quad \Lambda_{\text{CRC}} = \frac{3\pi}{GN_{\text{CRC}}},$$

where the second equality is the scale-certified display. This closes the same Einstein branch globally once the geometric scale $G_{\text{geom}} = \ell_{\star}^2$ is supplied. The capacity closure itself fixes the dimensionless product $\Lambda_{\text{CRC}} \ell_{\star}^2 = 3\pi/N_{\text{CRC}}$; the displayed static-patch length and curvature package uses the selected scale certificate. The observed cosmic age is a later FLRW comparison quantity rather than part of this theorem stack.

The D6 hypotheses are the cosmic record-capacity fixed point, its observed-branch de Sitter entropy readout, the de Sitter entropy relation, and the standard static-patch formulas. The local null-data route does not by itself determine the global capacity; that global value is fixed by the cosmic record-closure readback fixed point. This logic is compatible with the BW scaling branch but does not depend on the scaling-limit algebra being type I.

Input-free capacity closure. For capacity N , let $F(N)$ be the active cosmic record capacity read back by stable observers inside the OPH universe supplied with capacity N :

$$F(N) = \text{Cap}_{\text{read}}(\text{Obs}(\text{nf}(\mathfrak{U}_N))).$$

The cosmic record-closure capacity is the fixed point

$$N_{\text{CRC}} = F(N_{\text{CRC}}), \quad \Lambda_{\text{CRC}} = \frac{3\pi}{GN_{\text{CRC}}}.$$

If F is an OPH-derived contraction on the admissible capacity interval, then this fixed point is unique and stable. The count-density representation of the same target is obtained by letting Ω_N^{sc} be the terminal OPH normal forms that are repair-closed, observer/checkpoint-supporting, locally recovered-core closed, and whose own horizon record surface reads back capacity N . Since $\log \dim \mathcal{H}_{\partial, N} = N$, define the screen-normalized self-closure density

$$\Pi(N) = \frac{|\Omega_N^{\text{sc}}|}{\dim \mathcal{H}_{\partial, N}} = |\Omega_N^{\text{sc}}| e^{-N}.$$

The corresponding selector is

$$N_\star = \text{MAR} \arg \max_N [\log |\Omega_N^{\text{sc}}| - N].$$

Equivalently, with $\ell(N) = \log |\Omega_N^{\text{sc}}| - N$, the OPH-derived map

$$T_\eta(N) = N + \eta \ell'(N)$$

has a unique stable fixed point under the derivative-sign certificate for $H_N = \ell'$. Informally, this is the single screen size where the universe reads back its own boundary without deficit or slack. On the observed branch this fixed point is the de Sitter entropy capacity.

Capacity readout. The branch uses N_{scr} as the entropy capacity. The bare radius-squared ratio is $N_{\text{patch}} = (r_{\text{dS}}/\ell_P)^2$, and

$$N_{\text{scr}} = \pi N_{\text{patch}} = \frac{3\pi}{\Lambda \ell_P^2}.$$

The observed late-time scale gives $N_{\text{patch}} \simeq 1.05 \times 10^{122}$, $N_{\text{scr}} \simeq 3.31 \times 10^{122}$, and $\Lambda \ell_P^2 \simeq 2.85 \times 10^{-122}$.

What this solves vs. what it assumes. The model does **not** solve the classic “give me a unitary CFT at future infinity” problem. It does not aim there. It also does **not** prove that every refinement-stable MaxEnt branch lands in the static-patch geometric modular branch with emitted cap pair and standard modular action. What it does provide is a coherent route to **patch holography** in which de Sitter static patches are natural:

1. the fundamental object is a horizon screen in a static-patch description;
2. Λ is a capacity parameter tied to finite Hilbert-space dimension, not a locally reconstructible vacuum-energy term;
3. Einstein-like dynamics emerge up to Λg_{ab} ;
4. on the support-visible BW scaling branch, the realized scaling-limit cap modular action is geometric and may be outer on a non-type-I observer algebra.

BW-side boundary. The Lorentz side is closed at the support-visible scaling level by Theorem 4.2. The theorem intentionally avoids the false stronger route through a full-algebra unregularized common floor; the automorphism statement on the observer-facing geometric cap pair is the required static-patch content.

Many observers, one Λ . In this framework, each timelike observer is associated with a horizon patch rather than a single global description. The dimensionless de Sitter capacity relation is the shared global constraint across overlap-consistent descriptions; its SI Λ display uses the selected scale certificate.

Summary. The model gets de Sitter by moving the holographic screen from “infinity” to an observer’s horizon and by treating de Sitter entropy, i.e. finite screen capacity, as the global N_{CRC} closure datum. The input-free closure is the cosmic record-closure readback fixed point stated above, with the screen-normalized self-closure density as its count representation. It fixes the dimensionless curvature-capacity relation; the numerical SI static-patch display is scale-certified. The usual dS-holography obstacles are precisely the ones avoided by refusing the boundary-at-infinity viewpoint. The claim is a static-patch holography program with an explicit local/global split and an explicit extracted-geometric-subnet scaling boundary, without asserting a solved dS/CFT dual.

4.7 Gauge Reconstruction and Standard Model Structure

4.7.1 Edge sector category and gauge group reconstruction

At any fixed UV cutoff, edge-center completion provides finitely many visible seed charges and finite-dimensional intertwiners. Their rigid tensor closure is defined below and may have infinitely many simple classes. The local MaxEnt / collar-mixing package established earlier controls only fixed-cutoff recoverability, modular-support localization, and carried error terms on the realized branch. It is logically separate from the question whether zero-obstruction edge sectors survive refinement. The fixed-stage category is theorem-produced; the refinement/fiber ladder and cofinal compact-gauge witness require the explicit compact-gauge refinement receipt defined below.

On the ordinary or central-defect branch, path-independent movement of collar charges is supplied by overlap gluing. TransportabilityFromOverlapGluing constructs transport from overlap paths and proves the exact combined criterion: the central triangle class $[z]_\Sigma$ must vanish and at least one allowed strictification must have trivial represented sector holonomy. On the genuinely noncentral branch, $o_\Sigma^{(2)} = 0$ strictifies the higher associator, after which at least one allowed strict G_Σ -valued representative must have trivial represented holonomy. A nonzero $o_\Sigma^{(2)}$ remains a higher-gauge sector; an orbit for which every strict representative has residual loop action is also outside the ordinary path-independent DR sector. The full orbit q_Σ need not determine a unique ordinary H^1 class and is not itself the strictification test. Thus the overlap obstruction calculus classifies and routes sectors; it does not by itself select the Standard Model.

For the tensor construction at cutoff r , choose one common stagewise strict edge cocycle U_r^{str} from the allowed representatives (the ordinary branch uses its given strict system). Retain only seeds on which this same representative has trivial action around every closed overlap loop. Sectors requiring incompatible strictifications define alternative fixed-stage categories; they are not unioned before tensor closure.

Classification is not realization.

Proposition (obstruction neutrality of the Standard Model selection step). The ordinary trivial-holonomy condition, the combined central condition $[z]_\Sigma = 0$ plus a trivial-holonomy strictification, and the combined noncentral condition $o_\Sigma^{(2)} = 0$ plus an allowed strict representative with trivial represented G_Σ -holonomy are transportability conditions. After one common stage-wise strict representative is chosen, they permit an ordinary transportable bosonic sector category generated only by sectors trivial under that choice; they do not select

$$\frac{\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)}{\mathbb{Z}_6}.$$

On a cofinal tail carrying the compact-gauge refinement receipt, DR/Tannaka reconstruction returns $G = \text{Aut}_\otimes(\mathcal{F})$ for the constructed sector category and fiber functor. If that category is trivial, G is trivial; in general G is whatever compact group the tensor-fiber data reconstructs. The Standard Model quotient enters only after MAR is applied to a nonempty realized one-Higgs chiral sector package. QED.

Definition (fixed-cutoff tensor-generated sector category). Let $S_r = \{W_{\alpha,r} : P_{\alpha,r} \text{ is visible and has trivial holonomy under the common } U_r^{\text{str}}\}$. Define

$$\text{Sect}_r^{\text{bos}} = \langle \mathbf{1}, S_r, S_r^* \rangle_{\oplus, \otimes, \text{subobjects}, \text{isomorphism}} \subset \text{Rep}_{\text{fd}}(\widehat{K}_{\Sigma,r}).$$

This is the replete full additive Karoubi rigid tensor subcategory generated by the finitely many visible carriers. The original projectors $P_{\alpha,r}$ identify only one-collar seeds. The simple objects are all irreducible summands of finite tensor words in seeds and conjugates; they need not have

projectors in the original one-collar algebra, and there may be infinitely many of them even though every object and Hom space is finite-dimensional. A subobject $X \subset T$ is supported by its invariant idempotent in $\text{End}_{\widehat{K}_{\Sigma,r}}(T)$, or by the corresponding block in a supplied finite concatenated-collar realization of that tensor word.

Theorem (FixedCutoffBosonicSectorCategory). At every fixed regulator cutoff, on the ordinary or central-defect zero-obstruction branch and in the bosonic internal-gauge sector of the $3 + 1$ -dimensional EFT regime, the preceding construction is a semisimple rigid symmetric C^* -tensor category. Complete reducibility of finite-dimensional representations of the compact group gives semisimplicity and subobjects; the operator adjoint and norm give the C^* -structure; conjugate representations give duals; and the standard flip gives the bosonic symmetry. For a closed loop, U_r^{str} acts as the identity on every retained seed, hence as the identity on direct sums, tensor products, conjugates, and invariant subobjects. Thus the whole generated category has one monoidally coherent path-independent transport, rather than an objectwise choice of strictification. Collar concatenation realizes tensor words whenever the corresponding finite tensor-realization receipt is supplied. The canonical carrier-forgetful functor

$$F_r : \text{Sect}_r^{\text{bos}} \longrightarrow \text{Hilb}_{\text{fd}}$$

is faithful, $*$ -preserving, and symmetric strong monoidal. Fermionic signs, spinorial matter, and chirality are not part of this bosonic internal-gauge category; they belong to the later super-Tannakian or matter-sector lift.

Definition (compact-gauge refinement receipt). On a cofinal tail, first require a surjectivity/finite-extendability certificate for each finite-state restriction $\rho_{sr} : Q_s \rightarrow Q_r$. Only then is the commutative pullback $C(Q_r) \rightarrow C(Q_s)$, $f \mapsto f \circ \rho_{sr}$, injective. This does not produce a map of noncommutative collar algebras. For

$$\mathcal{A}_{\text{EC},r} = \bigoplus_{\alpha} M_{d_{\alpha,r}}(\mathbb{C}), \quad \mathcal{A}_{\text{EC},s} = \bigoplus_{\beta} M_{d_{\beta,s}}(\mathbb{C}),$$

the receipt separately supplies multiplicities $m_{\beta\alpha}^{rs}$ and unitaries V_{β}^{rs} , with $d_{\beta,s} = \sum_{\alpha} m_{\beta\alpha}^{rs} d_{\alpha,r}$, and defines

$$(j_{rs}(a))_{\beta} = V_{\beta}^{rs} \left[\bigoplus_{\alpha} (a_{\alpha} \otimes \mathbf{1}_{m_{\beta\alpha}^{rs}}) \right] (V_{\beta}^{rs})^*.$$

Injectivity requires every coarse α to occur. The whole center maps into the fine center exactly when each β has one coarse shadow; on that branch $j_{rs}(P_{\alpha,r}) = \sum_{\beta: \text{sh}_{sr}(\beta)=\alpha} P_{\beta,s}$. Composition requires $m_{\gamma\alpha}^{rt} = \sum_{\beta} m_{\gamma\beta}^{st} m_{\beta\alpha}^{rs}$ and coherent V 's. The same receipt supplies continuous surjections

$$\pi_{sr}^{\Sigma} : \widehat{K}_{\Sigma,s} \twoheadrightarrow \widehat{K}_{\Sigma,r}, \quad \pi_{tr}^{\Sigma} = \pi_{sr}^{\Sigma} \circ \pi_{ts}^{\Sigma},$$

and one common strict representative U_r^{str} at each stage. The pullbacks carry coarse generators into $\text{Sect}_s^{\text{bos}}$, intertwine U_r^{str} with U_s^{str} , preserve trivial action of every fine overlap loop without changing representative object by object, and agree with the displayed collar supports. The receipt also supplies compatible finite concatenated-collar realizations for every finite tensor word and invariant subobject projector. The finite-state consensus maps alone imply none of these matrix, center, compact-group, or tensor-realization data.

Theorem (RefinementFunctorAndFiberDescent). On a cofinal tail carrying that receipt, define

$$U_{rs} := (\pi_{sr}^{\Sigma})^* : \text{Sect}_r^{\text{bos}} \longrightarrow \text{Sect}_s^{\text{bos}}.$$

Then U_{rs} is fully faithful, symmetric strong monoidal, and $*$ -preserving; it preserves the tensor unit, duals, irreducibles, and the full zero-obstruction transport criterion, with $U_{st}U_{rs} = U_{rt}$. For the canonical forgetful fibers, $F_s U_{rs} = F_r$ literally on carrier Hilbert spaces and linear maps.

Proof. For $X = (H_X, \varrho_X)$, set $U_{rs}X = (H_X, \varrho_X \circ \pi_{sr}^\Sigma)$ and $U_{rs}(f) = f$. Surjectivity gives the exact identity

$$\mathrm{Hom}_{\widehat{K}_{\Sigma,s}}(U_{rs}X, U_{rs}Y) = \mathrm{Hom}_{\widehat{K}_{\Sigma,r}}(X, Y),$$

so pullback is fully faithful and a coarse simple cannot split. Pullback commutes with tensor products, unit, conjugates, duality maps, and the bosonic flip, while composition of the π 's gives composition of the U 's. The receipt intertwines the common stagewise strict representatives, so zero holonomy persists for the whole tensor-generated category without a sector-specific gauge change. Since pullback changes only the action, the comparison $F_s U_{rs}X = H_X = F_r X$ is the identity; naturality, monoidality, and refinement coherence are identity diagrams. The displayed block formula, not finite-set duality, separately proves the noncommutative algebra embedding and its center behavior. QED.

The $U(1)$ charge-one test. If $W_1(z) = z$ is visible and retained by the common stagewise representative U_r^{str} , the chosen category contains $W_n = W_1^{\otimes n}$ for $n \geq 0$ and $W_{-n} = (W_1^*)^{\otimes n}$ for $n > 0$. Charge two is $W_2 = W_1 \otimes W_1$; its subobject idempotent, equivalently the unique nonzero central projection in its intertwiner algebra, is $\mathbf{1}_{W_2} \in \mathrm{End}_{U(1)}(W_1^{\otimes 2}) \cong \mathbb{C}$, represented physically in the certified two-collar algebra, not necessarily in the original one-collar algebra. At the next certified refinement, $U_{rs}W_2 = ((\pi_{sr}^\Sigma)^* W_1)^{\otimes 2}$ remains simple and $F_s(U_{rs}W_2) = \mathbb{C} = F_r(W_2)$ with identity comparison. Thus the fixed collar remains finite while the tensor-generated simple skeleton may be infinite. A finite fusion/truncation alternative would require a separately stated associative fusion quotient.

Consequently, on a cofinal tail carrying the compact-gauge refinement receipt, the EFT branch has a directed ladder $(\mathrm{Sect}_r^{\mathrm{bos}}, U_{rs}, F_r)$ of tensor-generated bosonic edge-sector categories satisfying that zero-obstruction criterion, with fully faithful monoidal pullback functors and compatible forgetful fibers. Write

$$\mathrm{Sect}_\infty := \varinjlim_r \mathrm{Sect}_r^{\mathrm{bos}},$$

for the directed colimit retaining the sectors and intertwiners that persist in that certified system. Without the receipt, only the fixed-stage categories and forgetful functors have been constructed; Sect_∞ and $\mathrm{Aut}_\otimes(\mathcal{F})$ are conditional objects. The theorem below is neutral about whether Sect_∞ is trivial or nontrivial: if the realized branch furnishes only the tensor unit, the reconstructed compact group is the trivial group.

Persistence lemma. Write $U_{rs} : \mathrm{Sect}_r^{\mathrm{bos}} \rightarrow \mathrm{Sect}_s^{\mathrm{bos}}$ for the pullback functor on a receipt-certified refinement tail. If a realized zero-obstruction edge-sector class α_{r_0} appears on a sufficiently fine collar, has overlap-visible support, and has representatives $\alpha_s \in \mathrm{Sect}_s^{\mathrm{bos}}$ with $U_{st}\alpha_s \simeq \alpha_t$ for all later $t \succeq s$, then it determines a unique object $[\alpha] \in \mathrm{Sect}_\infty$. Fully faithful pullback along a surjective compact-group map cannot erase or split a coarse simple, and the receipt preserves its full zero-obstruction data. This is a persistence result conditional on the certified ladder, not a consequence of the finite-state refinement maps and not a proof that a nontrivial colimit exists.

This gives a refinement-limit bosonic tensor category Sect_∞ of edge charges:

- objects: zero-obstruction sector labels that persist in the assumed directed system,
- morphisms: intertwiners between sectors,
- tensor product: fusion by collar concatenation, $\alpha \otimes \beta = \bigoplus_\gamma N_{\alpha\beta}^\gamma \gamma$,

- duals: orientation reversal / charge conjugation $\alpha \leftrightarrow \bar{\alpha}$,
- symmetric braiding in the EFT regime (no anyonic statistics in 3+1D).

The monoidal refinement maps descend the tensor unit, associators, duals, and bosonic symmetry to the colimit, while the compatible stagewise forgetful carriers descend to a faithful bosonic fiber functor $\mathcal{F} : \text{Sect}_\infty \rightarrow \text{Hilb}_{\text{fd}}$. The fibers are finite-dimensional objectwise even though Sect_∞ can have infinitely many simple objects.

Theorem 6.1 (Receipt-conditional bosonic sector category and Tannaka/DR reconstruction). On the ordinary or central zero-obstruction bosonic EFT branch, suppose a cofinal tail carries the compact-gauge refinement receipt used by `RefinementFunctorAndFiberDescent`. Then the colimit Sect_∞ is a rigid symmetric C^* tensor category with faithful bosonic fiber functor $\mathcal{F}$. Let $\text{Aut}_\otimes(\mathcal{F})$ denote the group of unitary symmetric monoidal natural automorphisms of $\mathcal{F}$. Then

$$G := \text{Aut}_\otimes(\mathcal{F})$$

is a compact group and $\text{Sect}_\infty \simeq \text{Rep}(G)$. In particular G is unique up to isomorphism, and this group is a compact subgroup of a product of unitary groups.

Proof. The `MaxEnt/local-Gibbs/collar-mixing` package is not used here to prove nontriviality or to supply the refinement receipt. The fixed-cutoff categories are constructed by `FixedCutoffBosonicSectorCategory`, and on the certified tail the monoidal pullback functors and compatible forgetful fibers are constructed by `RefinementFunctorAndFiberDescent`. Since those functors preserve tensor products, unit, duals, and $*$ -structure, the directed colimit inherits a rigid symmetric C^* -tensor structure, and the compatible stagewise carrier spaces descend to a faithful bosonic fiber functor $\mathcal{F}$. Fix a small skeleton of Sect_∞ . For every object X , a unitary symmetric monoidal natural automorphism $\eta \in \text{Aut}_\otimes(\mathcal{F})$ has a unitary component $\eta_X \in U(\mathcal{F}(X))$, so

$$\text{Aut}_\otimes(\mathcal{F}) \hookrightarrow \prod_X U(\mathcal{F}(X)), \quad \eta \mapsto (\eta_X)_X.$$

For each intertwiner $f : X \rightarrow Y$, naturality gives the closed relation $\mathcal{F}(f)\eta_X = \eta_Y\mathcal{F}(f)$. For each pair X, Y , the monoidal structure isomorphism $J_{X,Y}$ of $\mathcal{F}$ gives the closed relation

$$\eta_{X \otimes Y} = J_{X,Y} (\eta_X \otimes \eta_Y) J_{X,Y}^{-1}, \quad \eta_{\mathbf{1}} = \text{id}_{\mathbb{C}}.$$

Hence $G = \text{Aut}_\otimes(\mathcal{F})$ is a closed subgroup of the compact product $\prod_X U(\mathcal{F}(X))$, so G is compact. The DR/Tannaka hypotheses are therefore satisfied on the constructed pair $(\text{Sect}_\infty, \mathcal{F})$, and the Doplicher–Roberts/Tannaka reconstruction theorem gives a symmetric C^* -tensor equivalence $\text{Sect}_\infty \simeq \text{Rep}(G)$ [35, 36]. If another compact group G' gave the same category through a faithful bosonic fiber functor, the induced symmetric tensor equivalence $\text{Rep}(G) \simeq \text{Rep}(G')$ compatible with the forgetful functors would identify both groups as the automorphism group of the same fiber functor. Thus $G \cong G'$, so the reconstruction is unique up to isomorphism. QED.

Definition 6.1r (DHR/DR realization hypotheses). The field-algebra step below uses the following net-realization package, stated explicitly because it is logically additional to the Tannakian data of Theorem 6.1:

1. **(R1) Observable net.** An isotonomous net $O \mapsto \mathcal{A}(O)$ of C^* -algebras over the directed set of small-region collar localizations of the EFT regime, with quasi-local algebra $\mathcal{A}$ and a faithful vacuum representation.
2. **(R2) Locality, duality, Property B.** Spacelike commutativity; Haag duality $\mathcal{A}(O')' = \mathcal{A}(O)''$ in the vacuum representation, or the explicitly weaker essential-duality variant if that is the version invoked; and Property B.

3. **(R3) Localized endomorphisms.** Each persistent class $[\alpha] \in \mathbf{Sect}_\infty$ is realized by an endomorphism ρ_O^α of $\mathcal{A}$ acting trivially on $\mathcal{A}(O_1)$ for every localization region O_1 disjoint from O .
4. **(R4) Transportability.** For every admissible pair of regions $O, \tilde{O}$ there exists a unitary $u \in \mathcal{A}$ with $u \rho_O^\alpha(\cdot) u^* = \rho_{\tilde{O}}^\alpha(\cdot)$. This is a net-level statement about unitary intertwiners. Vanishing collar/represented holonomy supplies path-independent collar transport at the categorical level and is a necessary input to this hypothesis, but it is not by itself equivalent to DHR transportability.
5. **(R5) Statistics and conjugates.** Each ρ^α has finite statistics with permutation-symmetric (bosonic) statistics operator matching the symmetry of $\mathbf{Sect}_\infty$, and a conjugate endomorphism realizing the categorical dual $\bar{\alpha}$.
6. **(R6) Completeness.** The assignment $[\alpha] \mapsto [\rho^\alpha]$ is an equivalence of symmetric tensor C^* -categories between $\mathbf{Sect}_\infty$ and the category of localized transportable finite-statistics endomorphisms of $\mathcal{A}$.

This paper constructs the categorical side: the fixed-cutoff categories, the refinement functors and fibers, and the colimit pair $(\mathbf{Sect}_\infty, \mathcal{F})$. It does not construct the net realization (R1)–(R6); in particular no observable net is specified here on which the persistent sectors act as localized transportable endomorphisms. (R1)–(R6) therefore enter as explicit hypotheses, and every fixed-point identity $\mathcal{A} = \mathcal{F}_{\text{net}}^G$ in this paper is conditional on them.

Corollary 6.1 (two-level gauge reconstruction on zero-obstruction sectors). The reconstruction conclusion splits into two levels with strictly different hypotheses.

(i) *Tannakian level (proved).* If in the small-region limit the edge sectors satisfy the zero-obstruction transport criterion of Theorem 3.4b and lie on a cofinal tail carrying the compact-gauge refinement receipt, then Theorem 6.1 yields a compact group $G = \text{Aut}_\otimes(\mathcal{F})$ and a symmetric tensor equivalence $\mathbf{Sect}_\infty \simeq \text{Rep}(G)$. This level uses only the constructed category and fiber functor and involves no observable net.

(ii) *DHR/DR field-net level (conditional).* If in addition the realization hypotheses (R1)–(R6) of Definition 6.1r hold, then the Doplicher–Roberts reconstruction theorem [35, 36] supplies a field algebra $\mathcal{F}_{\text{net}}$ with normal (here bosonic) commutation relations, carrying an action of G with field operators implementing every persistent sector, such that

$$\mathcal{A} = \mathcal{F}_{\text{net}}^G.$$

Proof. Level (i) is Theorem 6.1. For level (ii): under (R1)–(R6) the localized transportable finite-statistics endomorphisms of $\mathcal{A}$ form a rigid symmetric C^* -tensor category equivalent to $\mathbf{Sect}_\infty$ by (R6), hence to $\text{Rep}(G)$ by level (i). The Doplicher–Roberts theorem [35, 36], whose locality, duality, Property B, transportability, statistics, conjugate, and completeness inputs are exactly (R2)–(R6) on the net (R1), then produces the crossed-product field net $\mathcal{F}_{\text{net}}$, its field operators, and the identification $\mathcal{A} = \mathcal{F}_{\text{net}}^G$ with G as gauge group. Without (R1)–(R6) the displayed fixed-point identity is not asserted; the unconditional content of this corollary is level (i). QED.

Corollary 6.1a (The compact-gauge setup and realized witness). On the central branch, the categorical collar-transport input to Corollary 6.1 is the combined theorem-level condition $[z]_\Sigma = 0$ plus trivial represented holonomy for an allowed strictification; this condition feeds the sector-category construction and hypothesis (R4) of Definition 6.1r, but it is not itself the net-level DHR-transportability statement, which additionally requires localized endomorphisms and unitary transporters on an observable net. On the genuinely noncentral branch, strict ordinary transport requires both $\alpha_\Sigma^{(2)} = 0$ and at least one allowed strict representative with trivial represented

G_Σ -holonomy; a nonzero $o_\Sigma^{(2)}$ remains a higher-gauge fixed-cutoff sector, while an orbit for which every strict representative has residual loop action remains nontransportable for the ordinary DR corollary. The full orbit q_Σ classifies the crossed-module representatives but need not determine a unique ordinary H^1 class. `FixedCutoffBosonicSectorCategory` constructs the fixed-stage category. `RefinementFunctorAndFiberDescent` constructs the fully faithful pullback functors and compatible forgetful fibers only from the explicit compact-gauge refinement receipt. Thus DR/Tannaka reconstruction yields a compact group on the receipt-certified branch at the Tannakian level; the field-algebra identity $\mathcal{A} = \mathcal{F}_{\text{net}}^G$ additionally requires the DHR/DR realization hypotheses (R1)–(R6); MAR, not the cocycle calculus alone, selects the realized Standard Model branch.

Theorem (gauge-sector classification-selection factorization). On the OPH compact-gauge lane, conditional on a cofinal tail carrying the compact-gauge refinement receipt, the realized Standard Model claim factors as

$$\begin{aligned} \text{overlap/gluing data} &\longrightarrow \text{associator strictification test} \longrightarrow \text{strict-representative holonomy test} \\ &\longrightarrow \text{Sect}_\infty \longrightarrow G = \text{Aut}_\otimes(\mathcal{F}) \longrightarrow \mathfrak{S}_{\text{MAR}}. \end{aligned}$$

The first arrow computes and, where possible, removes the central or crossed-module associator defect. The second computes represented holonomy across the allowed strict representatives. The third keeps only sectors for which at least one such representative has trivial loop action. The fourth reconstructs the compact group from the persistent tensor category and fiber functor. These are classification and reconstruction steps. The final arrow is the realization/selection step: MAR acts on realized admissible sector packages, not on obstruction classes alone. QED.

4.7.2 Selecting the SM factors (derived from MAR)

Theorem 6.1 yields *some* compact G . Axiom 5 (MAR, Minimal Admissible Realization) acts next on admissible sector packages; on the explicit one-Higgs chiral matter branch below, it selects the realized Standard Model quotient.

Axiom 5 (MAR): Minimal Admissible Realization. Among all OPH-realizable sector packages $\mathfrak{S}$ consisting of the connected Lie gauge-sector image relevant in the low-energy EFT, its admissible light chiral matter content, and one Higgs doublet, and which are (i) loop-coherent / transportable (the central triangle class vanishes, or the noncentral higher associator is strictifiable, and at least one allowed strict 1-cocycle representative has trivial represented sector holonomy), (ii) anomaly-free, (iii) refinement-stable with light chiral matter, (iv) single-Higgs Yukawa-completable with one connected abelian charge factor acting nontrivially on the coupled carrier, (v) intrinsically quark-sector CP-capable, (vi) weak-sector UV-completable on that same one-Higgs branch, the realized low-energy package is the lexicographically minimal one under

$$C(\mathfrak{S}) = (\chi_{\text{cpl}}, N_{\text{nonab}}, N_c, N_g).$$

The complete formal statement, admissibility definitions, and proof route are given in the rest of Section 6.2.

Here χ_{cpl} is the **coupled edge capacity**: the dimension of the minimal unitary carrier containing a common irreducible block on which the admissible pseudoreal and complex nonabelian charge types both act nontrivially. This is intentionally stronger than the abstract minimal faithful representation dimension. For $S(U(3) \times U(2))$, the block-diagonal action on $\mathbb{C}^3 \oplus \mathbb{C}^2$ is faithful of dimension 5, but it is not coupled and therefore does not enter MAR. The object minimized

by MAR is the sector package $\mathfrak{S}$, not the bare tensor category by itself. MAR is thus an explicit structural-economy selector on the admissible class, not a theorem derived from the earlier axioms.

Definition 6.1b (MAR realization space and physical equivalence). Let $\mathfrak{A}_{\text{MAR}}$ be the set of isomorphism classes of finite low-energy sector packages

$$\mathfrak{S} = (G^0, \mathcal{R}_{\text{light}}, H, \mathcal{Y}, \mathcal{F})$$

on the ordinary or central zero-obstruction bosonic branch of Theorem 6.1, together with the explicit realized one-Higgs chiral matter package used below. Here G^0 is the connected Lie gauge-sector image, $\mathcal{R}_{\text{light}}$ is the light chiral matter representation family, H is one Higgs doublet, $\mathcal{Y}$ denotes the Yukawa-completable charge data, and $\mathcal{F}$ is the descended finite bosonic fiber functor on the retained transportable sector package. A package is MAR-admissible exactly when it satisfies the six clauses in Axiom 5: zero obstruction, anomaly cancellation, refinement stability with light chiral matter, one-Higgs Yukawa completability with one connected abelian charge factor acting nontrivially on the coupled carrier, intrinsic quark-sector CP capability, and weak-sector UV completability on the same one-Higgs branch.

Two packages are physically equivalent when they are related by a compact-group isomorphism and a fiber-compatible symmetric monoidal equivalence that preserve the observer-visible representations, Yukawa invariants, anomaly polynomial, hypercharge lattice up to the allowed $U(1)$ normalization, and the one-Higgs branch, modulo generation relabeling, charge-conjugation convention, gauge-center quotienting, implementation hiding, and inert ancillary stabilization.

Definition 6.1c (MAR order). For $\mathfrak{S} \in \mathfrak{A}_{\text{MAR}}$, define

$$C(\mathfrak{S}) = (\chi_{\text{cpl}}, N_{\text{nonab}}, N_c, N_g) \in \mathbb{N}^4,$$

where N_{nonab} is the number of connected nonabelian simple factors acting nontrivially on the coupled carrier, N_c is the dimension of the minimal intrinsically complex color-type role, and N_g is the generation count on the realized one-Higgs chiral branch. MAR orders packages by the lexicographic order on $C(\mathfrak{S})$, then quotients ties by the physical equivalence relation above.

Proposition 6.1d (well-founded MAR minima). Every nonempty MAR-admissible class has a nonempty set of MAR-minimal packages. More precisely, if $\mathcal{A} \subseteq \mathfrak{A}_{\text{MAR}}$ is nonempty, then the set $C(\mathcal{A}) \subseteq \mathbb{N}^4$ has a lexicographically least element. The MAR-minimal packages in $\mathcal{A}$ are exactly the packages whose complexity vector is that least element.

Proof. The lexicographic order on $\mathbb{N}^4$ is well-founded: first minimize χ_{cpl} , then N_{nonab} on that fiber, then N_c , then N_g . Each step minimizes a nonempty subset of $\mathbb{N}$. QED.

Proposition 6.1e (meaning of MAR uniqueness). MAR uniqueness is uniqueness of the observer-visible low-energy package modulo the physical equivalence relation in Definition 6.1b. It is not uniqueness of a microscopic regulator representative. The subsequent lemmas prove that, within the connected positive-dimensional Lie admissible class with one connected abelian factor and faithful action on the minimal coupled carrier, all MAR-minimal representatives realize the same connected SM gauge image, the same exact hypercharge lattice after normalization, the same structural electroweak force content, $N_c = 3$, and $N_g = 3$. Any remaining differences are gauge-center quotient choices, relabelings, or inert implementation data and therefore do not change the physical predictions recorded in D8–D9.

Note on transport obstruction. The gluing obstruction is tracked explicitly so the gauge lane states its branch conditions. TransportabilityFromOverlapGluing identifies the central criterion as $[z]_{\Sigma} = 0$ together with an allowed trivial-holonomy strictification, and the genuinely noncentral criterion as $o_{\Sigma}^{(2)} = 0$ together with at least one allowed strict representative having trivial represented G_{Σ} -holonomy. When $o_{\Sigma}^{(2)} \neq 0$, the sector is handled by crossed-module data

rather than by the ordinary DR field-algebra corollary; when $o_{\Sigma}^{(2)} = 0$ but every strict representative has residual holonomy, it also does not enter that corollary. The full orbit q_{Σ} classifies the crossed-module representatives but need not determine one ordinary H^1 class. `FixedCutoffBosonicSectorCategory` constructs the fixed-cutoff category. On a cofinal tail carrying the compact-gauge refinement receipt, `RefinementFunctorAndFiberDescent` constructs the refinement/fiber descent and the realized witness theorem supplies nonempty ordinary/central compact-gauge witness data. The realized Standard Model branch enters only after MAR is applied to the admissible one-Higgs sector packages.

What MAR derives. Product structure, minimal sector content, and coupled edge-capacity minimality are consequences of MAR applied to the admissible class:

- **Product structure:** follows from the minimal coupled carrier $\mathbb{C}^3 \otimes \mathbb{C}^2$, which enforces commuting color and weak actions.
- **Minimal sector content:** the pseudoreal doublet and complex triplet are the minimal nonabelian representations satisfying the admissibility conditions, while the connected abelian charge factor is an explicit admissibility input.
- **Coupled edge-capacity minimality:** this is the first component of MAR's complexity vector.

With MAR stated, the SM derivation proceeds via standard lemmas:

Lemma 6.2 (Product factorization implies product group). If $\text{Sect} \simeq \text{Rep}(G)$ and $\text{Sect} \simeq \text{Sect}_1 \boxtimes \text{Sect}_2$, then

$$G \cong G_1 \times G_2, \quad \text{Sect}_i \simeq \text{Rep}(G_i).$$

QED.

Lemma 6.3 (SU(2) from a pseudoreal doublet). Let H be a positive-dimensional compact connected Lie group with a faithful 2D pseudoreal unitary representation V . Then the semisimple part of H contains an $SU(2)$ factor acting as the fundamental doublet. Finite or disconnected counterexamples are not part of the theorem package, which only applies the lemma to the identity component on the relevant nonabelian image. QED.

Lemma 6.4 (SU(3) from an irreducible triplet). Let H be a positive-dimensional compact connected Lie group with a faithful irreducible complex 3D unitary representation W . Then the semisimple image contains an $SU(3)$ factor acting as the fundamental triplet. Finite or disconnected counterexamples are not part of the theorem package, which only applies the lemma to the identity component on the relevant nonabelian image. QED.

Lemma 6.5 (Connected abelian factor criterion). If the admissible sector package contains a connected abelian charge factor acting nontrivially on the coupled carrier, then the identity component of the abelianized reconstructed group contains a one-torus. Under the single connected abelian-factor admissibility condition, this factor is $U(1)$. QED.

Proposition 6.6 (physical group quotient). If the realized matter spectrum has hypercharges quantized in sixths, then the kernel acting trivially on all realized sectors is Z_6 , so

$$G_{\text{phys}} = \frac{SU(3) \times SU(2) \times U(1)}{Z_6}.$$

QED.

Proposition 6.6a (SM from MAR). Under Axiom 5 (MAR):

- By the definition of χ_{cpl} , the realized MAR package contains a light chiral pseudoreal weak-type role and a light chiral complex color-type role on a common coupled carrier; clause (iii) imposes refinement stability on that branch, and clause (iv) adds one connected abelian charge factor acting nontrivially on the same carrier together with one-Higgs Yukawa completeness.
- The minimal faithful pseudoreal representation is the doublet ($\chi = 2$), giving $SU(2)$. No intrinsically complex 2D irrep qualifies in the connected compact Lie class, because the connected derived image of any irreducible 2D unitary representation lies in $SU(2)$, so the nonabelian action is pseudoreal up to abelian twist. The first intrinsically complex case is therefore the triplet ($\chi = 3$), giving $SU(3)$.
- The minimal coupled carrier for both is $\mathbb{C}^3 \otimes \mathbb{C}^2$, giving coupled edge capacity $\chi_{\text{cpl}} = 6$.
- The block-diagonal faithful representation $\mathbb{C}^3 \oplus \mathbb{C}^2$ of $S(U(3) \times U(2))$ has dimension 5, but it is not coupled and therefore is not the MAR minimizer.
- The maximal compact subgroup of $U(6)$ acting on $\mathbb{C}^3 \otimes \mathbb{C}^2$ with commuting actions is $(SU(3) \times SU(2) \times U(1))/(\text{finite center})$.
- The commutant of $SU(3) \times SU(2)$ inside $U(6)$ is exactly $U(1)$, so any connected abelian factor acting nontrivially on the coupled carrier is necessarily a single $U(1)$, and no additional continuous factors appear without increasing χ_{cpl} .
- Product structure is not separately assumed: it follows from the tensor product structure of the minimal coupled carrier.

Combined with Proposition 6.6 (hypercharges quantized in sixths from the realized spectrum), this yields:

$$G_{\text{phys}} = \frac{SU(3) \times SU(2) \times U(1)}{Z_6}.$$

The full proof route is given in the remainder of Section 6.2.

Theorem 6.6aa (realized compact-gauge witness and physical UV landing). On the ordinary or central zero-obstruction bosonic branch, suppose a cofinal tail carries the compact-gauge refinement receipt. Then the OPH compact reference architecture admits an OPH-realizable cofinal heat-kernel edge-sector witness

$$\mathcal{W}_{\text{SM}} = \{Q_i, u_i^c, d_i^c, L_i, e_i^c, H\}_{i=1}^3$$

with

$$Q_i = (3, 2)_{1/6}, \quad u_i^c = (\bar{3}, 1)_{-2/3}, \quad d_i^c = (\bar{3}, 1)_{1/3}, \quad L_i = (1, 2)_{-1/2}, \quad e_i^c = (1, 1)_1, \quad H = (1, 2)_{1/2}.$$

The fixed-cutoff edge heat-kernel law gives $p_R(t) \propto d_R e^{-tC_2(R)}$, hence positive support for every finite-dimensional witness sector at finite t . The pullback functors constructed from the receipt carry those zero-obstruction labels cofinally. The witness is anomaly-free and one-Higgs Yukawa-complete by the hypercharge theorem, contains the weak-type and color-type roles used by MAR, and satisfies the CP-capability and weak-sector UV clauses that force $N_g = 3$ once $N_c = 3$. Therefore the MAR-admissible class is nonempty, and every OPH-admissible microscopic UV completion of the realized branch projects, modulo physical equivalence, to

$$\frac{SU(3) \times SU(2) \times U(1)}{Z_6}, \quad N_c = 3, \quad N_g = 3,$$

with the Standard Model hypercharge lattice. Microscopic regulator representatives remain underdetermined up to gauge presentation, implementation hiding, and inert ancillary stabilization. QED.

Theorem (Realized Standard Model branch). Assume the ordinary or central zero-obstruction bosonic branch; a cofinal tail carrying the compact-gauge refinement receipt; the fixed-cutoff bosonic sector category, refinement/fiber descent, and compact-group reconstruction above; a nonempty realized one-Higgs chiral MAR-admissible class witnessed by $\mathcal{W}_{\text{SM}}$; weak-type and intrinsically complex color-type roles on a common coupled carrier; one connected abelian charge factor acting nontrivially on that carrier; and the anomaly-free, one-Higgs Yukawa-complete, CP-capable, and weak-sector UV clauses of Axiom 5. Then MAR minima exist, and every MAR-minimal observer-visible package in that class is physically equivalent to

$$\left(\frac{\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)}{\mathbb{Z}_6}, \mathcal{R}_{\text{SM}}, H, \mathcal{V}_{\text{SM}} \right), \quad N_c = 3, \quad N_g = 3,$$

with

$$\mathcal{W}_{\text{SM}} = \{Q_i, u_i^c, d_i^c, L_i, e_i^c, H\}_{i=1}^3,$$

$$Q_i = (3, 2)_{1/6}, \quad u_i^c = (\bar{3}, 1)_{-2/3}, \quad d_i^c = (\bar{3}, 1)_{1/3}, \quad L_i = (1, 2)_{-1/2}, \quad e_i^c = (1, 1)_1, \quad H = (1, 2)_{1/2}.$$

The proof is the earlier well-founded MAR-minima proposition plus the weak/color classification lemmas, the minimal coupled-carrier result $\mathbb{C}^3 \otimes \mathbb{C}^2$, the $U(1)$ commutant, hypercharge theorem, three-color corollary, MAR-branch generation theorem, and the $\mathbb{Z}_6$ kernel proof. QED.

Corollary 6.6b (Three colors on the realized branch). On the realized one-generation matter package carried by the minimal coupled block of Proposition 6.6a, the color multiplicity is fixed by the same MAR derivation: the color factor is the fundamental triplet of $\text{SU}(3)$, so

$$N_c = 3.$$

The selector is inside the stated theorem stack. QED.

Corollary 6.6c (structural electroweak force and charges). On the realized one-Higgs branch, the electroweak gauge factor has Lie algebra

$$\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y.$$

The Higgs doublet $H = (1, 2)_{1/2}$ selects a nonzero neutral vacuum vector

$$\phi_0 = \frac{v}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad v \neq 0,$$

with

$$Q\phi_0 = (T_3 + Y)\phi_0 = 0,$$

so its vector stabilizer has unbroken generator

$$Q = T_3 + Y,$$

and the unbroken gauge factor is $U(1)_Q$. The charged weak generators give

$$W^\pm = \frac{1}{\sqrt{2}}(W^1 \mp iW^2),$$

while the neutral $\text{SU}(2)_L$ and $U(1)_Y$ gauge fields span the Z/A basis on the D10 quantitative branch. Explicitly,

$$e^{i\alpha T_3} e^{i\beta Y} \phi_0 = e^{i(\beta-\alpha)/2} \phi_0,$$

so exact vector invariance ties the phases by $\beta = \alpha$, locally, and leaves $U(1)_Q$. The projective ray $[\phi_0]$, however, is unchanged for independent α and β : its stabilizer is the two-torus $U(1)_{T_3} \times U(1)_Y$, modulo the inherited finite center. Projectivization therefore classifies the carrier ray but cannot by itself select the electromagnetic diagonal. Thus the recovered structural package contains the weak charged-current carriers $W^\pm$, the neutral weak carrier Z , the electromagnetic carrier A , and the exact charge operator $Q = T_3 + Y$. The mixing angle, v , and numerical W/Z masses belong to the D10 running/matching surface rather than to this recovered-core structural corollary. QED.

Corollary 6.6d (conditional Maxwell sector on the realized electromagnetic branch).

On the ordinary one-Higgs branch, let A_Q be the support-visible connection component on the unbroken electromagnetic factor $U(1)_Q$. The abelian restriction of the compact-gauge curvature gives

$$F_Q = dA_Q.$$

Assume in addition the Lorentzian low-energy Maxwell action

$$S_{\text{EM}}[A_Q, J_Q] = -\frac{1}{2g_Q^2} \int F_Q \wedge *F_Q + \int A_Q \wedge *J_Q,$$

the Euler-Lagrange equations are

$$dF_Q = 0, \quad d * F_Q = g_Q^2 * J_Q.$$

After canonical electromagnetic normalization, these are Maxwell's equations. The numerical value of g_Q , equivalently α_{em} , belongs to the Ward-projected electromagnetic readout. The compact-group reconstruction and connection label do not themselves supply this action or its nonzero kinetic coefficient. QED.

4.7.3 Refinement stability and unprotected relevant operators

The state-side refinement stability used later in the gauge branch is contained in the third OPH axiom through its refinement-closure clause. By that clause the same finite local constraint family is preserved across cutoffs, so the realized UV states lie in one common finite-dimensional MaxEnt family rather than in a new coupling space at every refinement step. The selected stable branch of this family is therefore not an extra state axiom beyond the MaxEnt branch itself, but the closure clause it relies on is a substantive renormalization condition: coarse-graining a finite-range Gibbs family generically generates interactions outside any fixed finite density list. The closure defect, the moment-matching I-projection realizing $R_{\ell \rightarrow L}$, and its trace-norm residual bound are quantified in Definition 2.6a and Lemma 2.6b; the clause is exactly the statement that the defect vanishes along the realized branch, and that vanishing remains assumed rather than proved. This state statement does not supply the compact-gauge refinement receipt.

Concretely, if

$$\omega_\ell(\lambda) = Z_\ell(\lambda)^{-1} \exp \left(- \sum_x \sum_a \lambda_a O_a(x) - \sum_b \mu_b Q_b \right),$$

then any refinement channel $\Phi_{\ell \rightarrow L}$ compatible with Axiom 3, including its refinement-closure clause, acts on the realized branch by an induced finite-dimensional map

$$\Phi_{\ell \rightarrow L}(\omega_\ell(\lambda)) = \omega_L(R_{\ell \rightarrow L}(\lambda)).$$

The resulting refinement-stable state branch is simply a trajectory or invariant subset of this finite-dimensional multiplier map. Accordingly, when sections speak of a refinement-stable directed colimit of sectors, persistence along this MaxEnt branch is only the state-side input. The additional

compact-gauge receipt must supply finite extendability, center-compatible block embeddings, coherent surjective compact-group maps, preservation of the full transport obstruction, and finite tensor realizations. The MaxEnt/mixing package supplies none of those data and does not decide whether the resulting certified colimit is trivial or nontrivial.

The consensus paper makes the state-side coarse-graining connection explicit. Given quotient normal-form maps n_r , obstruction maps h_r , and coarse-graining maps (ρ_{sr}, χ_{sr}) , reconciliation commutes with coarse-graining at the macroscopic readout scale whenever the two square defects

$$d_r^Q(\rho_{sr}n_s(x), n_r\rho_{sr}(x)), \quad d_r^H(\chi_{sr}h_s(x), h_r\rho_{sr}(x))$$

are controlled. Exact refinement naturality gives zero defect; approximate RG matching is theorem-grade only to the extent that these defects are bounded on the selected branch. This keeps the MaxEnt refinement map $R_{\ell \rightarrow L}$ connected to the reconciliation law without promoting arbitrary coarse-graining channels to OPH laws.

Relevant operators that are neither symmetry-forbidden nor retained in the selected constraint family are precisely the directions that try to push the flow off that stable branch.

Lemma 6.7 (refinement-stable MaxEnt branch forbids unprotected relevant operators). Assume the local finite-constraint MaxEnt/refinement branch of the third OPH axiom. Let $\mathcal{O}$ be a gauge-invariant Lorentz-scalar relevant deformation in the emergent EFT sense ($\Delta < 4$ in $3 + 1D$), allowed by symmetry and absent from the retained constraint family. Then the selected branch can keep the coupling of $\mathcal{O}$ at zero only if symmetry forbids that direction or the constraint family explicitly retains it. Otherwise generic refinement induces a nonzero component along that direction and the flow leaves the selected branch. This is a branch-persistence statement, not a universal entropy-ordering theorem for arbitrary off-branch phases.

Proof sketch. Linearize the induced refinement map $R_{\ell \rightarrow L}$ on the finite-dimensional multiplier space around the selected branch. A relevant operator gives an unstable direction with scaling exponent $y > 0$. If $\mathcal{O}$ is not fixed by symmetry and is not part of the retained constraint family, generic UV mismatch produces a nonzero component along that direction. Because $y > 0$, repeated coarse-graining amplifies the component and pushes the flow off the selected branch unless one fine-tunes it away at every scale or protects it by symmetry or by the declared constraint family. QED.

Corollary 6.8 (chirality selector). A gauge-invariant Dirac mass term is a relevant scalar. If both chiralities exist in conjugate representations, the mass term is allowed and will be generated under refinement unless symmetry-forbidden or explicitly retained as a protected constraint. Therefore keeping light fermions on the selected refinement-stable branch requires chiral matter content or an explicit protecting mechanism. QED.

4.7.4 Generation number from CKM CP capability and weak-sector completability (derived from MAR)

Anomaly cancellation is generation-by-generation, so it does not fix the number of generations. On the realized one-Higgs branch, the lower and upper bounds come from the CP-capability and weak-sector UV clauses declared in MAR, and MAR then selects the minimum.

Proposition 6.9 (The number of generations is $N_g = 3$). On the realized one-Higgs quark branch of Proposition 6.6a, with the derived $N_c = 3$ from Corollary 6.6b and with the CP-capability and weak-sector UV clauses contained in Axiom 5, the generation number is

$$N_g = 3.$$

Inputs.

1. **Intrinsic CKM CP capability** is part of the realized quark branch through clause (v) of Axiom 5.
2. **Weak-sector UV completability** is part of the same realized one-Higgs branch through clause (vi) of Axiom 5.
3. **MAR minimality** acts on the same realized branch once the first three complexity entries are fixed.
4. Use the derived $N_c = 3$ from Corollary 6.6b.

Step 1: CKM CP-capability lower bound. The number of physical CP-violating phases in an $N_g \times N_g$ CKM matrix is:

$$\#(\text{CP phases}) = \frac{(N_g - 1)(N_g - 2)}{2}.$$

- For $N_g = 1, 2$: this is 0 $\rightarrow$ **no intrinsic CKM CP capability**.
- For $N_g = 3$: this is 1 $\rightarrow$ **intrinsic CKM CP capability is available**.

So intrinsic CKM CP capability requires:

$$N_g \geq 3.$$

Step 2: SU(2) asymptotic freedom upper bound. The one-loop coefficient is:

$$b_{\text{SU}(2)} = \frac{22}{3} - \frac{1}{3}N_g(N_c + 1) - \frac{1}{6},$$

where the final $-1/6$ is the contribution of one complex Higgs doublet. Asymptotic freedom means $b_{\text{SU}(2)} > 0$, i.e.,

$$N_g(N_c + 1) < \frac{43}{2}.$$

With $N_c = 3$, we have $N_c + 1 = 4$, so:

$$4N_g < \frac{43}{2} \quad \Rightarrow \quad N_g \leq 5.$$

Combining: $3 \leq N_g \leq 5$.

Step 3: MAR selection. These are not extra post-MAR selectors: they are the numerical content of clauses (v) and (vi) of Axiom 5 on the realized one-Higgs branch. Given the allowed window $\{3, 4, 5\}$, MAR (fourth component of the complexity vector $C(\mathfrak{S})$) selects the smallest viable choice:

$$N_g = 3.$$

QED.

Why this is convincing.

- It predicts a **single integer**.
- It uses **two explicit MAR clauses** (intrinsic CKM CP capability and weak-sector UV completability) plus MAR's lexicographic minimality, rather than a separate post hoc admissibility menu.
- It is not a fit to a continuous number.
- Under the stated gauge-selection hypotheses, this is a derived result.

Corollary 6.9a (MAR-minimal SM package is unique up to physical equivalence). On the ordinary or central zero-obstruction bosonic branch, assume the MAR-admissible class is nonempty and contains the explicit realized one-Higgs chiral matter package used in Theorem 6.1 through Proposition 6.9. Then every MAR-minimal representative in that branch has the same observer-visible Standard Model package:

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \quad N_c = 3, \quad N_g = 3,$$

with the hypercharge lattice of Proposition 6.6 and the finite quotient fixed by Proposition 6.6. The only residual freedom is physical equivalence in the sense of Definition 6.1b.

Proof. Proposition 6.1d supplies the MAR minima. Lemmas 6.3–6.5 and Proposition 6.6a fix the minimal weak, color, coupled-carrier, and connected abelian roles at the least first three complexity entries. Corollary 6.6b fixes $N_c = 3$, Proposition 6.9 fixes $N_g = 3$, and the hypercharge and quotient propositions fix the visible charge lattice and $\mathbb{Z}_6$ kernel. The equivalence relation removes only relabelings, gauge-center conventions, and inert implementation data. QED.

4.7.5 Hilbert-space formulation of gluing data

Let $\{P_i\}$ be a good cover of the screen. For each patch, fix a representation

$$\pi_i : \mathcal{A}_i \rightarrow \mathcal{B}(\mathcal{H}_i).$$

For each overlap, choose a unitary intertwiner

$$U_{ij} : \mathcal{H}_j \rightarrow \mathcal{H}_i$$

such that for all $O \in \mathcal{A}_{ij}$,

$$\pi_i(O) = U_{ij} \pi_j(O) U_{ij}^\dagger.$$

Normalize $U_{ii} = 1$ and $U_{ji} = U_{ij}^\dagger$.

Lemma 6.10 (centrality on triple overlaps). On a triple overlap define

$$\Omega_{ijk} := U_{ij} U_{jk} U_{ki}.$$

For all $O \in \mathcal{A}_{ijk}$,

$$\Omega_{ijk} \pi_i(O) = \pi_i(O) \Omega_{ijk}.$$

Proof. Conjugation by U_{ki} sends $\pi_i(O)$ to $\pi_k(O)$, by U_{jk} to $\pi_j(O)$, by U_{ij} back to $\pi_i(O)$. Thus conjugation by Ω_{ijk} fixes $\pi_i(O)$, so Ω_{ijk} commutes with $\pi_i(O)$. QED.

Lemma 6.11 (gauge behavior). If $\tilde{U}_{ij} = V_i U_{ij} V_j^\dagger$ with V_i acting trivially on overlap observables, then

$$\tilde{\Omega}_{ijk} = V_i \Omega_{ijk} V_i^\dagger.$$

In particular, if Ω_{ijk} is central, its class is gauge invariant. QED.

4.7.6 Loop obstruction class (central defect)

Assume the defect is central, the overlap centers are identified with one fixed abelian unitary coefficient group Z_Σ , and overlap transport acts trivially on Z_Σ . With $U_{ji} = U_{ij}^{-1}$, define the implementer-level multiplier by

$$\Omega_{ijk} = U_{ij}U_{jk}U_{ki} = z_{ijk} \in Z_\Sigma,$$

equivalently $U_{ij}U_{jk} = z_{ijk}U_{ik}$. This is the abelian truncation of the full 2-group obstruction in Section 3.4. The multiplier must be defined at the implementer level: $\text{Ad}(z_{ijk})$ is the identity for central z_{ijk} and does not by itself record the defect.

Theorem 6.12 (central defect strictification and residual holonomy). The family $\{z_{ijk}\}$ is an untwisted Čech 2-cocycle with coefficients in Z_Σ , and its class $[z] \in \check{H}^2(N_\Sigma, Z_\Sigma)$ is gauge invariant. On any quadruple overlap P_{ijkl} ,

$$z_{jkl}z_{ikl}^{-1}z_{ijl}z_{ijk}^{-1} = 1.$$

A central edge rephasing by a cochain in $C^1(N_\Sigma, Z_\Sigma)$ removes every triangle defect iff $[z] = 0$. After such a rephasing, write U_{ij}^{str} for the resulting genuine edge 1-cocycle. Let $\mathcal{C}_\alpha$ be the full transported charge block, including its carrier and multiplicity/intertwiner data. Loop-coherent, endpoint-only transport exists on that block iff, in addition,

$$\text{Hol}_{\alpha, U^{\text{str}}}([\gamma]) = U_\gamma^{\text{str}}|_{\mathcal{C}_\alpha} = \mathbf{1}_{\mathcal{C}_\alpha} \quad \text{for every } [\gamma] \in \pi_1(|N_\Sigma|, i_*).$$

Thus strict path independence is equivalent to the combined conditions $[z] = 0$ and trivial represented holonomy for one allowed strictification.

Proof. The fixed-coefficient and trivial-action hypotheses place every multiplier in the same group Z_Σ . Associativity compares $(U_{ij}U_{jk})U_{kl}$ with $U_{ij}(U_{jk}U_{kl})$ on a quadruple overlap. Substituting $U_{ij}U_{jk} = z_{ijk}U_{ik}$ gives the displayed Čech cocycle identity. A vertex-frame change $U_{ij} \mapsto V_i U_{ij} V_j^\dagger$ conjugates z_{ijk} , as in Lemma 6.11, and therefore leaves a central multiplier unchanged. Separately, a central edge rephasing $U_{ij} \mapsto a_{ij}^{-1}U_{ij}$, with $a \in C^1(N_\Sigma, Z_\Sigma)$, replaces z by the corresponding Čech coboundary. Hence $[z] = 0$ is equivalent to removing all triangle multipliers by a central 1-cochain. This yields a strict 1-cocycle, not automatically endpoint-only transport. For two paths p, p' with the same endpoints, the remaining discrepancy is the holonomy of the closed loop $p'^{-1}p$; it vanishes for every pair exactly when the displayed holonomy representation on $\mathcal{C}_\alpha$ is trivial. Conversely, strict path independence makes every triangle comparison and every represented closed-loop action trivial. QED.

If overlap transport acts nontrivially on a local system of centers, the same comparison contains the transported factor $\varphi_{ij}(z_{jkl})$ and must be written with the corresponding twisted Čech differential; that branch is not the untwisted theorem stated here.

4.7.7 EFT reduction to anomaly cancellation

Assume ExtEFT: a low-energy 3+1D chiral gauge theory exists with group G . Then the obstruction class $[z]$ is structurally analogous to the EFT 't Hooft anomaly class and is expected to map to it after a separate anomaly-descent construction. That map lies outside this manuscript, so $[z] = 0$ is used here as the internal triangle-defect strictifiability condition rather than a proved equivalence to EFT anomaly cancellation; the separate trivial-holonomy condition of Theorem 6.12 is still required for transportability.

4.7.8 Hypercharge from anomaly freedom and Yukawas

Theorem 6.13 (Hypercharge from anomaly freedom and Yukawas). Assume gauge group $SU(N_c) \times SU(2) \times U(1)_Y$ and one generation of left-handed Weyl fermions (Q, u^c, d^c, L, e^c) , with a Higgs doublet H and Yukawa terms

$$QH u^c, \quad QH^\dagger d^c, \quad LH^\dagger e^c.$$

Then anomaly freedom and Yukawa invariance fix the hypercharges up to an overall normalization, yielding the Standard Model pattern for $N_c = 3$.

Proof. Yukawa invariance gives

$$Y_u = -(Y_Q + Y_H), \quad Y_d = -Y_Q + Y_H, \quad Y_e = -Y_L + Y_H.$$

Anomaly cancellation yields

$$\begin{aligned} SU(2)^2 U(1) : \quad N_c Y_Q + Y_L &= 0, \\ \text{grav}^2 U(1) : \quad 2N_c Y_Q + N_c Y_u + N_c Y_d + 2Y_L + Y_e &= 0. \end{aligned}$$

Solving gives

$$Y_L = -N_c Y_Q, \quad Y_H = N_c Y_Q, \quad Y_u = -(N_c + 1)Y_Q, \quad Y_d = (N_c - 1)Y_Q, \quad Y_e = 2N_c Y_Q.$$

With these relations, $SU(N_c)^2 U(1)$ and $U(1)^3$ anomalies vanish automatically. Fixing the normalization by $Q = T_3 + Y$ and $Q(\nu_L) = 0$ gives

$$Y_Q = \frac{1}{2N_c}.$$

For $N_c = 3$,

$$Y_Q = \frac{1}{6}, \quad Y_L = -\frac{1}{2}, \quad Y_e = 1, \quad Y_u = -\frac{2}{3}, \quad Y_d = \frac{1}{3}, \quad Y_H = \frac{1}{2}.$$

Without Yukawas, the cubic anomaly leaves two discrete branches (Y_u, Y_d exchange). Yukawa invariance selects the branch with a single Higgs doublet. QED.

Corollary 6.13a (Exact rational hypercharges). With the derived $N_c = 3$, the hypercharge assignments are uniquely fixed to exact rational values:

$$Y_Q = \frac{1}{6}, \quad Y_L = -\frac{1}{2}, \quad Y_u = -\frac{2}{3}, \quad Y_d = \frac{1}{3}, \quad Y_e = 1, \quad Y_H = \frac{1}{2}.$$

Why this is convincing.

- These are **exact rationals**, not approximate numbers.
- Their ratios are fixed by anomaly freedom + Yukawa invariance, and the absolute lattice is fixed by the standard normalization, with no continuous parameters to adjust.
- This high-precision set of numbers strongly constrains the realized matter package and matches observation exactly.

4.7.9 Witten anomaly on the realized color branch

Theorem 6.14 (Witten anomaly consistency on the realized color branch). On the realized one-generation package emitted by Proposition 6.6a and Corollary 6.6b, the global SU(2) anomaly is absent generation by generation.

Inputs.

1. The realized color factor is the SU(3) triplet from Corollary 6.6b.
2. The matter content per generation includes:
 - one left-handed quark doublet Q which is an SU(2) doublet and carries color,
 - one left-handed lepton doublet L which is an SU(2) doublet and color singlet.
3. **Witten's global SU(2) anomaly constraint** (Witten, 1982): the number of left-handed SU(2) doublets must be even.

Proof. Count SU(2) doublets per generation:

- Quark doublets: N_c copies (one per color),
- Lepton doublets: 1 copy.

Total doublets per generation:

$$N_c + 1.$$

Substituting the realized value $N_c = 3$ gives

$$N_c + 1 = 4.$$

Witten anomaly cancellation therefore requires an even number of doublets, and the realized branch satisfies it exactly:

$$4 \equiv 0 \pmod{2}.$$

So the realized triplet-doublet package is globally consistent generation by generation. QED.

Theorem-stack role. The D8 minimal-coupled-carrier derivation emits the specific value $N_c = 3$ and carries it into D9. Witten's anomaly is retained as a nontrivial consistency check on that emitted triplet-doublet package, not as a residual selector that upgrades oddness to 3.

Why this is convincing.

- It checks a **global anomaly constraint** on the emitted branch without introducing a new selector.
- The parity constraint is independent of continuous parameters, RG running, masses, or Yukawa values.
- It confirms that the realized SU(3) triplet plus lepton doublet package is globally consistent generation by generation.

4.7.10 Bond-dimension gatekeeping

In tensor-network or code realizations, gauge actions act on edge factors of size χ , so emergent compact gauge groups embed in $U(\chi)$. This shows a capacity constraint: accommodating SU(3) color and SU(2) weak factors shows $\chi \geq 6$ in the minimal case, consistent with the MAR-derived gauge group.

4.7.11 Classical carrier modes and the quantum-particle gate

An abstract compact group identifies a symmetry type, and a classical field equation identifies a classical dynamical branch. Neither datum alone constructs a propagating quantum particle. The following receipt separates those levels.

The upstream structural results remain separate receipts. Conditional on the compact-gauge refinement receipt, Theorem 6.1 constructs the refinement-limit bosonic edge-sector category and its compact group; a field-algebra realization with that group as a local gauge symmetry still requires the DHR/DR hypotheses (R1)–(R6) of Definition 6.1r. On the realized electroweak branch, a chosen nonzero neutral Higgs vacuum *vector* has stabilizer $U(1)_Q$, whereas its projective ray alone has the larger two-torus stabilizer recorded above. On the gravitational side, Theorems 4.3c–4.3e and 5.1 conditionally supply the observer-facing geometry, Lorentzian event-manifold, and classical Einstein branch. None of these structural outputs supplies the action Hessian or quantum data required below.

Definition 6.18 (carrier-mode and quantum-particle receipts). Fix a structural invariant speed $c_\star$. This is the common invariant null-cone conversion speed. Its representation as $299\,792\,458\text{ ms}^{-1}$ is exact by the SI definition of the metre and is not a prediction of that decimal magnitude. A *classical massless carrier-mode receipt* for a field q_X consists of: (C1) a stated background and phase; (C2) an explicit quadratic action with positive nonzero physical kinetic coefficient; (C3) a gauge fixing or constraint reduction with a finite physical projector Π_X ; and (C4) a positive reduced Hamiltonian. Separately, the action Hessian restricted to the physical subspace must have the form

$$K_X^{\text{phys}}(\omega, \mathbf{k}) = Z_X(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_X, \quad Z_X > 0.$$

Equivalently, the free reduced Green function exhibits the action-level pole

$$D_X^{\text{phys}}(\omega, \mathbf{k}) = \frac{i \Pi_X / Z_X}{\omega^2 - c_\star^2|\mathbf{k}|^2 + i0}.$$

This receipt proves a classical massless carrier mode and the absence of an algebraic mass term in that declared quadratic action. It does not yet prove a particle.

A *quantum-particle receipt* additionally requires: (Q1) a positive-energy vacuum quantization and a physical Hilbert space obtained by constraint reduction or BRST cohomology; (Q2) a physical two-point function whose Källén–Lehmann measure contains a positive-residue massless contribution

$$d\rho_X(\mu^2) = Z_X^{\text{pole}} \delta(\mu^2) d\mu^2 + d\rho_X^{\text{cont}}(\mu^2), \quad Z_X^{\text{pole}} > 0;$$

or an equivalent joint energy–momentum spectrum containing a positive-residue $p^0 > 0$, $p^2 = 0$ mass shell (at fixed $\mathbf{k}$, $\omega = c_\star|\mathbf{k}|$); and (Q3) the stability/asymptotic-state or LSZ hypotheses appropriate to the phase, including a deconfined asymptotic sector for a colored carrier. Only a receipt passing (C1)–(C4) and (Q1)–(Q3) licenses a quantum-particle claim. A reconstructed group, a connection label, or a classical field equation by itself fails this gate.

Theorem 6.18 (action-level modes on the Maxwell, pure-Yang–Mills, and Einstein branches). The following statements hold on the stated additional action and phase branches.

- i. **Electromagnetic branch.** Suppose the source-free ordinary electromagnetic vacuum $J_Q = 0$ has an unbroken $U(1)_Q$ connection with the Lorentzian Maxwell action

$$S_Q = -\frac{1}{4g_Q^2} \int F_{Q,\mu\nu} F_Q^{\mu\nu} d^4x, \quad 0 < g_Q^2 < \infty,$$

about the ordinary vacuum, with no Higgs, Stueckelberg, medium, or nonlocal quadratic mass operator. In radiation gauge the reduced variables are the two transverse components and

$$H_Q^{(2)} = \frac{1}{2g_Q^2} \int (|\mathbf{E}_T|^2 + |\mathbf{B}|^2) d^3x, \quad K_{Q,T} = g_Q^{-2} (\omega^2 - c_\star^2 |\mathbf{k}|^2) \Pi_T, \quad \text{rank } \Pi_T = 2.$$

Equivalently, covariant ξ -gauge gives

$$D_{\mu\nu}^Q(k) = \frac{-ig_Q^2}{k^2 + i0} \left(\eta_{\mu\nu} - (1 - \xi) \frac{k_\mu k_\nu}{k^2 + i0} \right),$$

whose longitudinal part decouples from a conserved current. Thus this branch has two classical massless electromagnetic carrier modes. Calling them photons requires the quantum-particle receipt.

- ii. **Pure Yang–Mills branch.** Suppose a compact group G is supplied with the Lorentzian two-derivative pure Yang–Mills action with $0 < g^2 < \infty$, expanded about the topologically trivial background in a gauge $\bar{A}_\mu = 0$, in a perturbative or deconfined phase with no Higgs condensate. Writing the perturbation as a_μ^a , its quadratic action is

$$S_{\text{YM}}^{(2)} = -\frac{1}{4g^2} \sum_a \int (\partial_\mu a_\nu^a - \partial_\nu a_\mu^a) (\partial^\mu a^{a,\nu} - \partial^\nu a^{a,\mu}) d^4x.$$

Lorenz gauge gives the standard covariant inverse, while Gauss' law and radiation gauge leave two transverse components per generator, with

$$H_{\text{YM}}^{(2)} = \frac{1}{2g^2} \sum_a \int (|\mathbf{E}_T^a|^2 + |\mathbf{B}^a|^2) d^3x \geq 0,$$

with strict positivity on nonzero reduced finite-energy modes of $\mathbf{k} \neq 0$ under the stated boundary conditions. Their quadratic transverse kernel and free Green function are

$$K_{\text{YM},T}^{ab} = \delta^{ab} g^{-2} (\omega^2 - c_\star^2 |\mathbf{k}|^2) \Pi_T, \quad D_{\text{YM},T}^{ab} = \frac{ig^2 \delta^{ab} \Pi_T}{\omega^2 - c_\star^2 |\mathbf{k}|^2 + i0}, \quad \text{rank } \Pi_T = 2$$

so there are $2 \dim G$ perturbative classical transverse modes and no hard mass in that quadratic expansion. It does not produce a gauge-invariant asymptotic gluon. In the confining QCD phase, and on a gauge-invariant Yang–Mills branch with a positive physical gap, the colored free pole need not occur in the physical spectrum.

- iii. **Pure Einstein branch.** Suppose, in addition to the derived classical Einstein equation, that the field-content branch is the two-derivative Einstein–Hilbert action with $G > 0$ about a Minkowski vacuum with $\Lambda = 0$, without higher-curvature kinetic terms, bimetric mixing, or extra scalar/vector fields. After de Donder gauge and residual-gauge reduction, the two transverse-traceless components obey

$$S_{\text{TT}}^{(2)} = \frac{1}{64\pi G} \int [(\partial_t h_{ij}^{\text{TT}})^2 - c_\star^2 (\nabla h_{ij}^{\text{TT}})^2] d^4x,$$

$$H_{\text{TT}}^{(2)} = \frac{1}{64\pi G} \int [(\partial_t h_{ij}^{\text{TT}})^2 + c_\star^2 (\nabla h_{ij}^{\text{TT}})^2] d^3x, \quad D_{ij,kl}^{\text{TT}}(k) \propto \frac{i\Pi_{ij,kl}^{\text{TT}}}{k^2 + i0}, \quad \text{rank } \Pi^{\text{TT}} = 2.$$

Thus the pure Einstein linearization has two classical massless spin-two wave modes on this background. On a curved Einstein background the relevant statement is instead about the Lichnerowicz operator and its boundary conditions; a Poincaré particle pole is not automatic. The present OPH derivation does not construct the metric-field quantization, physical graviton Hilbert space, or positive-residue two-point pole required by (Q1)–(Q3).

Proof. For (i), expand the Maxwell action to second order. Gauss' law removes the longitudinal canonical pair, leaving two transverse polarizations with the displayed positive Hamiltonian and dispersion $\omega^2 = c_\star^2 |\mathbf{k}|^2$. Inverting the gauge-fixed Hessian gives the displayed covariant Green function. For (ii), write $F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + O(A^2)$; the quadratic Hessian is one Maxwell Hessian per Lie-algebra generator, while the nonabelian interactions begin at cubic order. For (iii), the second variation of the Einstein–Hilbert action is the Fierz–Pauli kinetic form. De Donder gauge and its residual gauge freedom reduce it to the transverse-traceless action shown, with the $+$ and $\times$ polarizations. These Hessian calculations establish (C1)–(C4). None constructs the quantum data in (Q1)–(Q3), so no unconditional particle conclusion follows. $\square$

Corollary 6.18a (conditional particle interpretation). If a branch in Theorem 6.18 is also supplied with its quantum-particle receipt, its positive-residue $\delta(\mu^2)$ term defines a massless particle pole with the displayed physical polarizations. Without that receipt, this paper claims only the corresponding classical or perturbative carrier mode. In particular, the Tannaka group alone does not imply a Maxwell kinetic term or deconfined phase, and the classical Einstein equation alone does not imply a graviton state.

Hard-term versus physical-spectrum boundary. The vanishing quadratic parameters $\mu_{Q,\text{quad}}^2$, $\mu_{Y\text{M},\text{quad}}^2$, and $\mu_{\text{EH,TT}}^2$ in the displayed actions are action-level statements, not universal exact particle masses. Gauge or diffeomorphism redundancy can coexist with Higgs/Stueckelberg completions, plasma or medium self-energies, confinement, higher-derivative poles, bimetric sectors, or additional massive fields. Excluding those possibilities requires the declared field-content and phase hypotheses rather than the symmetry label alone.

4.7.12 Quotient-protected charge quantization

Theorem 6.19 (Charge quantization and no fractional color singlets). If the global gauge group is

$$G_{\text{phys}} = \frac{\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)}{Z_6},$$

as derived in Proposition 6.6, then

All color-singlet states have integer electric charge.

Equivalently: no stable isolated particles with charges like $\pm 1/3$ can exist as color singlets.

Proof. The Z_6 quotient identifies the center elements $(e^{\hat{2}\pi i/3}, -1, e^{\hat{i}\pi/3}) \in \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ with the identity. For a color-singlet state ($\tau = 0$), the $\text{SU}(3)$ factor acts trivially. The resulting identification requires the $\text{SU}(2) \times \text{U}(1)$ quantum numbers to satisfy

$$(-1)^{2j} \cdot e^{i\pi n/3} = 1,$$

where j is the $\text{SU}(2)$ spin and $n = 6Y$ is the integer hypercharge label. This gives $n \equiv -6j \pmod{6}$, i.e., $n \equiv 0 \pmod{6}$ for integer j and $n \equiv 3 \pmod{6}$ for half-integer j . Equivalently: Y is integer when j is integer, and Y is half-integer when j is half-integer.

After electroweak breaking, $Q = T_3 + Y$. For integer j , $T_3 \in \mathbb{Z}$ and $Y \in \mathbb{Z}$, so $Q \in \mathbb{Z}$. For half-integer j , $T_3 \in \mathbb{Z} + 1/2$ and $Y \in \mathbb{Z} + 1/2$, so $Q = (\text{half-integer}) + (\text{half-integer}) \in \mathbb{Z}$. In both cases, $Q \in \mathbb{Z}$. QED.

Experimental bound: No fractionally charged color-singlet particles have been observed. Three independent high-precision bounds confirm this:

1. **Neutrality of matter** (PDG 2024): The proton-electron charge sum satisfies

$$|q_p + q_e|/e < 1 \times 10^{-21},$$

confirming charge quantization to 21 decimal places.

2. **Fractional charge searches in bulk matter:** Silicone oil drop experiments limit fractionally charged particle abundance to

$$(\text{fractionally charged particles})/\text{nucleon} \lesssim 10^{-22}.$$

3. **Collider searches** (CMS, PRL 134, 2025): Exclusions for stable particles with $q \in [e/3, 0.9e]$ up to masses ~ 640 GeV (95% CL).

Existing searches are fully consistent with these structural consequences.

These are exact symmetry- or quotient-protected outputs: the empirical content is that violations are excluded up to available experimental sensitivity.

4.7.13 Coupling extraction from edge-sector probabilities

The edge-center completion (Theorem 2.3) yields sector probabilities p_α on collar boundaries. These probabilities encode the renormalized gauge coupling through a heat-kernel/Laplacian weighting law.

Abelian case (Z_n). For a Z_n gauge theory, the edge sectors are labeled by charge $q \in \{0, 1, \dots, n-1\}$. The correct "Casimir" eigenvalue is the Laplacian eigenvalue of the boundary random walk:

$$\lambda_q = 4 \sin^2 \left(\frac{\pi q}{n} \right).$$

Note: only in the limit $n \rightarrow \infty$ and $q \ll n$ does $\lambda_q \approx (2\pi q/n)^2 \propto q^2$. For finite n , the exact form is essential.

The sector probabilities follow a heat-kernel law:

$$p_q \propto e^{-t(\mu)\lambda_q},$$

where $t(\mu)$ is the "modular time" parameter encoding the scale. The extraction formula is:

$$t(\mu) = -\frac{\log(p_q/p_0)}{\lambda_q}, \quad g_{\text{ent}}^2(\mu) = \frac{t(\mu)}{2\pi}.$$

Consistency requires that t extracted from different charges q agrees; this has been verified numerically (see Section 6.14).

Electric-center measurement. The edge sectors are measured using the *electric-center* prescription. For a region A and boundary vertex $v \in \partial A$, define the restricted star operator:

$$Q_v^{(A)} = \prod_{\ell \in \text{star}(v) \cap A} X_\ell^{\pm 1},$$

where X_ℓ is the shift operator on link ℓ . The sector projectors are:

$$P_{v,q} = \frac{1}{n} \sum_{m=0}^{n-1} \omega^{-mq} \left(Q_v^{(A)} \right)^m, \quad \omega = e^{2\pi i/n},$$

and the probabilities are $p_{\{v,q\}} = \langle P_{\{v,q\}} \rangle$. This electric-center operator, built from X's rather than Z's, correctly captures the boundary gauge charge/flux that labels entanglement edge sectors.

Non-abelian generalization. For $SU(N)$ gauge theories, the edge sectors are labeled by irreducible representations with probabilities:

$$p_j \propto d_j e^{-t(\mu)C_2(j)},$$

where d_j is the dimension and $C_2(j)$ the quadratic Casimir. Extraction:

$$t(\mu) = -\frac{\log(p_j/p_0)}{C_2(j)}, \quad g_{\text{ent}}^2(\mu) = \frac{t(\mu)}{2\pi}.$$

Theoretical derivation. The fixed-cutoff same-overlap edge law is carried by the regulated microphysics package once the declared overlap-sector projectors, reversible thermal branch, and quasi-local local-Gibbs generator of Theorem 2.6 are in hand. On this synthesis surface, the point is to summarize that fixed-cutoff source package and to state the refinement/Peter–Weyl lift boundary explicitly rather than to relocate the leaf-level source surface.

Theorem 6.20 (Fixed-cutoff edge-sector law and refinement lift boundary). Under the OPH axioms, the fixed-cutoff overlap-gauge realization of Sections 2.3 and 3.2, and the quasi-local local-Gibbs form supplied by Theorem 2.6, the regulated same-overlap edge-sector probability distribution satisfies:

$$p_R = \frac{d_R e^{-t\lambda_R}}{\sum_{R'} d_{R'} e^{-t\lambda_{R'}}}$$

where λ_R is the Laplacian eigenvalue on the R-isotypic component and t is determined by the collar Gibbs parameter.

Proof.

Step 1 (Edge Hilbert space). From the fixed-cutoff overlap-gauge realization, the edge degrees of freedom at a boundary circle $\Sigma = \partial C$ live in a Hilbert space transforming under the gauge group G . Microscopically this is a finite-dimensional quantum-link edge space on the declared overlap interface; the effective representation-theoretic description used only for the refinement lift is modeled by $L^2(G)$. By the Peter–Weyl theorem [45]:

$$L^2(G) \cong \bigoplus_R V_R \otimes V_R^*$$

where V_R is the carrier space of irrep R .

Step 2 (Gauge invariance). The Gauss law constrains physical states. For an entanglement cut at Σ , the physical edge Hilbert space decomposes as $\mathcal{H}_{\{\text{edge}\}}^{\{\text{phys}\}} = \bigoplus_R W_R$ where W_R contains states with flux in representation R .

Step 3 (Natural Hamiltonian). From the local-Gibbs form, the MaxEnt generator restricted to edge modes takes the form $H_{\{\text{edge}\}} = \sum_R h_R P_R$ where P_R is the projector onto the R-sector. The key claim is that $h_R = \lambda_R$.

Justification: For a compact simple factor, the group Laplacian $\Delta_G = -\sum_a (T^a)^2$ is the unique bi-invariant second-order differential operator up to scale. For a product group $G = \prod_i G_i$, the most general bi-invariant second-order operator is a positive linear combination $\sum_i c_i \Delta_{G_i}$, with one coefficient per factor. Any other gauge-invariant local choice would require higher derivatives, violating locality. For finite groups, the Cayley graph Laplacian plays the same role: $\lambda_R = |S| - (1/d_R) \sum_{s \in S} \chi_R(s)$.

Step 4 (MaxEnt selection). MaxEnt selects the Gibbs state:

$$\rho_{\text{edge}} = \frac{1}{Z} e^{-tH_{\text{edge}}} = \frac{1}{Z} \sum_R e^{-t\lambda_R} P_R.$$

Step 5 (Sector probabilities). The probability of sector R is $p_R = \text{Tr}(\rho_{\text{edge}} P_R)$. The effective dimension for entanglement is d_R (not d_R^2) because we trace over one side of the cut. This gives:

$$p_R = \frac{d_R e^{-t\lambda_R}}{Z}.$$

QED.

Why the entropy rank is d_R (instead of d_R^2). The full edge space has dimension d_R^2 in sector R (from $V_R \otimes V_R^*$). Entanglement entropy, however, measures correlations *across* the cut. After tracing over one side, the reduced density matrix has effective rank d_R . Mathematically: in the Markov normal form, the edge factor on one side contributes $\log d_R$ to the entropy.

Scope. The fixed-cutoff same-overlap derivation is the theorem-bearing part of the edge-law package. The quasi-local local-Gibbs generator used here is derived from Theorem 2.6: if MaxEnt constraints are expectations of finitely many quasi-local operators, the entropy maximizer is automatically a Gibbs state with a quasi-local generator. The additional representation-theoretic step summarized here is the compact-group / Peter-Weyl refinement lift, together with its factorwise Laplacian identification. The fixed-cutoff same-overlap Casimir law is supplied on the microphysics surface; model-family uniqueness and large-edge continuations have their own claim boundaries.

Normalization anchor: 2D Yang-Mills. The parameter t can be exactly matched to a conventional coupling in 2D Yang-Mills, where the physical Hamiltonian is literally the group Laplacian:

$$H = \frac{g^2}{2} \Delta_G, \quad \Delta_G \chi_R = -C_2(R) \chi_R \quad \Rightarrow \quad E_R = \frac{g^2}{2} C_2(R).$$

Euclidean evolution for "time" A (the area of a cylinder in 2D YM) gives $\text{weight}(R) \propto \exp(-A E_R) = \exp(-g^2 A C_2(R)/2)$. Comparing with the heat-kernel expansion $K_t(U) = \sum_R d_R \chi_R(U) e^{-t C_2(R)}$ yields the exact identification:

$$t_{\text{phys}} = \frac{g^2 A}{2} \quad (\text{in 2D YM, no ambiguity}).$$

This shows that the Laplacian + MaxEnt $\rightarrow$ heat-kernel structure is exact in a solvable two-dimensional Yang-Mills case. The coefficient in front of C_2 is fixed. This normalization anchor is a two-dimensional edge-theory check, separate from the compact paper's conditional support-visible compact-gauge repair-gap theorem. In any regime where the edge theory reduces to an effective 2D YM with known "Euclidean thickness" A_{eff} :

$$g^2(\mu) = \frac{2}{A_{\text{eff}}(\mu)} \cdot \frac{\Delta_R(\mu)}{C_2(R)},$$

and the RHS must be R -independent. This R -independence is an internal precision consistency test; the formula itself is the normalization map that connects t to the conventional gauge coupling.

4.7.14 Numerical validation of the heat-kernel law

The heat-kernel/Laplacian weighting of edge sectors has been validated in explicit 2D Z_n gauge models on closed geometries.

Model. A 2×2 periodic lattice gauge theory (8 links) with Z_n link Hilbert spaces and Hamiltonian:

$$H = -K \sum_p \text{Re}(B_p) - h \sum_\ell \text{Re}(X_\ell) - \Gamma \sum_v \text{Re}(A_v),$$

where X_ℓ is the Z_n shift on link ℓ , B_p is the oriented plaquette operator (product of Z 's around plaquette p), and A_v is the oriented star/Gauss operator (outgoing X , incoming $X^\dagger$). With $K = 1$ and $\Gamma = 5$, the ground state satisfies $\langle A_v \rangle = 1$ at all vertices to numerical precision.

Region and edge operator. Region A consists of links whose tail has $x = 0$ ("half-lattice" cut). At each boundary vertex v , the electric-center edge charge is the restricted star $Q_v^{\wedge\{A\}} = \prod_{\ell \in \text{star}(v) \cap A} X_\ell^{\wedge\{\pm 1\}}$.

Results for Z_2 . With $\lambda_1 = 4\sin^2(\pi/2) = 4$:

h	p ₀	p ₁	t	g _{ent}
0.5	0.8266	0.1734	0.391	0.249
1.0	0.9612	0.0388	0.803	0.357
2.0	0.9917	0.0083	1.194	0.436

Z_2 has a single nontrivial Laplacian eigenvalue, so a one-parameter fit of t can always be made exactly for any admissible nontrivial weight. The Z_2 table is therefore an implementation and convention check, not a test of the heat-kernel form.

Results for Z_3 (symmetry-forced consistency check). With $\lambda_1 = \lambda_2 = 4\sin^2(\pi/3) = 3$:

h	p ₀	p ₁	p ₂	t(q=1)	t(q=2)	g _{ent}	m _{plaq}
0.2	0.4395	0.2803	0.2803	0.1500	0.1500	0.154	2.22
0.5	0.7509	0.1245	0.1245	0.5989	0.5989	0.309	1.75
1.0	0.9606	0.0197	0.0197	1.2956	1.2956	0.454	4.07
1.5	0.9851	0.0074	0.0074	1.6288	1.6288	0.509	7.06
2.0	0.9921	0.0039	0.0039	1.8440	1.8440	0.542	10.10

Both equalities in this table are forced by symmetry before any heat-kernel ansatz is imposed. The equality $p_1 = p_2$ is exact charge-conjugation symmetry in Z_3 , and the two nontrivial Fourier modes share the degenerate eigenvalue $\lambda_1 = \lambda_2 = 3$, so extracting t from $q = 1$ and $q = 2$ must give the same value for *any* conjugation-invariant distribution. The observed agreement ($t_{\{q=1\}} \approx 1.2956389318579$ versus $t_{\{q=2\}} \approx 1.2956389318521$ at $h = 1.0$, i.e. $\sim 10^{-14}$) therefore validates the implementation and the symmetry conventions only; it cannot distinguish the heat-kernel form from an arbitrary conjugation-symmetric distribution on Z_3 , and it is not counted as an independent confirmation of the heat-kernel law. The substantive overconstrained tests are the Z_5 and S_3 calculations below, which involve two distinct nonzero eigenvalues.

Region-choice robustness. At $h = 1$, the extracted g_{ent} is nearly independent of region size:

- 2 links (one vertex's outgoing links): $g_{\text{ent}} \approx 0.453$

- 4 links (half-lattice): $g_{\text{ent}} \approx 0.454$
- 6 links (three vertices): $g_{\text{ent}} \approx 0.453$

This locality confirms that the coupling is dominated by physics near the cut, not global book-keeping, exactly what is expected if this behaves like a local QFT observable.

Results for Z_5 (golden ratio test). The Z_5 case provides a stringent test because the Laplacian eigenvalues have a distinctive ratio involving the golden ratio $\phi = (1+\sqrt{5})/2$:

$$\lambda_q = 4 \sin^2 \left(\frac{\pi q}{5} \right), \quad \frac{\lambda_2}{\lambda_1} = \frac{\sin^2(72^\circ)}{\sin^2(36^\circ)} = \phi^2 \approx 2.618.$$

This ratio distinguishes the Laplacian law from naive alternatives: a linear model ($\lambda_q \propto q$) would predict ratio 2, while a quadratic model ($\lambda_q \propto q^2$) would predict ratio 4.

Direct diagonalization of the displayed Hamiltonian on the full 2×2 -torus link basis (5^8 states, $K = 1$, $\Gamma = 5$), measuring the restricted-star edge charge at a boundary vertex of the half-lattice region, gives:

h	Measured ratio $\ln(p_2/p_0)/\ln(p_1/p_0)$	Deviation from ϕ^2
0.50	2.876	9.9%
0.20	2.750	5.1%
0.10	2.671	2.0%
0.05	2.643	1.0%

In the weak-field limit $h \rightarrow 0$ the measured ratio converges to the golden ratio squared, and the stated limiting direction agrees with the tabulated sequence: the deviation decreases monotonically as h decreases. (An earlier version of this table came from a dual/flux-basis run whose tabulated parameter ran opposite to the h of the displayed Hamiltonian, so prose and table disagreed about the limit; the direct link-basis rerun above, reproducible with `code/edge_sectors/heat_kernel_holdout_validation.py`, removes that inconsistency. At large h the ratio instead drifts toward the perturbative order-counting value 2, which is why the weak-field direction is the declared convergence regime for this model.) The test is genuinely overconstrained: Z_5 has two distinct nonzero eigenvalues, t is fitted on the $q = 1$ sector alone, the $q = 2$ weight is a held-out prediction, and the deviation column reports the held-out residual. Convergence of that residual confirms that the vacuum entanglement spectrum encodes the precise geometric structure of the gauge group Laplacian.

Significance. This validates the mathematical law (sector probabilities weighted by Laplacian eigenvalues) in explicit 2D gauge-invariant models with non-flat sector distributions. Because the nontrivial Z_3 eigenvalues are degenerate, it is the Z_5 test (and the S_3 test below) that is structurally identical to $SU(2)/SU(3)$: two or more distinct nonzero eigenvalues overconstrain the slope, and held-out agreement confirms the mechanism works before jumping to nonabelian groups.

Results for S_3 (first nonabelian test). The abelian tests above use charge-sector projectors that reduce to Fourier modes. For nonabelian groups, the edge-sector projector must be generalized to character projectors:

$$P_{v,R} = \frac{d_R}{|G|} \sum_{h \in G} \chi_R(h^{-1}) Q_v^{(A)}(h),$$

where d_R is the dimension of irrep R , χ_R is its character, and $Q_v^{\wedge}\{(A)\}(h)$ is the restricted gauge action at boundary vertex v acting only on links in region A .

For S_3 (the smallest nonabelian group, order 6), there are three irreps: trivial ($d=1$), sign ($d=1$), and standard ($d=2$). The Cayley-graph Laplacian eigenvalues for the transposition generating set are:

$$\lambda_{\text{triv}} = 0, \quad \lambda_{\text{sign}} = 6, \quad \lambda_{\text{std}} = 3.$$

Exact reduction on one plaquette. For the single-plaquette model (4 links), imposing Gauss's law at all vertices means the physical wavefunction depends only on the plaquette holonomy's conjugacy class. Since S_3 has exactly 3 conjugacy classes, the gauge-invariant Hilbert space is 3-dimensional, spanned by the character states $\{|\chi_R\rangle\}$. In this basis, the edge-sector probabilities are exactly $p_R = |c_R|^2$ where $|\psi_0\rangle = \sum_R c_R |\chi_R\rangle$. This is an exact identity for the one-plaquette gauge-invariant sector.

The heat-kernel ansatz predicts $p_R \propto d_R \exp(-t \lambda_R)$. Extracting t independently from the sign and standard irreps provides an overconstrained test: the ratio $\lambda_{\text{sign}}/\lambda_{\text{std}} = 6/3 = 2$ is a parameter-free prediction. Results from a single-plaquette S_3 lattice gauge model ($K=1$, $\Gamma=5$):

h	p_{triv}	p_{sign}	p_{std}	t (sign)	t (std)	$\Delta t/t$	log-ratio
0.5	0.909	0.0013	0.089	1.09	1.01	8.4%	2.17
1.0	0.980	7.5×10^{-5}	0.020	1.58	1.54	2.8%	2.06
2.0	0.996	4.3×10^{-6}	0.004	2.06	2.04	1.0%	2.02
5.0	0.9993	1.0×10^{-7}	0.00066	2.68	2.67	0.3%	2.006
12	0.9999	3.0×10^{-9}	0.00011	3.27	3.27	0.1%	2.002
100	1.0000	6.1×10^{-13}	2.0×10^{-6}	4.69	4.69	0.009%	2.0002

The " $\Delta t/t$ " column shows the fractional difference $(t_{\text{sign}} - t_{\text{std}}) / \bar{t}$. The "log-ratio" column shows $\log(p_{\text{sign}}/p_0) / \log(p_{\text{std}}/(2 p_0))$, which should equal $\lambda_{\text{sign}}/\lambda_{\text{std}} = 2$ if the heat-kernel form holds exactly. Equivalently, t is fitted from the standard sector alone and the sign-sector weight is a held-out prediction; log-ratio minus 2 and $\Delta t/t$ are the held-out residuals. Unlike Z_3 , the two nonzero eigenvalues are distinct (6 versus 3), so no symmetry forces this agreement.

As h increases, both diagnostics converge: $\Delta t/t$ drops below 10^{-4} and the log-ratio approaches 2.000. This is exactly the expected behavior: finite-size corrections are largest at strong coupling; the heat-kernel form becomes exact as the perturbative regime is approached.

This gives a nonabelian validation of the edge-sector extraction mechanism. The structure (character projectors, Laplacian eigenvalues from the group's Cayley graph, overconstrained t extraction) is identical to the $SU(2)$ and $SU(3)$ continuation setup.

Parameter-free predictions for $SU(2)$ and $SU(3)$. The heat-kernel law yields exact, parameter-free ratio predictions that require no scheme matching. Define the "Casimir log-gap":

$$\Delta_R \equiv \ln\left(\frac{p_0}{d_0}\right) - \ln\left(\frac{p_R}{d_R}\right) = t C_2(R).$$

Ratios of Δ_R cancel all unknowns (t , partition function):

$$\frac{\Delta_{R_1}}{\Delta_{R_2}} = \frac{C_2(R_1)}{C_2(R_2)} \quad (\text{exact, parameter-free}).$$

$SU(2)$ predictions. Irreps labeled by spin j have $d_j = 2j+1$ and $C_2(j) = j(j+1)$. The framework predicts:

- $\Delta_1/\Delta_{1/2} = 2/(3/4) = 8/3 \approx 2.667$
- $\Delta_{3/2}/\Delta_{1/2} = (15/4)/(3/4) = 5$
- $\Delta_{3/2}/\Delta_1 = (15/4)/2 = 15/8 = 1.875$

SU(3) predictions. Irreps labeled by Dynkin indices (p,q) have $C_2(p,q) = (p^2 + q^2 + pq + 3p + 3q)/3$. Using the fundamental $\mathbf{3} = (1,0)$ with $C_2 = 4/3$ as the reference:

- $\Delta_8/\Delta_3 = 3/(4/3) = 9/4 = 2.25$
- $\Delta_6/\Delta_3 = (10/3)/(4/3) = 5/2 = 2.5$
- $\Delta_{10}/\Delta_3 = 6/(4/3) = 9/2 = 4.5$
- $\Delta_{15}/\Delta_3 = (16/3)/(4/3) = 4$
- $\Delta_{27}/\Delta_3 = 8/(4/3) = 6$

These are the SU(2)/SU(3) analogs of the Z_5 golden-ratio test: exact rational numbers fixed entirely by group theory, with no adjustable parameters.

Preliminary SU(3) results. A one-plaquette SU(3) "quantum link" model (finite truncated irrep basis, $n_max = 12$, $\kappa = 2$) has been used to extract t from 14 different irreps simultaneously. The results show internal consistency at the 1-3% level:

bare g^2	extracted t (mean $\pm$ std)	g_ent	gap
0.3	0.314 $\pm$ 0.0005	0.224	1.92
0.5	0.539 $\pm$ 0.0025	0.293	1.83
0.8	0.896 $\pm$ 0.012	0.378	1.72
1.0	1.144 $\pm$ 0.025	0.427	1.64

The standard deviation across irreps provides a built-in error estimate. This is a QCD proton-physics surrogate rather than a full proton calculation; it lacks dynamical quarks and operates on a single plaquette. It nevertheless demonstrates that the nonabelian extraction machinery produces self-consistent outputs without tuning.

Extracting the normalization factor A_eff . The 2D YM anchor (Section 6.13) gives $t = g^2 A / 2$, so the "effective Euclidean thickness" is

$$A_{eff} = \frac{2t}{g^2}.$$

Computing this from the SU(3) table:

bare g^2	extracted t	A_eff
0.3	0.314	2.093
0.5	0.539	2.156
0.8	0.896	2.240
1.0	1.144	2.288

Mean: $A_eff \approx 2.19$ with point-to-point scatter $\sim 4\%$.

Extrapolation to weak coupling. The systematic drift in A_eff shows fitting $A_eff(g^2) = A_0 + a \cdot g^2$. A weighted linear fit gives:

$$A_0 = 2.004 \pm 0.012$$

with $\chi^2/\text{dof} \approx 0.09$, indicating excellent consistency. This strongly shows that, in this toy UV completion, the "missing normalization" converges to $A_{\text{eff}} \rightarrow 2$ as $g^2 \rightarrow 0$.

The normalization factor behaves like a quasi-constant and extrapolates to a simple value (≈ 2) in the weak-coupling limit. This provides a concrete path to absolute coupling predictions: once A_{eff} is determined from microphysics, the conversion $g^2 = 2t/A_{\text{eff}}$ fixes the gauge coupling without additional free parameters.

Internal validation summary. The finite-group calculations separate into symmetry-forced implementation checks and genuinely overconstrained tests:

- **Z₂, Z₃:** implementation and convention checks. Z₂ has one nontrivial eigenvalue (one parameter, one datum); in Z₃ the equalities $p_1 = p_2$ and $t_{\{q=1\}} = t_{\{q=2\}}$ are forced by charge conjugation plus the degenerate eigenvalue $\lambda_1 = \lambda_2$, so the $\sim 10^{-14}$ agreement is expected from symmetry alone
- **Z₅:** overconstrained golden-ratio test ($\lambda_2/\lambda_1 = \phi^2$, two distinct nonzero eigenvalues), held-out residual 1.0% at $h = 0.05$, decreasing monotonically as $h \rightarrow 0$
- **S₃:** overconstrained nonabelian Casimir log-ratio test ($\lambda_{\text{sign}}/\lambda_{\text{std}} = 2$), held-out residual 0.01%
- **SU(3):** 14-irrep simultaneous extraction, internal scatter 1-3%

The Z₃ equality is forced by symmetry and is retained only as a regression check of the implementation; it is not an independent confirmation of the heat-kernel law. The substantive validation rests on the Z₅, S₃, and SU(3) tests, each of which fits t using fewer spectral sectors than it predicts, involves two or more distinct nonzero eigenvalues, and reports held-out residuals that converge in the declared regime of the corresponding model ($h \rightarrow 0$ for the Z₅ torus model; h large for the single-plaquette S₃ model). This provides internal validation of the mechanism "MaxEnt + Laplacian $\Rightarrow$ heat-kernel sector weights" before applying it to physical gauge groups. A self-contained held-out refit script for the finite-group models is provided at [code/edge_sectors/heat_kernel_holdout_validation.py](#); it fits t on a declared subset of sectors and prints held-out residuals for the remaining sectors.

4.7.15 Frozen IBM Quantum Cloud engineering archive

The former IBM Quantum Cloud program is frozen as an engineering and reproducibility archive. It checked whether OPH-motivated reduced-sector structures survived preparation and readout on real superconducting-qubit devices, and whether a programmed record-gated controller beat restricted controller nulls. Standard quantum mechanics predicts every prepared state and dynamic circuit used in that program. The runs therefore have no power to distinguish OPH from quantum mechanics, regardless of shot count or blinding quality. No new quantum-cloud campaign is promoted as OPH evidence until a source-closed observable with a different numerical QM prediction and an identifiable effect size is derived.

Detailed circuits, representative legacy derived summaries, the complete blinded-run receipts, and archived rerun instructions are public in the frozen project write-up [extra/IBM_QUANTUM_CLOUD.md](#) and code/data bundle [code/ibm_quantum_cloud/](#).

- **Stage 1 (Markov/recoverability benchmark).** On `ibm_marrakesh` and `ibm_fez`, the structured state reconstructs below both controls in conditional mutual information and above

both controls in Petz fidelity. On `ibm_marrakesh`, the observed ordering is $0.2309 < 0.3890 < 0.9474$ for CMI and $0.9297 > 0.8649 > 0.6066$ for Petz fidelity; on `ibm_fez`, it is $0.1498 < 0.4992 < 0.9166$ and $0.9479 > 0.8117 > 0.6449$. This reproduces the programmed ordering on two real backends; the Petz calculation is offline and the result is not theory-discriminating.

- **Z₃ hardware sanity check.** The two-sector Z_3 extraction (whose two-sector agreement is expected a priori from the degenerate eigenvalues and charge-conjugation symmetry, so this is an implementation check, not an overconstrained test) passes cleanly on hardware: across prepared $t = 0.30, 0.60, 0.90$, the mean extracted t is within about 0.02 of the target, the two independent extractions agree to within 0.0142, 0.0034, and 0.0043, and leakage stays below 0.1%. This shows that the reduced-sector preparation and readout path is internally coherent on-device before sharper ratio claims are interpreted.
- **Z₅ exact-ratio preparation.** The programmed target is $\Delta_2/\Delta_1 = \varphi^2 \approx 2.618$. The measured values span approximately 2.407 to 2.878, and the focused high-shot $t = 0.90$ interval lies below the exact target. Because the target amplitudes are directly prepared, QM predicts the same ideal ratio.
- **S₃ nonabelian layout diagnostic.** The first hardware run revealed a real layout-dependent bias. Reversing the qubit layout moved the mitigated ratio from 1.8724 to 2.0299, and a repeat returned 2.0657. Because the better mapping was selected after observing the bias, this is a post-hoc device diagnostic rather than a blinded confirmation.

These frozen runs document preparation, readout, layout, and feedback engineering on real devices. They add no evidence for OPH over standard quantum mechanics because the two descriptions make no different prediction for the implemented interventions. Their proper role is reproducibility and controller validation, not theory confirmation.

4.8 Claim Boundaries, Tests, and Scope

4.8.1 Classification of results and dependencies

The documentary node labels D1–D12 are used only to keep the dependency boundary compact. The phase split is:

$$\text{Phase I} = (D1\text{--}D5) \cup (D7\text{--}D9), \quad \text{Phase II} = D6 \cup D10, \quad \text{Phase III} = D12.$$

Phase	Nodes	Meaning
I	D1–D5, D7–D9	Recovered core: fixed-cutoff overlap repair and collar structure; the Lorentz/three-dimensional spatial-chart/null-modular/Einstein branch; receipt-conditional bosonic compact gauge reconstruction; the support-visible four-dimensional Euclidean Yang–Mills form and conditional repair-gap theorem on the compact-gauge continuum/transfer branch; realized-branch Standard Model quotient, exact hypercharge, structural electroweak force content, $N_c = 3$, and $N_g = 3$ under the explicit matter-package and admissibility inputs.
II	D6, D10	Quantitative closures: screen-capacity closure of the same Einstein branch and the P -driven electroweak/gauge-coupling branch. These depend on declared external or branch-specific inputs.
III	D12	Continuations: flavor details beyond the stated theorem surfaces, H^3 record-worldline stitch certificates and the event-manifold receipts E1–E4 of Theorem 4.3e, dark-sector proposals, conditional screen-spectrum and CMB/inflation-replacement kernels, H_0/S_8 and growth branches, heuristic baryogenesis branches, spectroscopy, hadrons, and string/worldsheet effective-description branches.

Phase I is the theorem-bearing core using the stated scaling, transport, and gauge theorem stack. Phase II supplies quantitative closures on top of that core. Phase III contains program branches whose extra ansätze are not part of the recovered-core theorem package. In particular, the conditional screen-spectrum branch reaches a MaxEnt Gaussian screen covariance only after the quotient-ensemble selector, quotient-normal-form scalar field, positive repair operator, operator-tilt, and fixed expected quadratic scalar-release-energy receipts pass. A primordial curvature spectrum also needs a screen-to-radial lift receipt, and physical TT/TE/EE spectra require finite covariant source, source-provenance, pooled-reducer, Boltzmann transfer, and frozen-likelihood gates. Low- ℓ CMB kernels, parity envelopes, inflation-replacement claims, and dark/anomaly H_0/S_8 growth modifications are continuation gates unless their finite-collar source functions, recipient-stress closures, refinement ladders, and immutable likelihood contracts are supplied.

No semantic promotion by relabeling. A finite OPH diagnostic does not become a physical observable merely because it is renamed. Capacity bookkeeping is not mass without an independent energy readout; a record archive is not radiation without a source, propagation, and detector channel; finite repair eigenvalues are not continuum normal modes without an operator and readout bridge; exact finite recovery is not a Page curve without physical radiation entropy and exterior time calibration; and a finite conditional-mutual-information score is not a geometric island, entanglement wedge, quantum extremal surface, or replica saddle. These are not rhetorical distinctions: deterministic relabeling preserves the same information ancestry, so physical promotion requires source-separated bridge objects, residual ledgers, negative controls, and frozen validation targets.

The effective low-energy summary is

$$\mathcal{L}_{\text{eff}}^{\text{OPH}} \approx \sqrt{-g} \left[\frac{1}{16\pi G} (R - 2\Lambda) + \mathcal{L}_{\text{SM}}^{\text{realized branch}} \right] + \sum_i \frac{c_i}{M_*^{\Delta_i-4}} \mathcal{O}_i.$$

Here $\mathcal{L}_{\text{SM}}^{\text{realized branch}}$ denotes the realized $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)/\mathbb{Z}_6$ sector. The higher-dimension terms absorb UV-sensitive and continuation-level corrections outside the recovered core.

4.8.2 Structural assessment

- **Dynamics:** the GR chain runs through geometry readout, geometric modular covariance, the null bridge, bounded-interval transport, fixed-cap generalized-entropy stationarity, fixed-

volume small-ball area variation, carried-remainder control, and the tensor upgrade. The scaling branch uses the support-visible BW scaling theorem on the prime geometric subnet.

- **Gauge structure:** compact gauge reconstruction uses the transportable bosonic sector package on a cofinal tail carrying the compact-gauge refinement receipt. MAR plus anomaly algebra and the explicit CKM/weak-sector clauses on the same one-Higgs branch select the realized Standard Model quotient, $N_c = 3$, $N_g = 3$, and the structural electroweak force content $SU(2)_L \times U(1)_Y \rightarrow U(1)_Q$.
- **Microscopic theory:** finite quantum-link style presentations give concrete fixed-cutoff examples. They do not select a unique microscopic representative.
- **Noncentral gluing:** the crossed-module higher-gauge package closes the fixed-cutoff topological branch. The realized zero-obstruction bosonic compact-gauge branch is carried by a separate theorem stack on the ordinary or central branch.

4.8.3 Known-force and charge coverage

The recovered stack does not leave any known long-range or Standard Model gauge force unassigned:

- **Gravity:** D3–D6 recover Lorentz kinematics, the three-dimensional observer-facing spatial chart, the null-stress bridge, and the Jacobson-type Einstein branch, with T_{ab} as the stress-energy source. On the additional pure Einstein–Hilbert/Minkowski action branch, Theorem 6.18 gives two classical transverse-traceless modes; a graviton quantum requires the separate particle receipt.
- **Strong interaction:** D8–D9 recover the $SU(3)_c$ color factor, the color triplet $N_c = 3$, quark color triplet/antitriplet assignments, and the eight gluon generators $(8, 1, 0)$. Confinement and hadron spectra are separate infrared QCD questions, not missing gauge-charge assignments.
- **Weak interaction:** D8–D9 recover the $SU(2)_L$ weak doublet structure, the one-Higgs branch, and the charged weak carriers $W^\pm$ from the broken $SU(2)_L$ generators. D10 supplies the quantitative running/matching readout for v and the W/Z validation rows.
- **Hypercharge and electromagnetism:** The hypercharge theorem, the $\mathbb{Z}_6$ quotient, and Corollary 6.6c fix the $U(1)_Y$ lattice, the unbroken generator $Q = T_3 + Y$, integer charge for color singlets, and the electromagnetic connection label A_Q . Corollary 6.6d gives $F_Q = dA_Q$, $dF_Q = 0$, and $d * F_Q = g_Q^2 * J_Q$ only after the low-energy Maxwell action is supplied. Theorem 6.18 then gives two classical massless transverse modes; a photon particle additionally requires its quantum-pole receipt. The Thomson-limit fine-structure endpoint is a D10 quantitative closure.

4.8.4 Testable Items and Phenomenological Branches

The detailed prediction surfaces live in the companion papers. The synthesis-level list is:

- **Structural and action-level tests:** exact color and electroweak charge assignments, exact hypercharge quantization, $N_c = 3$, $N_g = 3$, and no gauge-mediated proton decay on the product-group branch; plus the classical Maxwell, perturbative pure-Yang–Mills, and pure-Einstein quadratic carrier-mode tests of Theorem 6.18. Quantum particle poles are separate receipt-gated claims.
- **Information-theoretic gravity bound:** modular-additivity defects give explicit upper bounds on GR deviations wherever the Markov/mixing hypotheses apply.
- **Edge-sector tests:** Casimir-ratio predictions, including $\Delta_8/\Delta_3 = 9/4$ for the $SU(3)$ edge-sector benchmark, test the heat-kernel mechanism.

- **Quantitative particle checks:** the P -driven electroweak, Higgs/top, quark, charged, and neutrino lanes are compared on their declared surfaces in Ref. [5], with W/Z compare-only, Higgs/top exact on the declared D10/D11 surface, quarks selected-class exact, charged absolute masses closed as a corpus-level no-go, and the target-informed weighted-cycle neutrino candidate rejected by the correlated NuFIT 6.1 profile and by its open basis contract.
- **H3 record-token continuation:** Ref. [5] also contains a conditional H^3 record-worldline stitch theorem. It is a certificate for localized observer-visible tokens crossing real chart or partition interfaces after sector/gauge transport, not a derivation of particle species, masses, gauge charges, or scattering amplitudes. Its Lorentzian lift is the conditional event-manifold packet of Theorem 4.3e, under receipts E1–E4.
- **Continuation templates:** black-hole combs, PBH burst templates, dark-sector response laws, heuristic baryogenesis continuations, and critical-string lifts are not recovered-core tests.

4.8.5 Scope Boundaries

The recovered core does not derive the following:

- charged-lepton absolute masses on the available corpus, full flavor-labeled neutrino closure, CKM/PMNS closure, global public quark-frame classification beyond the selected class, or general Yukawa hierarchy;
- hadron masses and resonances, which require nonperturbative production computation;
- dark-sector response laws, heuristic baryogenesis continuations, strong-CP proposals, proton-spin fractions, and late-stage spectroscopy templates;
- inflation-replacement, CMB low- ℓ /parity kernels, H_0/S_8 growth modifications, and cosmological dark/anomaly Boltzmann kernels beyond the contract stated in Section 4.6.16;
- a primordial curvature spectrum or physical TT/TE/EE spectra from the screen-spectrum continuation unless the operator-tilt, scalar-release-energy, radial-lift, finite covariant source, source-provenance, pooled-reducer, transfer, frozen-likelihood, and no-data-use receipts are present;
- Page-curve/island closure, physical black-hole evaporation claims, PBH/LIGO comb claims, QNM/ringdown predictions, or critical-superstring completion;
- a theorem selecting equal-sized primitive observer carriers, fixed primitive observer capacity, or an isomorphic local cell algebra from $P_\star$ and N_{CRC} alone;
- a unique microscopic representative or a full fermionic/super-Tannakian gauge reconstruction.

N_{CRC} is the global capacity fixed point for the cosmological branch. P is selected on the Phase-II outer/inner closure branch and is not a Phase-I axiom. A contradiction in a Phase-III continuation retracts that continuation. A contradiction in D10 challenges the quantitative-closure branch. A contradiction in Phase I challenges the recovered-core claim set. $P_\star$ and N_{CRC} fix the canonical geometric cell area and the total equal-area cell count $K_{\text{cell}} = 4N_{\text{CRC}}/P_\star$ on that chart. They do not select a primitive observer algebra, a fixed observer capacity, or a unique k -cell carrier block.

4.8.6 Comparison with other unification approaches

Unified models attempting to tie together QFT, gravity, and SM structure tend to encounter a repeatable set of conceptual difficulties. This subsection examines how the observer-patch holography framework addresses these common pitfalls.

1. Subsystem factorization in gauge theory and gravity.

In gauge theories and gravity, the Hilbert space does not cleanly split as "inside $\otimes$ outside" across a cut. This infects entanglement entropy definitions, area terms, edge modes, and observable identification. Many unification attempts handwave this or patch it with conventions.

Resolution: The framework builds from a net of von Neumann algebras on patches plus overlap consistency. It does not start from naïve tensor factorization. The gauge-as-gluing + regulator package yields edge-center completion: a canonical block decomposition on collars where the center captures superselection data at the cut, and the state becomes (exactly or approximately) Markov across the collar. The entropy split $S(\rho_C) = S_{\text{bulk}} + \langle L_C \rangle$ follows from having a center with sector labels. This replaces the ad hoc "add an area term" move.

2. Modular Hamiltonian nonlocality.

Many entanglement-based gravity derivations depend on modular Hamiltonians that look like local stress-tensor charges (true only in special states/regions). In generic QFT states, modular Hamiltonians are nonlocal, making "first law of entanglement $\Rightarrow$ Einstein equation" arguments fragile.

Resolution: The Markov collar condition does heavy lifting: approximate Markov implies approximate modular additivity, with the defect controlled by conditional mutual information. This makes "modular locality" a controlled approximation. It is not treated as an assumption. On the extracted prime geometric subnet of Theorem 4.2, the controlled tangent-half-space comparison then locks modular flow to geometric dilations with rigid 2π normalization.

3. Lorentz invariance as a derived output.

Discrete microscopic models generally break Lorentz symmetry, and many unified proposals simply postulate Lorentz invariance in the IR.

Resolution: Lorentz kinematics are tied to geometric modular flow on caps. On the extracted prime geometric subnet, the support-visible BW scaling theorem gives cap-pair extraction, support-readable modular covariance, cap-normal BW framing, and modular flow as conformal transformations on S^2 . The compact cap-normal theorem then identifies a sky direction with $q(\Omega) = (1, \Omega)$, represents $C(\mathbf{c}, \alpha)$ by $n_C = (\cot \alpha, \csc \alpha \mathbf{c})$, proves the signed incidence formula, and gives $n_{gC} = \Lambda_g n_C$. Hence $\text{Conf}^+(S^2) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1)$. The associated observer-frame chart is $H^3 \simeq \text{SO}^+(3, 1)/\text{SO}(3)$, so the observer-facing spatial dimension on that branch is exactly three. A cap gives an H^3 plane/half-space, not a preferred observer point or populated bulk. No external spacetime symmetry axiom is added. The step from this frame kinematics to an actual 3+1 event spacetime is the conditional event-manifold packet (Theorem 4.3e): an event base of signature $(-+++)$ over which H^3 is strictly the frame fiber, valid exactly on the population/separation/frame/cocycle receipt branch, with countermodels showing $\dim H^3 = 3$ alone promotes nothing.

4. Dynamics beyond kinematics.

Many approaches produce emergent geometry/kinematics but stall at dynamics: why Einstein's equations (with the right coefficient) rather than some other geometric PDE?

Resolution: The framework combines the MaxEnt-selected fixed-cap generalized-entropy stationarity theorem, the derived $K_C = 2\pi B_C$ structure on the extracted geometric cap branch, the internal null modular bridge identifying the half-line generator with the local null-stress charge, and the internal small-ball bridge from the geometric cap generator. The E0 dependency discharge

states the scaling-limit regularity, bounded-interval kernel, remainder-control, and tensor-upgrade conditions needed by the Einstein branch. It does not rely only on "assume a UV CFT."

5. Gauge symmetry origin and compactness.

Most unification stories pick a gauge group and work out consequences. Emergent-gauge approaches sometimes produce noncompact groups or uncontrolled redundancies.

Resolution: Gauge symmetry is recast as redundancy in overlap identifications (gauge-as-gluing). The visible fixed-cutoff carriers generate a rigid tensor subcategory with the ordinary forgetful fiber. When the explicit compact-gauge refinement receipt supplies coherent surjective group pullbacks, block-algebra embeddings, and tensor realizations, the refinement-limit category is constructed and Tannaka-Krein / Doplicher-Roberts reconstruction yields a compact group G on that certified bosonic branch. "Gauge symmetry" names the gluing redundancy at the conceptual level. "Compact group" is the mathematical form compatible with finite-dimensional objectwise fibers; the finite-state refinement maps alone do not produce it.

6. Classical carrier modes versus quantum particles.

Writing a symmetry group or a classical Einstein equation is not enough to produce a kinetic term, a physical phase, a quantization, or a particle pole. Conversely, a displayed free quadratic pole does not by itself establish an interacting asymptotic particle.

Resolution: Definition 6.18 separates a classical carrier-mode receipt from a quantum- particle receipt. The stated Maxwell, pure-Yang-Mills, and pure-Einstein actions yield transverse or TT classical modes and action-level zero hard-mass parameters on their declared backgrounds and phases. Higgs/Stueckelberg completions, media, confinement, higher-derivative terms, and extra fields remain possible outside those branches; a photon, gluon, or graviton particle is claimed only when the physical Hilbert-space, positive-residue pole, and asymptotic/phase gates pass.

7. Global consistency, anomalies, and loop patching.

Building physics from local patches hits loop/holonomy problems: consistent gluing on a tree but obstructions around loops. These obstructions are often anomalies or global topological constraints.

Resolution: This is elevated to a first-class organizing principle: gluing data on overlaps define cocycles; central defects define a Čech obstruction class $[z]$, while noncentral defects define a full 2-group/crossed-module orbit together with the separate question whether it admits a strict G -valued 1-cocycle representative. Vanishing of $[z]$, or noncentral associator strictifiability, removes the triangle or higher associator defect; global endpoint-only consistency additionally requires trivial represented holonomy for at least one allowed strict representative. Anomalies become one form of failure to glue rather than a mysterious quantum pathology, without conflating the associator and ordinary loop obstructions.

8. Charge quantization without a GUT.

Without embedding into a simple GUT group, explaining charge quantization (why all isolated color singlets are integer charged) is awkward. Standard lore requires grand unification or monopoles.

Resolution: The framework leans on global group structure (the Z_6 quotient) and derives congruence/selection rules for allowed representations/hypercharges. This gives a structural explanation for integer-charged color singlets without introducing the proton-decay channel of simple-GUT models.

9. Coupling unification usually forces proton decay.

Traditional simple-group unification introduces leptoquark gauge bosons (X, Y) mediating proton decay. Experiment keeps pushing limits up, pressuring minimal GUTs.

Resolution: The retained D10 discussion is geometric/entropic only at the calibration level: shared edge diffusion data, heat-kernel weights, printed running/matching/threshold/scheme conventions, and extra calibration assumptions can mimic unification-style running without embedding

into a simple Lie group. If the reconstructed gauge group genuinely factorizes as a product (sector factorization selector), there are no mixed generators playing the X/Y role. "Unification-like couplings" and "group unification" therefore come apart.

10. Cosmological constant locality.

The cosmological constant problem is a graveyard of unified theories: local QFT estimates are enormous, and tiny observed Λ seems to demand absurd fine tuning.

Resolution: From null modular data, T_{ab} is reconstructed only up to ϕg_{ab} . Local consistency conditions and null focusing are blind to vacuum-energy shifts, so the Einstein equation is fixed only up to Λg_{ab} . The dimensionless Λ -capacity relation is global, tied to the static-patch capacity $\log \dim H_{tot}$, while the SI curvature value also uses the selected scale certificate. This resolves the conceptual tension: local microphysics *cannot* fix Λ by structural information-theoretic reasons.

11. UV infinities and nonrenormalizability.

Unified programs struggle to give sharp, finite microscopic definitions. Formal continuum structures, infinite entropies, and regularization dependence abound.

Resolution: The regulator construction uses local patch algebras that are type-I and finite-dimensional, with a MaxEnt branch whose generator is quasi-local and obeys a Lieb-Robinson bound. So the fixed-cutoff UV branch is interacting in the ordinary finite-range sense, and the fundamental degrees of freedom are finite and live on the screen. The framework does *not* claim a unique microscopic UV completion: physical uniqueness is only modulo gauge or implementation hiding together with inert ancillary stabilization. The genuinely noncentral topological branch is also closed at fixed cutoff by the higher-gauge crossed-module collar theorem, while the support-visible BW scaling branch is closed by theorem and the realized zero-obstruction bosonic compact-gauge branch is carried by its declared theorem stack.

12. Predictivity vs. parameter explosion.

Unified models often explode in parameters, sectors, or vacua, becoming hard to test directly because everything depends on choices.

Resolution: The framework compresses freedom into a "pixel area" (resolution) parameter and a total Hilbert space capacity (size) parameter, then derives structure from consistency (Lorentz form, the conditional Einstein branch, receipt-certified compact-group reconstruction, charge-quantization patterns, and action-level carrier modes only on explicitly selected phases). The explicit MAR selector then picks the SM factors and sector factorization on the admissible class discussed in Section 6.2. Quantum-particle poles remain a separate receipt rather than an exact-zero output of symmetry.

Structural pattern. The framework treats locality, Lorentz invariance, gauge symmetry, and gravity as consequences of consistency conditions among overlapping descriptions together with information-theoretic properties of states. Modular rigidity then supplies the familiar symmetry and dynamical structures.

Engineering deliverables. Certain problems can be stated as explicit closure tasks:

- Λ 's dimensionless capacity relation is structurally explained as a global capacity parameter; the input-free prediction is the self-closure fixed-point statement plus the selected OPH scale certificate for SI display
- A full microphysical derivation of geometric modular action is required

These are shared challenges across unification approaches. The framework provides an explicit map of where they live and what would resolve them.

5 Consensus, Defects, and Implementation Hiding

The fixed-cutoff consensus package admits an equivalent finite patch-net formulation. Local patch descriptions agree on overlaps, local recovery-derived repair rules remove mismatches, and the physical output is the schedule-independent quotient normal form together with the induced terminal expectation functionals on the declared physical observable algebras. This section records the fixed-point, defect, and gauge-quotient statements in that language.

The consensus lane also carries a simple finite-candidate law-selection model: once one equips candidate repair rules with a fitness functional, replicator dynamics gives a clean toy picture in which more successful reconciliation laws dominate a competing pool. That selection picture is useful for the overall synthesis picture, but in the consensus paper it is a finite-candidate monotonicity result, not part of the recovered-core gravity/gauge theorem surface, not a universality claim, and not a literal cosmological dynamics claim.

On its declared fixed-cutoff branch, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics* [3] imports four core D1 surfaces into this synthesis surface:

1. **Constraint-code firewall.** A bare finite overlap net is a finite constraint code: its codewords are exactly the globally consistent states $C = \Phi^{-1}(0)$. It is not automatically a QECC/topological code, and its graph min-cut does not determine code distance. Distance/min-cut, Knill–Laflamme resilience, spectral convergence, BFT liveness, and hardware speedup enter only through separate certificates.
2. **Asynchronous confluence.** For the declared accepted repair law, the local-fit contract makes Φ a Lyapunov functional, hence gives termination on the finite patch net. The fixed-cutoff union-collar gluing package supplies the local diamond on the physical quotient, and only that confluence condition together with repair completeness yields a unique schedule-independent normal form from a fixed initial quotient state. Same-boundary uniqueness requires the additional unique consistent extension condition in the preserved boundary/sector fiber.
3. **Cycle obstruction and higher-gauge defects.** On the abelian branch, global consistency holds exactly when cycle holonomy vanishes; on the genuinely noncentral branch, the crossed-module gluing orbit

$$q_\Sigma \in \check{H}^2(N_\Sigma, H_\Sigma \rightarrow G_\Sigma)$$

labels the full fixed-cutoff gluing orbit. It can admit multiple strict representatives related by H_Σ -valued edge changes and need not determine one ordinary G_Σ -valued 1-cocycle class. Associator strictifiability is the separate condition that this orbit admit a representative $(g^{\text{str}}, 1)$; strict endpoint-only transport additionally requires at least one allowed strict representative with trivial represented loop holonomy.

4. **Gauge quotient and observable-level confluence.** The repair law descends to the overlap-invariant quotient, and the induced terminal state on the declared physical observable algebras is unique there even when microscopic representatives differ by gauge or sector relabelings inside one quotient-local glued state.
5. **Record algebra and stability.** On the declared fixed-cutoff observer-accessible operator surface, central record projectors carry the Born/Lüders rules directly, while approximate record projectors inherit explicit $(\varepsilon, \delta_{\text{rec}})$ stability bounds on that same event surface. This is a theorem about the stated algebraic record interface, not a requirement that OPH first rebuild every mathematical ingredient from operational records alone.

The consensus paper is equally explicit about the imported repair-law data and what sits outside the fixed-cutoff theorem stack. The declared repair step includes the touched-overlap local-fit contract and the union-collar gluing package on the physical quotient; the branch conditions above that step are repair completeness and, on the Petz branch, the stated support/CPTP clause. On that same fixed-cutoff surface, exact normal-form computation is finite-state and decidable, automatic approximate stability is only collar-local through the splice and record estimates, and long-run noisy approximate consensus is theorem-grade only after a fair-block contraction certificate is supplied for the chosen exported patch-net family. That certificate boundary is not a dependency for the support-visible BW theorem, local Einstein branch, receipt-conditional compact-gauge reconstruction, or realized Standard Model branch.

6 Particle-Spectrum Branch

The particle branch follows the same logical order as *Deriving the Particle Zoo from Observer Consistency* [5]. One first fixes the compact-gauge classification basis, applies Minimal Admissible Realization (MAR) to the realized one-Higgs branch to identify the Standard Model quotient, and only then asks how transport-stable overlap data organize the observed particle families.

Structural carrier and particle-gate package. On the realized compact-gauge branch, the structural carrier roles are fixed by the compact gauge and MAR-admissibility chain:

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \quad N_c = 3, \quad N_g = 3.$$

This does not determine quantum-particle masses. The compact paper separately proves that explicit Maxwell, perturbative pure-Yang–Mills, and pure-Einstein quadratic actions have transverse or TT classical massless modes on their stated backgrounds and phases. Photon, gluon, or graviton particle language additionally requires a positive-energy physical quantization, positive-residue two-point pole, and the appropriate asymptotic/deconfinement receipt.

Proof route. The proof route is the one developed in *Recovering Relativity and the Standard Model from Observer Overlap Consistency* [2] and *Deriving the Particle Zoo from Observer Consistency* [5]. Under the explicit compact-gauge refinement receipt, coherent bosonic pullback ladder, symmetry, and forgetful-fiber conditions of the compact paper, the surviving transportable edge-sector category reconstructs a compact internal symmetry group. MAR then selects the realized admissible branch, anomaly freedom fixes the hypercharge lattice, the minimal coupled carrier fixes $N_c = 3$, and the same one-Higgs quark branch fixes $N_g = 3$. The particle branch builds on that structural skeleton and does not bypass it.

Sector	Claim tier	What is fixed here	Boundary to stronger claim
Carrier roles and classical modes	structural group/content theorem plus conditional quadratic-action theorem	realized quotient, $N_c = 3$, $N_g = 3$; two Maxwell transverse modes, $2 \dim G$ perturbative Yang–Mills modes, and two Einstein TT modes on stated branches	no zero-GeV particle row until the physical Hilbert-space, pole-residue, and phase gates pass
Electroweak bosons	validation branch + exact sidecar	target-free continuation rows plus the exact frozen-repair pair $M_W = 80.377 \text{ GeV}$, $M_Z = 91.18797809193725 \text{ GeV}$	claim ledger records the validation scope

Sector	Claim tier	What is fixed here	Boundary to stronger claim
Fine-structure endpoint	comparison coordinate + source audit trunk	measured endpoint $\alpha^{-1}(0) = 137.035999177(21)$ with CODATA-derived comparison pixel $P_C = 1.6309682094\dots$; source/root value $\alpha_{\text{root}}^{-1} = 136.9948351646\dots$; mixed-provenance no-hadron diagnostic $A_{\alpha_U}^{\text{fp}} = 137.0359595008\dots$; bookkeeping factor $C_{24,Q} = 1.00096478597323262538\dots$	source-only promotion requires a no-target-leak OPH-QCD spectral payload and same-scheme integrated endpoint certificate; neither the comparison pixel nor the residual supplies that source map
Higgs/top stage	exact source-only D11 split theorem on the declared D10/D11 running, matching, and threshold surface + compare-only exact sidecar	shared scalar ρ_{HT} , source-only split selectors, and the declared D11 Jacobian together with an exact Higgs/top inverse pair	the split theorem is closed on the declared quantitative surface; the inverse pair is compare-only; the auxiliary direct-top PDG codomain is compare-only and is closed as a corpus-limited no-go
Quark family	selected-class theorem + exact supporting surfaces	exact running quark sextet together with explicit exact forward Yukawas on the public physical quark frame class f_P	selected-class closure on f_P ; supporting exact surfaces and the separate mass bridge are carried by Ref. [5]; global classification of all public quark frame classes is closed as a corpus-limited no-go; this surface does not derive the physical strong-CP invariant or prove its vanishing
Charged leptons	continuation branch + exact witness	exact same-family charged triple together with exact same-carrier centered readback on a target-anchored witness	same-family-only on the exact witness; theorem lane carries the closed common-shift no-go, and the determinant-line lift applies only on theorem-grade physical charged data; the theorem-grade uncentered trace lift is not emitted
Neutrinos	rejected target-informed weighted-cycle candidate + diagnostic sidecars	frozen comparison coordinates only; source-level PMNS, physical ordering, Majorana phases, and absolute masses are unformed	the candidate fails the NuFIT 6.1 correlated profile; its family kernel is a template, its charged basis is open and nearly degenerate, and its historical shared-basis recovery is tautological
Hadrons	backend-gated nonperturbative continuation	stable-channel and readout architecture are defined	source-only masses require one production backend export bundle, then executed unquenching, runtime receipt, and production systematics

Claim-tier boundary. Structural carrier roles and the realized Standard Model branch are theorem-bearing outputs; classical propagation is conditional on the displayed action/background/phase receipts, and quantum-particle interpretation is a stronger open gate. The electroweak D10 branch, the Higgs/top stage, quark rows, charged-lepton rows, neutrino rows, and hadron rows sit on their stated quantitative-closure, continuation, or nonperturbative claim tiers. On the target-free electroweak D10 lane, the logical order is forward: the outer/inner closure relation fixes P , and then P fixes $\alpha_U(P)$, equivalently $t_U(P)$ and $t_{\text{tr}}(P)$, then the downstream scale data and public electroweak rows. Measured electroweak quantities are compare-only validation targets and do not feed back into the transmutation solve. Source payloads, same-scheme remainders, interval certificates, and empirical hadron records document public validation and endpoint bookkeeping; they document the fine-structure endpoint bridge but do not source-close the Ward-projected hadronic spectral transport needed to promote the public endpoint to a

source-only theorem.

The $\alpha_U(P)$ source proof record. The electroweak hierarchy used by the clock-scale branch depends first on a source readout of the unified diffusion coupling

$$\alpha_U(P_\star).$$

For a trial pixel P , define

$$M_U(P) = E_\star e^{-2\pi} P^{1/6}, \quad E_{\text{cell}}(P) = \frac{E_\star}{\sqrt{P}}.$$

On the realized color branch,

$$N_c = 3, \quad \beta_{\text{EW}} := N_c + 1 = 4,$$

and the transmutation scale for a candidate coupling a is

$$v(P, a) = E_{\text{cell}}(P) \exp\left[-\frac{2\pi}{\beta_{\text{EW}} a}\right].$$

The source running family is

$$\alpha_i^{-1}(\mu; P, a) = a^{-1} + \frac{b_i}{2\pi} \log\left(\frac{M_U(P)}{\mu}\right), \quad i = 1, 2, 3.$$

The beta-coefficient packet (b_1, b_2, b_3) , matching convention, threshold packet, and renormalization convention are part of the source proof record. They are declared before any comparison with measured electroweak or gravitational data.

The hypercharge coupling is

$$\alpha_Y(\mu; P, a) = \frac{3}{5} \alpha_1(\mu; P, a).$$

The source Z -scale is a fixed point of the source equations, not an inserted measured mass:

$$\mu_Z(P, a) = \frac{v(P, a)}{2} \sqrt{4\pi\alpha_2(\mu_Z; P, a) + 4\pi\alpha_Y(\mu_Z; P, a)}.$$

For $G = \text{SU}(2), \text{SU}(3)$, define the finite representation heat-kernel sums

$$Z_G(t) = \sum_R d_R e^{-tC_2(R)}, \quad \bar{\ell}_G(t) = \frac{1}{Z_G(t)} \sum_R d_R e^{-tC_2(R)} \log d_R.$$

The source heat-kernel parameters are

$$t_2(P, a) = 4\pi^2 \alpha_2(\mu_Z(P, a); P, a), \quad t_3(P, a) = 4\pi^2 \alpha_3(\mu_Z(P, a); P, a).$$

The unified-coupling residual is

$$\Phi_U(P, a) := \bar{\ell}_{\text{SU}(2)}(t_2(P, a)) + \bar{\ell}_{\text{SU}(3)}(t_3(P, a)) - \frac{P}{4}.$$

The source value is the zero

$$\Phi_U(P, \alpha_U(P)) = 0.$$

Definition 6.1 (No-hidden-free-variable admissibility). *The unified-coupling proof record is admissible only when the representation cutoffs, beta packet, matching prescription, threshold packet, and renormalization convention are frozen before comparison with measured electroweak or gravitational data:*

$N_2, \quad N_3, \quad (b_1, b_2, b_3), \quad \text{matching prescription}, \quad \text{threshold packet}, \quad \text{renormalization convention}.$

None of these may be selected by minimizing the error against

$$M_W, \quad M_Z, \quad \alpha_s(M_Z), \quad \sin^2 \theta_W, \quad v_{\text{exp}}, \quad G_{\text{exp}}.$$

Tuning any of these objects after comparison with observed electroweak or gravitational data makes the hierarchy row a calibration.

Theorem 6.2 (No-measured-input unified-coupling proof record). *Let*

$$\mathcal{R}_U = (P_\star, N_2, N_3, I_U, \Phi_U, K_U, a_U^{(0)}, r_U, m_U, \text{DAG}_U)$$

be an admissible unified-coupling proof record. Here N_2, N_3 are the finite representation cutoffs for SU(2) and SU(3), $I_U \subset \mathbb{R}_+$ is the declared physical interval for α_U , Φ_U is the heat-kernel closure residual, K_U is an interval-Newton or Krawczyk operator, $a_U^{(0)}$ is the displayed candidate, r_U is the residual bound, m_U is a lower derivative bound, and DAG_U is the dependency graph.

Assume

$$\begin{aligned} K_U(I_U) &\subset \text{int}(I_U), \\ |\Phi_U(P_\star, a_U^{(0)})| &\leq r_U, \\ |\partial_a \Phi_U(P_\star, a)| &\geq m_U > 0 \quad (a \in I_U), \end{aligned}$$

and assume that DAG_U contains no directed path from measured electroweak masses, measured low-energy gauge couplings, measured G , Planck area, Planck mass, measured Λ , or any equivalent gravity-calibrated or electroweak-calibrated scale. Then there is a unique source value

$$\alpha_U(P_\star) \in I_U$$

such that

$$\Phi_U(P_\star, \alpha_U(P_\star)) = 0,$$

and

$$|\alpha_U(P_\star) - a_U^{(0)}| \leq \frac{r_U}{m_U}.$$

Consequently, the electroweak hierarchy

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp \left[-\frac{2\pi}{(N_c + 1)\alpha_U(P_\star)} \right]$$

is no-measured-input on this proof record.

Proof. The residual $\Phi_U(P_\star, a)$ is built from the source-side pixel $P_\star$, declared representation cutoffs, source transmutation law, source running law, and source Z -scale fixed-point equation. The dependency condition excludes measured electroweak data and all gravity-calibrated scales. The

interval-Newton/Krawczyk inclusion proves existence and uniqueness of a zero of $\Phi_U(P_\star, a)$ inside I_U . The residual estimate and derivative lower bound give

$$|\alpha_U(P_\star) - a_U^{(0)}| \leq \frac{|\Phi_U(P_\star, a_U^{(0)})|}{m_U} \leq \frac{r_U}{m_U}.$$

Substitution into the declared transmutation law gives the displayed hierarchy. The dependency graph contains no measured G , Planck unit, measured M_Z , measured M_W , measured $\alpha_s(M_Z)$, or measured low-energy electroweak coupling ancestor, so the emitted hierarchy is source-side on $\mathcal{R}_U$. $\square$

Lemma 6.3 (Exponential sensitivity of the hierarchy row). *On the transmutation branch*

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp\left[-\frac{2\pi}{4\alpha_U}\right],$$

and on a clock row whose dominant scale dependence is inherited from $v/E_\star$, the induced gravity readout satisfies approximately

$$\frac{\partial \ln G}{\partial \alpha_U} = \frac{\pi}{\alpha_U^2}.$$

At $\alpha_U \simeq 0.0411$,

$$\frac{\partial \ln G}{\partial \alpha_U} \simeq 1.86 \times 10^3.$$

Therefore a certified relative gravity precision η_G requires

$$|\delta \alpha_U| \lesssim \frac{\eta_G}{1.86 \times 10^3}.$$

Proof. For $\beta_{EW} = 4$,

$$\ln(v/E_\star) = -\frac{1}{2} \ln P_\star - \frac{2\pi}{4\alpha_U}.$$

Thus

$$\frac{\partial \ln(v/E_\star)}{\partial \alpha_U} = \frac{2\pi}{4\alpha_U^2} = \frac{\pi}{2\alpha_U^2}.$$

Since the gravity row is quadratic in the clock scale,

$$G \propto \varepsilon_{Cs}^2,$$

and the clock scale inherits this dominant factor, one obtains

$$\frac{\partial \ln G}{\partial \alpha_U} = 2 \cdot \frac{\pi}{2\alpha_U^2} = \frac{\pi}{\alpha_U^2}.$$

$\square$

Remark 6.4 (Digits policy). *The exponential branch allows precision only to the extent that the source proof records carry interval bounds. The number of printed digits in G_{SI} may not exceed the precision certified by $\mathcal{R}_U$, $\mathcal{R}_{QCD}$, $\mathcal{R}_{flav}$, and $\mathcal{R}_{Cs}$. A display such as*

$$\varepsilon_{Cs} = 2\pi\nu_{Cs}\sqrt{\hbar G_{ref}/c^5}$$

is a calibration checksum unless the full source stack emits the same interval with a no- G dependency graph.

No- G clock hierarchy for the gravity scale. The numerical gravity row uses the dimensionless cesium clock gap

$$\varepsilon_{\text{Cs}} := \frac{\hbar\omega_{\text{Cs}}}{E_\star}, \quad E_\star := \frac{\hbar c}{\ell_\star}, \quad \omega_{\text{Cs}} = 2\pi\nu_{\text{Cs}}.$$

The size of this gap is a hierarchy problem for the OPH scale branch. The gap is represented as a factorized source-side readout. The source-only scale branch is accepted as no- G only when the gap factorizes through source-side dimensionless physics:

$$\mathcal{R}_\gamma = \mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\text{abs}} + \mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}} + \mathcal{R}_{\text{atom}}^{133\text{Cs}}.$$

Here $\mathcal{R}_U$ emits $\alpha_U(P_\star)$ and the electroweak scale, $\mathcal{R}_\alpha$ emits the electromagnetic coupling used by the atomic Hamiltonian, $\mathcal{R}_e^{\text{abs}}$ emits the electron absolute mass ratio, $\mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}}$ emits the cesium nuclear source packet, and $\mathcal{R}_{\text{atom}}^{133\text{Cs}}$ emits the hyperfine spectral gap from those dimensionless data.

At leading OPH source level, the electroweak hierarchy is generated by the branch-fixed transmutation law

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp\left[-\frac{2\pi}{(N_c + 1)\alpha_U(P_\star)}\right], \quad N_c = 3.$$

Thus the weak-to-UV ratio is generated from the dimensionless source data $P_\star$, N_c , and $\alpha_U(P_\star)$. The cesium gap then has the schematic source form

$$\varepsilon_{\text{Cs}} = \mathcal{H}_{\text{Cs}}\left(\alpha_\star, \frac{v}{E_\star}, \frac{\Lambda_{\text{QCD}}}{E_\star}, y_e, y_q, \text{nuclear data, atomic corrections}\right),$$

where all arguments are dimensionless source outputs on the declared branch. In leading hyperfine scaling this contains the familiar structure

$$\varepsilon_{\text{Cs}} \sim \mathcal{C}_{\text{Cs}} \alpha_\star^4 \frac{m_e^2}{m_p E_\star},$$

with relativistic, many-body, recoil, QED, finite-nuclear-size, and nuclear-moment corrections absorbed into $\mathcal{C}_{\text{Cs}}$.

Definition 6.5 (Cesium source packet). *A source-only cesium packet is*

$$\mathcal{D}_{\text{Cs}}^{\text{src}} = \left(\alpha_\star, \frac{m_e}{E_\star}, \frac{\Lambda_{\text{QCD}}}{E_\star}, \{y_q\}, \mathcal{N}_{133}, \mathcal{B}_{\text{atom}}^{\text{Cs}}\right).$$

$\mathcal{N}_{133}$ contains the source-side nuclear data for ^{133}Cs , including nuclear spin, magnetic moment, magnetization distribution, charge radius, recoil data, and finite-size data. $\mathcal{B}_{\text{atom}}^{\text{Cs}}$ contains the atomic many-body, relativistic, QED, recoil, and finite-nucleus correction package. Every object in $\mathcal{D}_{\text{Cs}}^{\text{src}}$ is dimensionless and has no dependency path from measured G , Planck area, Planck mass, Planck time, measured Λ , or an algebraically equivalent gravity-calibrated scale.

Definition 6.6 (OPH cesium Hamiltonian). *Given $\mathcal{D}_{\text{Cs}}^{\text{src}}$, define the dimensionless cesium Hamiltonian*

$$\hat{H}_{\text{Cs}}^{\text{OPH}} = \hat{H}_{\text{DC}} + \hat{H}_{\text{Breit}} + \hat{H}_{\text{QED}} + \hat{H}_{\text{recoil}} + \hat{H}_{\text{nuc}} + \hat{H}_{\text{manybody}}.$$

The hats indicate that every term is measured in units of $E_\star$, not in SI units and not in Planck units inferred from measured G .

Theorem 6.7 (No- G OPH clock-hierarchy theorem). *Assume the local pixel proof selects $P_\star$. Assume the unified-coupling proof record $\mathcal{R}_U$ of Theorem 6.2 emits $\alpha_U(P_\star)$ and*

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp \left[-\frac{2\pi}{(N_c + 1)\alpha_U(P_\star)} \right].$$

Assume the electromagnetic endpoint branch $\mathcal{R}_\alpha$ emits $\alpha_\star$, the charged-lepton absolute-scale branch $\mathcal{R}_e^{\text{abs}}$ emits

$$\frac{m_e}{E_\star} = \frac{y_e}{\sqrt{2}} \frac{v}{E_\star},$$

the QCD/nuclear branch $\mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}}$ emits $\mathcal{N}_{133}$, and the atomic branch $\mathcal{R}_{\text{atom}}^{133\text{Cs}}$ supplies isolated spectral enclosures

$$J_{F=3}, \quad J_{F=4}$$

for the source-only Hamiltonian

$$\hat{H}_{\text{Cs}}^{\text{OPH}} = \hat{H}_{\text{DC}} + \hat{H}_{\text{Breit}} + \hat{H}_{\text{QED}} + \hat{H}_{\text{recoil}} + \hat{H}_{\text{nuc}} + \hat{H}_{\text{manybody}}$$

with disjoint intervals and certified hyperfine gap

$$\varepsilon_{\text{Cs}} = \lambda_{F=4}(\hat{H}_{\text{Cs}}^{\text{OPH}}) - \lambda_{F=3}(\hat{H}_{\text{Cs}}^{\text{OPH}}).$$

Assume further that the combined dependency graph

$$\text{DAG}_{\text{Cs}} = \text{DAG} \left(\mathcal{R}_U, \mathcal{R}_\alpha, \mathcal{R}_e^{\text{abs}}, \mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}}, \mathcal{R}_{\text{atom}}^{133\text{Cs}} \right)$$

contains no directed path from

$$G_{\text{exp}}, \quad \ell_P^2 = \frac{\hbar G}{c^3}, \quad m_P = \sqrt{\frac{\hbar c}{G}}, \quad t_P = \sqrt{\frac{\hbar G}{c^5}}, \quad \Lambda_{\text{exp}}, \quad \frac{3\pi c^3}{\hbar G},$$

or from any algebraically equivalent gravity-calibrated scale. Then ε_{Cs} is a no- G hierarchy readout. Consequently

$$\gamma_\star = \frac{\ell_\star \nu_{\text{Cs}}}{c} = \frac{\varepsilon_{\text{Cs}}}{2\pi}$$

is non-circular, and

$$G_{\text{SI}} = \frac{c^5}{4\pi^2 \hbar \nu_{\text{Cs}}^2} \varepsilon_{\text{Cs}}^2$$

has no measured- G ancestor.

Proof. $\mathcal{R}_U$ emits the small weak ratio from $P_\star$, N_c , and $\alpha_U(P_\star)$, all of which are dimensionless branch data. $\mathcal{R}_\alpha$ emits the electromagnetic coupling used by the atomic Hamiltonian. $\mathcal{R}_e^{\text{abs}}$ emits the electron mass ratio. $\mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}}$ emits the cesium nuclear source packet. Therefore every coefficient of $\hat{H}_{\text{Cs}}^{\text{OPH}}$ has no measured- G ancestor. The atomic branch gives isolated spectral enclosures for the two hyperfine levels, so the gap

$$\varepsilon_{\text{Cs}} = \lambda_{F=4}(\hat{H}_{\text{Cs}}^{\text{OPH}}) - \lambda_{F=3}(\hat{H}_{\text{Cs}}^{\text{OPH}})$$

is well-defined and inherits the no- G dependency condition of the Hamiltonian.

Finally,

$$\varepsilon_{\text{Cs}} = \frac{\hbar(2\pi\nu_{\text{Cs}})}{\hbar c/\ell_\star} = \frac{2\pi\nu_{\text{Cs}}\ell_\star}{c},$$

so

$$\gamma_\star = \frac{\ell_\star \nu_{\text{Cs}}}{c} = \frac{\varepsilon_{\text{Cs}}}{2\pi}.$$

Substitution into $G_{\text{SI}} = c^3 \ell_\star^2 / \hbar$ gives the displayed expression for G_{SI} . The only non-display parent of the expression is the source-side ε_{Cs} readout, so the gravity row is non-circular under the stated hypotheses. $\square$

Remark 6.8 (Clock-certificate boundary). *The theorem above is not discharged by printing*

$$\varepsilon_{\text{Cs}} = 3.1139305134 \dots \times 10^{-33}.$$

That decimal is source-predictive only if it is emitted by

$$\mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\text{abs}} + \mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}} + \mathcal{R}_{\text{atom}}^{133\text{Cs}}.$$

If instead it is computed as

$$\varepsilon_{\text{Cs}}^{\text{cal}} = 2\pi \nu_{\text{Cs}} \sqrt{\frac{\hbar G_{\text{exp}}}{c^5}},$$

then it is a gravity-side calibration checksum, not a source-side clock prediction. The paper surface supplies the algebraic certificate shape and the $\mathcal{R}_U$ numerical witness. The full source-only $\mathcal{R}_{\text{Cs}}$ branch requires the source-only electromagnetic endpoint, charged-lepton absolute-scale gate, source-side cesium nuclear packet, and atomic spectral enclosure.

Remark 6.9 (Hadronic and nuclear certificate boundary). *The unresolved load-bearing pieces of the no- G SI gravity row are the hadronic/same-scheme endpoint payload in $\mathcal{R}_\alpha$ and the source-side cesium QCD/nuclear packet in $\mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}}$. The displayed endpoint and ε_{Cs} rows therefore fix those payloads for checksum and comparison purposes. They are not counted as source-only OPH emissions.*

The source-only completion route is a Ward-projected QCD/nuclear spectral certificate from a dedicated OPH optical-compute backend, or an equivalent nonperturbative backend with manifest provenance and systematics. Ordinary CPU/GPU runs in this paper are not the declared source certificate for that hadronic payload.

Screen-normalized self-consistency for total capacity. The global counterpart to the local P -closure is the cosmic record-closure capacity,

$$N_{\text{CRC}},$$

the active horizon capacity at which the universe can contain observers who read back the same boundary capacity that supports them. This synthesis paper records the derivation here, next to the pixel derivation and the selected no- G scale certificate, because the three roles must stay distinct. The local equation for P says that one screen cell has the same outside geometric reading and inside electromagnetic reading. The global equation for N_{CRC} says that the whole horizon has the same outside supplied capacity and inside observer-readable cosmic record capacity. The selected scale certificate supplies $\gamma_\star = \ell_\star \nu_{\text{Cs}} / c$, equivalently $B_\star = 3\pi / \ell_\star^2$, on its own observed branch.

The operational use of observation is reverse engineering, not fitting. Because OPH treats the realized universe as a closed mathematical fixed structure, approximate measurements can identify which basin or branch interval internal observers inhabit. The quantitative claim then requires the OPH map on that interval to be a self-map with a contraction, derivative-sign, strict-concavity, or

equivalent uniqueness proof. The observational readout supplies x_0 or the interval; the fixed-point theorem supplies $x_\star$. The exact $P_\star$ and N_{CRC} values must be computed as closure endpoints.

What $P_\star$ and $N_\star$ determine by themselves is the dimensionless geometry. With

$$N_\star = \frac{3\pi}{\Lambda_\star \ell_\star^2}, \quad P_\star = \frac{a_{\text{cell}}}{\ell_\star^2},$$

one obtains

$$\Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}, \quad \Lambda_\star a_{\text{cell}} = \frac{3\pi P_\star}{N_\star}.$$

If $\mathcal{B}_\ell := \Lambda_\star N_\star$, then

$$\mathcal{B}_\ell \ell_\star^2 = 3\pi, \quad \mathcal{B}_\ell a_{\text{cell}} = 3\pi P_\star.$$

Under the length rescaling $L \mapsto \lambda L$, both $P_\star$ and $N_\star$ stay fixed while $\mathcal{B}_\ell \mapsto \lambda^{-2} \mathcal{B}_\ell$. Thus the two closures do not determine an SI value for $\mathcal{B}_\ell$, $\ell_\star^2$, or G_{SI} without the selected scale certificate. When the selected no- G scale certificate supplies $\gamma_\star$, equivalently $B_\star$, the SI display is

$$\gamma_\star = \frac{\ell_\star \nu_{\text{Cs}}}{c}, \quad B_\star = \mathcal{B}_\ell = \Lambda_\star N_\star, \quad \ell_\star^2 = \frac{3\pi}{B_\star}.$$

The branch certificate used by the SI gravity row is

$$B_\star = 3.60787391468 \dots \times 10^{70} \text{ m}^{-2},$$

which gives

$$\ell_\star^2 = 2.61228030237 \dots \times 10^{-70} \text{ m}^2.$$

The rounded $N_{\text{scr}} \simeq 3.31 \times 10^{122}$ display is only a capacity-scale label; the exact G row uses the fixed-point value and the selected scale certificate. With the Planck-2018 curvature benchmark, using only the rounded capacity would shift the Newton row at the 8.4×10^{-4} level.

For a trial entropy capacity N , the outside global readout is the de Sitter curvature term

$$\Lambda_{\text{out}}(N) = \frac{3\pi}{GN}.$$

The G used in this capacity branch is the selected scale certificate $G_{\text{SI}} = c^3 \ell_\star^2 / \hbar$, carried in geometric units as $G_{\text{geom}} = \ell_\star^2$. That trial value defines a candidate OPH universe: horizon size, accessible records, dilution rate, checkpoint continuation, observer-supporting capacity, and late-time de Sitter boundary. Observers inside that candidate universe do not see an external parameter N directly. They infer boundary capacity from internal geometry, causal accessibility, stable records, and self-read closure. The corresponding capacity readback map is

$$F(N) := N_{\text{read}}(N),$$

where $F(N)$ is the active cosmic record capacity read back by observers inside the universe supplied with capacity N . The global closure equation is

$$\boxed{N_{\text{CRC}} = F(N_{\text{CRC}})}.$$

In words, N_{CRC} is the only global capacity at which the universe can read back its own boundary without deficit or slack.

At fixed cutoff r , the finite OPH machinery makes this map explicit. Let $\mathfrak{U}_{r,N}$ be the finite OPH universe candidate with supplied active boundary capacity N . The consensus branch supplies

a quotient normal-form map $\text{nf}_{r,N}$ on the declared terminating/confluent repair surface. Let Obs select the stable self-reading observer sector, and let Cap_{read} return the capacity reconstructed by that sector. Then the finite readback map has the schematic form

$$F_r(N) = \text{Cap}_{\text{read}}(\text{Obs}(\text{nf}_{r,N}(\mathfrak{U}_{r,N}))).$$

The refinement-limit map F is the cofinal limit of this finite readback construction when that limit exists on the admissible capacity interval.

The uniqueness proof is the direct global analogue of the local P -fixed-point proof. Let I be the admissible interval of active horizon capacities. If the OPH-derived readback map $F : I \rightarrow I$ satisfies

$$0 \leq F'(N) \leq \kappa < 1 \quad (N \in I)$$

and the observed branch satisfies $F(N_{\text{obs}}) = N_{\text{obs}}$, then N_{obs} is the unique stable fixed point. Indeed, if N_1 and N_2 are two fixed points, then

$$|N_1 - N_2| = |F(N_1) - F(N_2)| \leq \kappa |N_1 - N_2|.$$

Since $\kappa < 1$, this implies $N_1 = N_2$. Banach iteration $N_{k+1} = F(N_k)$ also converges to the same fixed point from any starting value in I .

The physical meaning of the contraction condition is that changing supplied raw capacity changes observer-readable active capacity by less than one-for-one. Extra Hilbert-space room does not automatically become public record capacity. It must be reachable, recordable, self-readable, stable under repair, coupled to observer continuation, and visible on the quotient. Capacity is physical capacity only insofar as the universe can read it.

The existing self-closing normal-form count is the finite-count representation of the same closure target. For a candidate entropy capacity N , let $\mathcal{H}_{\partial,N}$ be the observer-facing horizon record Hilbert space with $\log \dim \mathcal{H}_{\partial,N} = N$, and let $\Omega_{r,N}^{\text{sc}}$ be the terminal normal forms that are repair-closed, observer/checkpoint-supporting, locally recovered-core closed, and whose own horizon record surface reads back capacity N . Define

$$A_r(N) := |\Omega_{r,N}^{\text{sc}}|, \quad \Pi_r(N) := \frac{A_r(N)}{\dim \mathcal{H}_{\partial,N}} = A_r(N)e^{-N}.$$

The subtraction by N in

$$\log \Pi_r(N) = \log |\Omega_{r,N}^{\text{sc}}| - N$$

is not a tunable penalty; it is exactly division by the full screen Hilbert-space size e^N . Thus the count-density selector is

$$N_{r,\star} = \text{MAR} \arg \max_N \left[\log |\Omega_{r,N}^{\text{sc}}| - N \right].$$

In the refinement limit this becomes

$$N_{\star} = \text{MAR} \arg \max_N \left[\log |\Omega_N^{\text{sc}}| - N \right].$$

When the limiting pressure

$$\ell(N) := \log |\Omega_N^{\text{sc}}| - N$$

is differentiable, the count-density proof can also be written as the stationarity fixed point

$$T_{\eta}(N) := N + \eta \ell'(N), \quad T_{\eta}(N) = N \iff \ell'(N) = 0.$$

The proof certificate is the derivative-sign certificate for

$$H_N(N) := \ell'(N),$$

namely $H_N(N_0) = 0$ and $H'_N(N) < 0$ on the admissible interval, or the stronger concavity bound

$$-M \leq \ell''(N) \leq -m < 0.$$

This certificate is another way to express uniqueness of the global self-read closure. A numerical solver may locate a candidate root, but uniqueness and stability come from the OPH-derived readback or pressure certificate. Software checks, when used, check the finite grammar or transfer-function certificate; they are not a brute-force enumeration of cosmological normal forms and are not the proof by themselves.

The observed de Sitter read-off is

$$N_{\text{obs}} = S_{\text{dS}} = \frac{A_{\text{dS}}}{4\ell_P^2} = \frac{3\pi}{\Lambda\ell_P^2} \simeq 3.31 \times 10^{122}.$$

On the observed branch,

$$N_{\text{CRC}} = N_\star = N_{\text{obs}}, \quad \Lambda_{\text{CRC}}\ell_\star^2 = \frac{3\pi}{N_{\text{CRC}}}, \quad \Lambda_{\text{CRC}} = \frac{3\pi}{GN_{\text{CRC}}}.$$

The last equality is the scale-certified display. The fixed-point certificate makes this value the unique stable capacity of the self-reading solution we inhabit, not a freely tunable creator-set parameter. Informally, N_{CRC} is the single horizon capacity at which the outside total horizon capacity and the inside observer-accessible public record agree.

Outer/inner self-consistency for the pixel ratio. This synthesis paper is the canonical place where the local pixel ratio P is defined, so the compact and particle papers can cite one derivation rather than duplicate it. The construction has six steps.

1. **Outer pixel ratio.** The outer-side local screen datum is the dimensionless area ratio

$$P := \frac{a_{\text{cell}}}{\ell_\star^2}.$$

Here $\ell_\star^2 = 3\pi/B_\star$ is supplied by the selected scale certificate. After that scale is emitted, it is displayed in SI language as the Planck area.

2. **Exact equilibrium benchmark.** The total/bulk/edge hierarchy has one exact self-similar balance point. Writing

$$x(C) := \frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = 1 + \frac{\langle L_C \rangle}{S_{\text{bulk}}(C)},$$

the self-similar balance condition

$$\frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = \frac{S_{\text{bulk}}(C)}{\langle L_C \rangle}$$

gives $x^2 - x - 1 = 0$ and hence $x = \varphi = (1 + \sqrt{5})/2$. The realized branch must sit near, but not exactly at, this point because exact equilibrium is too symmetric to support durable records, structure, and dynamics.

3. **Outer detuning variable.** On the declared Phase-II closure surface we parametrize the outer-side detuning by

$$\alpha_{\text{ext}}(P) := \frac{P - \varphi}{\sqrt{\pi}}, \quad \text{equivalently} \quad P = \varphi + \alpha_{\text{ext}}(P)\sqrt{\pi}.$$

4. **Source-only D10 anchor from the same pixel.** Let $I \subset \mathbb{R}_+$ be the physical pixel interval and let

$$F_{D10} : I \longrightarrow \mathcal{D}_{D10}$$

denote the declared D10 source map. For any trial $P \in I$, the forward D10 lane emits

$$P \longmapsto \alpha_U(P) \longmapsto (t_U(P), t_{\text{tr}}(P)) \longmapsto (t_2(P), t_3(P), v(P)) \longmapsto A_Z(P) = \alpha_{\text{em}}^{-1}(m_Z^2; P).$$

5. **Ward-projected Thomson transport.** Let T_Q be the declared Ward-projected $U(1)_Q$ transport operator from the m_Z -anchor to the Thomson limit. This is the endpoint functional that reads the fixed point as $\alpha^{-1}(0) = 137.035999177(21)$. Define

$$A_T(P) := T_Q(A_Z(P), F_{D10}(P)) = \alpha_{\text{Th}}^{-1}(P).$$

The inner coupling used in the outer/inner closure is the coupling, not the inverse coupling:

$$\alpha_{\text{in}}(P) := \frac{1}{A_T(P)}.$$

6. **Closure.** The realized pixel ratio is the fixed point for which the outer detuning equals the inner observation coupling:

$$\alpha_{\text{ext}}(P) = \alpha_{\text{in}}(P).$$

Equivalently,

$$H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_T(P)} = 0, \quad \text{or} \quad P = \varphi + \frac{\sqrt{\pi}}{A_T(P)} = \varphi + \alpha_{\text{in}}(P)\sqrt{\pi}.$$

Writing

$$G(P) := \varphi + \frac{\sqrt{\pi}}{A_T(P)},$$

the closure problem is the fixed-point condition $G(P) = P$. On any physical interval where this induced map is a self-map and a contraction, the closure root is locally unique. Equivalently, the solver can use

$$H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_T(P)}$$

and accept the unique zero of H on the declared branch interval. Observer-facing data may localize that interval. The measured endpoint cannot replace the closure solve.

The closure interpretation assigns one screen cell two coupled roles. On the outer side it is one cell of the screen whose displacement above the exact self-similar equilibrium φ sets the geometric detuning. On the inner side the same cell emits the smallest electromagnetic observation step available on the particle branch. The pixel ratio P is the size of that cell in units of the scale-certificate area $\ell_\star^2$, and the realized P is the value at which those outer and inner readings agree.

The fixed-point equation is the mathematical form of that consistency condition. The public root is

$$P_\star \simeq 1.6309682094, \quad \alpha^{-1}(0) = 137.035999177(21).$$

The value is forced on the declared branch because the same screen cell must be one object with two compatible descriptions. The geometric detuning fixes the outer side, the Ward-projected Thomson endpoint fixes the inner electromagnetic side, and the fixed point is the sole shared value. The source/root audit value is 136.994835.... Combining it with $\alpha_U(P_C)$, evaluated at the CODATA-derived comparison pixel, gives the mixed-provenance no-hadron diagnostic $A_{\alpha U}^{\text{fp}} = 137.0359595008\dots$; it is not a source-only prediction. A zero-momentum laboratory measurement sees the dressed $U(1)_Q$ current after charged-lepton vacuum polarization, confined-quark/hadron spectral transport, and same-scheme endpoint matching. The claim table records the source-audit trunk, endpoint residual, and interval-certificate disclosures. Once the fixed point $P_\star$ is selected, every downstream quantity is propagated forward from that same $P_\star$. A separate hardware note reports an optical-cavity check of the same fixed-point geometry; this is treated as corroborating engineering evidence. The same closure surface also admits an inverse observational use case in which observers infer P and total screen capacity from inside the world.

One clean refinement studies the same closure directly at zero momentum. That route would close the electromagnetic row without transporting from the m_Z anchor. The present paper keeps the fixed-point idea itself simple and treats that zero-momentum promotion as a separate quantitative task.

Non-hadron output surface. For quick orientation, the non-hadron output surface consists of receipt-gated classical carrier modes with no promoted quantum mass rows, the electroweak exact sidecar, the Higgs/top exact sidecar, a same-family charged witness, and the selected-class quark theorem with exact forward Yukawas on f_P . The rejected weighted-cycle neutrino candidate is retained as a comparison and falsification record, outside the exact-output surface. This synthesis paper uses those facts as lane markers. Ref. [5] carries the exact values, supporting surfaces, diagnostic sidecars, and per-sector caveats.

Extended derivation. The structural-to-family route and the full per-sector claim-tier audit are developed in *Deriving the Particle Zoo from Observer Consistency* [5].

7 Screen Microphysics, Records, and Observer Continuation

The screen-microphysics lane is the engineering side of the paper corpus. It asks whether some microscopic model might exist in principle and gives one explicit regulated carrier architecture: a federation of finite observer patches with echosahedral multi-port interfaces, recurrent toroidal subchannels, exposed overlap data, records, repair instruments, and observer-facing interfaces all made concrete at fixed cutoff. A spherical screen is used there only as an observer-facing regulator chart for support-visible cuts, not as a claim that the universe is a literal spherical quantum computer.

This lane has five theorem-bearing components. First, the reference architecture itself is explicit. Second, the regulated patch-net embedding theorem is closed on its fixed-cutoff object/local-interface surface, and the edge-sector thermal/Casimir branch carries an explicit fixed-cutoff law with one compact lift above that theorem surface. Third, the fixed-cutoff measurement/Born-rule package is closed on its declared operator and record-event surface. Fourth, the fixed-cutoff Bell/CHSH package is closed on the declared two-wing surface. Fifth, the checkpoint/restoration

observer package is closed. The downstream boundaries on this lane are packet-level quotient closure where one wants an autonomous exported dynamics, compact-group/Peter–Weyl bookkeeping on the D10/compact surfaces, fair-block contraction certificates for long-run noisy approximate consensus on exported packet nets, and model-family universality. The support-visible BW / geometric-modular / local-Einstein closure lives on the companion recovered-core papers rather than on this fixed-cutoff engineering lane. A reconstruction of Hilbert, C^* - or von Neumann algebra structure, Born probabilities, trace structure, and entropy from operational records alone remains a separate question outside this synthesis lane.

Reference-architecture takeaway. The synthesis-level point is that the regulated screen-side architecture supplies a concrete habitat in which local observables, overlap observables, record registers, and synchronization maps can all be written without pretending to identify a unique final UV completion. Hardware evidence is not imported by this synthesis theorem surface unless it appears in a public, hash-stable OPH evidence bundle. The OPH-FPE simulator repository is the corresponding finite-run software surface for emitted receipts and replayable OPH experiment bundles [10].

Imported theorem from Ref. [4] (Fixed-cutoff record algebra and Born–Lüders package). For each completed compare/write/verify slice of the regulated microphysics, the declared pointer and overlap-sector projectors generate a finite commutative central record algebra

$$\mathcal{Z}_{\text{rec}}.$$

Every observer-accessible event E in that algebra has probability

$$\mathbb{P}(E) = \text{Tr}(\rho P_E),$$

and conditioning on E gives the operational post-measurement state

$$\rho|_E = \frac{P_E \rho P_E}{\text{Tr}(\rho P_E)}.$$

If a practical readout instead uses projectors Q_a with commuting central reference projectors $\hat{Q}_a$ on the same declared slots and

$$\delta_{\text{rec}} = \max_a \|Q_a - \hat{Q}_a\|,$$

then

$$\|[Q_a, Q_b]\| \leq 4 \delta_{\text{rec}},$$

and any accessible-state perturbation $\|\tilde{\rho} - \rho\|_1 \leq \varepsilon$ changes each declared elementary record-event probability by at most $\varepsilon + \delta_{\text{rec}}$.

Imported theorem from Ref. [4] (Fixed-cutoff Bell / CHSH package). On the declared fixed-cutoff physical observable algebra, fix commuting left/right wing subalgebras, central binary setting registers, binary projective readouts on each wing, and a physical source state on the joint wing algebra. Then the compare slice carries the joint law

$$p(a, b \mid x, y) = \text{Tr}(\rho_{LR} P_{a|x}^{(L)} P_{b|y}^{(R)}),$$

the local marginals are independent of the remote setting, the Bell correlators satisfy

$$E(x, y) = \text{Tr}(\rho_{LR} A_x B_y),$$

and the CHSH combination obeys

$$|E(0,0) + E(0,1) + E(1,0) - E(1,1)| \leq 2\sqrt{2}.$$

If the source family contains an explicit two-qubit branch with the stated Pauli readouts, the same fixed-cutoff surface saturates $2\sqrt{2}$ exactly on that branch. The source-state input is explicit: the theorem derives the Bell law and the Tsirelson bound from the stated fixed-cutoff operator surface, while Bell-pair preparation and two-qubit factor structure are stated branch conditions.

Imported theorem from Ref. [4] (Checkpoint/restoration and observer backup). For an observer patch P_O , let the checkpoint data consist of the observer-facing record algebra, the accessible local state, the future update schedule, and the externally visible overlap interface data. Exact restoration reproduces the full future law of observer-accessible events, while an ε -accurate restoration changes that future law by at most ε in total variation.

Observer-facing conclusion. The theorem-level content is modest but sharp: the paper corpus supplies a fixed-cutoff measurement interface, a fixed-cutoff checkpoint/restoration package, and an operational observer-identity criterion on observer-accessible event algebras. Stronger substrate-selection or strange-loop closure claims are outside the recovered theorem package.

Extended derivation. The fixed-cutoff edge-law, measurement, Born-rule, Bell/CHSH, and checkpoint/restoration packages are developed in *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH* [4].

8 Worksheet/String Branch

The synthesis paper carries the string/worldsheet branch on the same theorem boundary used across the paper corpus: a genuine theorem-level bridge from the edge-sector theorem surface to the two-dimensional Yang–Mills partition function, followed by a controlled large- N_{edge} effective-description theorem on the stated branch. This two-dimensional heat-kernel partition statement and declared-branch large-edge effective description sit beside the compact paper’s conditional support-visible compact-gauge four-dimensional Yang–Mills form and repair-gap theorems, whose continuum identification and gap transport require the separately named receipts.

Imported theorem (Heat-kernel edge-sector bridge). Import the fixed-cutoff same-overlap edge-law package from *Federated Echosahedral Screen Microphysics* [4]. Under the same overlap-gauge realization and local-Gibbs/MaxEnt edge branch, the edge-sector weights satisfy

$$p_R(t) \propto d_R e^{-tC_2(R)}.$$

Consequently the edge partition function matches the standard two-dimensional Yang–Mills heat-kernel form. The compact paper carries OS/Wightman-strength continuum reconstruction for the four-dimensional Euclidean Yang–Mills form and the OPH mass-gap claim separately on the support-visible compact-gauge branch.

Imported proof route. The microphysics source surface carries the fixed-cutoff thermal/Casimir law on the declared overlap interface. Peter–Weyl decomposition [45] then identifies the resulting sum with the standard heat-kernel expression for two-dimensional Yang–Mills on the compact-group refinement lift.

Compact-carrier theorem. Writing the closed edge partition function as

$$Z_{\text{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)},$$

the compact carrier states the exact bridge itself as a named theorem: Peter–Weyl identifies it with the compact-group heat kernel at the identity,

$$Z_{\text{edge}}(t) = K_t(1).$$

The Chapman–Kolmogorov gluing law for K_t is the precise sense in which collar sewing matches the two-dimensional Yang–Mills heat-kernel surface before any large- N_{edge} continuation is invoked. The compact carrier is a partition-identity theorem on its stated branch. The four-dimensional OPH claim identifies the compact-gauge branch with the Euclidean Yang–Mills form and applies the repair-dynamics theorem on that same support-visible branch.

Controlled large- N_{edge} branch. Fix a distinct large- N_{edge} realization, with $N_{\text{edge}} \neq N_c = 3$, and the fixed- τ variable

$$\tau = tN_{\text{edge}}$$

on a compact window I . If the resulting edge free energy satisfies the compact paper’s large- N_{edge} criterion, with remainder control on I , then the Gross–Taylor rewriting [44] is a controlled theorem-level worldsheet effective description of the edge dynamics on that branch. Critical-superstring claims, worldsheet CFT closure, anomaly cancellation, and full massless-spectrum matching are outside the recovered core theorem package.

Higher-gauge continuation slot. The genuinely noncentral crossed-module gluing data on cuts provide the natural continuation-level slot for B -field / gerbe data in any string-style reorganization of the OPH edge sector. This is a structural analogy only; it is not an open/closed-string boundary formalism or a critical-string theorem.

9 Cross-Lane Theorem Boundary and Continuation Directions

The paper corpus has the following theorem boundary:

1. The fixed-cutoff collar, higher-gauge, consensus, record, Bell/CHSH, checkpoint, and restoration packages are theorem-bearing on their stated finite-regulator surfaces.
2. Physical UV uniqueness is quotient-level and stable under inert ancillas. OPH fixes terminal values on declared physical observables, not a unique microscopic representative.
3. The Lorentz/null-modular/Einstein chain is closed on the support-visible refinement/scaling branch by the finite cap-normal BW scaling theorem on the prime geometric subnet: fixed-collar Markov replacement with carried r_{FR} , δ^{M} , and regularized modular remainders, support-readable modular covariance, support-order faithfulness, BW framing, held-out oriented cross-ratio rigidity, and independently normalized geometric 2π -KMS convergence make the cap modular automorphism geometric. The compact cap-normal theorem then identifies S^2 with projective future null rays $q(\Omega) = (1, \Omega)$, represents oriented round caps by $n_C = (\cot \alpha, \csc \alpha \mathbf{c})$, proves the signed incidence formula, and obtains $n_{gC} = \Lambda_g n_C$ and $H^3 \simeq \text{SO}^+(3, 1)/\text{SO}(3)$. This is not a finite-cell Lorentz-invariance claim, a one-shot upgrade from small CMI to exact Markov geometry, a proof that bare finite consensus produces the cap-normal certificate, or a populated/neutral-bulk claim.

4. Record-populated observer-facing H^3 is a separate conditional theorem: calibrated record-conditioned modular cap responses localize a quotient-visible token only when they factor through one point-local H^3 source on a compact domain and the finite cap frame has $\alpha > 0$. Exact data identify one point; finite noisy data produce a localization ball, and a unique finite point requires a positive residual gap Δ_{loc} . This does not derive particle species, stress-energy, or chart-blind neutral bulk.
5. The cosmological constant branch uses the cosmic record-closure fixed point $N_{\text{CRC}} = F(N_{\text{CRC}})$ and fixes the dimensionless relation $\Lambda_{\text{CRC}} \ell_\star^2 = 3\pi/N_{\text{CRC}}$ on the same Einstein branch. Once the selected scale certificate supplies $G_{\text{geom}} = \ell_\star^2$, this is the familiar display $\Lambda_{\text{CRC}} = 3\pi/(G_{\text{geom}} N_{\text{CRC}})$. Its count-density representation is $N_\star = \text{MAR} \arg \max_N (\log |\Omega_N^{\text{sc}}| - N)$. On the observed branch this readback fixed point is the de Sitter entropy capacity.
6. Compact gauge reconstruction uses the theorem-produced combined transportability criterion and fixed-cutoff tensor-generated bosonic sector category. On a cofinal tail carrying the explicit compact-gauge refinement receipt, the bosonic refinement ladder and compatible forgetful fibers reconstruct a compact group from sectors with strictifiable central or higher-associator defects and at least one allowed trivial-holonomy strict representative. That is the classification stage. The receipt-certified MAR-admissible witness data plus the explicit one-Higgs matter package then select the realized Standard Model quotient, exact hypercharge, $N_c = 3$, and $N_g = 3$.
7. The particle branch is sector-split: electroweak W/Z is a validation pair, Higgs/top is exact on its declared quantitative surface, selected-class quarks close on their declared public frame class, the weighted-cycle neutrino candidate is rejected and has no theorem surface, the physical strong-CP invariant is not emitted by this surface, charged-lepton source landing from P to physical charged data is a corpus-limited no-go boundary, the auxiliary direct-top PDG row is compare-only, and source-only hadron masses require a production OPH backend.
8. The string/worldsheet branch is a controlled heat-kernel continuation branch. The critical-string selector is a separate gate on top of that edge-language branch.

Local unification boundary. At the synthesis level, one local unification surface is explicit and one uniform theorem tier for c , G , W , Z , and H is absent. The declared local input P governs the electroweak/Higgs trunk and fixes the matching between cell area and edge entropy. The gravity normalization is emitted by the selected scale certificate, while the invariant causal speed c is structural from the Lorentz branch and gains its SI label only on the local readout package.

Quantity	Best value on the paper surface	Common paper-surface chain	Caveat
c_*	299792458 m/s by SI definition	structural Lorentz/BW branch $\rightarrow$ common invariant null cone/speed $\rightarrow$ SI unit convention	the common cone and dimensionless speed equality are structural; the decimal magnitude is not predicted
G	$6.674299995910528 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ on the stated local extension surface	$\gamma_* = \ell_* \nu_{\text{Cs}}/c$, $B_* = 3\pi/\ell_*^2$, $a_{\text{cell}} = P\ell_*^2$, $\ell_{\text{shared}} = P/4$, $G_{\text{SI}} = c^3\ell_*^2/\hbar$	the pixel fixes the shared cell/edge identity and cancels in the Newton area law; the strict classical-regime clause is explicit
M_W	target-free continuation row 80.37700001539531 GeV; exact compare-only sidecar 80.377 GeV	$P \rightarrow \alpha_U(P) \rightarrow$ $(t_U(P), t_{\text{tr}}(P)) \rightarrow v(P) \rightarrow$ target-free electroweak repair chart $\rightarrow M_W$	compare-only validation sidecar; public validation records are the certified P root, source spectral measure payload, same-scheme remainder, and interval certificate
M_Z	target-free continuation row 91.18797807794321 GeV; exact compare-only sidecar 91.18797809193725 GeV	same target-free electroweak trunk	same compare-only validation caveat
M_H	exact D11 split-theorem pair $M_H = 125.1995304097179 \text{ GeV}$, $m_t = 172.352353288312 \text{ GeV}$	same P -driven electroweak core $\rightarrow$ D10 repair chart $\rightarrow$ ρ_{HT} and source-only split selectors $\rightarrow$ D11 Jacobian	Higgs row is exact on the declared running, matching, and threshold surface; the inverse sidecar is compare-only; the auxiliary direct-top PDG row is a compare-only codomain with a corpus-limited no-go boundary

The exact-release package on that surface is local and finite: the strict classical-regime clause and the target-free electroweak chart identity for W/Z . The familiar-unit display split on that same surface is

$$L_{\text{loc}} = \sqrt{a_{\text{cell}}} \hat{L}(P), \quad t_{\text{loc}} = \frac{\sqrt{a_{\text{cell}}}}{c} \hat{T}(P),$$

$$E_{\text{loc}} = \frac{\hbar c}{\sqrt{a_{\text{cell}}}} \hat{E}(P), \quad \Theta_{\text{loc}} = \frac{\hbar c}{k_B \sqrt{a_{\text{cell}}}} \hat{\Theta}(P),$$

with dimensionless $\hat{L}, \hat{T}, \hat{E}, \hat{\Theta}$. The gravity-side shared-edge identity fixes the cancellation of P , while the local SI readout comes from ℓ_*^2 . Meters, seconds, GeV, and Kelvin are downstream display conventions built from the structural c row and the familiar constants $\hbar, k_B$.

Companion papers. Detailed derivations appear in Refs. [2, 3, 4, 5, 8, 9].

10 Conclusion

This paper presents the full framework on one synthesis surface. The five-axiom basis, the derived gravity and gauge branches, the consensus formulation, the particle-spectrum branch, the regulated screen-microphysics lane, the fixed-cutoff measurement and observer package, and the controlled large- N_{edge} worldsheet effective-description branch appear on one common synthesis surface.

The corpus is uneven but concrete. It has a sharp fixed-cutoff local picture, a derived recovered core for relativity and Standard-Model structure, real particle-side quantitative outputs, and a usable microscopic engineering lane. The scope boundaries are equally explicit: the E0 bounded-interval kernel in the null-stress branch, the cosmic record-capacity fixed point for the global branch constant, microscopic representative uniqueness, public validation artifacts for the D10 electroweak rows, strong CP, full flavor/CKM closure, charged-lepton source landing from P_* , and source-only hadron masses without a production OPH backend. Refs. [2, 3, 4, 5] are the depth surfaces for sector-by-sector proofs, quantitative details, and continuation-level material. The local cross-lane numerical surface is explicit enough to summarize concisely: the fixed point P_* organizes the bosonic mass trunk and the gravity-side shared edge-entropy identity beneath the emitted local G

row, while the same familiar-unit package reads meters and seconds from the single ruler $\sqrt{a_{\text{cell}}}$ together with the structural invariant speed c , and reads GeV and Kelvin from the inverse local ruler through $\hbar$ and k_B .

A Candidate Microphysics Reference Architecture

The synthesis paper keeps the observer-facing theorem package in the main text, but the concrete federated patch-carrier architecture belongs in an appendix. The theorem-bearing fixed-cutoff source surface for the edge heat-kernel / Casimir leaf is the microphysics companion paper; the appendix records only the synthesis-level restatement imported from that surface. The reference architecture used here is a federation of finite observer patches with echosahedral multi-port interfaces, boundary-fixed patch algebras, boundary-fixed overlap algebras on collars, declared shared readout packets, local repair instruments, and record registers exposed on the observer-facing interface. A spherical screen is a regulator chart for support-visible cuts, not a literal substrate claim.

Theorem (Regulated patch-net embedding). On a fixed finite patch federation and declared finite overlap cover, the regulated microphysics construction realizes an explicit finite patch-net layer: patch vertices are genuine finite observer patches, overlap edges are genuine collars or port interfaces, packet fields are actual readouts of declared overlap observables, and allowed repair branches are concrete local channels on the same finite regulator.

Proof sketch. All patch and overlap algebras are finite-dimensional at fixed cutoff, and the declared packet fields are readouts of explicit overlap observables in those algebras. The candidate repair menu is implemented by local finite-support channels on the same neighborhoods. Therefore the abstract finite patch-net syntax is embedded into an explicit regulator object rather than introduced as a separate formal layer.

Theorem (Fixed-cutoff edge heat-kernel branch). On the thermalized overlap branch of the same regulated architecture, the stationary sector law is

$$\pi_\beta(\alpha) \propto d_\alpha e^{-\beta C_2(\alpha)},$$

and along refinement ladders converging to a compact-group Peter–Weyl decomposition [45] the same weights converge cylinderwise to the heat-kernel/Casimir law used in the OPH edge-sector branch.

Proof sketch. The regulated proposal kernel is chosen so that the weighted proposal symmetry $d_\alpha q_{\alpha\beta} = d_\beta q_{\beta\alpha}$ holds. Detailed balance with the Casimir acceptance rule then fixes the stationary distribution in finite dimension, and the compact-group heat-kernel law is the corresponding refinement lift.

Extended derivation. The full fixed-cutoff architecture, validation package, and theorem stacks are developed in *Federated Echosahedral Screen Microphysics* [4].

B Interpretive Epilogue: State-and-Law Habitat

This appendix records the OPH state-and-law habitat available inside the framework. A strange-loop closure theorem is outside this appendix.

The OPH inputs used here are:

1. the patch net $P \mapsto A(P)$ of von Neumann algebras together with isotony and overlap restriction maps;
2. the global state on the inductive-limit algebra, whose restrictions witness a nonempty overlap-consistent local sector;
3. the finite Axiom-3 MaxEnt/refinement branch, which supplies a common finite family of gauge-invariant local constraint observables across cutoffs;
4. on the fixed-cutoff branch, the derived finite type-I presentations and compact boundary gluing groups used elsewhere in the OPH package.

The law slot is encoded by the finite expectation-value coordinates of the retained Axiom-3 constraint family. That choice furnishes a compact convex state-and-law habitat for the framework.

Theorem (Internal state-and-law habitat theorem). Fix finitely many screen patches $P_1, \dots, P_N$ in the standard OPH setup. For each i , let

$$M_i := A(P_i), \quad M_{ij} := A(P_i \cap P_j),$$

where M_i is the patch von Neumann algebra and $M_{ij} \subset M_i, M_j$ is the overlap algebra. Let

$$S_i := S(M_i) \subset M_i^*$$

be the full state space of M_i , endowed with the weak* topology $\sigma(M_i^*, M_i)$.

Choose a finite family of self-adjoint gauge-invariant local constraint observables

$$\mathcal{O} = \{O_1, \dots, O_m\}$$

from the Axiom-3 MaxEnt branch, with each O_a supported in one of the chosen patches; write $i(a)$ for an index such that $O_a \in M_{i(a)}$.

Define the overlap-consistent state sector

$$X_{\text{ov}} := \left\{ (\omega_1, \dots, \omega_N) \in \prod_{i=1}^N S_i : \omega_i|_{M_{ij}} = \omega_j|_{M_{ij}} \text{ for all } i, j \right\}.$$

Define the OPH law-coordinate map

$$c : X_{\text{ov}} \rightarrow \mathbb{R}^m, \quad c(\omega_1, \dots, \omega_N) := (\omega_{i(1)}(O_1), \dots, \omega_{i(m)}(O_m)).$$

Let

$$X_{\mathcal{O}} := \left\{ (\omega_1, \dots, \omega_N, \ell) \in \left(\prod_{i=1}^N S_i \right) \times \mathbb{R}^m : (\omega_1, \dots, \omega_N) \in X_{\text{ov}}, \ell = c(\omega_1, \dots, \omega_N) \right\}.$$

Then:

1. the ambient product

$$E := \left(\prod_{i=1}^N M_i^* \right) \times \mathbb{R}^m$$

is a Banach space for any product norm;

2. $X_{\mathcal{O}}$ is nonempty and convex;
3. $X_{\mathcal{O}}$ is norm-closed in E ;
4. $X_{\mathcal{O}}$ is compact in the product topology

$$\tau := \left(\prod_{i=1}^N \sigma(M_i^*, M_i) \right) \times \text{Euclidean topology on } \mathbb{R}^m;$$

Proof. Each M_i is a von Neumann algebra in the OPH patch net, hence a Banach space, so each dual M_i^* is Banach. A finite product of Banach spaces is Banach, hence so is E .

For each i , the state space

$$S_i = \{\omega \in M_i^* : \omega \geq 0, \omega(1) = 1\}$$

is convex and weak* compact by Banach–Alaoglu, because it is a weak* closed subset of the dual unit ball. For every pair (i, j) , restriction along $M_{ij} \subset M_i$ and $M_{ij} \subset M_j$ defines affine weak* continuous maps

$$r_{ij} : S_i \rightarrow S(M_{ij}), \quad r_{ji} : S_j \rightarrow S(M_{ij}).$$

Hence X_{ov} is an intersection of affine equalizer sets, so it is convex and τ -closed inside $\prod_i S_i$. Since $\prod_i S_i$ is compact and convex in the product weak* topology, X_{ov} is compact and convex as well.

Nonemptiness comes from the OPH global state on the inductive-limit algebra: its restrictions

$$(\omega|_{M_1}, \dots, \omega|_{M_N})$$

belong to X_{ov} because the restrictions agree on every overlap by construction.

Each coordinate $\omega_{i(a)} \mapsto \omega_{i(a)}(O_a)$ is affine and weak* continuous, since evaluation at a fixed algebra element is weak* continuous. Thus c is affine and τ -continuous. The graph map

$$G : X_{\text{ov}} \rightarrow E, \quad G(x) := (x, c(x)),$$

is affine and continuous, so its image $X_{\mathcal{O}} = G(X_{\text{ov}})$ is convex and τ -compact.

To see norm-closedness in E , let $(x_n, c(x_n)) \rightarrow (x, \ell)$ in norm. Norm convergence implies weak* convergence on each coordinate, so $x \in X_{\text{ov}}$ because X_{ov} is weak* closed. Since c is weak* continuous, $c(x_n) \rightarrow c(x)$, while norm convergence in $\mathbb{R}^m$ also gives $c(x_n) \rightarrow \ell$. Hence $\ell = c(x)$, so $(x, \ell) \in X_{\mathcal{O}}$.

Finally, $X_{\mathcal{O}}$ is a compact convex subset of the locally convex topological vector space E equipped with τ . $\square$

The additional inputs needed for a strange-loop closure theorem are:

1. a closure map T built from internal operations such as overlap repair, Axiom-3 MaxEnt reprojection, collar recoverability, and record-sector coarse-graining;
2. a proof that the stronger “observer-supporting” or “selected-world” criteria carve out a nonempty T -invariant observer-supporting subset of $X_{\mathcal{O}}$;

3. any uniqueness, contraction, or Lyapunov-type stability estimate for that closure map.

The habitat theorem supplies only the ambient Banach/compact-convex setting. It does not define a canonical internal self-map, it does not identify any nonempty invariant observer-supporting subset, and it does not prove uniqueness or stability. In particular, nonemptiness and compact-convexity of the ambient habitat do not by themselves imply the existence of a nonempty invariant observer-supporting sector. This does not erase the exact finite packet branch proved on the consensus surface: there the quotient normal-form map pushes forward to a continuous affine idempotent self-map of the finite packet simplex, with fixed points exactly the packets supported on quotient normal forms. Any further strange-loop reading of the ambient habitat as a closure story for existence is interpretive only and is not part of the recovered theorem package.

B.1 Additional Problem Closures

Using the phase split of Section 4.8.1, the broader closure claims read as follows. The table also records appendix-tier closure stories so they cannot float outside the official claim ledger. In this table, “Proved corollary of earlier theorems” means proved under the dependencies named in the row, including declared branch conditions and cited companion-paper theorem surfaces.

Claim	Status	Boundary
Quantum gravity consistency	Proved corollary of earlier theorems	Lorentz and Jacobson-type Einstein branches follow from the support-visible BW scaling theorem together with the stated null-bridge and fixed-cap conditions.
UV completion / microscopic uniqueness	Separate scope boundary	Fixed-cutoff theorem packages exist, but a unique microscopic representative is not identified. The support-visible BW scaling theorem and the receipt-certified realized combined-zero-obstruction bosonic compact-gauge branch (central or higher-associator strictification plus an allowed trivial-holonomy strict representative) are closed on their declared companion-paper theorem surfaces; the separate boundary is microscopic representative uniqueness rather than a missing recovered-core lift.
Measurement problem	Separate scope boundary	Ref. [4] closes the fixed-cutoff central-record / Born-Lüders interface, but a full philosophical or continuum-level measurement closure is not derived here.
Cosmological principle and horizon homogeneity	Separate scope boundary	MaxEnt symmetry inheritance is suggestive only; an independent derivation of cosmological isotropy and homogeneity sits outside the recovered-core theorems.
Observer-relative modular time	Proved corollary of earlier theorems	On the BW_{S^2} geometric branch, observer-relative time is read from the cap modular automorphism parameter of the observer's accessible algebra/state pair. This is the D3 reading only.
Global / operational problem of time	Separate scope boundary	OPH does not prove a separate operational-clock theorem or a global solution of the problem of time.
Black-hole information paradox	Separate scope boundary	Edge-center sectorization and recoverability motivate an internal resolution template, but the full black-hole-information branch sits outside Phase I.
D6 cosmological-capacity closure at the cosmic record fixed point	Proved corollary of earlier theorems	At the fixed point $N_{\text{CRC}} = F(N_{\text{CRC}})$, with observed-branch readout $N_{\text{CRC}} = S_{\text{dS}}$, the same Einstein branch fixes the dimensionless relation $\Lambda_{\text{CRC}} \ell_{\star}^2 = 3\pi/N_{\text{CRC}}$; the static-patch SI display additionally uses the selected OPH scale certificate, and $N_{\star}$ is the count-density representation of the same closure.
Bare cosmological-constant problem from local null data	Separate scope boundary	The local null-data route determines the Einstein branch only modulo Λg_{ab} and does not derive the screen-capacity identification from bare OPH axioms.
Magnetic monopole expectation from simple GUT groups	Proved corollary of earlier theorems	The realized product-group branch $SU(3) \times SU(2) \times U(1)/\mathbb{Z}_6$ removes the standard simple-GUT symmetry-breaking monopole channel.
Proton stability against gauge-mediated decay	Proved corollary of earlier theorems	The realized product-group branch removes the standard simple-GUT gauge-mediated proton-decay channel.
Proton spin fraction	Separate scope boundary	The proton-spin claim requires nonperturbative QCD completion beyond Phase I.
Dark matter phenomenology	Separate scope boundary	Modular-anomaly response laws are conjectural beyond the recovered gravity chain.
Baryon asymmetry scale	Separate scope boundary	Suppression-counting estimates are not a substitute for a derived out-of-equilibrium mechanism.
Three generations	Proved corollary of earlier theorems	$N_g = 3$ is recovered on the realized Minimal Admissible Realization branch.
Downstream flavor structure, Koide, and charged-lepton fits	Separate scope boundary	Companion flavor constructions require extra ansätze and sit outside the recovered core.
Edge-to-2D-YM / controlled large- N_{edge} worldsheet branch	Proved corollary of earlier theorems	On the stated overlap-gauge, local-Gibbs/MaxEnt, compact-group, and large- N_{edge} conditions, the edge-sector heat-kernel bridge and controlled Gross–Taylor worldsheet effective description are theorem-level. This is a two-dimensional partition and worldsheet-effective branch, separate from the four-dimensional compact-gauge theorem.
Support-visible compact-gauge Yang–Mills form and mass gap	Conditional theorem under explicit receipts	The finite compact-gauge cylinder system has a proof-bearing projective weak-* / GNS extraction. Identification with the four-dimensional Euclidean Yang–Mills state and the equality $\Delta_{\text{YM}} = \Delta_{\text{rep}}$ additionally require the renormalized Yang–Mills, finite ground-state-transform/cross-fiber, transfer/vacuum, OS-regularity/noncollapse, and uniform-gap receipts; extraction alone does not transport the gap.
Critical superstring / worldsheet CFT lift	Separate scope boundary	Critical-superstring claims, worldsheet CFT closure, anomaly cancellation, and full massless-spectrum matching require extra ingredients outside the recovered core.
Why anything exists / strange-loop closure	Separate scope boundary	Appendix B provides only the habitat theorem and fixed-point setting. The OPH closure map, invariant observer-supporting sector, and uniqueness/stability proofs are not part of this theorem package.
Fixed-cutoff check-	Proved corollary of earlier	Ref. [4] proves same-interface checkpoint/restoration and

C Observer Continuation and Backup

This appendix uses the fixed-cutoff checkpoint/restoration/error/identity theorem package proved in Ref. [4] and summarizes its algebraic interface in observer language. It does not define a strange-loop closure map on an invariant observer-supporting sector or establish uniqueness and stability for such a map. Nor does it upgrade the fixed-cutoff theorem stack to continuation across re-designed environments: the proved package is same-interface restoration on observer-accessible event algebras with explicit error control.

C.1 Observer as Algebraic Pattern

Here, an observer is represented by

$$O = (P, \mathcal{A}(P), \rho, R),$$

where P is a screen patch, $\mathcal{A}(P)$ is its local algebra, ρ is the local state, and R is the record algebra. Records are carried exactly by central record projectors and, on practical readout surfaces, by approximately commuting projectors in overlap centers, so they are shareable without violating no-cloning constraints for generic quantum states.

Ref. [4] proves a fixed-cutoff observer-facing measurement interface: a finite central record algebra, Born probabilities for its event projectors, and Lüders conditioning on that same commuting record algebra. This appendix uses that fixed-cutoff measurement package in observer language. If a practical readout uses projectors that are δ_{rec} -close in operator norm to that central reference algebra and the accessible state is perturbed by at most ε in trace norm, each declared elementary record-event probability shifts by at most $\varepsilon + \delta_{\text{rec}}$.

C.2 Markov Collar Factorization

For a collar tripartition A - B - D , the exact Markov normal form is used only when $I(A : D|B) = 0$ holds literally, or along a controlled fixed-collar family for which the exact-Markov replacement modulus $\delta_{A:B:D}^{\text{M}}(\varepsilon) \rightarrow 0$. In that exact or controlled limit one obtains

$$\rho_{ABD} = \bigoplus_{\alpha} p_{\alpha} \rho_{Ab_L^{\alpha}}^{(\alpha)} \otimes \rho_{b_R^{\alpha}D}^{(\alpha)}.$$

The sector label α is classical center data. This decomposition is the mathematical basis for extracting an interior observer state with a controlled boundary interface. Small CMI by itself supplies a recovered comparison state; it is not silently promoted to an exact Markov state.

C.3 Checkpoint and Restoration Map

A checkpoint is the tuple

$$\mathcal{C} = (R, \alpha, \rho_{\text{int}}^{(\alpha)}),$$

with $\rho_{\text{int}}^{(\alpha)}$ the interior state after fixing α . Given a compatible target environment state $\sigma_{\text{env}}^{(\alpha)}$, a restored state is

$$\rho_{\text{new}}^{(\alpha)} = \rho_{\text{int}}^{(\alpha)} \otimes \sigma_{\text{env}}^{(\alpha)}.$$

For approximate Markov collars, recovery is controlled by the standard bound

$$\|\rho_{ABD} - (\text{id}_A \otimes \mathcal{R}_{B \rightarrow BD})(\rho_{AB})\|_1 \leq 2\sqrt{\ln 2 \varepsilon}.$$

This gives quantitative trace-distance control on the recovered comparison state under finite CMI (finite collar error). The exact HJPW splice form instead requires literal exact Markovity or the separate fixed-collar replacement modulus tending to zero, in each case together with the Markov-split alignment hypothesis of Section 2.3 identifying the HJPW factors with the preselected edge factors. Interpreting either control as observer continuation requires additional modeling assumptions beyond the bound itself.

The fixed-cutoff microphysics claim is stronger than the bare recoverability estimate written above. In Ref. [4], exact checkpoint restoration preserves the full future law of observer-accessible events, while an ε -accurate restoration changes that future law by at most ε in total variation. The stronger strange-loop / substrate-transfer closure story is outside this theorem.

C.4 Physical Meaning

At fixed cutoff, Ref. [4] proves a checkpoint/restoration theorem and backup corollary for observer-accessible event algebras. The proved package stops at same-interface backup/restoration with explicit future-law error control; stronger continuation and substrate-selection claims are outside it. This appendix states the observer-facing meaning of that theorem and its algebraic prerequisites. The stronger substrate-selection, strange-loop closure, uniqueness, and stability package is separate from the fixed-cutoff theorem.

D Cosmology, Horizons, and Modular-Anomaly Continuations

This appendix carries the cosmology- and continuation-facing technical statements for the synthesis paper. They do not enlarge the recovered core; they make the structural continuation boundary explicit on the synthesis surface.

D.1 Observer Registry and Clock-Naturality Certificate Contract

For finite exported observer systems, a theorem-faithful evidence bundle separates executor metadata from observer semantics. It contains:

- (i) a semantic history DAG or labeled poset $H_O = (E_O, \preceq_O, \ell_O)$, with event keys computed from semantic payload, observer token, visible footprint, and semantic parents;
- (ii) a provenance envelope containing worker ID, repair iteration, retry count, queue position, timestamp, and packet latency, none of which may enter the semantic event key unless declared as physical input;
- (iii) local observer registry groupoids with disjoint patch-observer, cap-observer, and future-observer namespaces, explicit birth-event classes, continuation arrows, split/merge lineage arrows, and overlap functors preserving those data;
- (iv) a global registry descent certificate: cocycle agreement, trivial monodromy, no duplicate global observer identifiers, no mixed patch/cap/future namespace collisions, and no reuse of local anchor-patch identifiers as local observer indices;
- (v) observer-algebra extraction data

$$\text{ObsAlg}(\widehat{q}_{\text{nf}}) = (\mathcal{A}_O, \omega_O, \mathcal{R}_O, \dots)$$

whose outputs are state-preserving isomorphic for history-equivalent or implementation-equivalent terminal augmented normal forms;

- (vi) a support-cap chart map J_O natural with those observer-algebra isomorphisms; and
- (vii) a clock instrument Θ_O with affine reparameterization and residual bound for state, calibration, and record-readout errors.

Theorem D.1 (Finite observer-clock certificate soundness). *On a finite exported model, suppose the augmented repair relation on $\hat{Q} = \{(q, [H])\}$ terminates, terminal states are exactly the consistent physical states with completed semantic records, every augmented critical pair has a history-coherent join, implementation steps are linearizable to semantic commits or stutters, registry descent passes the groupoid cocycle and monodromy checks, and the observer-algebra/chart/clock maps satisfy the state-preserving and affine-residual certificates above. Then worker count, queue order, retry delivery, scheduler counters, and packet latency that is not declared physical input do not change the observer-readable history class, registry class, or clock law beyond the declared affine clock residual.*

Proof. Termination plus the augmented local diamond gives confluence by Newman’s lemma for terminal history classes rather than only physical quotient states. Linearizability says every implementation execution projects to the same semantic relation up to stutters, so executor metadata cannot create new semantic events. Registry descent identifies the same global observer groupoid across local presentations. State-preserving observer-algebra extraction and natural support-cap charts transport the modular flow along the same observer identity, while the clock instrument certificate bounds the operational readout after the declared affine reparameterization. $\square$

Remark D.2 (Falsifiers). *The certificate fails if semantic event identifiers depend on worker IDs, timestamps, queue positions, repair iteration counters, or retries; if duplicate retry delivery creates two visible record commits; if two namespaces collide in the global registry; if registry cocycles or lineage arrows fail on overlaps; if histories from two executors are not labeled-poset isomorphic; or if clock residuals exceed the declared bound. These are public evidence-bundle failures that exceed field-name mismatches.*

D.2 Cosmological Principle

Lemma 4.1 (MaxEnt symmetry inheritance). Let G act on the screen by automorphisms α_g . If the constraint set is G -invariant and von Neumann entropy is G -invariant, then the MaxEnt optimizer ω can be taken G -invariant. If the optimizer is unique, it is automatically G -invariant.

Theorem 4.2 (Isotropy from $\text{SO}(3)$ -invariant constraints). Under $\text{SO}(3)$ -invariant constraints and MaxEnt uniqueness, the reference state satisfies

$$\omega \circ \alpha_g = \omega \quad \forall g \in \text{SO}(3).$$

The corresponding stress tensor has perfect-fluid form:

$$\langle T_{ab} \rangle = \rho u_a u_b + p(g_{ab} + u_a u_b).$$

Theorem 4.3 (Schur-type homogeneity). Let (Σ, h_{ij}) be a three-dimensional Riemannian manifold. If at every point $p \in \Sigma$, the curvature tensor is $\text{SO}(3)$ -invariant, then

$$R_{ijkl}(p) = K(p)(h_{ik}h_{jl} - h_{il}h_{jk}),$$

and the second Bianchi identity forces $\nabla_m K = 0$, so K is constant.

Corollary 4.4. If OPH supplies isotropy for all observers, spatial geometry is a constant-curvature space form: S^3 , $\mathbb{R}^3$, or H^3 .

Theorem 4.5 (FLRW emergence). With the semiclassical Einstein equation from recoverable generalized entropy / entanglement equilibrium, a perfect-fluid stress tensor from $\text{SO}(3)$ -invariant MaxEnt, and positive Λ from finite screen capacity, the emergent geometry is FLRW:

$$ds^2 = -dt^2 + a(t)^2 \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right),$$

with Friedmann equations

$$H^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3}, \quad \dot{H} - \frac{k}{a^2} = -4\pi G(\rho + p).$$

D.3 Horizon-Problem Control

Theorem 6.1 (Anisotropy bound from Markov control). For a tripartition A - B - D with collar width δ , the Markov error satisfies

$$I(A : D \mid B) \lesssim c \cdot |\partial C|_{\text{UV}} \cdot e^{-\delta/\xi},$$

where ξ is the correlation length.

Corollary 6.2. For bounded observable O with $\|O\| \sim 1$,

$$|\Delta\langle O \rangle| \leq 2\sqrt{\ln 2 \cdot \varepsilon}.$$

On the CMB anisotropy target $\delta T/T \lesssim 10^{-5}$, this gives the familiar ε -scale bound used in the horizon-problem bookkeeping:

$$\varepsilon \lesssim 3.61 \times 10^{-11} \text{ bits}, \quad \frac{\delta}{\xi} \gtrsim \ln \left(\frac{c \cdot |\partial C|_{\text{UV}}}{\varepsilon_{\text{max}}} \right) \approx 24.$$

With $\xi \approx \sqrt{P} \ell_\star \approx 1.28 \ell_\star$, where $\ell_\star$ is displayed as the usual Planck length after the selected scale certificate, the corresponding collar benchmark is

$$\delta_{\text{CMB}} \approx 31 \ell_\star.$$

Theorem 6.3 (Homogeneity as the MaxEnt default). If MaxEnt constraints are uniform across UV cells with no marked points, the MaxEnt state is invariant under triangulation automorphisms. In the continuum limit this gives $\text{SO}(3)$ invariance.

Reading rule. The cosmological-principle and horizon-homogeneity package is a branch-local theorem package on the realized symmetric MaxEnt branch. The exact theorem-level content is:

1. $\text{SO}(3)$ -invariant constraint data plus MaxEnt uniqueness force an isotropic reference state and a perfect-fluid stress tensor.
2. If the same isotropy condition holds for all observers, the Schur/Bianchi argument forces constant-curvature spatial slices.

3. Combined with the semiclassical Einstein branch and positive Λ from finite screen capacity, the metric takes FLRW form.
4. Markov control supplies an explicit patch-overlap anisotropy bound and the collar benchmark used in the CMB homogeneity bookkeeping.

The package does not prove from the bare OPH axioms alone that the realized cosmological branch must satisfy $\text{SO}(3)$ -invariant constraint data, that MaxEnt uniqueness holds in every cosmological sector, or that the observed universe saturates the displayed CMB benchmark.

D.4 Conditional Screen-Spectrum Continuation

Theorem 6.4 (Conditional OPH screen spectrum). Let $(\mathcal{S}_r)$ be a cofinal finite spherical OPH screen system with a schedule-independent quotient-normal-form scalar q_r , removal of the background and dipole sector, and a target-free positive quadratic repair operator K_r . If $K_r \rightarrow K$, the finite scalar release energy satisfies $E_r/d_r \rightarrow A_q$, and local MaxEnt is imposed at fixed expected quadratic release energy, then q_r converges in finite harmonic distributions to the centered Gaussian screen field with covariance $A_q K^{-1}$. Exact per-sample release energy would instead give a microcanonical ellipsoid. If the certified repair-scale measure is $t^{\theta/2} dt/\Gamma(1 + \theta/2)$, then

$$K_{\text{asy}} = (-\Delta_{S^2})^{1+\theta/2}, \quad C_{\ell, \text{asy}}^q = A_q [\ell(\ell + 1)]^{-1-\theta/2},$$

as the large- ℓ asymptotic model. The theorem-grade finite- ℓ conformal-shell family uses the normalized gamma-ratio precision

$$\kappa_\ell(\theta) = \frac{\Gamma(\ell + 2 + \theta/2)}{\Gamma(\ell - \theta/2)},$$

with

$$C_\ell^q = A_q \frac{\Gamma(\ell - \theta/2)}{\Gamma(\ell + 2 + \theta/2)}.$$

If q is additionally certified as the thin-shell pullback of a homogeneous curvature field with $\Delta_\zeta^2(k) = A_\zeta(k/k_\star)^{-\theta}$, this same operator supplies the exact finite- ℓ lift; the pure fractional Laplacian is only the asymptotic scaffold.

Receipt boundary. The screen theorem is a Phase III conditional continuation. It requires independent receipts for the quotient-normal-form scalar readout, the positive repair operator, the operator tilt, and the fixed expected quadratic scalar release energy. Fawzi–Renner/Markov-collar observable control supplies upper bounds on release observables; it is not an amplitude equality and does not by itself determine A_ζ . A primordial curvature spectrum additionally needs a screen-to-radial lift certificate with the exact Bessel kernel, radial prior/null-space report, and forward-projection residual. No physical TT/TE/EE claim follows without independent Boltzmann transfer and likelihood closure.

D.5 Cosmological-Constant / Screen-Capacity Closure

Theorem 6.5 (D5–D6 local/global closure stack). On the gravity lane, assume the local Einstein branch has been recovered only modulo Λ_{gab} , the cosmic record-capacity fixed point

$$N_{\text{CRC}} = F(N_{\text{CRC}}),$$

its observed-branch de Sitter entropy readout

$$N_{\text{CRC}} = S_{\text{dS}},$$

the standard de Sitter entropy relation

$$S_{\text{dS}} = \frac{A_{\text{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

and the standard de Sitter static-patch formulas

$$r_{\text{dS}} = \sqrt{\frac{3}{\Lambda}}, \quad t_{\Lambda} = \frac{r_{\text{dS}}}{c}.$$

Then:

1. local null data determine the Einstein branch only modulo Λg_{ab} ;
2. with the selected scale certificate, the same branch has the global display

$$G_{ab} + \frac{3\pi}{GN_{\text{CRC}}} g_{ab} = 8\pi G \langle T_{ab} \rangle;$$

3. the same D6 closure fixes the entropy and dimensionless capacity relations

$$S_{\text{dS}} = N_{\text{CRC}}, \quad A_{\text{dS}} = 4GN_{\text{CRC}}, \quad r_{\text{dS}} = \sqrt{\frac{3}{\Lambda}}, \quad t_{\Lambda} = \frac{r_{\text{dS}}}{c};$$

4. the observed cosmic age is a downstream FLRW benchmark rather than an additional theorem output.

So the cosmological-constant package is one local/global theorem stack rather than a split local argument plus an unrelated global readout.

Scope boundary. The D6 hypotheses are the local Einstein branch, the cosmic record-capacity fixed point $N_{\text{CRC}} = F(N_{\text{CRC}})$, its observed-branch de Sitter entropy readout, the de Sitter entropy relation, and the standard static-patch formulas. The local null-data route does not by itself determine the global capacity; that value is fixed by the readback closure. The capacity closure fixes $\Lambda_{\text{CRC}} \ell_{\star}^2 = 3\pi/N_{\text{CRC}}$; the SI static-patch scale additionally requires the selected scale certificate.

Input-free capacity closure. The zero-input global capacity is the cosmic record-closure fixed point. For candidate capacity N , let $F(N)$ be the active horizon capacity read back by stable observers inside the OPH universe supplied with capacity N :

$$F(N) = \text{Cap}_{\text{read}}(\text{Obs}(\text{nf}(\mathfrak{U}_N))).$$

The target is

$$N_{\text{CRC}} = F(N_{\text{CRC}}), \quad \Lambda_{\text{CRC}} = \frac{3\pi}{GN_{\text{CRC}}}.$$

Equivalently, before the selected scale certificate is applied, the target fixes $\Lambda_{\text{CRC}} \ell_{\star}^2 = 3\pi/N_{\text{CRC}}$. The screen-normalized density of terminal self-closing observer normal forms is the count representation of the same target. Let Ω_N^{sc} be the terminal OPH normal forms that are repair-closed,

observer- and checkpoint-supporting, carry the recovered local package, and whose own horizon record surface reads back capacity N . Since $\log \dim \mathcal{H}_{\partial, N} = N$, define

$$\Pi(N) := \frac{|\Omega_N^{\text{sc}}|}{\dim \mathcal{H}_{\partial, N}} = |\Omega_N^{\text{sc}}| e^{-N}.$$

The corresponding Minimal Admissible Realization selector is

$$N_\star = \text{MAR} \arg \max_N [\log |\Omega_N^{\text{sc}}| - N].$$

Equivalently, with $\ell(N) = \log |\Omega_N^{\text{sc}}| - N$, the OPH-derived stationarity map

$$T_\eta(N) = N + \eta \ell'(N)$$

has a unique stable fixed point under the derivative-sign certificate stated in the synthesis section. Informally, this is the single balance point where the universe reads back its own boundary without deficit or slack. On the observed branch,

$$N_{\text{CRC}} = N_\star \simeq 3.31 \times 10^{122}.$$

This rounded display locates the capacity scale; precision rows use the fixed-point value together with the selected scale certificate.

Capacity normalization. The branch uses N_{scr} as the de Sitter entropy capacity. The bare horizon ratio is

$$N_{\text{patch}} = \left(\frac{r_{\text{dS}}}{\ell_P} \right)^2,$$

so

$$N_{\text{scr}} = \pi N_{\text{patch}} = \frac{3\pi}{\Lambda \ell_P^2}.$$

For the observed late-time scale, $N_{\text{patch}} \simeq 1.05 \times 10^{122}$ and $N_{\text{scr}} \simeq 3.31 \times 10^{122}$. These are benchmark displays, not high-precision inputs for the Newton row.

D.6 Black-Hole Structural Package and Continuation Boundary

The AMPS-style information trilemma assumes:

1. an outgoing mode B entangled with an interior partner A (smooth horizon),
2. late radiation B entangled with early radiation R (unitarity), and
3. monogamy of entanglement.

The false assumption in this setting is the naive tensor factorization

$$\mathcal{H} \stackrel{?}{=} \mathcal{H}_{\text{inside}} \otimes \mathcal{H}_{\text{outside}} \otimes \mathcal{H}_{\text{radiation}}.$$

Theorem 7.1 (Edge-center black-hole decomposition). On the fixed-cutoff edge-center collar branch, assume the black-hole collar state lies in the exact Markov class or in the idealized exact-recoverability limit used elsewhere in the paper stack. Then the collar decomposition gives

$$\rho_{A_\delta B_\delta D_\delta} = \bigoplus_\alpha p_\alpha \left(\rho_{A_\delta b_L^\alpha} \otimes \rho_{b_R^\alpha D_\delta} \right).$$

The glue between inside and outside is the edge-sector label α in the center. Given α , inside and outside factorize.

Theorem 7.2 (Hawking/KMS normalization). On the geometric modular branch where the horizon generator carries the standard 2π KMS normalization, the outside algebra is in a KMS state at inverse temperature

$$\beta = \frac{2\pi}{\kappa}$$

with respect to the horizon Killing generator.

Corollary 7.3 (Recoverability-style interior encoding). On the same fixed-cutoff collar carrier, when $I(A_\delta : D_\delta \mid B_\delta) \leq \varepsilon$, there exists a recovery channel $\mathcal{R}_{B_\delta \rightarrow A_\delta B_\delta}$ such that

$$\|\rho_{A_\delta B_\delta D_\delta} - (\mathcal{R}_{B_\delta \rightarrow A_\delta B_\delta} \otimes \text{id}_{D_\delta})(\rho_{B_\delta D_\delta})\|_1 \leq 2\sqrt{\ln 2 \cdot \varepsilon}.$$

So the interior collar data are encoded in the exterior collar together with the shared cut data, rather than supplied by an independent tensor factor.

Corollary 7.4 (Discrete area spectrum and Schwarzschild transition identity). The same edge-center package carries a discrete area operator

$$L_C = \sum_{\alpha} (\log d_{\alpha}) P_{\alpha}, \quad A_{\alpha} = 4G_{\text{geom}} \log d_{\alpha} = 4\ell_{\star}^2 \ln d_{\alpha},$$

and for a Schwarzschild sector transition $d \rightarrow d'$ one has

$$\Delta(Mc^2) = k_B T_H \ln(d'/d).$$

On the emission branch $d' = d/k$ with $k > 1$, the emitted line energy magnitude is

$$E_{\text{emit}} = k_B T_H \ln k.$$

Scope boundary. The retained structural theorem package is exactly Theorem 7.1, Theorem 7.2, Corollary 7.3, and Corollary 7.4. Integer-transition combs, QNM selectors, Page-type linewidth estimates, PBH burst templates, Kerr/LIGO horizon spectroscopy templates, and any Page-curve or island closure are outside that structural core. Those items require continuation inputs such as an integer-transition selection rule, a semiclassical evaporation-power model, a QNM/transition identification, greybody matching, or an explicit island prescription.

The same no-relabeling rule applies here. A finite capacity register is not an exterior mass, an archive record is not Hawking radiation, a finite repair spectrum is not a GR quasinormal-mode spectrum, and an exact finite recovery threshold is not a Page time unless the corresponding independent physical bridge supplies source-separated readout, calibration, residuals, controls, and frozen validation targets.

D.7 Modular-Anomaly Continuation

Theorem 9.1 (Modular additivity defect). Define the collar modular additivity defect by

$$\Delta K_{\delta} := K_{ABD} - K_{AB} - K_{BD} + K_B.$$

Then

$$\langle \Delta K_{\delta} \rangle_{\omega} = -I(A : D \mid B)_{\omega}.$$

This is the precise structural identity underneath the modular-anomaly dark-sector continuation. By itself it does not close a dark-matter theorem.

The corresponding continuation packages the anomalous contribution into the modified Einstein equation

$$G_{ab} + \Lambda g_{ab} = 8\pi G (\langle T_{ab} \rangle + \langle T_{ab}^{\text{anom}} \rangle),$$

with rest-frame normalization

$$\langle T_{00}^{\text{anom}} \rangle := \frac{15}{8\pi^2} \frac{\delta \langle K_C^{\text{anom}} \rangle}{\ell^4}.$$

On that continuation surface the anomaly gravitates, is dark by construction, is covariantly conserved, and is effectively classical because it is organized by central/recoverability data.

If one tentatively identifies the imported D6 static-patch radius r_{dS} as the unique IR scale available to the continuation, the natural benchmark acceleration is

$$a_0^{(\text{OPH})} := \frac{15}{8\pi^2} c^2 \sqrt{\frac{\Lambda}{3}} = \frac{15}{8\pi^2} \frac{c^2}{r_{\text{dS}}} \approx 1.03 \times 10^{-10} \text{ m/s}^2,$$

close to the empirical MOND scale. That numerical proximity is only a benchmark unless a separate source/response law and a controlled galaxy-scale limit are derived. An illustrative MOND/RAR-style response would read

$$g_{\text{obs}} \approx g_b + \sqrt{a_0 g_b},$$

but that functional form is not derived from the OPH axioms.

Reading rule. The anomaly term gravitates, is dark by construction, and is covariantly conserved, but the synthesis surface keeps the entire dark-sector package on the D12 continuation tier rather than promoting any MOND/RAR-style law, galaxy-rotation template, or cluster phenomenology to theorem claim tier. The imported D6 static-patch scale fixes only the benchmark normalization. The missing inputs are the controlled nonrelativistic limit, derived sign and closure control for the effective anomaly contribution, a derived source/response law, proof of the relevant dominant galaxy-scale source, lensing and cluster phenomenology, cosmological abundance and structure-formation checks, and environment/stability bounds. Those are continuation-level branches above the identity written here.

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Data Availability declaration: The paper stack, source manuscripts, public release PDFs, simulation code, and implementation notes are available through the OPH project repository and linked public release surfaces. Some hardware and prototype evidence requires a sanitized ancillary audit bundle before external release because raw logs may include device configuration, operational metadata, or private infrastructure details. The intended audit bundle contains source hashes, simulator configs, finite-receipt JSON, verifier code, hardware acceptance records, calibration logs, and selected run traces sufficient to replay the stated implementation claims.

Code Availability declaration: The OPH simulator, finite-consensus harnesses, visualization payload schemas, and public paper build sources are maintained in the OPH project repositories. Submission artifacts include a standalone source file and a source bundle so that the manuscript can be rebuilt without relying on private local paths.

Ethics declaration: not applicable.

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Lane	Output	Role	Support boundary
Carrier modes	two transverse Maxwell modes; 2 dim G perturbative Yang–Mills modes; two Einstein TT modes	conditional quadratic-action theorem	no zero-GeV particle row until the physical Hilbert-space, pole-residue, and phase gates pass
	80.377		
W/Z	91.18797809193725 GeV	weak-sector benchmark pair	benchmark scope is explicit
	$\alpha_{\text{root}}^{-1} = 136.9948351646\dots$		
Fine structure	$A_{\alpha U}^{\text{fp}} = 137.0359595008\dots$	endpoint provenance bookkeeping	source/root audit, mixed-provenance comparison diagnostic, and measured endpoint are separate coordinates;
	$\alpha^{-1}(0) = 137.035999177(21)$		source-only promotion requires the OPH-QCD backend
	125.1995304097179		
Higgs/top	172.35235532883115 GeV	declared electroweak quantitative relation	auxiliary direct-top PDG comparison is a benchmark check in the available derivation
	0.00051099895		
	0.1056583755		
Charged leptons	1.7769324651340912 GeV	empirically anchored same-family exact witness	source landing from P to physical charged data lacks a theorem-level uncentered trace lift in the available derivation
	$u = 0.00216, d = 0.00470$		
	$s = 0.0935, c = 1.273$		
Quarks	$b = 4.183$	selected-frame exact theorem	exact only on the selected public quark frame class; global public-frame classification has a no-go boundary in this paper; the physical strong-CP invariant is not derived here
	$t = 172.35235532883115$ GeV		
	$\sin^2 \theta_{23} = 0.582056$		
Neutrinos	$\delta_{\text{CP}} = -54.419^\circ$	rejected target-informed template candidate	NuFIT 6.1 gives $\Delta\chi^2 = 20.12$ with the tabulated atmospheric likelihood and 18.44 without it, above the two-parameter 3σ contour value 11.83; the shared-basis recovery was tautological, the charged basis is open, and no source-only PMNS or absolute-mass row is emitted [11]
	$\Delta m_{21}^2/\Delta m_{32}^2 = 0.03072111$		
Hadrons	no first-principles prediction emitted; empirical closure route separate	OPH-QCD backend absent; empirical closure policy stated	source-only rows require source QCD parameters, quotient ensemble, Ward current ledger, two-current/higher-point/transition spectral exports, same-scheme remainder, and no-target-leak